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Olivier Dubray

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**Design and characterization of transmitter
architectures using silicon ring resonator
modulators for high bit rate communications**

Devant le jury composé de :

Erasme, Didier Professeur
Poon, Joyce Professeur
Peucheret, Christophe Professeur
O'Connor, Ian Professeur
Menezo, Sylvie Docteur

Télécom ParisTech
University of Toronto
FOTON/ENSSAT
Institut Nanotechnologies de Lyon
CEA-Leti

Président
Rapporteur
Rapporteur
Directeur de thèse
Encadrant de thèse

Abstract :

Over the past decade, with the diversification of connected devices (PCs, Tablets, TVs and Smartphones), the Internet ecosystem has drastically extended. Today, 80 % world traffic is concentrated in the data centers where the data rate, the size and the cost is still growing. To address such scaling issues as bandwidth density, energy consumption and cost of the interconnects inside the data centers, the development of new optical transmitters is critical.

The aim of this thesis is to propose and evaluate transmitter architectures using silicon photonics technology to address next 400 Gbit/s data rate standard over up to 2 kilometer links. The selected electro-optical modulator is the silicon ring resonator modulator which has substantial benefits: low footprint, low energy consumption and enables dense multiplexing.

The optical transmitter architectures evaluations were successively optimized: from the active junction to the complete optical transmitter. This study identified the performances trade-offs impacted by the ring resonator modulator parameters. A compact model was generated to physically optimize the component in a reduced simulation time.

Then, using the compact model, two transmitter architectures were studied based on classical architecture. Both are based on eight ring resonator modulators arranged in series modulating eight different wavelengths. The difference is the modulation format: the first one is electrically modulated at 50 Gbaud in 2-levels pulse amplitude modulation (PAM-2) and the second one at 25 Gbaud 4-levels pulse amplitude modulation (PAM-4). The two solutions fit the 400 Gbit/s performances demand with the use of the same trade-offs.

Finally, new transmitter architectures were proposed to generate PAM-4 modulation. Unlike the previous architecture, they have in input two parallel bit streams which are optically combined to generate the PAM-4 modulation. The first solution is based on two silicon ring resonator modulator arranged in series. This architecture was validated through 30 Gbit/s transmission characterizations with only $1 V_{pp}$. A second solution was then proposed, based on two silicon ring resonator modulators arranged in parallel in a Mach Zehnder interferometer. In the same way, transmission characterizations at 30 Gbit/s with $1.2 V_{pp}$ allows this architecture to be validated.

Keywords :

Optical Transmission, 400 Gbit/s, Silicon Photonics, Pulse Amplitude Modulation, Wavelength Dense Multiplexing, Ring Resonator Modulator

Résumé :

Depuis une dizaine d'années, avec la diversification des appareils connectés (PCs, Tablettes, TVs et Smartphones), l'écosystème Internet s'est drastiquement étendu. Aujourd'hui, 80 % du trafic IP mondial est concentré dans les centres de données. Pour répondre aux problématiques d'échelle des centres de données en terme de densité de bande passante, de consommation énergétique et de coût des interconnexions, le développement de nouveaux émetteurs optiques est critique.

L'objectif de cette thèse est de proposer et évaluer différentes architectures d'émetteur en photonique sur silicium afin de répondre à la prochaine norme de débit de 400 Gbit/s sur des transmissions de 2 kilomètres. Le modulateur électro-optique sélectionné est le modulateur silicium en anneau résonant. Il possède des avantages non négligeables: faibles dimensions, faible consommation énergétique, et il permet un multiplexage dense en longueurs d'ondes.

Durant cette thèse, l'optimisation de l'émetteur optique a été faite de manière progressive: de la jonction active à l'émetteur complet. Cette première étude a identifié les différents compromis à faire sur les paramètres du modulateur en anneau afin de définir ses performances. Un model compact du modulateur a été créé, permettant d'optimiser le composant avec un temps de simulation très réduit.

Puis, tout en utilisant le modèle compact, deux architectures d'émetteur ont été étudiées basées sur une architecture classique de la littérature. Elles sont basées sur la mise en série de 8 modulateurs en anneau, chacun modulant une longueur d'onde différente. La différence entre les deux architectures vient du format de modulation utilisé: la première est modulée avec des signaux sur deux niveaux électriques avec des signaux sur 2 niveaux électriques au format d'impulsions en amplitude (PAM-2) à 50 Gbaud ; alors que la deuxième est modulée avec des signaux sur 4 niveaux électriques au format d'impulsions en amplitude (PAM-4). Les deux solutions répondent aux demandes de performances de la norme 400 Gbit/s avec les mêmes points de compromis de fonctionnement.

Finalement, de nouvelles architectures d'émetteur optique ont été proposées permettant de réaliser la modulation PAM-4. Contrairement aux solutions précédemment étudiées, ces architectures utilisent en entrée deux flux de bits en parallèle qui sont optiquement combinés pour générer en sortie une modulation PAM-4. Une première solution est basée sur la mise en série de deux modulateurs en anneaux. Cette architecture a été validée avec des caractérisations de transmission optique à 30 Gbit/s avec seulement $1 V_{pp}$ de tension de commande. Une seconde solution a ensuite été proposée, elle est basée sur l'utilisation de deux modulateurs mis en parallèle dans un interféromètre Mach Zehnder. De la même manière, des caractérisations de transmission à 30 Gbit/s avec $1.2 V_{pp}$ de tension de commande ont permis de valider le fonctionnement de l'architecture.

Mots-clés :

Transmission optique, 400 Gbit/s, Photonique sur silicium, Modulation d'impulsions en amplitude, Multiplexage en longueur d'ondes, Modulateur en anneau résonant

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Chapter 1

Context and motivation

Over the past decade, with the diversification of connected devices (PCs, Tablets, TVs, Smartphones), the Internet ecosystem has drastically extended. The number of connections is growing faster than Internet users. The connection demand is increasing with new Internet applications like web storage, file sharing or Video-On-Demand (figure 1.1) [1]. Additionally, more and more computing applications are moving into the cloud. Cloud services may include data storage and software as a service - where applications are not run locally by the user, but in data-centers - and inputs/outputs are transferred over the networks. This requires a very high data rate for file-transfer between users and data-centers, between data-centers, but also within the data-centers. According to [2], global data center IP traffic will nearly triple between 2015 and 2020, with 77% of the traffic concentrated in data centers themselves. In this context, high data rate communications are required to meet the demand, and optical communication techniques might be the answer.

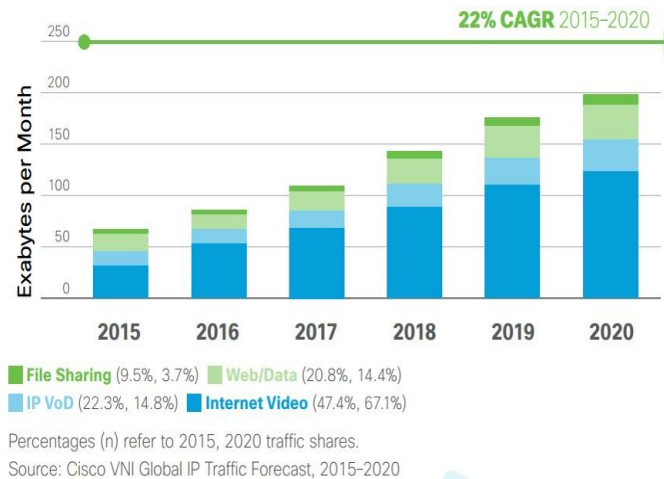


FIGURE 1.1: Cisco forecast for global IP traffic growth from 2015 to 2020 from [1]

This thesis is focused on optical communication segment for data-centers. These optical communications are point to point communications ranging between 500 meters and 10 kilometers at high data rate. At the beginning of this thesis, optical communication standards for 100 Gbit/s bit rate were defined by different groups like IEEE 802.3 group and Multi Source Agreement group (MSA). Then, during the thesis, they have started to study the next optical standards for 400 Gbit/s bit rate [3].

In the following, we will first describe the current state of optical communications for data centers at 100 Gbit/s data rate. We will then detail the evolution pathways to the 400 Gbit/s data rate.

1.1 100 Gbit/s optical communications

For achieving high data rate, optical communication standards play with mostly four main optical characteristics, which are :

- The number of symbols per second
- The number of bits per symbol
- The number of optical fiber pairs
- The number of wavelengths (wavelength division multiplexing (WDM))

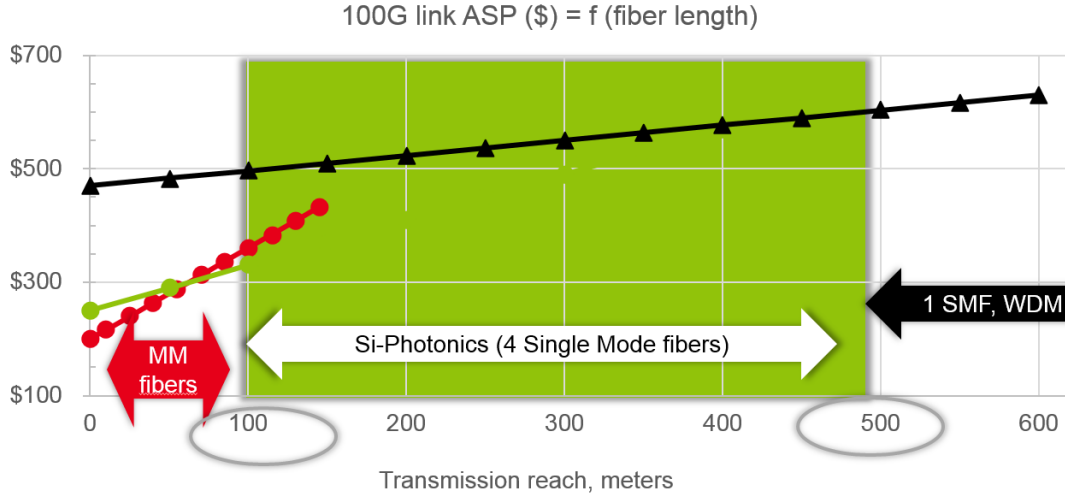
As the communication distance is small, for data center communications, optical transceivers with direct detection are preferred to high cost coherent transceivers and no optical amplifier are implemented in the optical transmission link.

Table 1.1 reports the main standards for 100 Gbit/s optical communications. All standards reported are characterized by one bit per symbol (binary pulse amplitude modulation PAM-2) [4–6]. They are divided into three segments depending on the reach. For small reach, up to 500 m, the data rate is achieved by increasing the number of optical fibers, multi mode fiber (MMF) or single mode fiber (SMF). Then, for higher reach, the data rate is achieved by multiplexing 4 or 10 different wavelengths.

TABLE 1.1: IEEE and MSA ethernet nomenclature and specifications for different optical transmission lengths. *MSA standards, Multi mode fiber (MMF), Single mode fiber (SMF)

Nomenclature	Data rate per lane [Gbit/s]	Optical fiber type	Number of optical fibers	Number of wavelengths
Up to 500m				
100GBASE-SR4	25	MMF at 0.85 μm	4	1
100GBASE-SR10	10		10	1
100GBASE-PSM4 *	25	SMF at 1.3 μm	4	1
Up to 2km				
100GBASE-10x10 *	10	SMF at 1.55 μm	1	10 (Δλ=7.2nm)
100GBASE-CWDM4 *	25	SMF at 1.3 μm	1	4 (Δλ=20nm)
100GBASE-CLR4 *	25		1	4 (Δλ=20nm)
Up to 10km				
100GBASE-10x10 *	10	SMF at 1.55 μm	1	10 (Δλ=7.2nm)
100GBASE-LR4	25	SMF at 1.3 μm	1	4 (Δf=800GHz)

Today, in function of the cost and the reach, three different technologies share the 100 Gbit/s market : vertical cavity surface emitting laser (VCSEL) technology, silicon photonic technology and indium phosphide (InP) technology. This segmentation is driven by the cost of optical fibers and transceivers according to Menezo et al. [7] and represented in figure 1.2.



Oct 21, 2016 | Sylvie Menezo, OSA meeting, Rochester

FIGURE 1.2: 100 Gb/s standards versus transmission reach from [7]

Transmitters based on vertical cavity surface emitting laser (VCSEL) technology represent one of the solutions satisfying the data rate demand at low costs [8–10]. However, they cannot be used for more than 500 m transmission lengths due to multimode nature of VCSEL. The market players decided to use VCSEL solutions for 0 to 100 m transmissions.

For 100 to 500 m, transceivers based on silicon photonic technology have emerged as the lowest cost solution respecting the MSA 100GBASE-PSM4 (figure 1.2). The aggregate 100 Gbit/s is achieved with 4 single mode fibers in parallel at 1.31 μm (25 Gbit/s per fiber). Two products are already commercially available : LUX42604 QSFP28 proposed by Luxtera in 2015, and 100G PSM4 QSFP28 proposed by Intel in 2016 with integrated hybrid laser [11, 12]. The market price for silicon photonic transceivers is 3\$ per Gbit/s [7].

For transmission distances up to 10 km, optical transceivers based on indium phosphide (InP) technology remain predominant. The 100G Coarse WDM-4 QSFP28 optical module from Finisar is based on 4x25 Gbit/s distributed feedback lasers (DFB) [13]. The market price of InP optical transceivers is lower than 5\$/Gbit/s [7]. The silicon photonic technology begins competing with InP optical transceivers for up to 10 km. For example, in 2016, Intel proposed a optical Coarse WDM-4 100Gbit/s transceiver based on silicon photonic technology [12].

In this thesis, we focus on the silicon photonic technology. This technology allows the integration of all the components required for a transceiver on the same technology platform. It

has the advantage of reducing the cost production and allowing a higher transceiver complexity. We will use this technology within the frame of the next 400 Gbit/s standard and we will target reach higher than 500 m to replace the [InP](#) technology.

1.2 400 Gbit/s optical communications

For the 400 Gbit/s data rate, different ways are under considerations by the IEEE group or [MSA](#) group. 400 Gbit/s standards will be a combination between data rate per channels, the number of wavelengths and the order of modulation (Fig. 1.3) [14]. One can note that a higher modulation format, the 4-levels pulse amplitude modulation ([PAM-4](#)), is considered to replace 2-levels pulse amplitude modulation ([PAM-2](#)). According to the number of wavelengths and the modulation format, the baud rate will be different and so the required bandwidth of the optical modulator will be different.

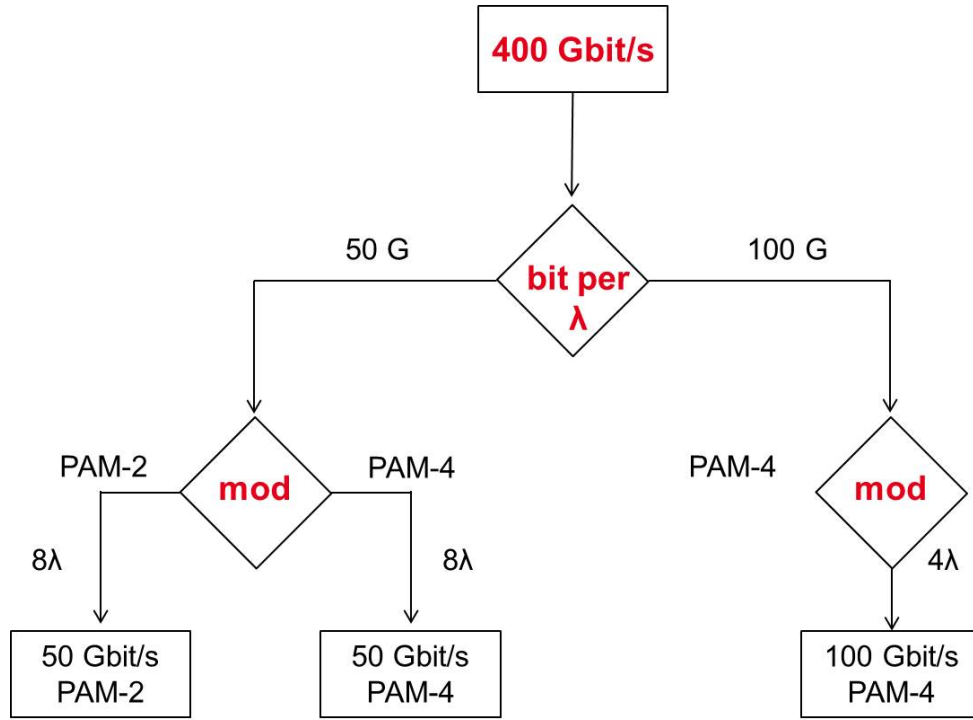


FIGURE 1.3: Possible architectures for 400 Gbit/s standard [14]. mod stands for modulation.

Table 1.2, based on [15–19], summarizes the different possible specifications of the 400 Gbit/s standards. [PAM-4](#) solutions are chosen for SR16, DR4, FR8 and LR8 standards. One can note that the 400GBASE-DR4 is equivalent to the MSA 100GBASE PSM4 standard. While no commercial product is available, the choice is not permanent and depends essentially on the vendors. Another architecture is proposed by the Open Compute Project consortium [19]. Their idea is to base their solution on an adaptive [WDM](#) architecture between 4 and 32 optical channels with 200 GHz spacing. This solution does not consider the [PAM-4](#) modulation format, but rather it keeps on a [PAM-2](#) modulation format. The Open Compute Project considers the

C band (1.55 μm) in order to keep a large available wavelength band between 1511 and 1562 nm. For this thesis, we will not fix any wavelength and we will work at 1310 nm and 1550 nm.

TABLE 1.2: 400 Gbit/s IEEE Ethernet specifications

Nomenclature	Distance	Data rate per lane [Gb/s]	Optical fiber type	Number of optical fibers	Number of wavelengths	Modulation
400GBASE-SR16	100m	25	MMF at 0.85 μm	16	1	PAM-2
400GBASE-DR4	500m	50	SMF at 1.3 μm	4	1	PAM-4
400GBASE-FR8	2km	25	SMF at 1.3 μm	1	8 ($\Delta f=800\text{GHz}$)	PAM-4
Open Optics Solution	2km	25	SMF at 1.5 μm	1	16 ($\Delta f=200\text{GHz}$)	PAM-2
400GBASE-LR8	10km	25	SMF at 1.3 μm	1	8 ($\Delta f=800\text{GHz}$)	PAM-4

Our aim is to propose transmitter architectures using silicon photonic technology for optical transmission inside data-centers with a reach up to 2 kilometers. In this thesis, the binary (PAM-2) and quaternary (PAM-4) modulation formats will be used. We decided to fix the number of multiplexed wavelengths to 8 with a 50 Gbit/s data rate per wavelength. The standard which is closed to our targeted characteristics transmitter is the 400GBASE-FR8 standard. We will use this standard as a basis for our study. In the following section, we report the 400GBASE-FR8 transmitter specifications.

1.2.1 400GBASE-FR8 specifications

Several characteristics will be used to evaluate the performances of an optical transmitter. They are :

- The total average output power P_{out}
- The optical modulation amplitude (OMA) at the output of the transmitter. It is the difference between the highest modulated optical power and the lowest modulated optical power at the output of the transmitter. ($\text{OMA}=P_{out1}[\text{mW}] - P_{out0}[\text{mW}]$).
- The extinction ratio (ER). It is the ratio between the highest modulated optical power and the lowest modulated optical power. ($\text{ER}=P_{out1}[\text{mW}]/P_{out0}[\text{mW}]$)

Figure 1.4 represents the different optical power at the output of the transmitter for a PAM-2 modulation format. For a PAM-4 modulation format, the previous characteristics stay valid. Two optical powers have to be added between the highest (P_{out1}) and the lowest (P_{out0}) optical powers in a way that the four levels are equidistant to each other.

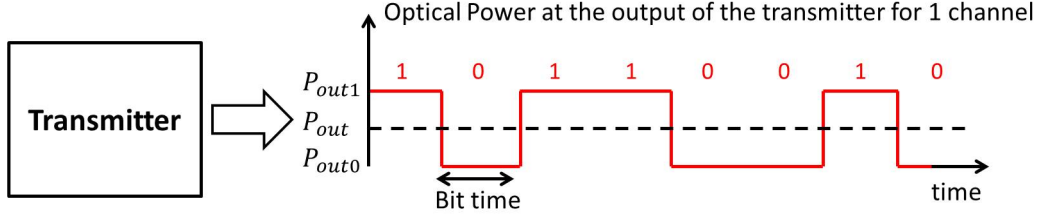


FIGURE 1.4: Optical powers at the output of the transmitter according to a PAM-2 modulation

To evaluate the quality of the optical link, the bit error rate (BER) is used. The BER is the ratio between the number of received bits that have been read with a wrong value compared to the total number of transmitted bits. Today, the targeted BER is 10^{-12} in optical communication links (*i.e.* in average one error over 10^{12} datas). However, electrical pre- and post-treatments such as forward error correction (FEC), including Reed-Solomon correctors, allow the BER requirement to be relaxed.

For the 400 GBASE-FR8, the values of the characteristics is reported in table 1.3. These values will be in the manuscript the targeted performances values.

TABLE 1.3: 400GBASE-FR8 specifications

Description	IEEE 400GBASE-FR8
Total average power P_{out} [dBm]	13.2
OMA Transmitter [dBm]	[0 to 5.5]
Extinction Ratio ER [dB]	≥ 4.5
Operating BER with a FEC	$2 \cdot 10^{-4}$

For our transmitter, we decide to not fix the modulation format PAM-2 or PAM-4. These two modulations will be evaluated with our proposed transmitter. In this following sections, we will compare briefly the two modulation formats.

1.2.2 Binary and quaternary pulse amplitude modulations explanation

The multilevel amplitude modulation is a pure amplitude modulation and referred as M-ary pulse amplitude modulation (PAM-M). It consists of coding the bits on M different levels of intensity amplitude. Figure 1.5 shows an ideal PAM-2 and PAM-4 modulation.

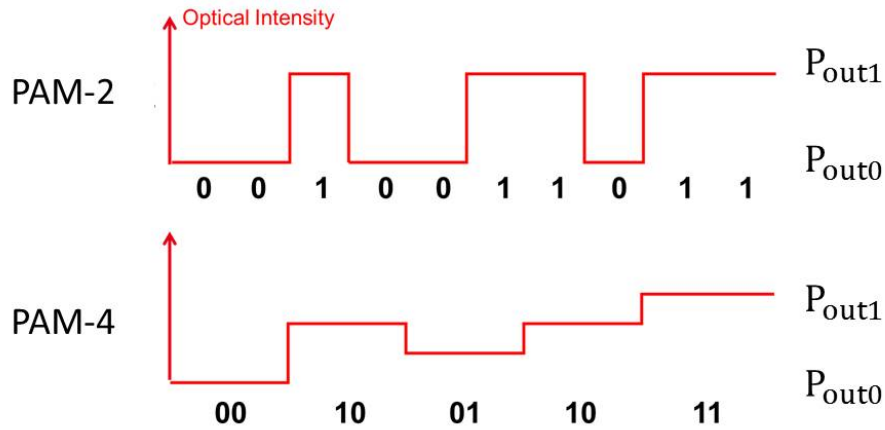


FIGURE 1.5: Ideal PAM-2 and PAM-4 modulations

As we can see the number of symbols per period is multiplied by 2 between the two modulations (fig.1.5). This means that for a same baud rate, PAM-4 modulation codes twice more data. But in return, the four levels have to be equally spaced which asks linear electrical drivers and linear electro-optical modulators.

On the receiver side, multilevel modulation signal are measured in direct detection with a photo-detector followed by a transimpedance amplifier (TIA) and a (M-1) levels decision circuit. Two supplementary decision thresholds are required to demodulate a PAM-4 modulated signal compared to a PAM-2 modulated signal as shown in figure 1.6. This will bring more complexity in the receiver part.

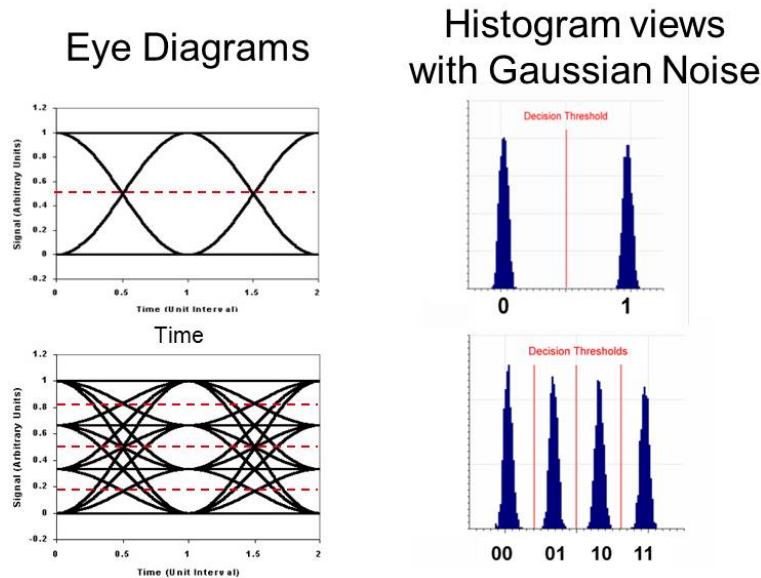


FIGURE 1.6: Eyes diagrams and histogram views of PAM-4 and PAM-2 modulations. Red lines represent the decision thresholds

Considering a noisy gaussian model, we can link the BER to the signal to noise ratio (SNR) for the two modulation types according to the equation 1.1 (generalize equation for PAM-M modulation) [20].

$$\text{BER}_{\text{PAM-M}} = \frac{M-1}{M \cdot \log_2(M)} \text{erfc} \left(\frac{Q}{(M-1)\sqrt{2}} \right) \quad (1.1)$$

with M the number of levels (in our case 2 or 4) and Q is quality factor related to the SNR by equation 1.2.

$$\text{SNR} = \frac{Q^2(M^2 - 1)}{6(M - 1)^2} \quad (1.2)$$

In order to achieve a BER of 10^{-12} , the signal to noise ratio of the PAM-4 compared to the PAM-2 modulation has to be increased by 4 dB as shown in figure 1.7.

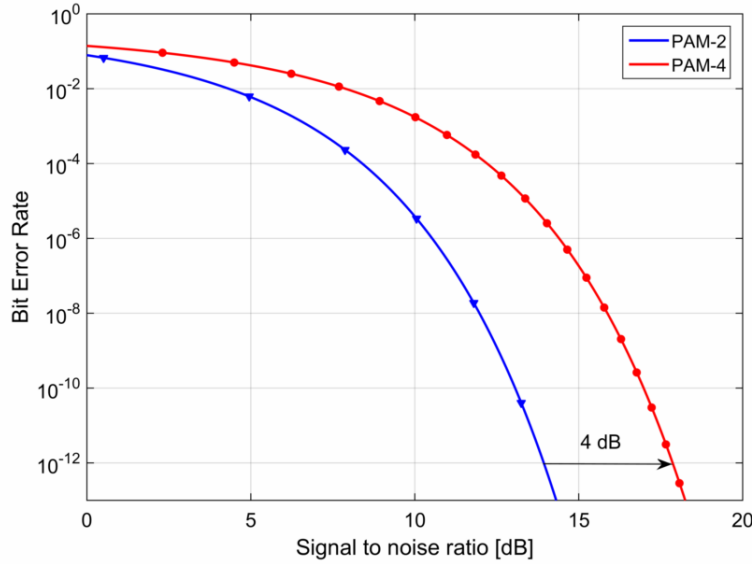


FIGURE 1.7: Bit error rate versus signal to noise ratio for PAM-2 and PAM-4 modulation formats

1.3 Contribution and organization of this thesis

The aim of this work is to study transceiver architectures with silicon photonic technology for 400 Gbit/s transmissions. The thesis essentially focus on transmitters composed of silicon ring resonator modulators (RRMs). This is a very promising electro-optical modulator with many benefits: low size, low power consumption and low cross-talk in WDM architecture. In addition, it has been already demonstrated at 50 Gbit/s in PAM-2 modulation [21]. However, it has a major drawback : a high sensitivity to temperature and fabrication variability. Our aim is to optimize ring resonator modulator performances and evaluate how it performs for the 400 GBASE ethernet standard. Especially, we target the performances required by the 400 GBASE-FR8.

In this context, the manuscript is divided into 5 chapters :

First, chapter 2 describes the principle of operation and the state of the art of the typical silicon photonic components implemented in an optical transceiver. It is divided in four parts: 1) propagation in the silicon waveguide, 2) passive components as couplers and multiplexers, 3) opto-electrical converters, 4) electro-optical converters. In each part, lumped-element models based on the physical characteristics of the components are presented. At the end of the chapter, a first optical power budget of a transmitter based on ring resonator modulators is reported. It is composed of 8 ring resonator modulators arranged in series. Each **RRM** modulates a different wavelength at 50 Gbit/s. In order to respect an **OMA** at the output of the transmitter higher than 0 dBm, **RRM** has to exhibit an extinction ratio higher than 8 dB and an insertion loss lower than 3 dB. These values will be used as a basis for the **RRM** optimization.

Chapter 3 is dedicated to the ring resonator modulator. The aim of this chapter is to present the transient **RRM** model, and to highlight some important considerations :

- the **RRM** modulation bandwidth depends on the position of the **RRM** resonance with respect to the continuous wave laser.
- the **RRM** resonance can be shifted due to its self-heating depending on the input optical power. This effect requires a careful compensation of continuous wave laser.
- the double-bus **RRM** structure allows us to geometrically achieve the critical coupling conditions.

Chapter 4 is dedicated to the validation of the **RRM** model with wafer level measurements. We present the extraction of each **RRM** parameters from measurements : waveguide group index and effective index, waveguide loss, coupling coefficients and their dependencies towards the applied voltage. The electric and electro-optical bandwidth is also measured depending on the position of the **RRM** resonance wavelength with respect to the laser wavelength. In addition, the effect of frequency chirping induced by the modulator is characterized through small signal frequency response measurements over a dispersive channel and with direct detection.

Chapter 5 is concerned with the **RRM** optimization starting from simulations of the PN junction, followed by **RRM** optimization. This study allows us to build a **RRM** compact model which identifies the optimum **RRM** parameters and operating point depending on the Q factor. This compact model is then used to know if the **RRM** can perform the 400GBASE-FR8 required performances. Two transmitters are compared using 50 Gbaud **PAM-2** or 25 Gbaud **PAM-4**. The two solutions fit the 400 Gbit/s performances demand with the use of the same trade-offs. However, **PAM-2** transmitter exhibits higher margin loss and should be preferred.

In the chapter 6, two transmitters architecture employing **RRMs** were proposed to generate **PAM-4** modulation in another way. Unlike the previous chapter, **PAM-4** modulation is generated by combining two binary bit streams. The first solution is based on two silicon ring resonator modulators arranged in series. This architecture is validated through 30 Gbit/s **PAM-4**

transmission characterizations with only 1 V_{pp} . A second solution is then proposed, based on two silicon ring resonator modulators arranged in parallel in a Mach Zehnder interferometer. In the same way, transmission characterizations at 30 Gbit/s PAM-4 with 1.2 V_{pp} allow this architecture to be validated.

Chapter 2

Silicon photonic building blocks

2.1 Introduction

In this chapter, we review the building blocks of an optical communication system with silicon photonic technology. An overview of an optical transceiver is shown in figure 2.1. On the transmitter side, the optical intensity is provided by an external laser coupled to the silicon chip with a single polarization grating coupler (also called a 1D grating coupler). The laser can be replaced by a heterogeneous integrated laser avoiding grating coupler insertion loss. The light is transmitted through silicon waveguides to an electro-optical converter (electro-optical modulators) and it is modulated according to an electrical signal generated by an electrical driver. The electrical driver can be complex, like PAM-4 driver. The modulated light is then coupled by a single polarization grating coupler in an optical fiber. In order to multiplex different wavelengths, an optical multiplexer can be added to the transmitter side. At the receiver side, a splitting polarization grating coupler (also called 2D grating coupler) is used to re-couple the light in the silicon chip. The splitting polarization grating coupler allows retrieval of the light with any polarization state.

Then, for wavelength multiplexed signals, a de-multiplexer is used where each output is linked to an opto-electrical converter (integrated photo-diodes). The received electrical signal is subsequently amplified and analyzed.

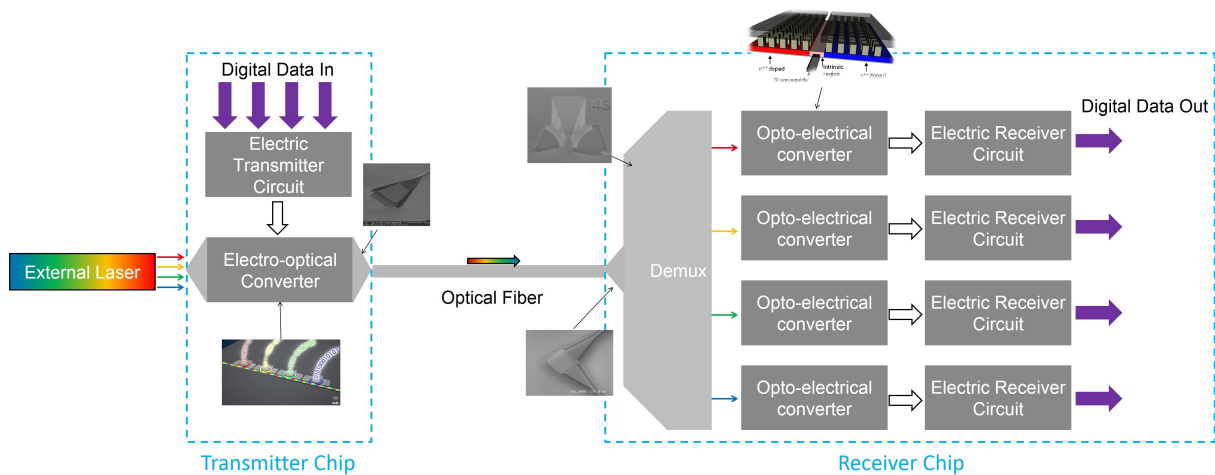


FIGURE 2.1: Optical transmission scheme

Two bands of wavelength are used : C band and O band. The C band is between 1530 and 1565 nm and known for low optical fiber loss and for erbium doped fiber amplifiers used in long haul transmission. The O band, between 1260-1360 nm, exhibits higher optical fiber loss but with no chromatic dispersion. Standard single-mode fiber SMF-28 characteristics are shown in table 2.1 [22].

TABLE 2.1: Single mode fiber main characteristics

	C band	O band
Wavelength [nm]	1530-1565	1260-1360
Attenuation [dB/km]	0.2	0.5
Chromatic Dispersion [ps/nm/km]	15-18	1

This chapter is divided into five main parts : 1) the optical propagation into silicon waveguides, 2) passive components, 3) opto-electrical converters, 4) eletro-optic converters, and 5) optical link power budget. The goal is to provide silicon photonic component building blocks with the required performance to design a transceiver. The state of art will be focused on the performance achieved in main fabrication platforms : OpSIS-IME, IMEC, STmicroelectronics and CEA-LETI. As this work uses essentially STmicroelectronics and CEA-LETI platforms, discussions and component optimizations are explained with these technologies constraints.

2.2 Propagation in silicon waveguide

Low loss silicon waveguides are key components for optical transceivers. Interconnections of a few centimeters are common. They are alternately composed of straight lines and bends. In this section, we report silicon waveguide characteristics for interconnections, but also for silicon waveguides compatible with electrical connections for electro-optical modulators.

2.2.1 Transverse electrical single mode waveguides

Among all silicon waveguides available, we chose to work with the rib waveguide because it is one of the most compatible with the other passive and active silicon photonic components. Its parameters are the width, height, and the slab height as described in figure 2.2. The width of the waveguide is chosen in order to achieve a transverse electric (TE) single mode waveguide. With regard to the CEA-LETI technology, the height is 220 nm for 1.55 μm applications and 300 nm for 1.31 μm applications. The slab height is fixed by the technology at 100 nm for the 220 nm height waveguide and 150 nm or 50 nm for the 300 nm height waveguide. These distances are fixed by the CEA-LETI process flow and cannot be changed. In the following, the rib-waveguide will be described according to this nomenclature : rib height/slab height/waveguide width in nanometers.

The waveguide loss depends on the technology platform. The side-wall scattering is at the heart of the major loss. Table 2.2 reports the rib-waveguide loss for different technology platforms. To resume, it can vary from 1 to 6 dB/cm. The loss of the waveguide is typically

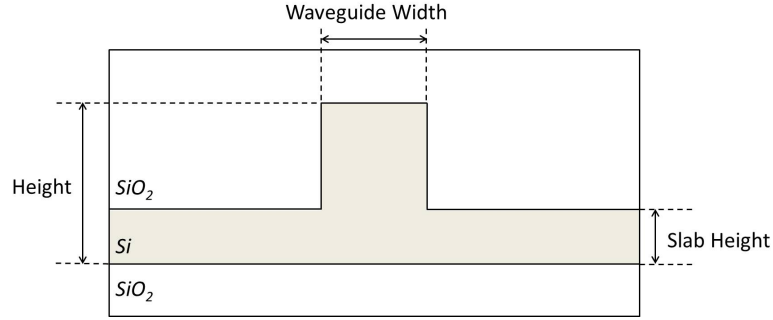


FIGURE 2.2: Schematic representation of rib waveguide structure

expressed in dB/cm, but can be converted to intensity absorption coefficient α in $[m^{-1}]$ as expressed in equation 2.1.

$$\alpha[m^{-1}] = \frac{\alpha[dB/cm] \cdot \ln(10) \cdot 10^2}{10} \quad (2.1)$$

TABLE 2.2: Rib-loss waveguide for different silicon photonics platform

	Waveguide Height/Slab/Width	Wavelength [nm]	rib-waveguide loss [dB/cm]
OpSIS-IME, Galland [23]	220/90/500 nm	1550	1.5
IMEC	220/160/650 nm	1550	1
Francken [24]	220/160/580 nm	1310	1.5
STMicroelectronics	306/166/320 nm	1310	1.8
Boeuf [25]	306/166/400 nm	1490	1.4
CEA-LETI, Szelag [26]	300/150/400 nm	1310	1.5
	300/50/320 nm		5

For routing connections and ring resonators, bend waveguides have to be considered. The bend loss is dominated by two factors : the loss due to the radiation in the rib waveguides and the loss due to the mode mismatch between the bend waveguide and the following straight waveguide. The bend loss grows exponentially with decreasing radius. Using Lumerical MODE software, figure 2.3 a) represents the simulated bending loss due to the radiation for 220/100/400, 300/150/400 and 300/50/320 rib waveguides. The smaller the slab, the smaller the radius that can be achieved without bend loss. The radius has to be chosen in order to not exceed 1 dB/cm bend loss. For the 300/150/400 waveguide, the limit radius will be 13 μm . For ring resonators, we will prefer the 300/50/320 waveguide as the limit radius is 2.7 μm .

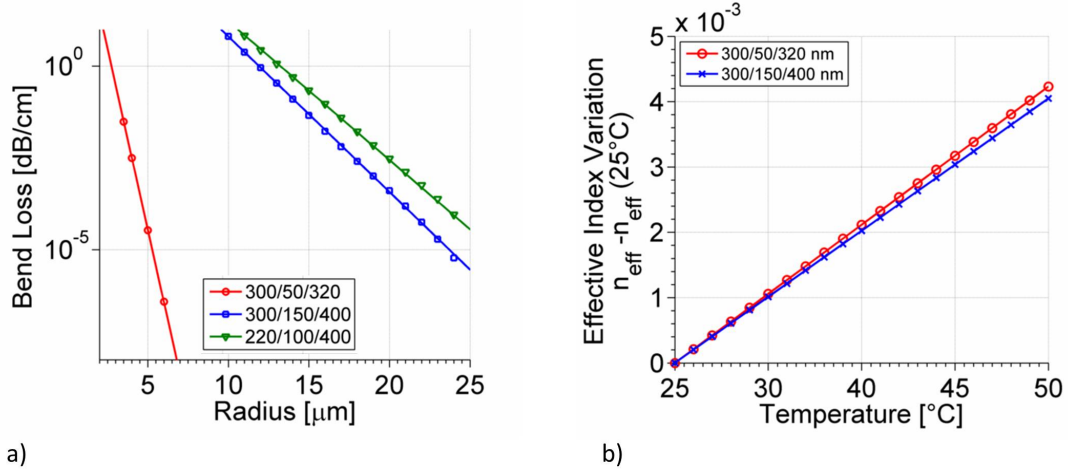


FIGURE 2.3: a) Simulated bend loss versus the radius at 1.55 μm for the 200/100/400 nm rib waveguides and at 1.31 μm for the 300/150/400 nm and 300/50/320 nm rib waveguides, b) Simulated effective index variation versus the temperature

The effective index of the rib waveguide exhibits a dependency on the wavelength and the temperature. The wavelength dependency is represented by the group index n_g as shown in equation 2.2.

$$n_g = n_{eff} - \lambda \cdot \frac{\partial n_{eff}}{\partial \lambda} \quad (2.2)$$

where λ is the wavelength, n_{eff} is the effective index. The effective index and the group index of our rib waveguide structures are reported in table 2.3 [27].

For the temperature effect, the variation of the silicon refractive index (n_{Si}) is quite similar at 1.31 μm and at 1.55 μm (equation 2.3).

$$\begin{cases} \frac{\partial n_{Si}}{\partial T} = 1.87 \cdot 10^{-4} K^{-1} \text{ at } 1500 \text{ nm} \\ \frac{\partial n_{Si}}{\partial T} = 1.93 \cdot 10^{-4} K^{-1} \text{ at } 1310 \text{ nm} \end{cases} \quad (2.3)$$

where T is the temperature. The refractive index of SiO_2 is 6.3 times less impacted by temperature than that of silicon. With Lumerical MODE software, we simulated the effective variation against the temperature for our rib structure (Figure 2.3 b)). The effective index variation increases linearly with the temperature.

Equation 2.4 represents a compact model of the rib-waveguide where the optical output wave E_{out} is related to the optical input wave E_{in} adding a loss and phase shift factors.

$$\begin{cases} n_{eff}(\lambda, T) = n_g(\lambda) + \lambda \cdot \frac{\partial n_{eff}}{\partial \lambda} + \frac{\partial n_{eff}}{\partial T} \cdot (T - T_0) \\ E_{tout} = E_{in} \cdot e^{-\frac{i2\pi n_{eff}(\lambda, T)}{\lambda} \cdot L} \cdot e^{-\frac{\alpha}{2} \cdot L} \end{cases} \quad (2.4)$$

where L is the length of the waveguide and T_0 is a reference temperature (generally 300 K). Table 2.3 sums up the simulated parameters for the 220/100/400 nm, 300/150/400 nm and 300/50/320 nm rib waveguides.

TABLE 2.3: Effective index and group index for the three available technology at 25 °C

Technology platforms	Wavelength [nm]	n_{eff}	n_g	$\frac{\partial n_{eff}}{\partial T}$ [K ⁻¹]
300/150/400 nm	1310	2.94	3.92	$2.02 \cdot 10^{-4}$
300/50/320 nm	1310	2.62	4.26	$2.11 \cdot 10^{-4}$
220/100/150 nm	1550	2.50	3.87	

2.2.2 Multimode waveguide

The rib-waveguide has too much loss to be implemented in centimeters-scale connections for global routing on-chip. Wider waveguides would be a solution to reduce the interaction between the mode and sidewall roughness. However, these waveguides are multi-mode. They cannot be used in bends because superior modes will be radiatively removed leading to important loss. The achieved loss is under 0.3 dB/cm, which is a 10-fold decrease compared to that of rib-waveguides. Table 2.4 reports three papers reporting multi-mode loss measurements.

TABLE 2.4: Rib-loss waveguide for different silicon photonics platform

Technology platform		Wavelength	rib-waveguide loss
		[nm]	[dB/cm]
Dong [28]	250/50/2000 nm	1550	0.274 ± 0.008
Li [29]	300/80/3000 nm	1550	0.026
Reboud [30]	300/150/2500 nm	1310	0.18 ± 0.02
Szelag [26]	300/150/1800 nm	1310	0.15

2.3 Passive Components

2.3.1 Optical coupler between waveguides and fibers

The coupling between the optical fiber core and the optical mode confined in the waveguide is an important part of the transceiver. The optical fiber core has a diameter of 9 μm which is almost 22 times larger than the waveguide (300/150/400 nm). The coupling needs to be low loss and broadband in terms of wavelength. The position of the wavelength where the coupling loss is minimum also has to be as stable as possible.

Two coupling techniques are used: edge-coupling and grating coupling. Edge coupling uses spot-size converters. In [31, 32], their edge coupling achieves relatively low loss (under 0.8 dB loss). This approach needs post processing to resolve polishing and etching of the facet as well as high-resolution optical alignment. The second solution grating couplers allows wafer level measurements. The grating coupler is a periodic structure, which diffracts the light from the waveguide to free space as shown in figure 2.4. Single-polarization grating couplers are used at the transmitter side. Polarization splitting grating couplers are used at the receiver side, as the light polarization coming from the fiber is not controlled. It allows to retrieve the total output power, regardless of the polarization. Figure 2.5 shows tilted scanning electron microscopy (SEM) pictures of single-polarization and polarization-splitting grating couplers.

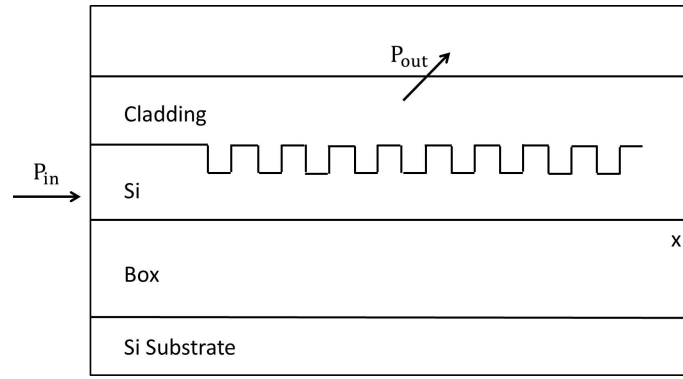


FIGURE 2.4: Scheme of cross section grating coupler structure

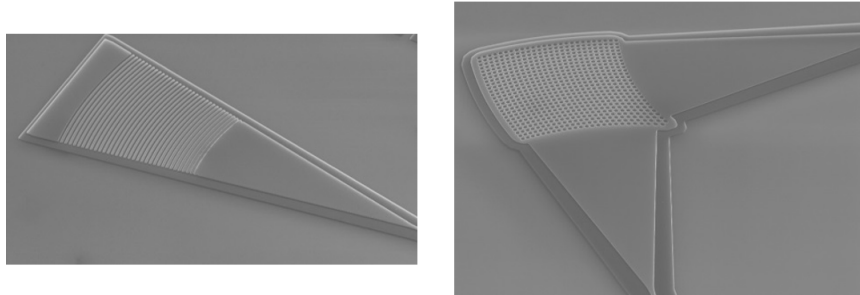


FIGURE 2.5: Tilted SEM pictures of single polarization and polarization splitting grating couplers.

The methodology to design a grating coupler is explained in [27]. In table 2.5, we report the grating coupler characteristics of the main fabrication platforms. Different processes are in development to decrease the loss of grating couplers. They are based on multi-layer staking below the grating coupler to retrieve the light diffracted in the substrate [25, 33, 34].

2.3.2 Splitters and combiners

Various splitters and combiners are used in the silicon photonic technology platform. The main components are the Y-junction and the multimode interferometer (MMI). Their role is to

TABLE 2.5: Gratings couplers characteristics for different silicon photonics platform

	Technology platform	Wavelength [nm]	Peak insertion Loss [dB]	Bandwidth
Single polarization grating coupler				
OpSIS-IME, Galland [23]	220/160 nm	1550	3.1	50 nm at 1.5 dB
IMEC Francken [24]	220/160 nm	1550	1.5	29 nm at 1 dB
STMicroelectronics Boeuf [25]	306/166 nm	1310	2.2	
	306/166 nm	1490	3.2	
CEA-LETI, Szelag [26]	300/150 nm	1310	2	26 nm at 1 dB
Splitting polarization grating coupler				
STMicroelectronics Boeuf [25]	306/166 nm	1310	3.25	
	306/166 nm	1490	3.1	
CEA-LETI, Szelag [26]	300/150 nm	1310	<4 dB	24 nm at 1 dB

split the light power equally from one waveguide into two waveguides or to combine light from two waveguides into one.

Operating as a splitter, the input optical intensity I is divided into two equal output optical intensities. Thus, each output intensities are equal to an intensity $I_{output} = \frac{I_{input}}{2}$. Operating as a combiner, there is total output intensity recombination if the two inputs optical fields are in phase and coherent. The optical field at the output of the combiner is described by equation 2.5.

$$E_{out\ combiner} = \frac{1}{\sqrt{2}} \cdot (E_{in1} + E_{in2}) \quad (2.5)$$

In reality, these components are not perfect. They exhibit excess loss and imbalance between the two output ports. Table 2.6 reports the loss and imbalance of MMI and Y junction for different platforms. The imbalance traduces the fact the intensities at the output of the splitter are not exactly equals. These components can be used in an interferometer structure.

The non-achievement of a perfect combine/splitter whitout imbalance or excess loss in the entire wavelength span is clearly a drawback. In [35], metamodeling optimization is used to build a new robust shape. Their adiabatic coupler achieves 0.36 dB insertion loss over a wavelength span of 32.7 nm.

TABLE 2.6: 1x2 MMI loss and imbalance for different silicon photonics platform

	Component	Wavelength [nm]	Insertion loss [dB/cm]	Imbalance [%]
OpSIS-IME, Galland [23]	Y Junction	1550	0.3	
CEA-LETI, Szelag [26]	MMI	1310	0.1	3

2.3.3 Wavelength multiplexers and demultiplexers

In silicon photonics, many solutions for wavelength multiplexers and demultiplexers are valid. These include: arrayed waveguide grating (AWG), echelle grating, cascaded Mach-Zehnder interferometer (MZI), and ring resonator filter (RRF). (De)-multiplexer need to be wavelength aligned with each channel, with low insertion loss, and with low channel crosstalk. The AWG is the most common (de)-multiplexer. It exhibits around 3 dB insertion loss and 15 dB channel cross-talk [27]. Large inter-wavelength spacing can be achieved between 4.56 nm (800 GHz) and 20 nm [26] at the cost of centimetre-long component. Furthermore, it is very sensitive to temperature variation. Proposed solutions in [36, 37] are reported to produce athermal AWGs. MZI and echelle grating multiplexers have almost the same performance level, and the same thermal, issue as the AWG [38] [39]. The main benefit of these three components is a broadband response, which allows standards with 20 nm wavelength spacing to be targeted.

The ring resonator filter is another alternative with a very small size. Figure 2.6 represents a typical receiver architecture with RRFs [40, 41]. Each ring collects one wavelength in its drop port where a photo-diode is connected. As the opposed to the AWG, where the goal is to obtain a flat channel, independent of process and temperature variations, each RRF is tuned to align its resonant wavelength to the laser wavelength and set the channel always to the minimum insertion loss.

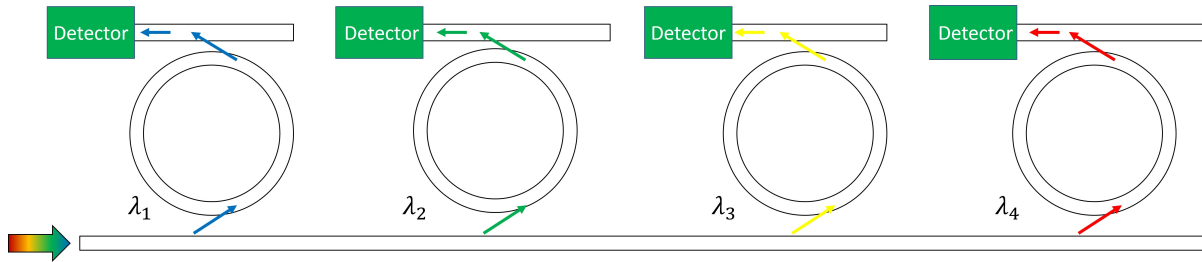


FIGURE 2.6: WDM receiver based on ring resonator filters

Ring resonator filter: principle of operation

The ring resonator filter is composed of one input port and two output ports as described in figure 2.7. The input optical field E_{in} is partially coupled with a coupling coefficient κ_1 to the ring and recoupled in the drop port E_{dout} with a coupling coefficient κ_2 .

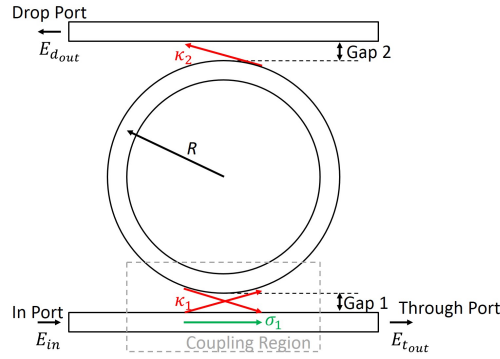


FIGURE 2.7: Schematic of ring resonator filter

Based on a scattering matrix approach [42–44], the transfer functions of the RRF are expressed in equation 2.6.

$$\begin{cases} T_{t_{out}}(\lambda) = \left| \frac{E_{t_{out}}}{E_{in}} \right|^2 = \frac{\sigma_1^2 + \sigma_2^2 e^{-2a} + 2\sigma_1\sigma_2 e^{-a} \cos(\phi(\lambda))}{1 + \sigma_1^2 \sigma_2^2 e^{-2a} - 2\sigma_1\sigma_2 e^{-a} \cos(\phi(\lambda))} \\ T_{d_{out}}(\lambda) = \left| \frac{E_{d_{out}}}{E_{in}} \right|^2 = \frac{(1 - \sigma_1^2) \cdot (1 - \sigma_2^2) \cdot e^{-a}}{1 + \sigma_1^2 \sigma_2^2 e^{-2a} - 2\sigma_1\sigma_2 e^{-a} \cos(\phi(\lambda))} \end{cases} \quad (2.6)$$

where $\sigma_1 = \sqrt{1 - |\kappa_1|^2}$ is the transmission coefficient between the through waveguide and the ring ; $\sigma_2 = \sqrt{1 - |\kappa_2|^2}$ is the transmission coefficient between the drop waveguide and the ring. They are related to amplitude fields and so are comprised between 0 and 1. e^{-a} and ϕ are, respectively, the field attenuation and the phase shift along one trip over the ring expressed by equation 2.7.

$$\begin{cases} a = \frac{\alpha_0}{2} \cdot 2\pi R \\ \phi(\lambda) = \frac{2\pi n_{eff}(\lambda)}{\lambda} \cdot 2\pi R \end{cases} \quad (2.7)$$

where R is the radius, n_{eff} is the effective index of the optical mode depending on n_g according to equation 2.2, α_0 is the intensity absorption coefficient.

Figure 2.8 represents the transfer function for a RRF with its main spectral characteristics. When ϕ is a multiple of 2π , the resonator is on resonance leading to a resonant wavelength defined by equation 2.8.

$$\lambda_{res}(m) = \frac{2\pi R}{m} \cdot n_{eff}(\lambda_{res}) \quad (2.8)$$

where m is the resonance order.

So at the resonant wavelength, the drop port intensity is maximum, while the through port intensity is minimum. The distance between two resonant wavelength is called the free spectral range (FSR) (equation 2.9). The FSR decreases with the radius decreasing the available channel spacing.

$$\text{FSR} = \lambda_{res}(m) - \lambda_{res}(m - 1) \simeq \frac{\lambda^2}{n_g \cdot 2\pi R} \quad (2.9)$$

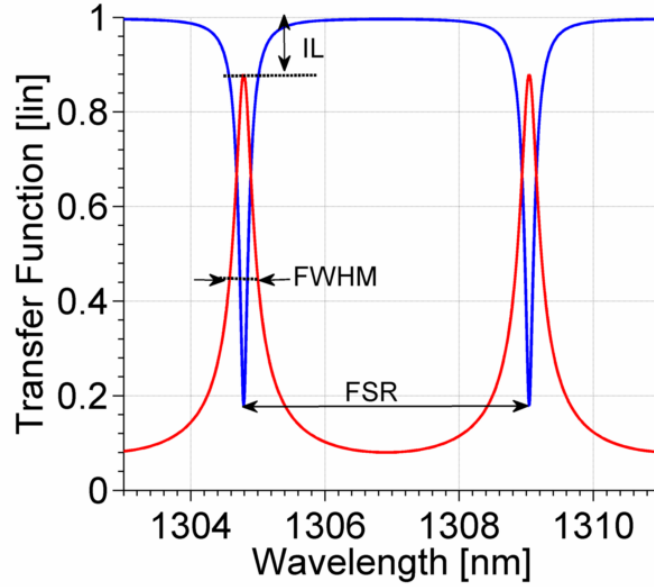


FIGURE 2.8: Optical spectrum at the through (blue) and drop port (red)

The channel is characterized by its Q factor, which is directly related to the full width at half maximum **FWHM** ($Q = \lambda_{res}/FWHM$) and therefore the selectivity of the resonance (equation 2.10).

$$Q \simeq \pi \cdot 2\pi R \cdot \frac{n_g}{\lambda_{res}} \cdot \frac{\sqrt{\sigma_1 \sigma_2 e^{-a}}}{1 - \sigma_1 \sigma_2 e^{-a}} \quad (2.10)$$

And the channel is characterized by its insertion loss expressed by equation 2.11.

$$IL = \frac{(1 - \sigma_1^2)(1 - \sigma_2^2)e^{-a}}{(1 - \sigma_1 \sigma_2 e^{-a})^2} \quad (2.11)$$

The minimum insertion loss is achieved at critical coupling ($\sigma_1 = \sigma_2 e^{-a}$). With negligible loss ($e^{-a} = 1$), insertion loss at the drop port tends to zero.

The choice of the Q factor is clearly very important. A high Q factor allows the selected wavelength to be collected precisely. However, if the Q factor is too high, it cuts the higher frequencies of the collected modulated signal and introduces signal distortion.

The ring resonator filter is a temperature sensitive component. A heater placed above the ring can be added to move the resonant wavelength adding static energy consumption. This tunability allows to have minimum insertion loss to be achieved, whatever the channel wavelength position. In order to obtain a flatter frequency response, a resonator filter with multiple rings in parallel can be set up. However, in this approach each ring has to be tuned to the desired wavelength [45,46].

2.4 Opto-electrical Converters

Today, fabrication platforms propose photo-detectors based on germanium lateral PIN structures. Figure 2.9 shows a schematic view and a SEM view of an integrated photo-diode from [47]. The germanium photo-diode is composed of an intrinsic region between P- and N-doped regions. Input photons are collected by the intrinsic region and then converted into hole/electron pairs, which are then collected in the P and N regions creating an electrical current. A junction length of 15 μm is sufficient to saturate the photo-diode. The responsivity and the bandwidth of the photo-detector vary essentially with the width of the intrinsic region. A larger intrinsic region leads to higher responsivity, but lower bandwidth due to greater junction capacitance. Table 2.7 reports the characteristics of lateral PIN germanium photo-diodes for the main silicon photonics platforms. The performance levels are quite impressive, with electro-optical bandwidth higher than 50 GHz and responsivity around 0.8 A/W. The responsivity can be further improved with avalanche photodiodes, which remain compatible with CMOS fabrication processes [48]. Equation 2.12 expresses a simple photo-diode model, which can be used in circuit simulations.

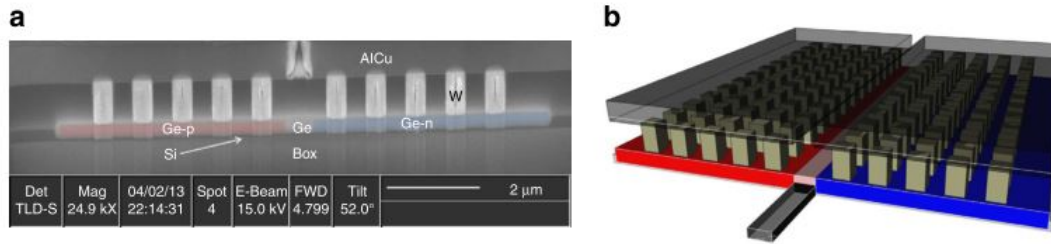


FIGURE 2.9: Lateral PIN on Si photodiode integrated at the end of a Si waveguide a) SEM view b) Schematic view (red P implantation, blue N implantation [47])

$$I_{out} = R_s \cdot P_{in} \cdot (1 - e^{-\frac{t-t_0}{\tau_{ph}}}) + I_{noise} \quad \text{for } t > t_0 \quad (2.12)$$

where I_{out} is the current at the output of the photo-diode, P_{in} is the received optical power, the photo-diode bandwidth is given by $f_c = 1/(2\pi\tau_{ph})$, R_s is the responsivity, I_{noise} is the noise intensity. The minimum optical power which can be measured is limited by the dark current I_d . The dark current is the electrical current generated by a photo diode in absence of any light.

At the output of the photo-detector, a transimpedance amplifier (TIA) amplifies and converts the current signal into a voltage signal with gain in V/A (or Ω). The most simple TIA is an operational amplifier in inverting configuration where the gain is set by the resistance R_{TIA} (figure 2.10).

The noises sources that have to be considered are the shot noise and the thermal noise expressed by equations 2.13 and 2.14. The shot noise comes from the fact that electron-hole pairs that result from absorption of photons are generated at random times. This creates instantaneous fluctuations in the electrical current generated by a photo-diode. As the shot noise depends on the received optical power, its value depends on the received optical intensity level. The thermal noise is generated by thermally induced vibration of free electrons in the

TABLE 2.7: Ge Electro-absorption photo-detector characteristics for different silicon photonics platform

	Wavelength [nm]	Responsivity [A/W]	Bandwidth [GHz]	Dark current I_d [nA]
OpSIS-IME, Galland [23]	1550	0.74	58	4
IMEC Francken [24]	1550	0.88	>50	<50
STMicroelectronics Boeuf [25]	1310	0.98	>20	10
CEA-LETI, Szelag [26]	1310	0.6	40	1

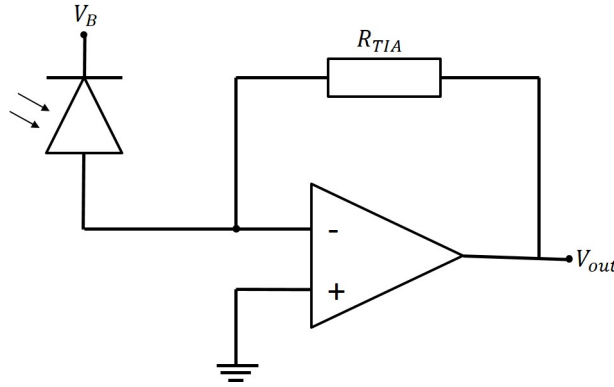


FIGURE 2.10: Electrical TIA schematic representation

semiconductor, essentially in the resistance of the TIA. When a TIA is characterized, its noise is generally represented by the measurement of its noise equivalent power (NEP). Equation 2.15 represents the noise calculation depending on NEP. In this case, the thermal noise is included in the NEP. So, the total noise is given by equation 2.16.

$$\begin{aligned} \text{Shot Noise : } V_{sh1} &= \sqrt{2q(I_d + R_s \cdot P_{00}) \cdot \delta f \cdot G^2} \\ V_{sh2} &= \sqrt{2q(I_d + R_s \cdot P_{11}) \cdot \delta f \cdot G^2} \end{aligned} \quad (2.13)$$

where q is the elementary charge equals to $1.602 \cdot 10^{-19}$ C, I_d is the dark current, P_{00} and P_{11} are respectively the lowest and the highest optical power, G is the transimpedance amplifier gain, V_{sh1} and V_{sh0} are the voltage shot noise for the highest and lowest levels, and δf is the receiver bandwidth.

$$\text{Thermal Noise : } V_{th} = \sqrt{\frac{4k_b \cdot T \cdot \delta f \cdot \text{Gain}^2}{G}} \quad (2.14)$$

where k_b is the boltzmann constant equals to $1.38 \cdot 10^{-23} \text{ J.K}^{-1}$, T is the temperature, V_{th} is the voltage thermal noise.

$$\text{TIA Noise : } V_{TIA} = \sqrt{(\text{NEP} \cdot R_s)^2 \cdot \delta f \cdot G^2} \quad (2.15)$$

where V_{TIA} is the TIA noise.

$$\begin{aligned} V_{noise1} &= \sqrt{V_{sh1}^2 + V_{TIA}^2} \\ V_{noise0} &= \sqrt{V_{sh0}^2 + V_{TIA}^2} \end{aligned} \quad (2.16)$$

where V_{noise1} and V_{noise0} are the voltage noise at the highest and lowest level

2.5 Electro-optic Converters

This section is intended to describe the operating principle of electro-optical modulators in silicon photonics technology. The electro-optical conversion consists of translating a variation of an electrical voltage into a variation in optical intensity. We will focus our discussion on the Mach-Zehnder modulator and the ring resonator modulator.

2.5.1 Electro-optic modulation in silicon photonic

Modulation Principles

In silicon, the effective refractive index and the absorption coefficient of the silicon waveguide can be varied by changing the concentration of free-carriers (holes and electrons) [49]. This effect is called the plasma dispersion effect. A semi-conductor junction architecture inside a silicon waveguide is used to change the concentration of free-carriers in order to change the refractive index and the absorption coefficient. The refractive index Δn and absorption coefficient $\Delta \alpha$ variations are described by equation 2.17.

$$\begin{aligned} \Delta n \text{ at } 1550 \text{ nm} &= -8.8 \cdot 10^{-22} \Delta N - 8.5 \cdot 10^{-18} \Delta P^{0.8} \\ \Delta \alpha \text{ at } 1550 \text{ nm} &= 8.5 \cdot 10^{-18} \Delta N + 6 \cdot 10^{-18} \Delta P \\ \Delta n \text{ at } 1310 \text{ nm} &= 6.2 \cdot 10^{-22} \Delta N - 6 \cdot 10^{-18} \Delta P^{0.8} \\ \Delta \alpha \text{ at } 1310 \text{ nm} &= 6 \cdot 10^{-18} \Delta N + 4 \cdot 10^{-18} \Delta P \end{aligned} \quad (2.17)$$

where ΔN and ΔP are the carrier densities of electrons and holes [cm^{-3}]. $\Delta \alpha$ is expressed in [cm^{-1}].

Three types of junction are used in the literature: PN junction based on carriers depletion, PIN junction based on carrier injection, and capacitive junction based on carrier accumulation [50]. Figure 2.11 represents the three different junctions architectures in a waveguide. The PN junction has to be placed in reverse bias configuration: the space charge region varies according to the voltage. As opposed to this, the PIN junction has to be placed in direct bias configuration in order to inject electrons and holes into the waveguide. For the capacitive junction, an oxide is placed inside the waveguide and with direct biasing, and electrons and holes concentrate on the oxide walls. These junction architectures are also evaluated in interleaved configurations [51].

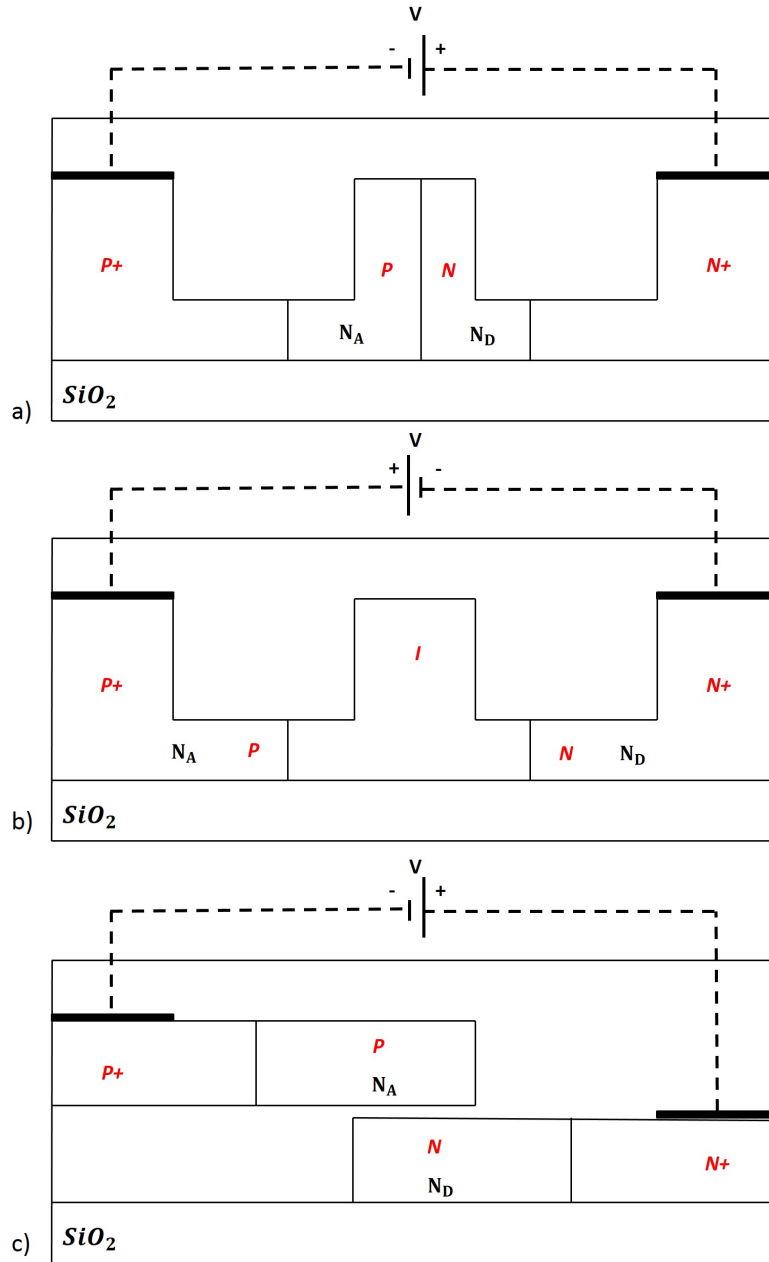


FIGURE 2.11: Junction architectures a) PN junction b) PIN junction and c) capacitive junction

The plasma dispersion effect is not used in direct modulation, because the variation of the absorption coefficient variation is too weak. This is why, the junction is placed inside an interferometer structure (e.g. Mach Zehnder interferometer or ring resonator). The refractive index variation leads to a variation in effective index Δn_{eff} , which is transformed into a phase variation. This phase variation with the interferometer is converted into an optical power variation.

2.5.2 Figures of Merit of the junction

Two figures of merit allow the comparison between different junction architectures. They are the overall optical loss (expressed in dB/cm) and the efficiency of the junction. The latter is expressed by $V_\pi L_\pi$ or by $\Delta\phi$ given generally between 0 V and a fixed voltage V_m . $V_\pi L_\pi$ represents the voltage required to produce a π phase shift with a junction length of L_π . Also, $\Delta\phi$ represents the phase shift between 0V and V_m with regard to the junction length (equation 2.18). If the junction efficiency increases, the phase shift increases and $V_\pi L_\pi$ decreases. In the next chapters, $\Delta\phi$ will be mainly given for V_m equals to 2 V.

$$V_\pi L_\pi [V.cm] = \frac{|V_m| \cdot \lambda [\mu m] \cdot 10^{-4}}{2 \cdot \Delta n_{eff}} \quad (2.18)$$

$$\Delta\phi [^\circ/mm] = \frac{360 \cdot \Delta n_{eff}(V_m)}{\lambda [\mu m] \cdot 10^{-3}} = \frac{180 \cdot V_m}{V_\pi L_\pi [V.mm]}$$

2.5.3 Electrical model

The junction architecture is also described by its electrical model. In this thesis, we will detail the electrical model for the PN junction only. But PIN electrical model can be found in [52] and for the capacitive junction in [53].

The simplest model of the PN junction is a resistance R in series with a capacitance C , as shown in the figure 2.12. The capacitance depends on the width of the space charge region W , which depends on the electron and hole doping concentrations N_A and N_D respectively, and can be expressed by equation 2.19.

$$W = \sqrt{\frac{2\epsilon}{q} \cdot \left(\frac{1}{N_A} + \frac{1}{N_D}\right) \cdot (V_{bi} - V_m)} \quad (2.19)$$

$$C = \frac{\epsilon \cdot L_{active} \cdot H_{guide}}{W}$$

$$V_{bi} = \frac{k_b \cdot T}{q} \cdot \ln\left(\frac{N_A N_D}{n_i^2}\right) \quad (2.20)$$

where k_b is the Boltzmann constant, q is the electron charge constant, T is the temperature, V_{bi} is the built-in voltage which is expressed in equation 2.20, L_{active} is the length of the active zone of the modulator, H_{guide} is the height of the waveguide, ϵ is the silicon permittivity and n_i is the intrinsic carrier concentration of the silicon, equal to 10^{10} cm^{-3} at $T=300\text{K}$.

The resistance of the PN junction is the sum of the resistances of each doped region. Each resistance segment i is expressed by equation 2.21.

$$R_i = \frac{\rho W_i}{L_{active} \cdot H_i} \quad (2.21)$$

where ρ is the resistivity, L_{active} is the active length and W_i and H_i are respectively the width and the height of the resistive segment.

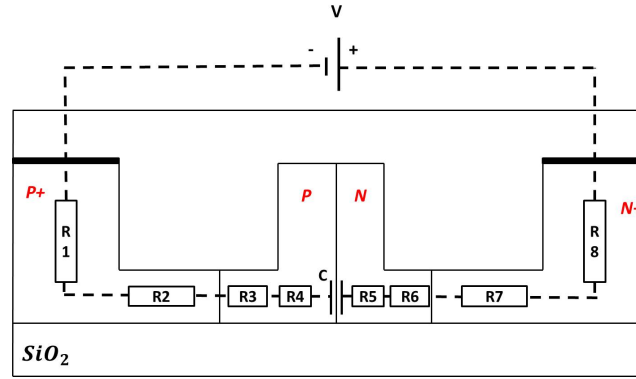


FIGURE 2.12: Electric scheme of the PN junction)

The RC equivalent model gives a first approximation of the electrical bandwidth limitation $f_{elec} = 1/2\pi RC$. When the absolute voltage increases, the width of the space charge region increases, which leads to a decrease in the capacitance and consequently an increase in the electrical bandwidth. Note that the electrical bandwidth does not depend on the active length L_{active} .

In table 2.8, figures of merit, associated with the electrical parameters of the three junctions are reported with acceptor and donor implantations of $1 \cdot 10^{18} \text{ cm}^{-3}$. The PIN junction exhibits a very slow electro-optical bandwidth (around 1 GHz) due to slow process of electrons and holes diffusion. High electro-optical bandwidth is essential for high data rate communications. With the use of pre-emphasis techniques (such as pre-feed forward equalizer (FFE)), high frequencies are pre-amplified, and compensate for the lack of electro-optical bandwidth. This technique allows 25 Gbit/s optical communication to be achieved, but at the expense of driver complexity and power consumption [52]. The capacitive junction allows better efficiency to be achieved than the PN junction at the expense of higher loss and higher capacitance, which leads to lower electrical bandwidths.

TABLE 2.8: Order of magnitude of principal figure of merits for the PN, PIN and capacitive junction with $N_A = N_D = 1 \cdot 10^{18} \text{ cm}^{-3}$

	PN junction	PIN junction	Capacitive Junction
Phase shift efficiency [$^\circ/\text{mm}$]	49	7100	62
Global Loss [dB/cm]	26	6	50
Access resistance $\Omega \cdot \mu\text{m}$	4.7	-	3.3
Capacitance fF/ μm	0.36	-	1.38
Electrical Bandwidth [GHz]	53	1	34

A summary of the state of the art of modulators will be put forward in section 2.5.4 for Mach Zehnder modulators and in section 2.5.5 for ring modulators.

2.5.4 Mach-Zehnder Interferometer Modulator

Mach-Zehnder Principle of operation

The Mach Zehnder modulator is represented in figure 2.13. The incident field is split into two waveguides and recombines with a multimode interferometer MMI, or directional coupler. In each arm, a junction architecture is placed and changes the phase of the electrical fields. At the output, destructive or constructive interference will be created and so changes the output intensity.

The input field E_{in} is split into both waveguides as electrical fields E_1 and E_2 . The electrical field in arm 1 (E_1) (resp. arm 2 (E_2)) propagates into a waveguide of length L_1 (resp. L_2) with phase constant $\beta_1 = 2\pi n_{eff1}/\lambda$ (respectively $\beta_2 = 2\pi n_{eff2}/\lambda$) and intensity attenuation constant α_1 (respectively α_2). The relation between the electric field at the input E_{in} and that before the recombiner E_{out1} (resp. E_{out2}) is given by equation 2.22.

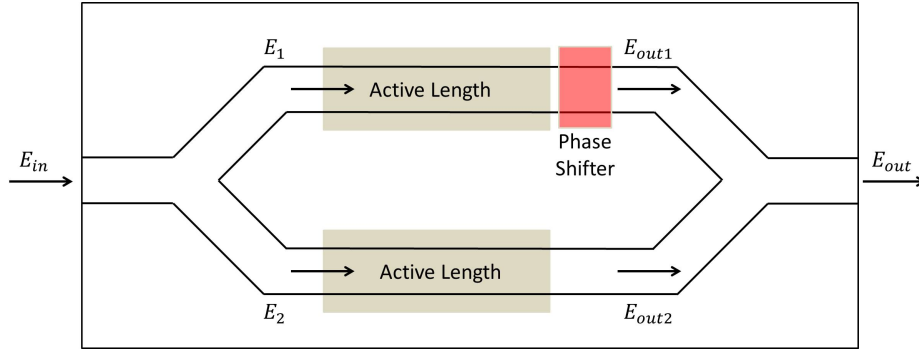


FIGURE 2.13: Scheme of the Mach Zehnder modulator

$$E_{out1} = E_{in} \frac{1}{\sqrt{2}} \cdot e^{-\frac{i2\pi n_{eff1}}{\lambda} L_1} \cdot e^{-\frac{\alpha_1}{2} L_1} \quad (2.22)$$

$$E_{out2} = E_{in} \frac{1}{\sqrt{2}} \cdot e^{-\frac{i2\pi n_{eff2}}{\lambda} L_2} \cdot e^{-\frac{\alpha_2}{2} L_2}$$

After the recombiner, the output electrical field with the associated optical intensity is given by equation 2.23.

$$E_{out} = \frac{1}{\sqrt{2}}(E_{out1} + E_{out2}) = \frac{E_{in}}{2} \left(e^{-\frac{i2\pi n_{eff1}}{\lambda} L_1} \cdot e^{-\frac{\alpha_1}{2} L_1} + e^{-\frac{i2\pi n_{eff2}}{\lambda} L_2} \cdot e^{-\frac{\alpha_2}{2} L_2} \right) \quad (2.23)$$

The MZM is said to be loss-balanced if $\alpha_1 = \alpha_2$ and length-balanced if $L_1 = L_2$. For a length-unbalanced MZM, the output intensity varies sinusoidally with the wavelength. The period is represented by the free spectral range FSR in wavelength (equation 2.24).

$$\text{FSR} = \frac{\lambda^2}{n_g |L_1 - L_2|} \quad (2.24)$$

Mainly, the **MZM** is set to be length-balanced in final broadband utilization. A loss-unbalanced **MZM** is not wanted because it introduces excess insertion loss. Hence, the two arms both have the same active length.

The output light is modulated by changing the effective index in the arms ($n_{eff1} = n_{eff} + \Delta n_{eff1}$ and $n_{eff2} = n_{eff} + \Delta n_{eff2}$ with for example a PN junction leading to equation 2.25 in field for a symmetrical **MZM** (with loss variation neglected). One notes that the optical field is modulated both in phase and in amplitude.

$$E_{out} = \frac{E_{in}}{2} \cdot e^{-\alpha \frac{L}{2}} \cdot \underbrace{\cos\left(\frac{\pi(\Delta n_{eff1} - \Delta n_{eff2})L}{\lambda}\right)}_{\text{amplitude modulation}} \cdot \underbrace{\exp\left(-\frac{i\pi(\Delta n_{eff1} + \Delta n_{eff2})L}{\lambda}\right)}_{\text{phase modulation}} \quad (2.25)$$

General Case

In the general case, the two arms are driven with two different voltages $u_1(t)$ and $u_2(t)$. Considering the effective index to be linearly related to the voltage, one can write the relation between the effective index variation and the driving voltage by equation 2.26.

$$\Delta n_{eff} = \frac{\lambda \cdot u(t)}{2V_{\pi}L} \quad (2.26)$$

With this assumption, equation 2.25 is rewritten in equation 2.27.

$$\begin{aligned} E_{out} &= \frac{E_{in}}{2} \cdot e^{-\alpha \frac{L}{2}} \cdot \cos\left(\frac{\pi(u_1(t) - u_2(t))}{2V_{\pi}}\right) \cdot \exp\left(-\frac{i\pi(u_1(t) + u_2(t))}{2V_{\pi}}\right) \\ &= \frac{E_{in}}{2} \cdot e^{-\alpha \frac{L}{2}} \cdot \cos\left(\frac{\pi(V_1 - V_2) \cdot u(t)}{4V_{\pi}} + \frac{\pi V_{bias}}{2V_{\pi}}\right) \cdot \exp\left(-\frac{i\pi(V_1 + V_2)u(t)}{4V_{\pi}}\right) \end{aligned} \quad (2.27)$$

where $u_1(t)$ and $u_2(t)$ are the voltage signals applied on the upper and bottom phase modulation sections. We assume that $u_1(t) = V_1 \cdot u(t)/2 + V_{bias}/2$ and $u_2(t) = V_2 \cdot u(t)/2 - V_{bias}/2$.

In order to set the Mach Zehnder modulator in a linear regime, it has to be biased at the quadrature point $\pi/2$. At the quadrature point, the intensity modulation and the phase modulation are shown in equation 2.28.

$$\begin{aligned} I_{out}(t) &= \frac{I_{in}}{2} \cdot e^{-\alpha L} \cdot (1 - \sin\left(\frac{\pi(V_1 - V_2) \cdot u(t)}{2V_{\pi}}\right)) \\ \phi(t) &= \frac{\pi(V_1 + V_2) \cdot u(t)}{4V_{\pi}} \end{aligned} \quad (2.28)$$

where $I_{out}(t) = |E_{out}(t)|^2$ is the output intensity and $\phi(t) = \text{Arg}(E_{out})$ is the output phase.

The quadrature point can be achieved with a bias voltage $V_{bias} = V_{\pi}/2$ but also with a phase shifter in one arm, which brings a $\pi/2$ phase shift. Figure 2.14 represents the optical intensity and the phase as a function of the effective variation Δn_{eff1} with active length of 4 mm, at $\lambda = 1.31 \mu\text{m}$ and $V_2 = 0$ (one arm modulated)). In this case, the required voltage V_1 to achieve V_{π} is 6.25 V, with a $V_{\pi}L_{\pi}$ of 2.5V.cm.

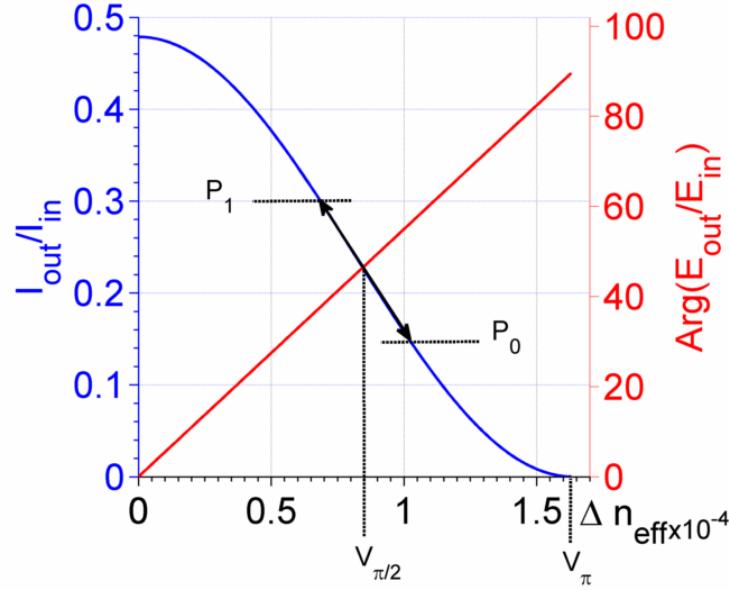


FIGURE 2.14: Transfer Function of one arm modulated MZM against Δn_{eff} with $L=4\text{mm}$ and $\lambda = 1.31\mu\text{m}$

The residual phase modulation is undesirable. It will impact the transmission performance after a few kilometers in a dispersive optical fiber. The chirp parameter α_{MZM} assesses the chirp impact in an optical transmission. It is defined by [54] and expressed in equation 2.29. In a case of sinusoidal voltages, the chirp parameter is expressed as a function of V_1 and V_2 in equation 2.30.

$$\alpha_{MZM} = \frac{\partial \phi / \partial t}{\partial I / \partial t} \cdot 2I \quad (2.29)$$

$$\alpha_{MZM} = \frac{V_2 - V_1}{V_1 + V_2} \quad (2.30)$$

Two different operating configurations can be identified : **MZM** modulated with one single arm and the dual drive push-pull **MZM**. For **MZM** modulated with one single arm, V_2 equals 0 and α_{MZM} tends to -1. For dual drive push-pull **MZM**, V_2 equals $-V_1$ and the chirp parameter α_{MZM} equals 0.

Until now, we have considered the phase modulation due to the . But, for the PN junction, the absorption coefficient variation also impact the phase modulation. In fact, this variation leads to a variation of the intensity coefficient α in equation 2.23 creating a phase modulation. In [55], the authors measured the residual chirp due to PN junction at 1545.4 nm and 20 Gbit/s and concluded that its effect is negligible for a few kilometers of propagation. As both arms are modulated in opposite directions, the dual drive push pull (**DDPPMZM**) is twice as efficient than the single arm modulated **MZM** ; or it can be designed with half the length to produce the same efficiency. On the other hand, it requires two driving voltages rather than one.

In [56,57], the authors used another architecture called a single-drive-push-pull architecture (SDPPMZM). This architecture connects two PN junctions in the center of the Mach-Zehnder. The DC voltage is applied in the center of the P doping region and the voltage swing is applied to the two N doping regions (figure 2.15). The two active regions are in parallel and observe opposite voltage swings. This architecture reduces the electrical drive power while preserving the push-pull configuration.

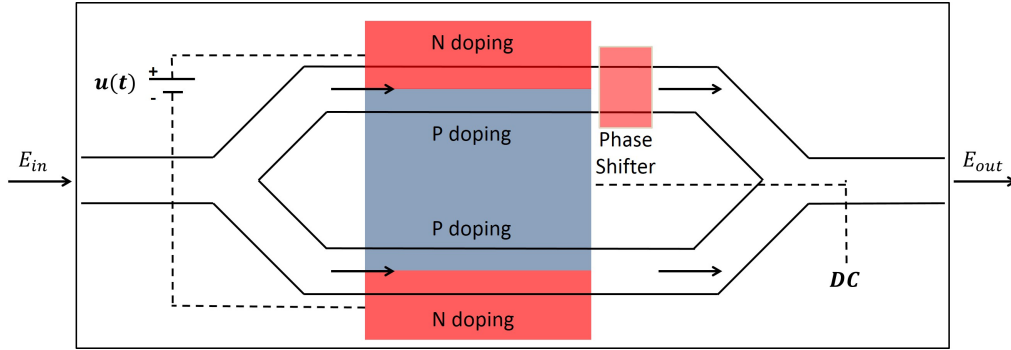


FIGURE 2.15: Scheme of the single drive push pull mach zehnder modulator

Electro-optical bandwidth

A travelling wave Mach Zehnder modulator (TW-MZM) has a junction length of a few millimetres (between 1 to 6 mm). Its electrodes can not be considered to be lumped, as electrical microwaves propagate along the electrode. The PN junction limitation in section 2.5.3 is not sufficient to describe the electrical limitation of the MZM. Radio frequency (RF) simulations have to be carried out to obtain the electrical microwave limitations (with for example HFSS software). The electro-optical bandwidth of TW-MZM is limited by three factors:

- the microwave attenuation of electrode transmission lines. It should be as low as possible.
- the microwave group velocity mismatch with optical group velocity.
- the impedance mismatch of the electrode transmission lines with the driver and termination impedance. Back reflections will be created for poor impedance matching.

The S-parameters of the various transmission lines architectures are intensively studied in many papers [57–60]. Bad electrode design will drastically reduce the electro-optical limitation. For example in [61], the electro-optical response has notches due to unwanted microwave mode propagation, which reduce the electrical bandwidth. Slow wave electrodes are also under analysis to achieve a better match between group velocities [62]. As RF electrodes design is a complex study, the authors in [63–65] also use segmented electrodes (S-MZM). This allows multiple lumped elements to be considered instead of transmission lines. This architecture requires multiple discrete drivers for each segment. Today, state of the art approaches use driver integration to control and modulate the MZM [63,66–68]. In [68], Temporiti et al. carried out the only link demonstration at 50 Gbit/s with push pull MZM and flip-chipped driver.

Mach-Zehnder modulator state-of-the-art

Table 2.9 details the state-of-art of silicon Mach Zehnder modulators. Various driving configurations are reported : travelling wave or lumped ; single drive or dual drive or segmented drive; push pull or not ([TW-DDPPMZM](#),[TW-SDPPMZM](#),[L-DDPPMZM](#)). With respect to the figures in the papers, when insertion loss is reported without voltage bias, we add 3 dB to set the [MZM](#) at the quadrature point. All demonstrations are made in [ASK](#) without [DSP](#). The targeted [BER](#) are not reported for all demonstrations, only open eye diagrams are shown.

TABLE 2.9: Summary of the state-of-art of silicon mach zehnder modulators. * are with drivers

	Architectures	Junction Type	$V_{\pi}L_{\pi}$ [V.cm] (at V)	IL [dB] (Length)	3 dB EO-BW [GHz]	Max bitrate [GBaud]	ER [dB] (V_{pp})
Carrier accumulation							
Webster [53] 2014	L-DD PPMZM	SISCAP	<0.2 (-)	5.6 (0.4mm)	-	40	8 ($1V_{pp}$)
Carrier injection							
Baba [69] 2015	L-DD PPMZM	PIN	0.31 (120 μ A)	3.5 (0.060mm)	17 (120 μ A)	25	2.7 ($2.8V_{pp}$)
Carrier depletion							
Marris-Morini [70] 2014	TW-SD MZM	PN	2.2 (-)	7.9 (0.95mm)	26 (-)	40	- ($7V_{pp}$)
Ding [58] 2014	TW-DD PPMZM	PN	2.62 (1V)	7.5 (3.5mm)	27 (1V)	40	4 ($1.6V_{pp}$)
Xu [61] 2014	TW-SD MZM	PN	1.8 (0V)	5.8 (1mm)	30 (3V)	40	4.1 ($3.5V_{pp}$)
Patel [57] 2015	TW-SD PPMZM	PN	2.8 (7V)	6.8 (4.25mm)	38 (2V)	60	3.8 ($4.8V_{pp}$)
Samani [59] 2015	TW-SD PPMZM	PN	3.15 (-)	7.5 (4.2mm)	21 (-)	55	4.3 (-)
Ferroti [60] [71] 2016	TW-SD PPMZM	PN	2.4 (2.5)	6.3 (4mm)	16 (1.25V)	25	4.7 ($2.5V_{pp}$)
Temporiti [68] [72] 2016	TW-DD PPMZM	PN	2.5 (2.5V)	4.89 (3.15mm)	-	56	2.5 ($1.6V_{pp}$)

2.5.5 Ring Resonator Modulator

Ring Resonator Modulator Principle operation

The second silicon photonic modulator is the ring resonator modulator (RRM). Its structure is identical to the ring resonator filter (section 2.3.3) with an active length inside the ring. Figure 2.16 represents a schematic view of the single-bus and two-bus RRM. The modulated output signal is at the through port. The drop port allows the operating point to be controlled against temperature and fabrication deviations. Since an active junction is placed inside the ring, an electro-optical effect can be produced by a modulating voltage V_m along the active length L_{active} and thus resonant wavelengths are shifted. In order not to change the coupling coefficient between the buses and the ring, the active length does not overlap with the coupling zones.

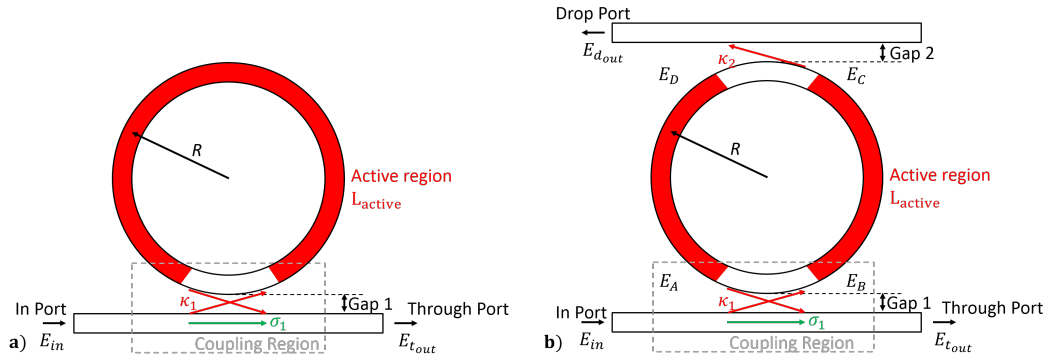


FIGURE 2.16: a)One-bus RRM Scheme b)Two-bus RRM Scheme

The effective index and the absorption coefficient vary with the voltage and impact the phase shift $\phi(V_m)$ and the field attenuation $a(V_m)$ along one trip around the ring. Their expressions are given by equations 2.31 and 2.32. Equations 2.6 and 2.7 of the steady state model of ring resonator remain valid for the RRM with a voltage dependency.

$$a(V_m, \lambda) = \frac{\alpha_0}{2} \cdot 2\pi R + \Delta\alpha(V_m) \cdot L_{active} \quad (2.31)$$

$$\phi(V_m) = \frac{2\pi n_{eff}(\lambda, V_0)}{\lambda} \cdot 2\pi R + \frac{2\pi \Delta n_{eff}(V_m)}{\lambda} \cdot L_{active} \quad (2.32)$$

where R is the radius, $n_{eff}(\lambda, V_0) = n_g + \lambda \cdot \frac{\partial n_{eff}}{\partial \lambda}$ is the effective index of the optical mode at V_0 , $\Delta n_{eff}(V_m)$ is the effective variation with the voltage applied to the active junction, α_0 is the intensity absorption coefficient and $\Delta\alpha(V_m)$ is the variation of the absorption coefficient against the voltage on the active junction. These parameters depend on the junction definition as explained in section 2.5.1.

The steady state model can be simplified by considering that the wavelength is close to the resonant wavelength ($|\lambda_{laser} - \lambda_{res}| < FSR/2$). Further, the transfer function of the ring modulator can be approximated by a Lorentzian function (Equation 2.33) [73].

$$T_{out}(\lambda) = 1 - \left(1 - \frac{1}{ER_{on-off}}\right) \cdot \frac{1}{1 + 4 \cdot \left(Q \cdot \frac{\lambda - \lambda_{res}}{\lambda_{res}}\right)^2} \quad (2.33)$$

where ER_{on-off} is the maximum extinction ratio between a complete on and off resonance.

For ring modulators, the efficiency of PN junction is also expressed in pm/V which expresses how much the resonance moves depending on the applied voltage. This value does not allow comparison between [RRMs](#) with different [FSR](#).

Figure 2.17 represents the transfer function of a ring modulator at the through port for two voltage values. One can note that the insertion loss and the extinction ratio will depend on the relative position between the resonant wavelength and the laser wavelength. As this component is the main component used in the manuscript, chapter 3 is dedicated to the ring modulator models allowing comparison between measurements and simulations in chapter 4.

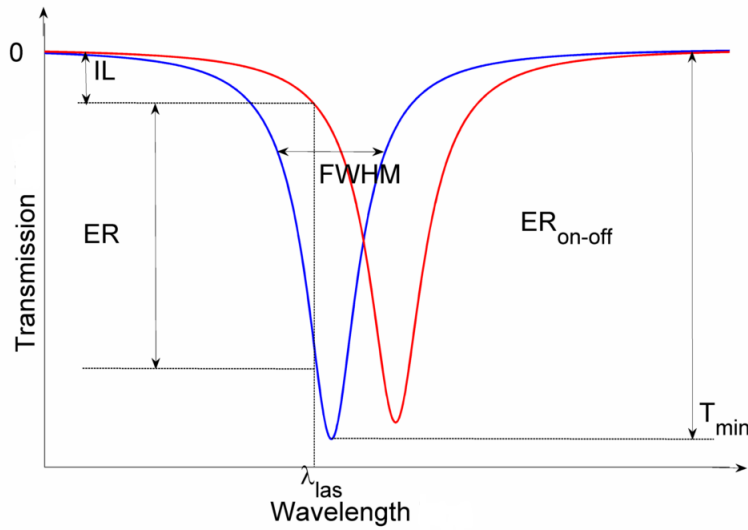


FIGURE 2.17: Optical transmission spectrum for two voltage V_0 and V_m

Electro-optical bandwidth

The ring resonator modulator is a lumped component. It is mainly limited by two contributions: an electrical contribution due to the junction architecture (fully detailed in section 2.5.3) and an optical contribution due to the photon lifetime τ_p . The photon lifetime is defined as the time needed for intensity (number of photon) to decay by a factor 1/e. Considering that the RRM is set to the critical coupling and the laser wavelength is set to the resonant wavelength, the optical bandwidth is approximated by the resonant linewidth, given by equation 2.34 [74].

$$f_{opt} = \frac{1}{2\pi\tau_p} = \frac{c}{Q \cdot \lambda} \quad (2.34)$$

where c is the speed of light in a vacuum. Through equation 2.34, we can observe that high Q factor directly reduces the optical bandwidth. Figure 2.18 reports the optical bandwidth against the Q factor at 1.31 μm and 1.55 μm . This approximation gives a first evaluation of the optical bandwidth but it has to be completed by a transient simulation of the RRM (see chapter 3).

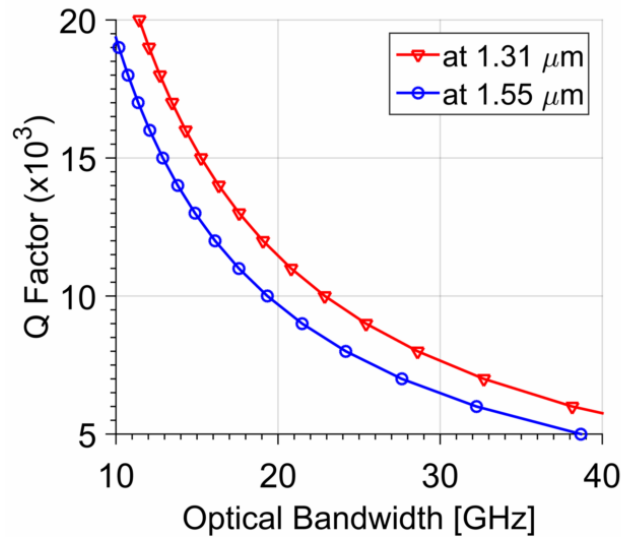


FIGURE 2.18: Q factor against optical bandwidth at 1.31 μm and 1.55 μm

Ring Resonator Modulator State of the art

Table 2.10 details the state-of-art of ring resonator modulators. This table highlights ring modulators with achieved data rate higher than 40 Gbaud. A very good performance was obtained by Pantouvaki et al. in [21] with the first demonstration at 56 Gb/s. Today, RRM s at lower data rates (between 10 to 25 Gbaud) are implemented inside complete electro-optical chip with WDM implementation [40,75].

TABLE 2.10: Summary of the state-of-art of ring resonator modulators (* with pre-emphasis driving voltages)

	Junction Type	Radius [μm] (FSR)	Q factor	$\Delta\lambda$ ($V_\pi L_\pi$) [V.cm]	IL [dB]	EO - BW [GHz]	Max bitrate [GBaud]	ER [dB] (V_{pp})
Carrier injection								
Baba [76] 2013	PIN	eq.10.7 (8.46nm)	4300	1.15nmV ⁻¹ (0.001)	5.2	1 (0.92V)	50*	4.58 (1.87)
Carrier depletion								
Xiao [77] 2012	Zigzag PN	eq.10.6 (-)	8015	16pmV ⁻¹ (1.7)	-	25 (-)	40	3.01 (3)
Li [78] 2012	PN	5 (-)	4000	28pmV ⁻¹ (-)	-	28 (1V)	40	7 (2)
Malacarne [79] 2014	PN	30 (3.25nm)	7000	10pmV ⁻¹ (3)	2	15 (0V)	30	9.4 (3.75)
Timurdogan [80] 2014	Vertical PN	4.8 (-)	6500	250pmV ⁻¹ (-)	0.92	21 (0V)	44	8 (2.2)
Pantouvaki [21] 2015	PN	5 (19.3nm)	3500	62.9pmV ⁻¹ (0.48)	2.4	38 (0V)	56	4 (2.5)

2.6 Evaluation of silicon photonics transmitter

In the previous sections, we detailed the principal silicon photonics components needed for a transceiver with their associated performances (silicon waveguides, fiber to silicon couplers, splitters, combiners, wavelength multiplexers, ring resonator filters, lateral germanium photo-detectors, Mach-Zehnder and ring resonator modulators). In the manuscript, transceivers will be based on ring resonator modulators and filters.

In the following section, we detail a case study of an optical transmitter represented in figure 2.19. The goal is to evaluate a preliminary optical design with silicon photonics components in order to be compliant with the IEEE 400GBASE-FR8 standard. This study will be used as preliminary RRM requirement specifications.

As a reminder, this IEEE 400GBASE-FR8 standard targets an optical link transmission at 400 GBit/s with optical fiber length of 2 km. In the worst case, the optical modulation amplitude OMA has to be at least 0 dBm with a minimum extinction ratio of 4.5 dB at the output of the transmitter (see section 1.2.1 for details concerning the standard).

The considered transmitter is composed of eight ring resonator modulators arranged in series, where each modulates a different wavelength. Each ring resonator modulator is PAM-4 modulated at 25 Gbaud. The transmitter achieves an aggregated bit rate of $8 \cdot 2 \cdot 25 = 400$ Gbit/s.

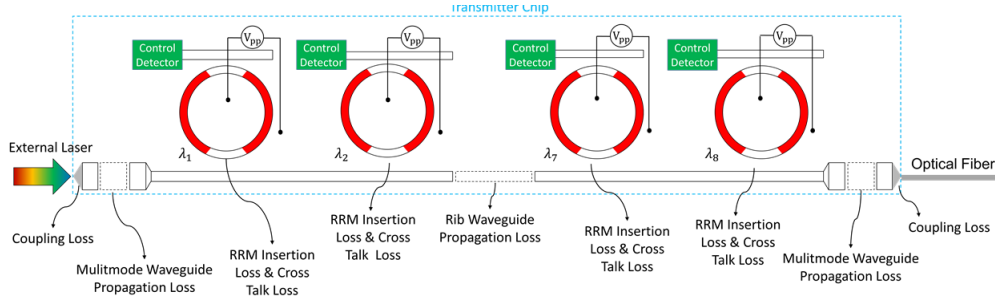


FIGURE 2.19: Schematic of the considered optical transmitter made from silicon photonic components essentially ring resonators modulators

In this part, we focus the study on the transmitter part. The goal is to determine the acceptable insertion loss IL_{RRM} and extinction ratio ER_{RRM} of the ring resonator modulators in order to achieve at least an OMA of 0 dBm at the output of the transmitter. The optical budget is reported in table 2.11. The laser input is set to 10 dBm per wavelength. The incurred total loss (*i.e.* input-output coupling loss + propagation loss + packing loss) inside the optical transmitter are evaluated at 5.7 dB. Multimode waveguides are used to connect input (output) grating coupler to the input (output) of the 8 ring resonator modulators. According to our experience and due to building and experimental issues, 3 centimeters of multimode waveguide are necessary between the optical fibers and the electrical connections.

The lowest optical power level at the output of the transmitter is defined as P_{00} and the highest optical power level as P_{11} . The OMA at the output of the transmitter is the difference between P_{11} and P_{00} . It depends on the insertion loss and the extinction ratio of the ring resonator modulators. No impairments due to electro-optical bandwidth limitation or signal distortion of the ring resonator modulators are considered. In this study, we consider that the adjacent RRM s do not affect themselves. There are enough wavelength spaced, *i.e.* the FSR is sufficient [40,75,81,82].

Figure 2.20 represents the OMA at the output of the transmitter against the insertion loss IL_{RRM} and the extinction ratio ER_{RRM} . RRM insertion loss lower than -2 dB matches the target 0 dBm OMA at the output of the transmitter. Further, with -3 dB insertion loss, the extinction ratio has to be higher than 8 dB. This transmitter study will be the basis of the RRM performance optimization in chapter 5. Signal distortion and bandwidth limitation will also be evaluated with transient ring resonator modulator simulations using the complete model described in chapter 3.2.

TABLE 2.11: Transmitter optical power budget

	Loss [dB]	Optical Power [dBm]
External Laser Output Power		10
Input Grating Coupler Loss	2	8
RRM Insertion loss	IL_{RRM}	$8-IL_{RRM}$
Propagation loss (3 cm multimode-waveguide)	0.7	$7.3-IL_{RRM}$
Output Grating Coupler Loss	2	$5.3-IL_{RRM}$
Packaging Loss	1	$4.3-IL_{RRM}$
Max level 11 P_{11}		$4.3-IL_{RRM}$
Min level 00 P_{00}		$4.3-IL_{RRM}-ER_{RRM}$
OMA Transmitter		$10\log_{10}(P_{11}[mW] - P_{00}[mW])$

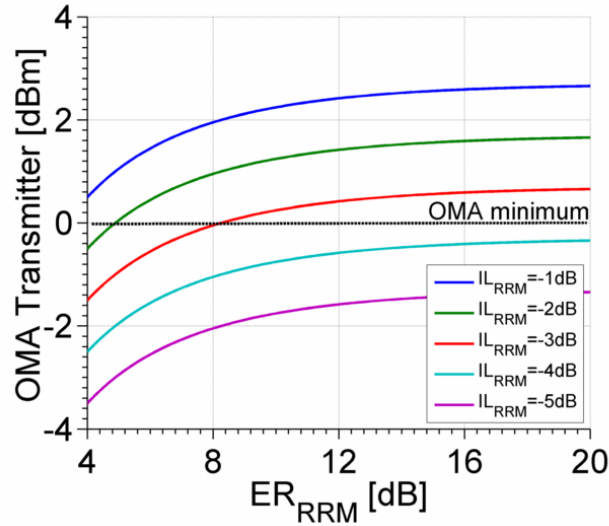


FIGURE 2.20: OMA at the output of the transmitter against the insertion loss and the extinction ratio of the RRM with input laser power of 10 dBm and 6.3 dB coupling and propagation loss

2.7 Conclusion

The basic models for the building block of an optical transceiver with silicon photonic components were presented in this chapter. They are silicon waveguides, couplers, splitters and combiners, wavelength (de)-multiplexers, photo-detectors and electro-optical modulators.

On the transmitter side, the electro-optical modulator is the most important component, that converts electrical data into optical data. The optical modulation is made with electro-refraction

principles, converting refractive index modulation (with PIN, capacitive, or PN junction) into phase modulation. The optical intensity modulation is then obtained by implementing the active region within interferometer structures (*e.g.* mach zehnder modulator or ring resonator modulator).

The Mach Zehnder modulator is the most common modulator. This modulator achieves today interesting performance levels. Transmission at 56 Gbit/s with 1.6 V_{pp} push-pull driving was demonstrated [68]. However, the ring resonator modulator appears as a very promising modulator. It has two main advantages: a very small size (about 10 μm radius), a sharp transfer function which allows WDM with low cross-talk penalty to be made with RRM's arranged in series. It achieves today high performance levels, with a first demonstration at 56 Gbit/s with 2.5 V_{pp} driving voltage [21]. This is why, the RRM is modulator studied in this thesis.

The figures of merit defining the RRM performance levels are the extinction ratio, the insertion loss and the electro-optical bandwidth. At a fixed driving voltage, these three parameters depend essentially on the Q factor and on the laser wavelength position with respect to the RRM resonant wavelength.

A first optical budget of an optical transmitter was defined. Without taking into account signal distortion and cross-talk, the RRM needs to achieve 8 dB of extinction ratio and less than 3 dB of insertion loss, allowing the 400GBASE-FR8 to be respected (OMA higher than 0 dBm at the output of the transmitter). We will develop a transient RRM model allowing us to take into account signal distortion, cross-talk and bandwidth limitation. The RRM will be then studied to transmit PAM-4 data at 25 Gbaud and PAM-2 data at 50 Gbaud .

Chapter 3

Silicon Ring Resonator Modulator

3.1 Introduction

This chapter is divided into two parts. The first part describes the transient model of the ring resonator modulator. This model allows the simulation of [RRM](#) limitations due to electro-optical bandwidth limitation, signal distortion and chirp effect.

The second part is dedicated to [RRM](#) tuning, in order to set the [RRM](#) to the correct operating point. Control loops with heaters will be discussed.

3.2 Ring resonator modulator transient model

As seen in section [2.5.5](#), the ring resonator modulator behavior results from combination of an electrical contribution due to the PN junction and an optical contribution due to the resonator. In this section, we will first describe the transient optical model, and then we will take the electrical contribution into considerations.

3.2.1 Optical Model

Two different transient models of [RRM](#) are reported in the literature. The first is based on the energy amplitude resolution of the resonator [83] and the second is based on coupled mode resolution [84]. We have chosen to use the second approach for reasons of flexibility. In fact, while the energy amplitude resolution approach is less time consuming, it can only resolve simple ring modulator. The coupled mode based modeling approach allows the non linearity, photon lifetime and chirp modulation of the [RRM](#) to be taken into account.

In the following, the transient model based on the coupled mode resolution is described. The time resolution step is defined by the time taken by the light to propagate over halfway around the ring $\delta t = n_g \cdot 2\pi R / (2 \cdot c)$, where n_g is the group index, R the ring radius and c the speed of light. As the voltage V_m is time dependent, the effective index $n_{eff}(V_m(t))$ and the absorption coefficient $\alpha(V_m(t))$ also depend on time. Equations [2.32](#) and [2.31](#) are reused for the transient model by adding this time dependency. The electromagnetic field at the through port $E_{t_{out}}$ is related to the electromagnetic fields inside the ring E_A, E_B, E_C, E_D as described in figure [3.1](#) b) according to equations [3.1](#), [3.2](#), [3.3](#).

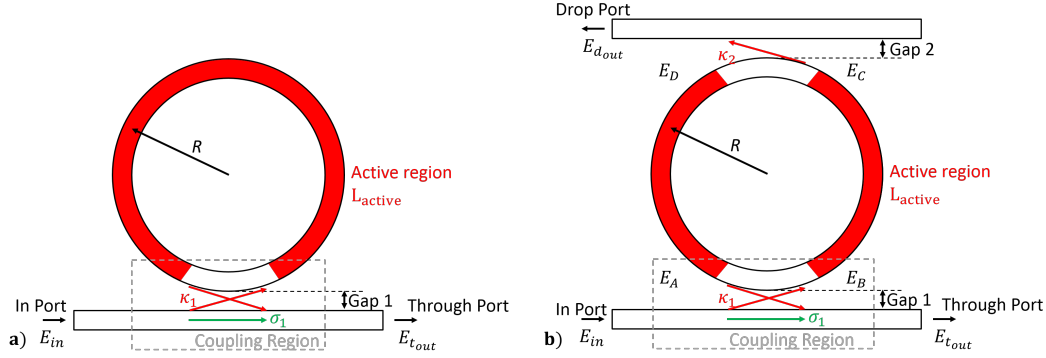


FIGURE 3.1: a) Single-bus RRM Scheme b) Two-bus RRM Scheme

$$\begin{cases} E_A(t) = e^{-\frac{\alpha(t-\delta t)}{2}} \cdot e^{-i \cdot \frac{\phi(t-\delta t)}{2}} \cdot E_D(t - \delta t) \\ E_B(t) = -i\kappa_1 \cdot E_{in} + \sigma_1 \cdot E_A(t) \\ E_C(t) = e^{-\frac{\alpha(t-\delta t)}{2}} \cdot e^{-i \cdot \frac{\phi(t-\delta t)}{2}} \cdot E_B(t - \delta t) \\ E_D(t) = \sigma_2 \cdot E_C(t) \end{cases} \quad (3.1)$$

where κ_1 and κ_2 are the amplitude coupling coefficient at the through port and drop port, and σ_1 and σ_2 are the amplitude transmission coefficients at the through port and drop port, respectively.

$$E_{t_{out}}(t) = -i\kappa_1 \cdot E_A(t) + \sigma_1 \cdot E_{in}(t) \quad (3.2)$$

$$E_{d_{out}}(t) = -i\kappa_2 \cdot E_C(t) \quad (3.3)$$

The propagation time within one-half ring, δt , is significantly lower than the modulation time. For example, a ring with a 10 μm radius and a group index of 4.2, the propagation time is 0.1 ps which is significantly lower than the modulation time of 20 ps for 50 Gbit/s data rate. Under this condition, the transient simulation result is obtained by iterative resolution of the system 3.1 at each $t = t + \delta t$. In addition, between each δt a slight variation of the optical fields is considered allowing the validation of the resolution. Note that the resolution of the system 3.1 stays valid for the single-bus ring resonator modulator with $\sigma_2 = 1$.

3.2.2 Comparison between approximated optical bandwidth and simulated optical bandwidth

The optical bandwidth can be simulated with the previously reported transient model (Eq. 3.1, 3.2, 3.3) with small-signals simulations. An ideal sinusoidal voltage signal is applied to the modulator. For each frequency, the OMA is simulated. The optical bandwidth is defined as the frequency at which the OMA is reduced by -3 dB from its value at low frequencies. The optical bandwidth depends on the relative position of the laser wavelength with the resonant wavelength of the RRM, but also depends on the peak-to-peak voltage applied to the modulator.

We simulate the optical bandwidth for different peak-to-peak voltage values and different laser wavelength positions. The simulation setup uses a ring resonator modulator with typical characteristics as described in table 3.1. Figure 3.2 a) reports the transfer function of the ring modulator. In this simulation, the laser wavelength is shifted from 40 pm to the resonant wavelength within the range $(\lambda_{lasermax}; \lambda_{lasermin})$. Figure 3.2 b) shows the OMA variation with frequency for different peak-to-peak voltage values. For this figure, the wavelength of the laser is set to the resonant wavelength when no signal is applied. We can notice that a small peak to peak voltage reduces the optical bandwidth. Figure 3.3 a) and b) reports the optical bandwidth against the position of the laser wavelength with the corresponding OMA at low frequency. The optical bandwidth increases when the separation between the laser wavelength and the resonant wavelength increases.

In the literature, to evaluate the optical bandwidth it is common to use the following approximation, linked to the Q factor, $f_{opt} = c/(Q \cdot \lambda)$. However, this approximation corresponds to the worst case: low peak-to-peak voltage and $\lambda_{las} = \lambda_{res}$. For this ring modulator, the Q factor is 13000 which corresponds to an optical bandwidth approximation of 17.6 GHz.

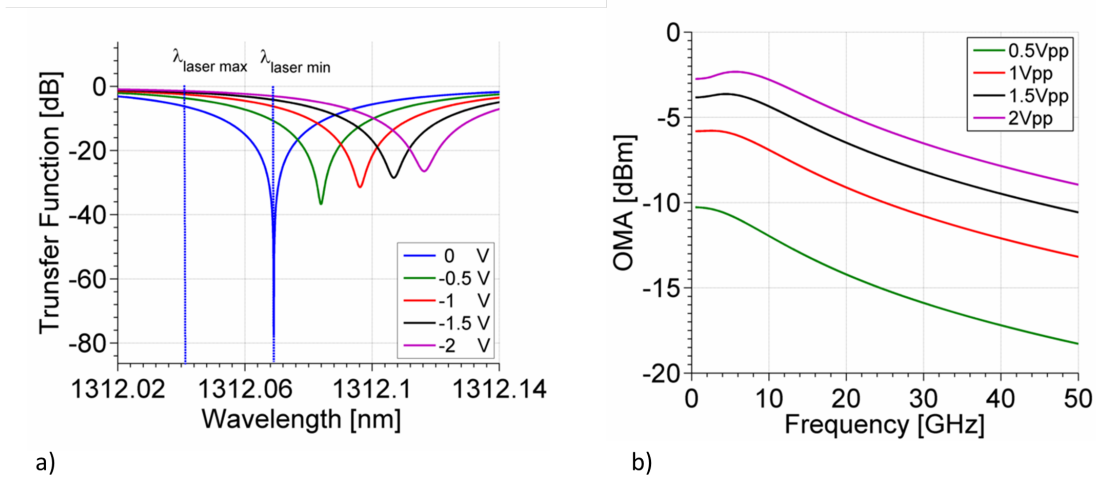


FIGURE 3.2: a) Transfer Function of the ring resonator modulator and position of laser b) Optical Modulation Amplitude against the frequency for different peak to peak voltage ($\lambda_{laser} = \lambda_{res}$)

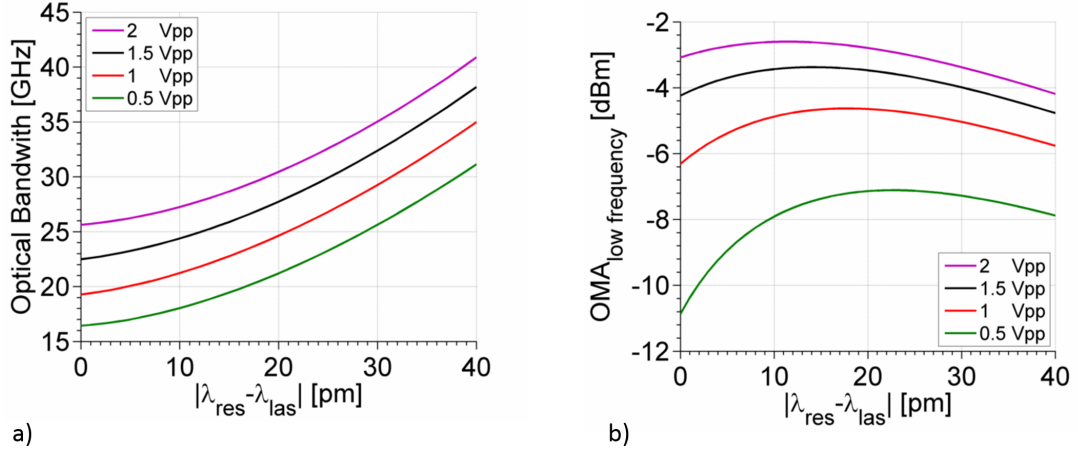


FIGURE 3.3: a) Optical bandwidth against the relative position of the laser wavelength and the resonant wavelength for several peak to peak voltage b) Corresponding low frequency OMA against the relative position of the laser wavelength and the resonant wavelength

TABLE 3.1: Ring Resonator Parameters

Radius [μm]	10	Losses [dB/cm]	23.7
Silicon Waveguide Type	W230/H300/M50	PN Efficiency [$^\circ/\text{mm}$] ([0V,-2V])	47.4
Coupling Regime	Critical Coupling	PN Efficiency [$\text{V}\cdot\text{cm}$] ([0V,-2V])	0.76
λ_{ref} [nm]	1310	Effective index variation $\frac{\partial n_{\text{eff}}}{\partial V}$ [1/V]	$1.07 \cdot 10^{-5}$
$n_{\text{eff}}(\lambda_{\text{ref}})$	2.613	Loss variation $\frac{\partial \alpha}{\partial V}$ [m^{-1}/V]	28.29
n_g	4.26		

3.2.3 Electrical Limitation

To complete the ring resonator modulator model, an electrical limitation has to be added. As seen in section 2.5.3, the PN junction can be modeled by an R-C circuit. Hence, a low-pass filter with an electrical pole equal to $f_{\text{elec}} = 1/2\pi RC$ is applied to the driving voltage where R is the access resistance and C is the junction capacitance. The driving voltage result is then an input for the optical ring resonator resolution as shown in figure 3.4.

In order to obtain rapidly a first approximation of the electro-optical bandwidth, one can use the approximation given by equation 3.4. For a ring resonator modulator, the electrical bandwidth is in most cases higher than the optical bandwidth (higher than 25 GHz) and is thus not the limiting factor. This ring resonator modulator model will be used in the next sections for RRM design and transmitter design.

$$f_{3\text{dB}} = \frac{f_{\text{opt}} \cdot f_{\text{elec}}}{\sqrt{f_{\text{opt}}^2 + f_{\text{elec}}^2}} \quad (3.4)$$

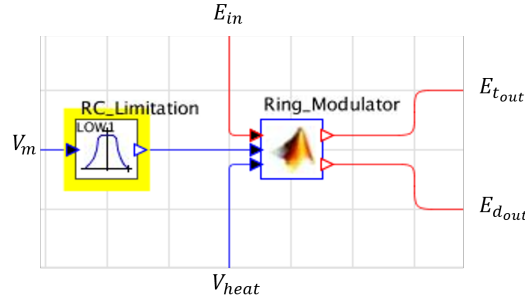


FIGURE 3.4: Silicon Ring Resonator Modulator Scheme Simulation

3.3 Ring resonator modulator tuning

The main issue of the ring resonator modulator is to maintain the position of its resonant wavelength in order to fix the operating point and to maintain the optical communication. Indeed, the resonant wavelength drifts with the temperature and changes with process variability. For a ring resonator modulator with 300 nm rib height, 50 nm slab height and 320 nm waveguide width (300/50/320 nm), the drift of the resonant wavelength due to the temperature variation is evaluated at 0.1 nm/°C. The amount of the resonant wavelength variation due to process variation is estimated between 1.1 to 0.67 nm for waveguide dimension variation of 1 nm. Table 3.2 summarizes simulated variation of the resonant wavelength against temperature variation, rib and slab height variations and waveguide width variation with FDTD simulations.

We can evaluate the standard deviation of rib height, slab height and waveguide width as 10 nm. This leads to a resonant wavelength standard deviation due to process variation of 14 nm [85]. This means that it will be extremely difficult to obtain identical RRM s at the wafer scale. For the temperature variation, the optical chip can be subjected to a maximum variation between 30°C and 40°C leading to a standard deviation of the resonant wavelength of 3 to 4 nm. This study shows that the resonator is clearly unstable and needs a feedback loop controller to maintain the wavelength response of the ring resonator aligned to the emission wavelength of the laser. The RRM monitoring has to be carried out regularly as the temperature can drift in a few milliseconds.

TABLE 3.2: Resonant wavelength variation against temperature and process variabilities

300/50/320 nm	Temperature	Rib Height	Slab Height	Waveguide width
λ_{res} variation	0.1 nm/°C	1.1 nm/nm	0.67 nm/nm	0.75 nm/nm

3.3.1 Heater

The easiest way to control the position of the resonant wavelength is to use a heater section placed near the RRM. It consists of a metallic or a doped section where the electrical power flowing into the section is converted to thermal power through Joule effect, and heats the ring resonator. As seen in section 2.2, a temperature increase results in a linear change in

the effective index, and moves the wavelength resonance to a higher wavelength. This effect is represented by the multiplication between $\partial n_{eff}/\partial T$ and $\partial T/\partial P_{heat}$ where P_{heat} is the electrical power flowing into the heater section. A temperature dissipation time constant can be evaluated between $\tau_{heat} = 10$ and $100\mu s$. Equation 3.5 describes the effective index variation according to the heater parameters.

$$\Delta n_{eff}(V_{heat}(t)) = \frac{\partial n_{eff}}{\partial T} \cdot \frac{\partial T}{\partial P_{heat}} \cdot P_{heat}(t) \cdot (1 - e^{-\frac{t-t_0}{\tau_{heat}}}) \quad \text{for } t > t_0 \quad (3.5)$$

where the electrical power consumption is $P_{heat}(t) = \frac{V_{heat}(t)^2}{R_{heat}}$.

With modal simulation (section 2.2), the parameter $\partial n_{eff}/\partial T$ is evaluated, and with numerical finite element temperature resolution the parameter $\partial T/\partial P_{heat}$ is obtained (for example with Comsol software simulations) [86]. One can note that the latter parameter is very dependent on the ring resonator structure and the exact position of the heater, and more specifically inside or above the ring. The wavelength variation due to the electrical power into the heater can be measured : $\partial \lambda/\partial P_{heat}$ generally expressed in nm/mW. The heater efficiency can also be given in mW/FSR. This coefficient is calculated with equation 3.6 related to $\partial P_{heat}/\partial \lambda$. Typical values of heater efficiency are around 50 mW/FSR.

$$\text{Efficiency [mW/FSR]} = \frac{\partial P_{heat}}{\partial \lambda} \cdot \text{FSR} \cdot 10^{-3} \quad (3.6)$$

3.3.2 Feedback loops

The control of the resonant wavelength is achieved with the heater. A feedback loop is necessary to adjust in real time the voltage applied across the heater. This controller can be done in two different ways, as represented in figure 3.5, depending on the presence of the drop port. For the two-bus RRM, a photo-detector is connected to the drop port output and measures the modulated optical power or the average optical power. An electronic chip then takes a decision to increase or to decrease the voltage applied to the heater. In the same way, for the single-bus RRM, the decision is made with the measure of the optical output power at the through port retrieved by a coupler and measured by a photo-detector. This structure leads to excess loss around -0.22 dB by an additional 95/5 % coupler. It is not adapted to a WDM transmitter where RRM are placed in series. Indeed, the photo detector will mix all wavelengths, making feedback control impossible. In this case, the solution is to tune each RRM in turn, switching off all undesired laser wavelengths.

The best solution is to use a drop port, which does not add excess loss and, as it is selective in wavelength, does not lead to impairments in a WDM transmitter. One can note that the junction length is smaller for the two-bus RRM than for the single-bus RRM as represented in figure 3.5. This reduction avoids interfering with the active junction in the coupling zone. We can wonder then if the single-bus RRM has better performance than the two-bus RRM. A performance comparison is reported in the following section 3.3.4.

Many authors have chosen the drop port structure to actively control the ring resonator modulator wavelength position. The first feedback loop controller is based on the comparison between the measured average optical power at the drop port with an average optical power

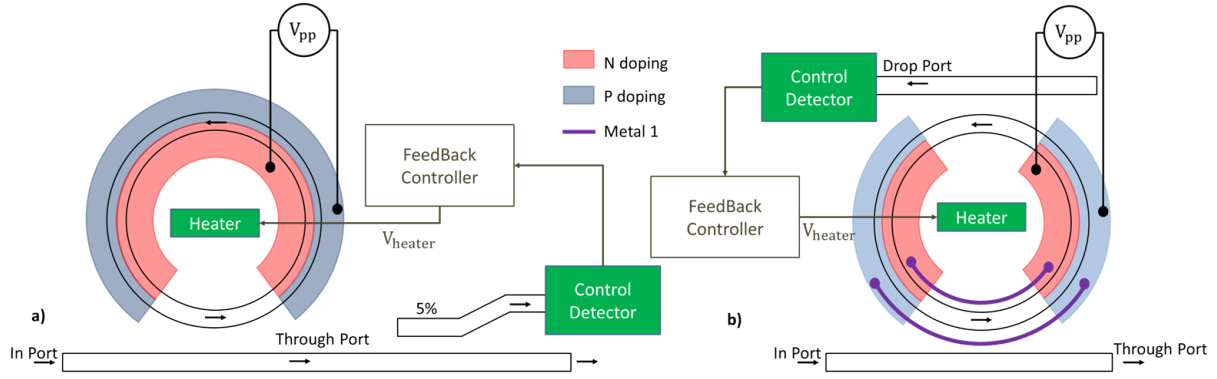


FIGURE 3.5: Schematic of a feedback loop for ring resonator modulator a) for the single-bus ring resonator modulator b) for the two-bus resonator modulator

reference [75, 87, 88]. Figure 3.6 represents two common procedures to find the reference average power. As the increase of the ring temperature is linearly proportional to the heater electrical power, it is preferable to vary linearly the electrical power instead of varying linearly the heater voltage.

The first possible procedure is composed of two steps. It allows the resonant wavelength and the laser wavelength to be superposed. The first step consists of tuning the voltage applied on the heater and find a maximum optical power at the drop port with 'off' modulation voltage applied to the modulator (step 1). Then, a first random sequence is applied to the modulator to measure and save the reference average power (step 2). This operating point has the best extinction ratio but can exhibit very high insertion loss.

The second possible procedure is composed of three steps. It allows to fix the insertion loss at the output of the through port. The first step consists of tuning the voltage applied on the heater and find the maximum optical power at the drop port with 'on' modulation voltage applied to the modulator (step 1). Then, the voltage applied to the heater is reduced to find the optical power at the drop port equal to $P = P_{max}(1 - 10^{(IL/10)})$ where IL is the desired insertion loss at the through port (step 2). Then, a first random sequence is applied to the modulator to measure and to save the reference average power (step 3).

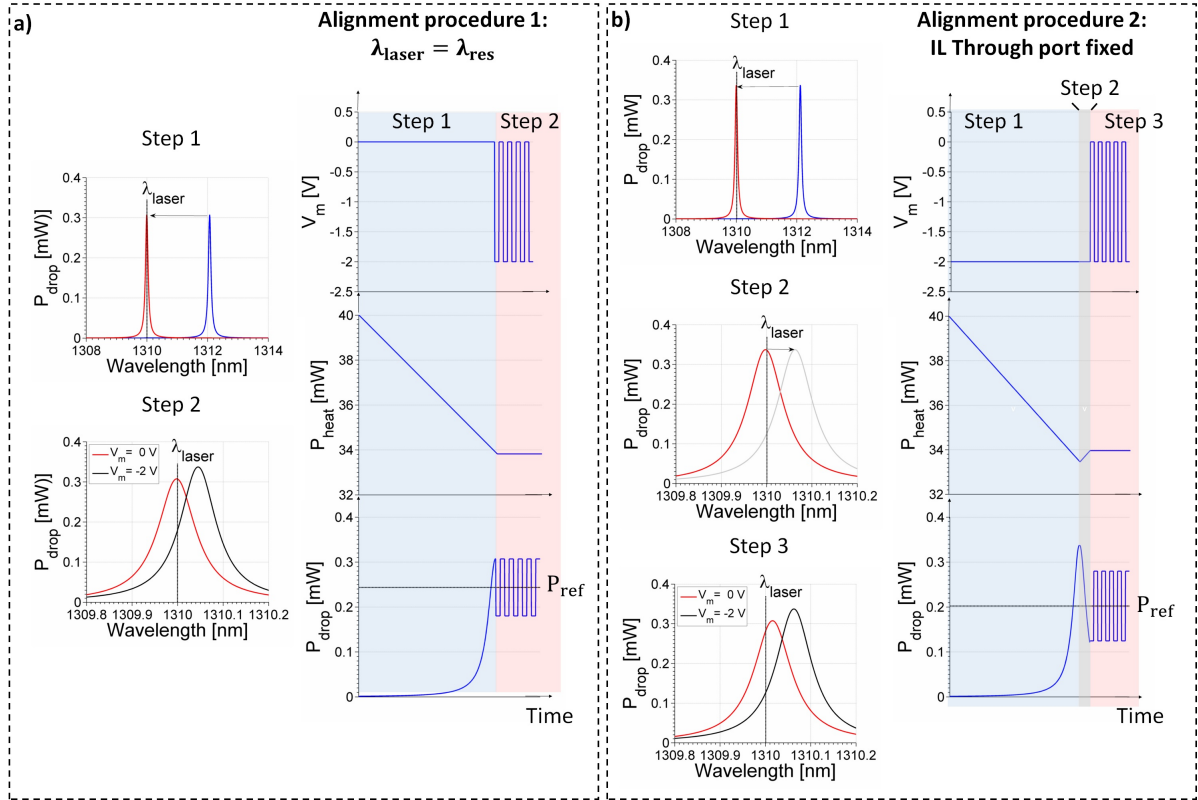


FIGURE 3.6: a) Alignment procedure to superpose the resonant wavelength and the laser wavelength b) Alignment procedure to fix the insertion loss at the output of the through port. The ring resonator modulator has the characteristics of table 3.1 with a heater efficiency of 50 mW/FSR. The input optical power at the through port is set to 1 mW.

A second feedback loop controller is used in [82,89–91], it is not based on the average optical power but it is based on the comparison of the **OMA** measured with the drop port photo-detector and a reference **OMA**. The **OMA** is defined as the difference between the power at the 'on' level minus the the power at the 'off' level. This monitoring is based on the presence of a maximum **OMA** at the through port according to the wavelength. This maximum coincides with a minimum **OMA** at the drop port as shown in figure 3.7. The maximum **OMA** ensures the best compromise between insertion loss and extinction ratio, but it does not ensure low insertion loss. Hence, this is the best operating point for **RRM** but not necessarily the best operating point for the complete transmitter. Furthermore, this solution is more precise but requires a wavelength controller as fast as the modulation rate, which can increase the power consumption.

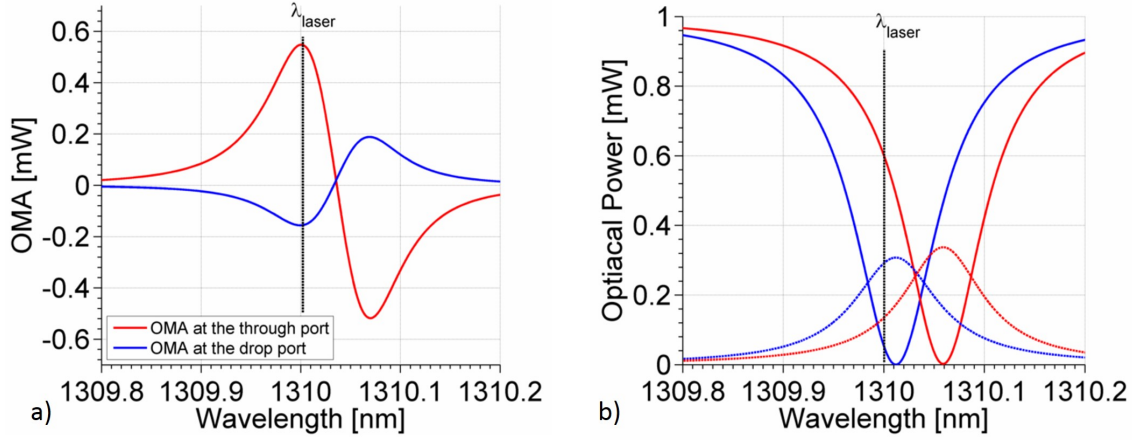


FIGURE 3.7: a) OMA at the through port and the drop port against the wavelength. The OMA maximum is tuned to the laser wavelength b) Corresponding optical power at the through port in lines and at the drop port in dotted for the two different level 0 V ('off' level blue) and -2 V ('on' level red). The ring resonator modulator has the characteristics of table 3.1 with a heater efficiency of 50 mW/FSR. The input optical power at the through port is set to 1 mW.

Table 3.3 summarizes the power consumption of the feedback loops in the literature. The power consumption is mainly due to the injected power in the heater. The comparison between the different papers is difficult because the level control varies between the different papers depending on the original alignment between the resonator wavelength and the laser wavelength. In the worst case, the needed heater power is the power needed to move the resonance wavelength by one FSR.

TABLE 3.3: Circuit power consumption for ring resonator modulator monitoring

	Power Consumption of the feedback loop [mW]	Power consumption of the heater [mW]	Total power consumption of the controller [mW]
Zheng [87] Opt.Expr.2014	0.2	14 (range 4.25 nm)	14.2
Agarwal [89] ECOC2014	145	-	-
Tanaka [88] ECOC2015	-	45.8 (range 5.6 nm)	-
Sun [91] VLSI2015	0.195	1.18 (range 2.82 nm)	1.375

Another alternative for feedback loops is to use a proportionate to absolute temperature sensor (PTAT) [92, 93]. The basic principle of the PTAT sensor is to use a physical parameter that is linearly proportional to the temperature. The proposed solution uses the difference between the voltage drop of two forward-biased diodes operating at different current densities. This difference is linearly proportional to the absolute temperature. Indeed, the relationship between the current and the voltage in a forward-bias is simply described by equation 3.7. Hence, the voltage difference between two forward-biased diodes with different sizes is expressed by equation 3.8. Figure 3.8 represents the proposed solution for the diodes layouts composing the PTAT sensor.

$$V_D = \frac{nk_B T}{q} \ln\left(\frac{I_D}{I_S}\right) \quad (3.7)$$

where k_B is the Boltzmann constant, T is the absolute temperature, q is the charge of electron, I_S is the reverse bias saturation current and n is fabrication constant between 1 and 2.

$$V_{D21} = V_{D1} - V_{D2} = \frac{nk_B T}{q} \left(\ln\left(\frac{I_{D2}}{I_{S2}}\right) - \ln\left(\frac{I_{D1}}{I_{S1}}\right) \right) = \frac{nk_B T}{q} \ln\left(N \frac{I_{D2}}{I_{D1}}\right) \quad (3.8)$$

where N is the ratio of diodes corresponding to the ratio of their reverse saturation currents. The higher the value of N , the higher the sensitivity of the PTAT sensor.

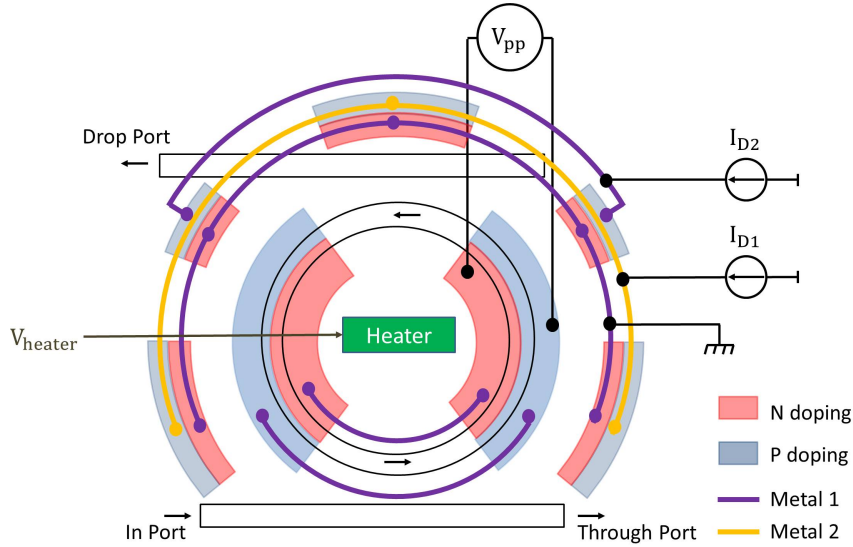


FIGURE 3.8: Schematic of a feedback loop for ring resonator modulator a) for the single-bus ring resonator modulator b) for the two-bus resonator modulator.

In [92], the PTAT sensor allows control of the ring resonator against a temperature variation of $3.2^\circ/\text{C}$, and should work for a wide range of temperature variations between $25^\circ/\text{C}$ and $150^\circ/\text{C}$. The main issue is that the feedback controller needs to know the PTAT voltage V_{D21} , where the ring modulator is well aligned. So, in the same way as previously, a preliminary tuning step is necessary to set up the feedback controller. This can be done by adding a photo-detector at the drop port. This sensor is more sensitive to small temperature variations than other solutions.

3.3.3 Self-heating in the ring resonator modulator

The feedback loop is necessary to control the operating point in real time. But, it can be disturbed by undesirable effects such as the self-heating effect. In PN depletion ring modulators, non-linear phenomena appear when a high optical power circulates in the ring. Due to mainly free carrier absorption (FCA), high optical power heats the ring, which causes resonance run-away [94]. This effect can ruin the ring resonator monitoring. That is why, it has to be taken into account. As it is a thermal phenomenon, its time constant τ_d is around $1 \mu s$. So, the self-heating will not impact the data modulation [94] [95]. A model used in [95] [96] describes the self-heating effect. It consists of an effective index dependency on the optical power in the ring (equation 3.9). σ_{self} is the thermal coefficient of the self-heating effect ; the extracted value for our measurements is $4.3 \cdot 10^{-3} W^{-1}$ (see the measurement section 4.2.6).

$$\Delta n_{eff}(t) = \sigma_{self} |E_A|^2 \cdot (1 - e^{-\frac{t-t_0}{\tau_d}}) \quad \text{for } t > t_0 \quad (3.9)$$

Some effects are worth highlighting. For the optical spectrum of the ring, the self-heating has enough time to be stable and produces a sharp resonance on the rising edge opposite to the sweep direction (figure 3.9 a) and b)); an hysteresis loop exists if the optical power is high enough to produce an optical bi-stability. The same identical process can be seen by changing the resonant wavelength position with a voltage applied to the heater [97]. This is why it is preferable to decrease the electrical power on the heater to find the operating point in order to reproduce a increasing wavelength sweep [91].

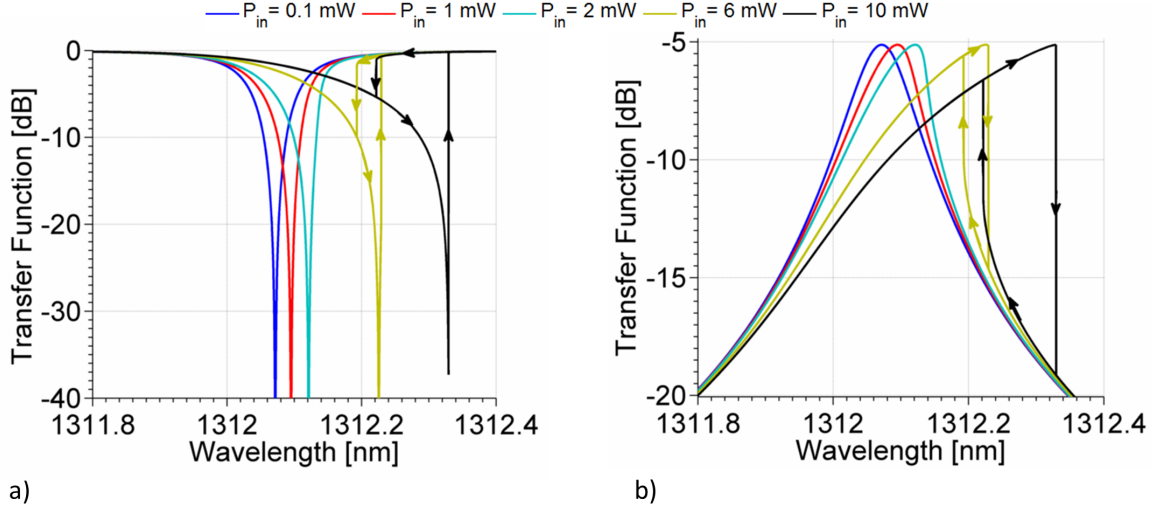


FIGURE 3.9: a) Transfer Function of the ring modulator for different input power at the through port ($\sigma_{self} = 0.04 W^{-1}$), b) Transfer Function of the ring modulator for different input power at the drop port. The ring resonator modulator has the characteristics of table 3.1 with $\sigma_{self} = 4.3 \cdot 10^{-3} W^{-1}$)

This effect has to be taken into account when the operating point is chosen. Indeed, the optical spectrum of the ring modulators is generally measured at low optical power, and the data transmission at high optical power. This leads to an operating point error. According to

our simulations, the optical bi-stability due to self-heating stays acceptable for optical power under 10 mW.

3.3.4 Comparison between the single-bus structure and the two-bus structure

We have seen that the drop port is necessary for the ring resonator monitoring. Its presence leads to a decreased junction length compared to the single-bus RRM. Indeed, there is no active junction in the coupling region. The active length is reduced for the two-bus-structure by a factor of approximately 5.6% considering that the only 320° of the ring is doped. On the other hand, the drop coupling allows compensation of the loss into the ring and so the ring resonator can be set to operate at the critical coupling condition in an easy way ($\sigma_1 = \sigma_2 e^{-a}$).

For the single-bus ring resonator modulator, the only parameter that can be varied in order to achieve the critical coupling condition is the coupling coefficient $\sigma_1 = \sqrt{1 - \kappa_1^2}$. This means that for a given Q factor, there exists only one couple (κ_1 , loss) capable of achieving the critical coupling condition. Equation 3.10 reports the relation between κ_1 , loss and Q factor at the critical coupling condition. Figure 3.10 a) shows the ER_{on-off} versus the loss with the corresponding coupling factor in figure 3.10 b) for different Q factors for the single-bus structure. The critical coupling condition is achieved when ER_{on-off} tends to infinity.

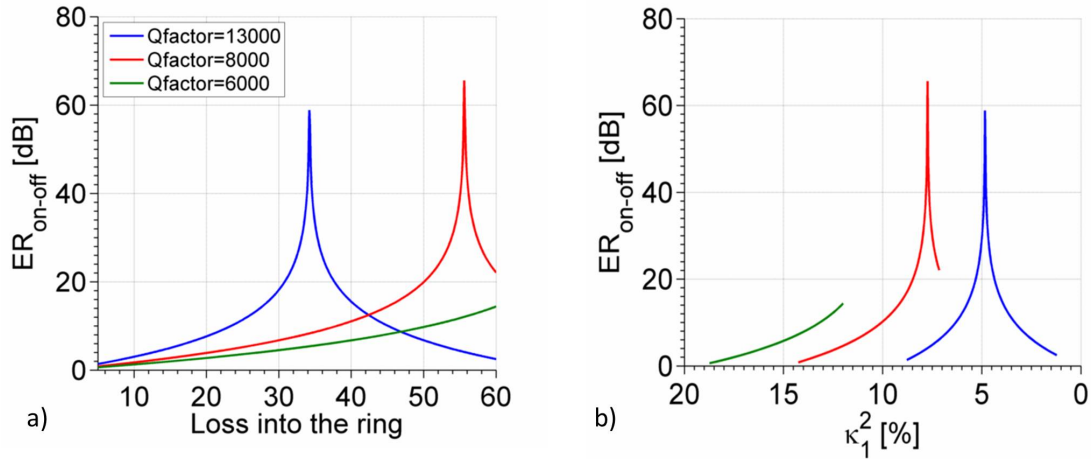


FIGURE 3.10: a) ER_{on-off} against the loss into the ring for a single-bus RRM ; b) ER_{on-off} against the coupling factor for a single-bus RRM. The ring resonator modulator has a radius of 10 μm . The two figures represent two different representation, so there are coupled.

$$\text{single-bus structure: } Q_{ccc} = \pi \cdot 2\pi R \cdot \frac{n_g}{\lambda_{res}} \cdot \frac{e^{-a_{ccc}}}{1 - (e^{-a_{ccc}})^2} \quad \text{where } \sigma_1 = e^{-a_{ccc}} \quad (3.10)$$

For the two-bus ring modulator, the coupling coefficient at the drop port κ_2 allows compensation of the loss and to achieve the critical coupling condition for a large range of Q factor. If the Q factor is too high, κ_2 tends to zero. This is why, there is, depending on the Q factor and κ_2 , a maximum loss beyond which the critical coupling condition cannot be achieved. This

condition is given by equation 3.11. The Q factor zones, where the critical coupling condition is respected, are represented in figure 3.11.

$$\text{two-bus structure: } Q_{ccc} < \pi \cdot 2\pi R \cdot \frac{n_g}{\lambda_{res}} \cdot \frac{\sigma_2 e^{-a_{ccc}}}{1 - (\sigma_2 e^{-a_{ccc}})^2} \quad \text{where } \sigma_1 = \sigma_2 \cdot e^{-a_{ccc}} \quad (3.11)$$

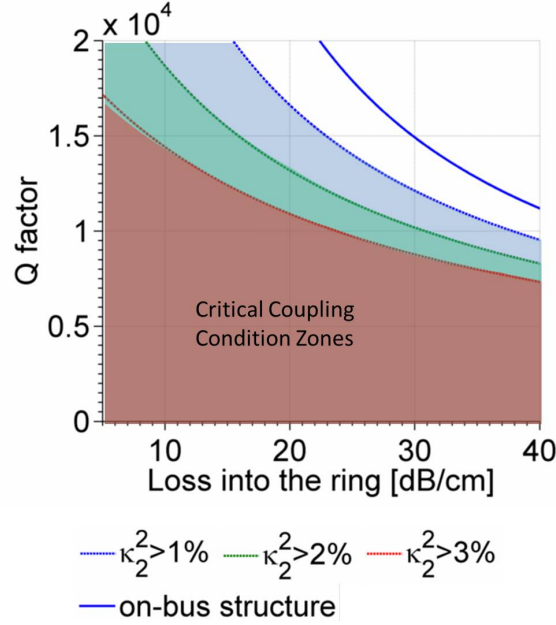


FIGURE 3.11: Q factor against the loss into the ring. Three zones of critical coupling condition are reported depending to the minimum admissible coupling coefficient at the drop port. The ring resonator modulator has a radius of 10 μm .

As the active junction length for the two-bus RRM is lower than for the single-bus RRM, we expect higher insertion loss for the two-bus RRM. For the comparison, with the same example as used previously (table 3.1), we set the Q factor to a typical value of $Q = 13000$ ($f_{opt} = 17.6$ GHz) and the loss into the ring at 23.7 dB/cm. In this case, the single-bus ring modulator cannot be set to the critical coupling condition as shown previously, but the two-bus RRM can. In order to compare the two structures, we simulate their transfer functions at 0 and -2 V. Figure 3.12 a) shows simulated transfer functions with a junction efficiency of 30 °/mm. In a first phase, the laser wavelength is set to the resonant wavelength for both (point 1) and we extract their insertion losses. The simulation is then repeated for different junction efficiencies and the resulting curves are reported in figure 3.12 b) in blue and red. The single-bus ring modulator shows lower insertion loss than the two-bus ring modulator but one has to notice that the extinction ratio is infinite for the two-bus ring modulator, as it is set to the critical coupling condition.

In a second phase, the wavelength of the laser is tuned for the two-bus-ring modulator in order to achieve the same extinction ratio than the single-bus ring modulator. A process example is shown in figure 3.12 a), where the laser wavelength moves from point 1 to point 2. In this case figure 3.12 b) in green, the two-bus ring modulator has lower insertion loss than the single-bus

ring modulator. So, the two-bus structure is more interesting than the single-bus structure since it allows more flexibility and better performance.

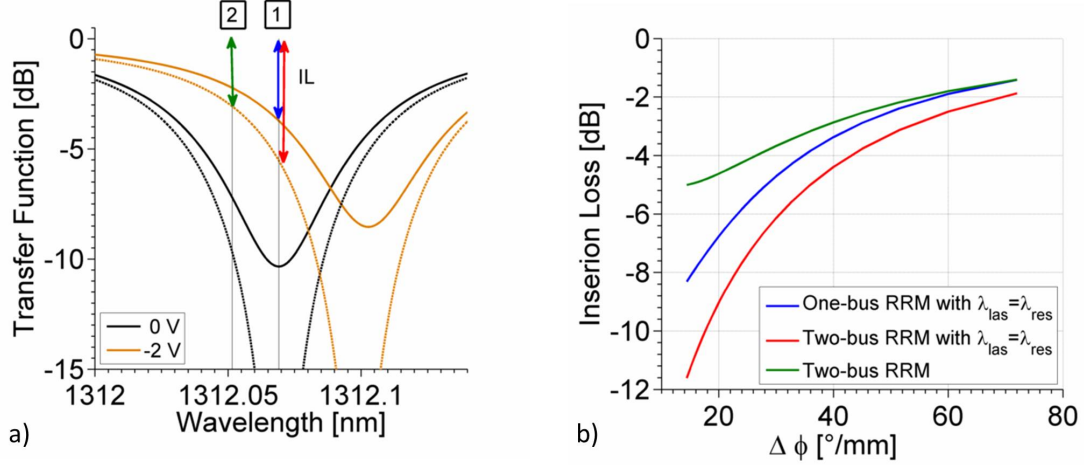


FIGURE 3.12: a) Transfer function of the single-bus ring resonator (lines) and the two-bus ring resonator (dotted) with same Q factor equals to 13000 and a phase shift equals to 30 °/mm. In order to keep the same extinction ratio, the laser wavelength is tuned between the point 1 and the point 2. b) Comparison of the insertion loss against the phase shift between the single-bus and the two-bus ring resonator modulator. The ring resonator modulator has the characteristics of table 3.1.

3.4 Conclusion

In this chapter, we have described the transient model for the ring resonator modulators. The transient model allows simulation of all effects in the ring modulator: the non-linearity, self-heating, optical bandwidth, electrical bandwidth and heater contribution. The two-bus ring resonator modulator is the structure of choice for the modulator. This structure allows a feedback control to be implemented in order to deal with temperature drift, and it also allows the critical coupling condition to be targeted whatever the value of the Q factor.

Many parameters describe the RRM responses. In the following chapters, only two characteristics in the steady state will be important: the extinction ratio and the insertion loss. For the electro-optical bandwidth, which is the main transient characteristic, it can be simply described by equation 3.4 with some assumptions.

Chapter 4

Silicon Ring Resonator Measurements

4.1 Introduction

This chapter is dedicated to the validation of the [RRM](#) model through static and dynamic measurements. Methodologies developed for [RRM](#) parameter extraction through measurements will be described. As we have seen in chapter 3, [RRM](#) parameters can be divided into three parts: the resonator parameters, the PN junction parameters, and the parameters in transient utilization. Table 4.1 summarizes these parameters.

TABLE 4.1: Ring resonator parameters

Resonator parameters			
Group Index	n_g	Loss	roughness loss α_r
Effective Index	$n_{eff}(0V)$		bending loss α_b
Coupling Factors	κ_1	Self heating coefficient	σ_{self}
	κ_2		
PN junction parameters			
Effective index variation against voltage	$\Delta n_{eff}(V)$	PN junction loss	$\alpha_j(0V)$
Loss variation against voltage	$\Delta\alpha(V)$		
Dynamic parameters			
Optical Bandwidth	f_{opt}		
Electrical Bandwidth	f_{elec}		
Electro-optical Bandwidth	f_{3dB}		

Measured [RRMs](#) were fabricated in the 200 nm CEA-LETI platform. Two different radii (10 and 15 μm) are available with a wide range of gap values. The [RRMs](#) are built with a 220/100 nm rib waveguide which includes a PN junction inside the 400 nm-wide waveguide (220/100/400 technology platform). These are single-bus ring structures so without drop waveguide ($\kappa_2 = 0$). Due to the technology properties, static and dynamic [RRMs](#) measurements are carried out at 1.55 μm .

4.2 DC measurements

4.2.1 DC characterization setup

The DC characterization setup is reported in figure 4.1. It makes use of a tunable laser source controlled by a CT400 tester which enables a wavelength sweep in a few seconds with 1 pm resolution, a polarization controller and an electrical source. The measured test structure consists of a **RRM** between two grating couplers for the optical power coupling inside and outside the chip. For correct measurements, the input light has to be correctly **TE** polarized, and the optical power at the input of the **RRM** has to be lower than 100 μW to avoid self-heating effects (see section 3.3.3).

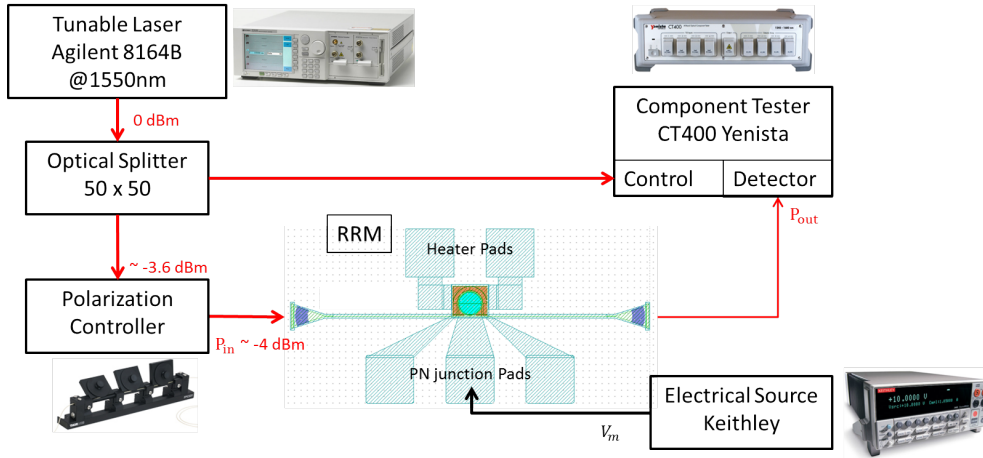


FIGURE 4.1: RRM experimental setup

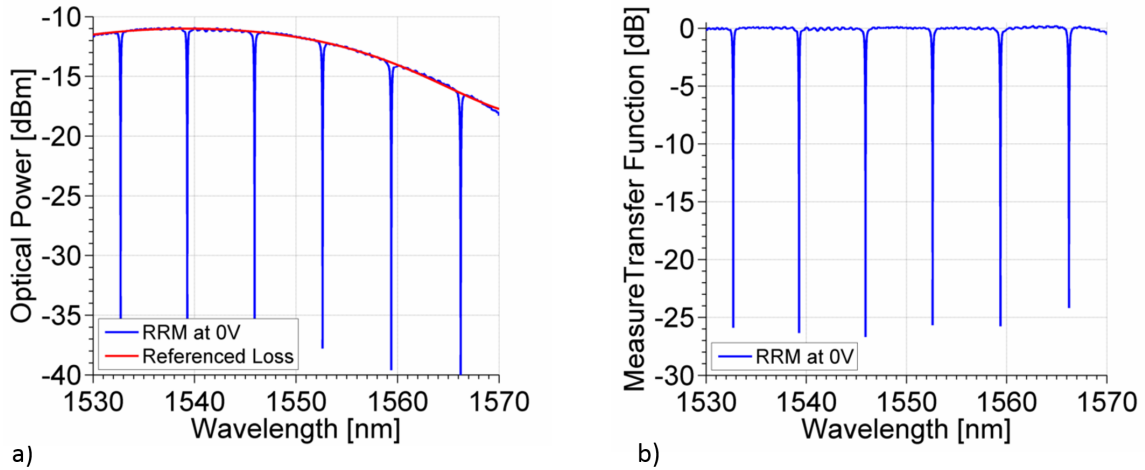


FIGURE 4.2: Non referenced measurement a) and referenced measurement b) of 15 μm radius RRM with 450 nm gap

The **RRM** optical transfer functions against the voltage applied on the PN junction were measured over the whole wafer. A measured optical transfer function example is reported

in figure 4.2 a). We can see that this measurement is the multiplication between the RRM transfer function and the transfer functions of the input and output grating couplers. In order to remove the grating couplers' contributions from the measured optical transfer functions shown in figure 4.2 a), the measurement of the transfer function of a grating coupler reference structure is performed. This reference structure is composed of two grating couplers linked by a small waveguide without any components inside. The figure 4.2 b) shows the transfer function of the RRM free of the grating couplers contributions.

We will now detail the static parameter extraction process from measured optical transfer functions of ring resonator modulators (table 4.1). On each die, we measured five ring resonator modulators with different geometries (radius and gap). In order to get the process variability, the measurements were repeated on 21 dies across the wafer.

4.2.2 Effective index and group index

The first parameters which can be extracted are the group index and the effective index of the ring resonator. They are obtained from the measurement of the FSR and the resonant wavelength λ_{res} (equation 4.1 and 4.2).

$$\text{FSR} = \frac{\lambda_{res}^2}{n_g 2\pi R} \quad (4.1)$$

$$\lambda_{res}(m) = \frac{2\pi R}{m} \cdot n_{eff}(\lambda_{res}) \quad (4.2)$$

where m is the order of resonance.

The comparison of 21 RRMs geometrically identical has to be made at the same order of resonance, m . Unfortunately, as the ring resonator modulator is a very sensitive component, the resonant wavelength variability across the wafer is greater than the FSR. This prevents determining which resonant wavelength corresponds to which order of resonance. A solution is proposed in [98] and [99]. It consists in plotting the group index in function of the resonant wavelength (an example is shown in figure 4.3 a)). Each downward diagonal line corresponds to an order of resonance. This relation originates from the physical variations (*i.e.* thickness, width) that change the effective index of the waveguide and simultaneously affect the group index. So with this methodology, we are able to compare the 21 geometrically identical RRMs. We apply this methodology to our measurements in figure 4.3 b) for the RRMs with a radius of 15 μm and a gap of 450 nm. Unfortunately, downward diagonal lines are less evident because the resonant wavelength variability is very high. We will focus our comparison around 1547 nm, where there seems to be one order of resonance.

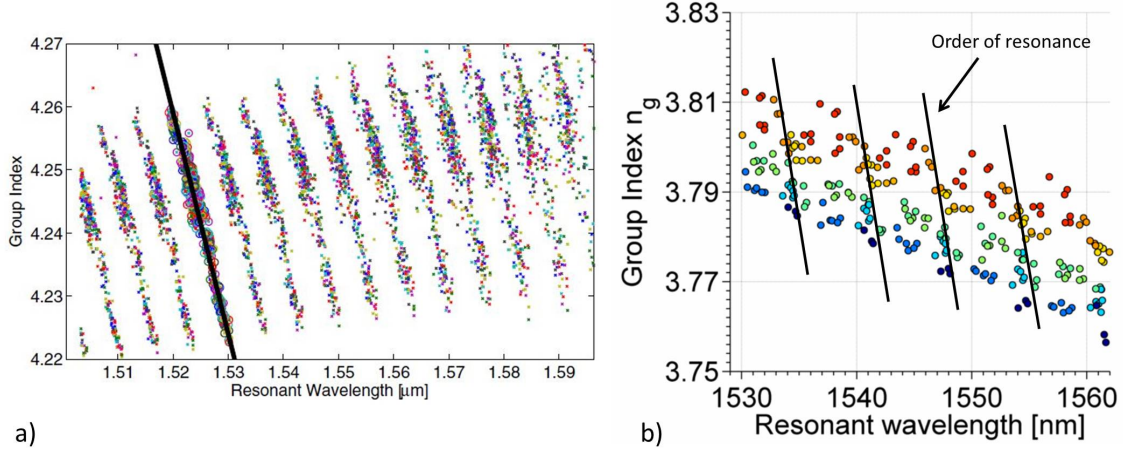


FIGURE 4.3: Extracted group index versus resonance wavelength for geometrically identical resonators across the wafer a) from [98] and b) for our measurements for a radius of 15 μm and a gap of 450 nm

Table 4.2 reports the measurements and extractions for different ring resonator modulators with their standard deviation among 21 dies on a 200 mm wafer. In theory, modal simulations give an effective index of $n_{eff}=2.497$ and a group index of $n_g=3.87$ which are very close to the measurement results.

TABLE 4.2: Measurements of the resonant wavelength and the FSR with the associated group index and effective index extraction. These values are the medians $\pm$ standard deviations among 21 dies on a 200 mm wafer (Ext. : Extracted value, Sim.: Simulated value)

Radius [μm]	Gap [nm]	λ_{res} [nm]	FSR [nm]	n_g		n_{eff}	
				Extr.	Sim.	Extr.	Sim.
10	350	1548.4 ± 2.7	$9.97 \pm 4e^{-2}$	$3.78 \pm 7e^{-3}$		$2.464 \pm 4e^{-3}$	
10	450	1547.5 ± 3.1	$9.96 \pm 4e^{-2}$	$3.78 \pm 7e^{-3}$		$2.462 \pm 5e^{-3}$	
15	350	1547.5 ± 1.9	$6.64 \pm 2e^{-2}$	$3.79 \pm 8e^{-3}$	3.84	$2.479 \pm 3e^{-3}$	2.49
15	450	1547.2 ± 1.6	$6.63 \pm 2e^{-2}$	$3.79 \pm 7e^{-3}$		$2.479 \pm 3e^{-3}$	
15	550	1546.8 ± 1.8	$6.64 \pm 2e^{-2}$	$3.79 \pm 7e^{-3}$		$2.478 \pm 3e^{-3}$	

The absolute position of the resonant wavelength (and therefore the effective index) is difficult to technologically stabilize. An effective index standard deviation of $\sigma = 3 \times 10^{-3}$ is sufficient to vary the resonant wavelength over the entire FSR [100]. As explained in section 3.3, the wavelength variations are due to the variation of the width of the waveguide, the slab height and the SOI height. According to [99], the waveguide width variation is suspected to be the major contributor to the effective index variation. In [101], authors improve the wavelength absolute position by a factor of 27% using rings with adiabatic bends. Even if, technology improvements may allow the absolute wavelength position to be stabilized, there is still a temperature dependency (0.1 nm/ $^{\circ}\text{C}$) that has to be controlled in real time (see feedback loops in section 3.3).

4.2.3 Coupling factor and loss

The second parameters which can be extracted from the measured transfer functions are the coupling factor κ_1 and the loss inside the ring. In order to validate our extraction process, we will compare the extracted values from the RRM measurements with simulations, but also with direct measurements of these parameters on simple tests structures. We will begin with the direct measurements and the simulations of the coupling factor and the loss inside the ring. Then, the extraction process of the coupling factor and loss through RRM's transfer function measurements is presented.

Coupling factor simulations and direct measurements

The coupling factor test structure consists of a waveguide coupler with the same radius and gap than for RRM (a layout example is shown in figure 4.4 a)). The coupling factor is measured using equation 4.3, which links the coupling factor to the measured output optical powers figures.

$$\kappa_1^2 = \frac{P_{dout}}{P_{in}} \quad (4.3)$$

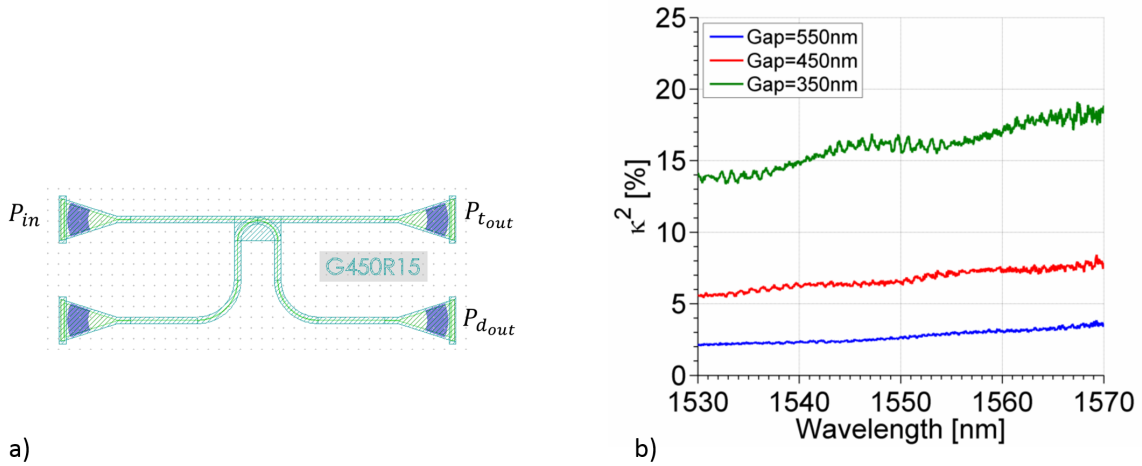


FIGURE 4.4: a) Layout example of coupling factors tests structures. b) Measured coupling factor versus wavelength for a ring radius of 15 μm

Coupling factor test structures with a gap between 150 and 550 nm for two radii (10 and 15 μm) are available in the wafer. The measured coupling factor is taken at 1.55 μm . It slightly increases with the wavelength because the effective index increases as a function of the wavelength (figure 4.4 b)). The measured values are reported in table 4.3 as measured values.

The simulations of the coupling factors are then made with FDTD simulations of the same referenced structures at 1.55 μm . The simulated values are reported in table 4.3 as simulated values.

Global loss simulations and direct measurements

The global loss is composed of three different contributions : loss due to the roughness of the waveguide (which is inherent to the fabrication processes), the loss due to the curvature of the ring (which depends on the ring radius), and the loss due to the carriers in the PN junction. The three contributions are measured separately with classical tests structures. These are waveguides with different lengths. By measuring the optical output variation in function of the length, the slope gives the loss in dB/mm (example in figure 4.5 for rib doped spirals) [102]. Measured values are reported in the table 4.3 as measured values.

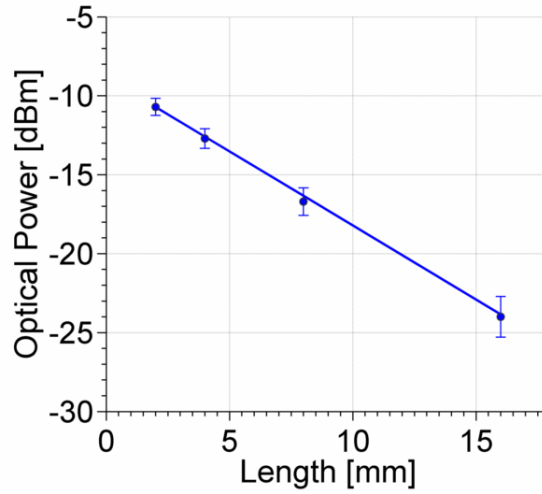


FIGURE 4.5: Median of the optical power and its standard deviation for different doped spirals. The slope gives the loss in dB/mm

For the loss simulations, the three contributions are simulated separately. The bending loss is evaluated with modal simulation as in section 2.2.1. The doping loss simulation is carried out with electrical simulations coupled with modal simulations [103]. Unfortunately, the roughness loss cannot be simulated, as it is very dependent on the fabrication platform. The simulated values reported in the table 4.3 for the global loss correspond to the sum of the simulated bending and doping losses.

Coupling factor and global loss extraction and comparison

The RRM resonance quality and depth are directly linked to the loss inside the ring and the coupling factor. Indeed, the Q factor and the ER_{on-off} are directly impacted by these two parameters (4.4 and 4.5).

$$Q = \pi \cdot 2\pi R \cdot \frac{n_g}{\lambda_{res}} \cdot \frac{\sqrt{\sigma_1 e^{-a}}}{1 - \sigma_1 e^{-a}} \quad (4.4)$$

$$ER_{on-off} = \left(\frac{\sigma_1 + e^{-a}}{\sigma_1 - e^{-a}} \right)^2 \cdot \left(\frac{1 - \sigma_1 e^{-a}}{1 + \sigma_1 e^{-a}} \right)^2 \quad (4.5)$$

where σ_1 is the amplitude transmission coefficient related to the coupling transmission $\kappa_1 = \sqrt{1 - \sigma_1^2}$ and a is the field attenuation related to the intensity absorption coefficients α_0 due to the roughness loss, α_b due to the bend loss and $\alpha_j(0V)$ due to the charges loss by $a = 1/2 \cdot (\alpha_0 + \alpha_b + \alpha_j(0V)) \cdot 2\pi R$ where R is the ring radius.

Process extraction of the coupling factor and the loss inside the ring from the measured transfer function of the **RRM** is simple. We numerically search the couple $[\kappa_1, \text{loss}]$ which fulfills at the same time the measured Q factor and the ER_{on-off} values.

Table 4.3 reports the various figures (extracted, directly measured and simulated) for Q factor and ER_{on-off} with the associated coupling coefficient and the global loss inside the ring. These values are given with their standard deviations against 21 dies in a 200 mm wafer for resonances close to 1547 nm.

TABLE 4.3: Measurement of the quality factor (Q) and the ER_{on-off} with the associated coupling factor (κ_1^2) and global loss extraction. These values are the medians $\pm$ the standard deviation among 21 dies on an 200 mm wafer (Ext. : Extracted value, Meas. : Measured value, Sim.: Simulated value; with * under-coupled regime , without * over-coupled regime)

R [μm]	Gap [nm]	Q factor	ER_{on-off} [dB]	κ_1^2 [%]			Global Loss [dB/cm]		
				Extr.	Meas.	Sim.	Extr.	Meas.	Sim.
10*	350	4800 $\pm$ 300	-20.3 $\pm$ 4.5	10.2 $\pm$ 0.7	10.2	10.6	63 $\pm$ 6.6	33	31
10	450	7400 $\pm$ 450	-8.5 $\pm$ 0.9	3.8 $\pm$ 0.1	4.1	4.1	64 $\pm$ 5.7		
15*	350	6500 $\pm$ 260	-6.1 $\pm$ 0.4	15.7 $\pm$ 0.6	15.5	15.7	24 $\pm$ 1.6		
15	450	12100 $\pm$ 500	-24.0 $\pm$ 2.5	6.3 $\pm$ 0.2	6.4	6.1	26 $\pm$ 1.4	-	17
15	550	17400 $\pm$ 930	-7.9 $\pm$ 0.5	2.2 $\pm$ 0.1	2.1	2.3	28 $\pm$ 1.7		

As explained in section 3.3.4, for a given loss in a single-bus ring resonator modulator, there exists only one coupling coefficient leading to the critical coupling condition ($\sigma_1 = e^{-a}$ and $ER_{on-off} = +\infty$). Ring resonator modulators are close to the critical coupling regime for a gap equal to 450 nm for 15 μm radius, and a gap equal to 350 nm for 10 μm radius. The coupling factor and loss comparison are also plotted in figures 4.6 and 4.7, respectively. The coupling factor decreases exponentially with the gap. For a fixed gap, the higher the radius, the higher the coupling zone and consequently the higher the coupling factor. As seen in figure 4.6, very good correlation is obtained between simulated, measured and extracted values for the coupling factor (with an error lower than 0.1 %). For the global loss, it decrease exponentially with the radius due to the decreasing bending loss. Unfortunately, it stays some errors between simulated, measured and extracted loss. Further investigations with more test structures are required to completely validate the extraction process.

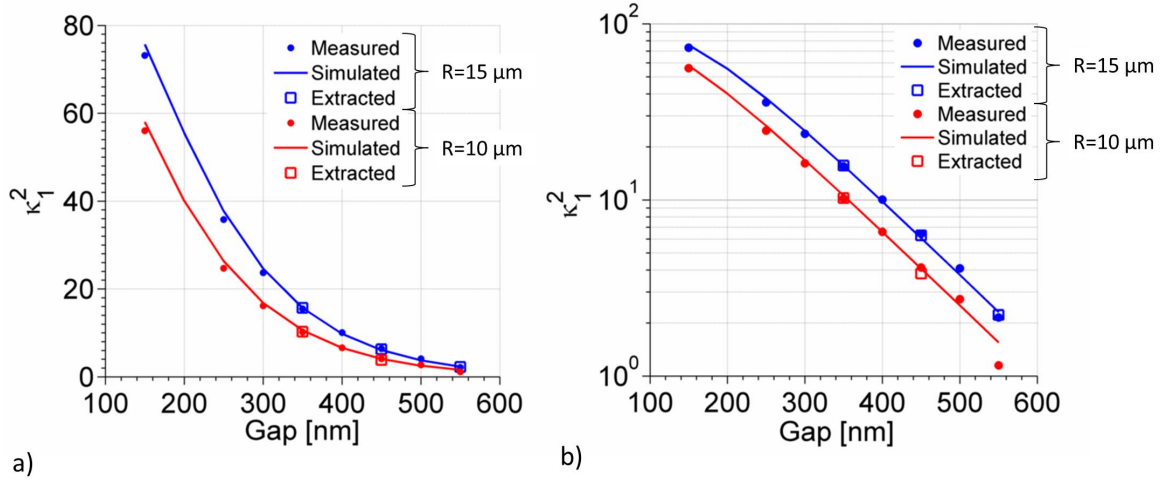


FIGURE 4.6: Comparison between the median of the measured coupling factor (points), the associated FDTD simulation (lines), and the extracted coupling factor from ring optical spectra (squares) at 1550 nm against the gap for different radii a) in a linear scale and b) in a logarithmic scale

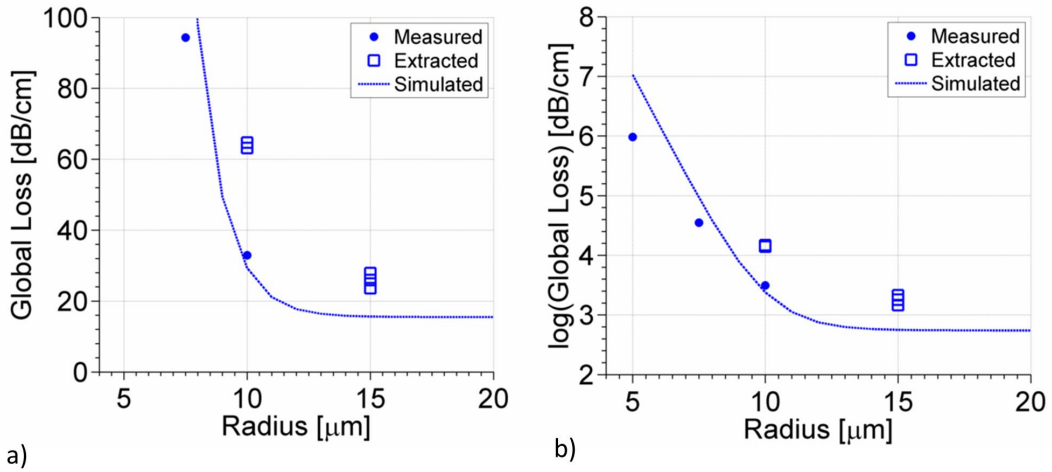


FIGURE 4.7: Comparison between the median of the measured global loss from test cells (points), modal simulation (dotted-line), and the extracted global loss from ring optical spectra (squares) against the radius of the ring a) in a linear scale and b) in a logarithmic scale

4.2.4 Effective index and absorption coefficient variation against voltage

We now detail the extraction of the effective index and the absorption coefficient due to the modulation. The variation of the resonant wavelength ($\lambda_{res}(V_m)$) and the variation of the minimum of transmission ($T_{min}(V_m)$) against the applied voltage depict the modulation effect. A referenced measured optical spectrum is reported in figure 4.8 a).

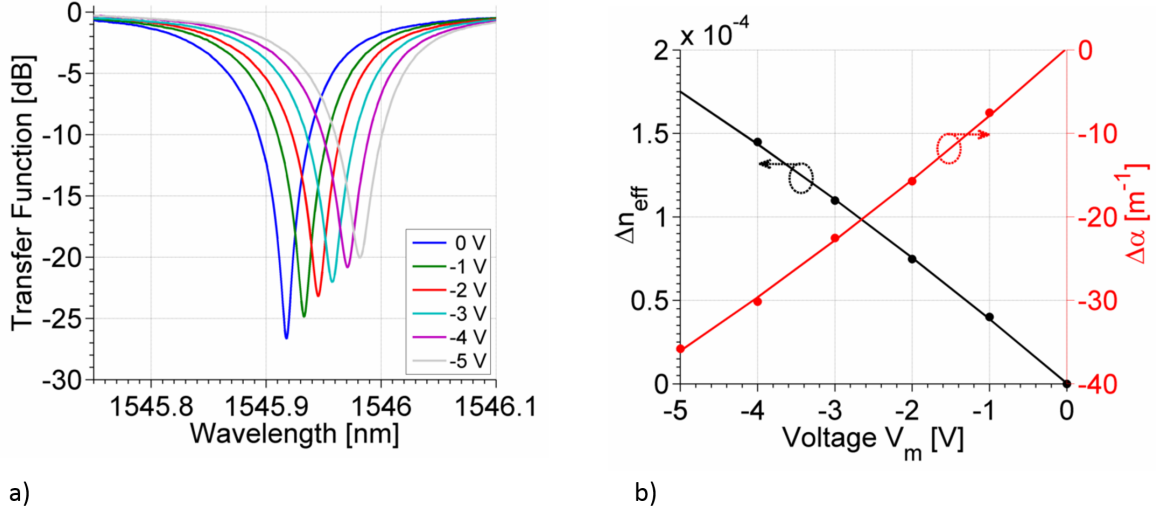


FIGURE 4.8: a) Reference measurement of 15 μm radius RRM with 450 nm gap for different voltages applied to the PN junction b) Extracted index variation and absorption coefficient variation against the voltage applied to the PN junction (points). The straight lines correspond to second order polynomial approximations

The effective index variation is linked to the resonant wavelength variation by equation 4.6. The absorption coefficient variation is extracted with the variation of the minimum of transmission at the through port with equation 4.7, where two cases must be considered: over- and under-coupled regimes derived from equation 2.6.

$$\Delta n_{eff}(V_m) = \frac{m \cdot \lambda_{res}(V_m) - n_{eff}(\lambda_{res}(V_m), 0V) \cdot 2\pi R}{L_{active}} \quad (4.6)$$

$$\begin{cases} \text{overcoupled regime : } \Delta \alpha(V_m) = \frac{1}{L_{active}} \cdot \left(-\log\left(\frac{\sigma_1 + \sqrt{T_{min}(V_m)}}{1 + \sigma_1 \sqrt{T_{min}(V_m)}} \right) - \frac{\alpha_0}{2} \cdot 2\pi R \right) \\ \text{undercoupled regime : } \Delta \alpha(V_m) = \frac{1}{L_{active}} \cdot \left(-\log\left(\frac{-\sigma_1 + \sqrt{T_{min}(V_m)}}{-1 + \sigma_1 \sqrt{T_{min}(V_m)}} \right) - \frac{\alpha_0}{2} \cdot 2\pi R \right) \end{cases} \quad (4.7)$$

In figure 4.8 b), the effective index variation and the absorption coefficient variation are reported and correspond to the extraction for the measurement of figure 4.8 a). One can note that the effective index variation and the absorption coefficient do not varying linearly with the voltage and are better approximated with second order polynomials (fig.4.8 b), lines). This effect demonstrates the non-linearity of the PN junction (see section 2.5.1).

The resonant wavelength is red shifted when the voltage increases, which characterizes the PN junction efficiency. This variation ($\partial \lambda_{res} / \partial V$) is expressed in pm/V and, it is very useful because it is measured directly. However, it also contains the effective index variation due to the dispersion. The best method is to calculate the $V_\pi L_\pi$ efficiency or the efficiency in $^\circ/\text{mm}$ with equation 4.8 (see section 2.5.2).

$$V_\pi L_\pi = \frac{-V_m \cdot \lambda_{res}(V_0)}{2 \cdot \Delta n_{eff}(V_m)} \quad (4.8)$$

Table 4.4 presents the efficiency measurements of the PN junction for the different **RRMs**. PN junctions are identical for all ring resonator modulators, only their lengths vary for the two different radii. The efficiency is evaluated at 2 V.cm and seems to be stable over the whole wafer. This value is very close to the simulated value of 1.8 V.cm [103]. We are aware that the efficiency is lower than the state of the art. A 4 dB/cm loss decrease is observed at -2 V which is coherent with the junction depletion (fewer carriers so less loss).

TABLE 4.4: Measurement of the modulation efficiency and the effective index and absorption coefficient variations against the applied voltage on the PN junction. These values are the median $\pm$ standard deviations among 21 dies on a 200 mm wafer

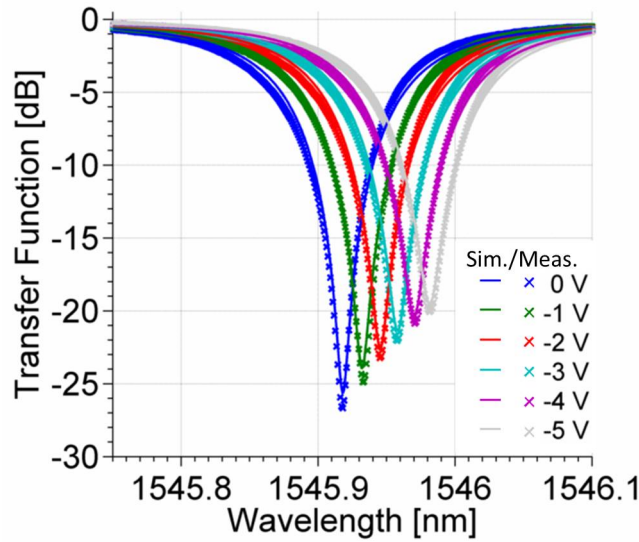
Radius [μm]	Gap [nm]	Efficiency [pm/V]	$V_\pi L_\pi$ at -2V [V.cm]	Efficiency at -2V [$^\circ/\text{mm}$]	Δn_{eff} at -2 V	$\Delta\alpha$ at -2 V [m^{-1}]	$\Delta\alpha$ at -2V [dB/cm]
10	350	14 ± 0.8	2.0 ± 0.1	18 ± 1	$7.7e^{-5}$ $\pm 5e^{-6}$	-18 ± 6.3	-3.7 ± 1.3
10	450	14 ± 0.9	2.0 ± 0.1	18 ± 1	$7.1e^{-5}$ $\pm 5e^{-6}$	-21 ± 6.4	-4.3 ± 1.3
15	350	14 ± 1.7	2.0 ± 0.2	18 ± 2	$7.8e^{-5}$ $\pm 8e^{-6}$	-21 ± 2	-4.2 $\pm .48$
15	450	14 ± 0.7	1.9 ± 0.1	19 ± 1	$8.0e^{-5}$ $\pm 4e^{-6}$	-17 ± 3	-3.4 ± 0.6
15	550	15 ± 0.7	1.9 ± 0.1	19 ± 1	$8.3e^{-5}$ $\pm 4e^{-6}$	-22 ± 3	-4.4 ± 0.5

4.2.5 Comparison between measurement and simulation with extracted parameters

Throughout the previous sections, we have extracted from measurements the following parameters: the group index n_g , the effective index n_{eff} , the coupling factor κ_1 , the global loss, the modulation efficiency $\Delta n_{eff}(V_m)$ and the loss variation $\Delta\alpha(V_m)$ for different **RRM** radii and gaps. For one ring modulator, we compare the complete measured optical spectrum with the simulated optical transfer function using extracted parameters (summarized in table 4.5). A very good agreement between measurements and simulations is shown in figure 4.9.

TABLE 4.5: Extracted parameters from measurement of 15 μm radius RRM with 450 nm gap used for a measurement/simulation comparison

Group Index	n_g	3.79
Effective Index	$n_{eff}(\lambda_{res}, 0V)$	2.4
Resonant Wavelength [nm]	$\lambda_{res}, 0V)$	1545.9
Global Loss [dB/cm]		27.3
Coupling Factor [%]	κ_1^2	6.4
Effective Index Variation	$\partial n_{eff}/\partial V$	$-3.9 \cdot 10^{-5}$
	$\partial^2 n_{eff}/\partial^2 V$	$-3.6 \cdot 10^{-7}$
Loss Variation	$\partial \alpha/\partial V$	8.3
	$\partial^2 \alpha/\partial^2 V$	$1.9 \cdot 10^{-1}$

FIGURE 4.9: a) Comparison between referenced measurement (crosses) and simulation with extracted parameters (lines) of 15 μm radius RRM with 450 nm gap for different voltage applied to the PN junction

4.2.6 Self-heating

The self-heating coefficient of the RRM is evaluated with spectrum measurement, as previously described, by varying the optical power input [104]. This measurement is only done for one ring resonator modulator with a radius of 15 μm and a gap of 450 nm. The laser optical power is increased from 3 to 12 mW (figure 4.10). The light undergoes 5.38 dB loss due to the instruments and the coupling before it enters the through port of the RRM.

From the measurements reported in figure 4.10, we notice that the resonant wavelength is red shifted when the input optical power (P_{in}) increases. This resonant wavelength variation is directly related to an effective index variation (Δn_{eff}) as previously seen for the PN junction. Hence, at the resonance wavelength, we can deduce a proportional coefficient between the effective index and the input optical power. As self-heating occurs within the ring (section

3.3.3), then to extend the relation to other wavelengths, we have to consider the optical power in the ring ($|E_A|^2$) instead of the input optical power. In our case, this relation is $|E_A|^2 = 18.13 \cdot P_{in}$. Therefore, the self-heating coefficient σ_{self} is equal to $4.3 \cdot 10^{-3} [W^{-1}]$

In figure 4.10, we also compare the measured and simulated RRM transfer functions with the extracted self-heating coefficient. A very good agreement between simulation and measurements allows us to consider the self-heating modeling to be validated.

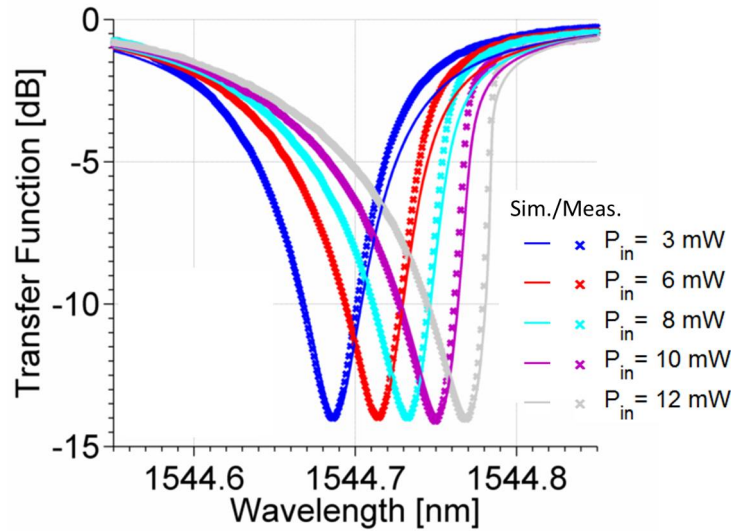


FIGURE 4.10: a) Comparison between referenced measurement (crosses) and simulation with extracted parameters (lines) of 15 μm radius RRM with 450 nm gap for different input optical powers

4.3 AC Characterization : small signal analysis

In order to evaluate the RRM transient response, AC characterization is carried out with a Vector Network Analyzer (VNA). This measurement allows the evaluation of both the electrical and electro-optical bandwidths.

4.3.1 AC characterization setup

The measurement setup is reported in figure 4.11. The VNA generates a sinusoidal voltage signal swept in frequency from a 100 MHz to 35 GHz (port 1). This voltage signal is then applied to the PN junction of the modulator, producing a sinusoidal optical power output signal. The modulated light is collected with a high speed photo-detector linked to the other VNA input port (port 2).

We have to be careful that the electro-optical S21 parameter does not represent the optical amplitude modulation (OMA). Indeed, the OMA is proportional to the S21 parameter according to equation 4.9. An important point is that the -3 dB OMA bandwidth corresponds to the -6 dB S21 bandwidth. The photo-diode frequency response is measured before the test in order to remove its contribution.

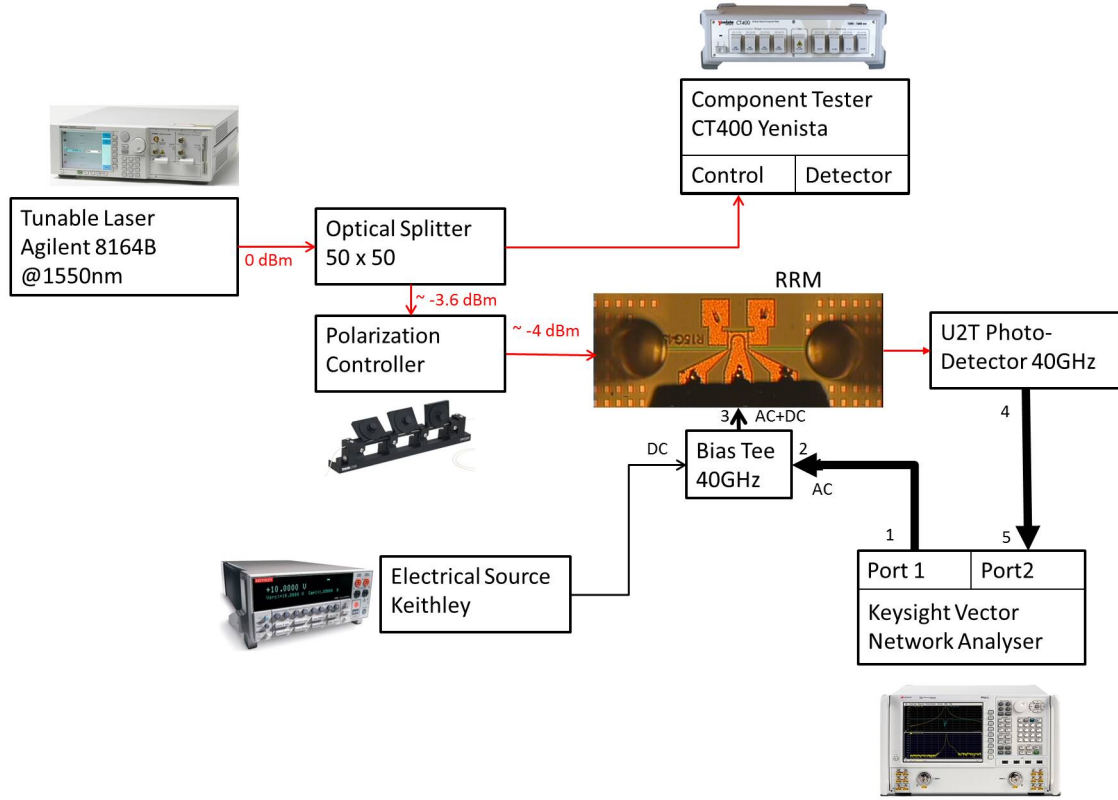


FIGURE 4.11: Measurement setup of AC characterisation of Ring Modulators

$$\begin{aligned}
 S21_{dB} &= 10\log_{10}\left(\left|\frac{\Delta V_{out}}{\Delta V_{in}}\right|^2\right) \\
 &= 10\log_{10}\left(\left|\frac{R_s \cdot Gain \cdot \Delta P_{out}}{\Delta V_{in}}\right|^2\right) \\
 &= 10\log_{10}\left(\left|\frac{R_s \cdot Gain}{\Delta V_{in}}\right|^2\right) + 2 \cdot 10\log_{10}(OMA)
 \end{aligned} \tag{4.9}$$

where ΔV_{in} and ΔV_{out} are the voltage amplitudes at ports 1 and port 2 respectively of the **VNA**, R_s and $Gain$ are the responsivity and the gain of the high speed photo-detector, ΔP_{out} is the amplitude variation of the optical power received by the photo detector which corresponds to the **OMA**.

A process calibration has to be followed precisely to obtain measurements that are as accurate as possible. This calibration allows removal of the electrical cable response between port 1 of the **VNA** and the electrical probe. The different calibration steps are the followings :

- Cal 1 : two-port VNA cable calibration with constructor tools. This calibration measures the S-parameters between the points 1 - 2 linked directly to 4 - 5 in figure 4.11.
- Power calibration just before the probe at point 3. This calibration ensures that the same electrical power is applied to the modulator, whatever the frequency.

- Cal 2 : one-port calibration with a calibration substrate. This calibration measures the S-parameters between the point 1 and the probe.
- A subtraction between Cal 1 and Cal 2 has to be done in order to remove Cal1 (which would otherwise be counted twice)
- The subtraction result is applied to the VNA and allows us to consider that all electrical cable contributions are removed.

We will now detail the S-parameter measurement for one ring resonator modulator with a radius of 15 μm and a gap of 450 nm. This RRM is the same as in section 4.2.5.

4.3.2 Electrical bandwidth

With the measurement of the S11 parameter, the electrical bandwidth is extracted using a circuit model as in [21, 105]. The circuit model is represented in figure 4.12 with Advanced Design System (ADS) tools. V_{AC} and the 50 Ohm terminal simulate the electrical sweep. The access resistance and the junction capacitance are expressed by R_s and C_j . In order to take into account parasitic resistances and capacitances, we add the following parameters: C_p to model the capacitance between the electrodes through the top dielectrics, R_{si} and C_{ox} to model the current path through the BOX and Si layers. Figure 4.13 compares the measured S11 with the simulated S11 with the electrical circuit of figure 4.12.

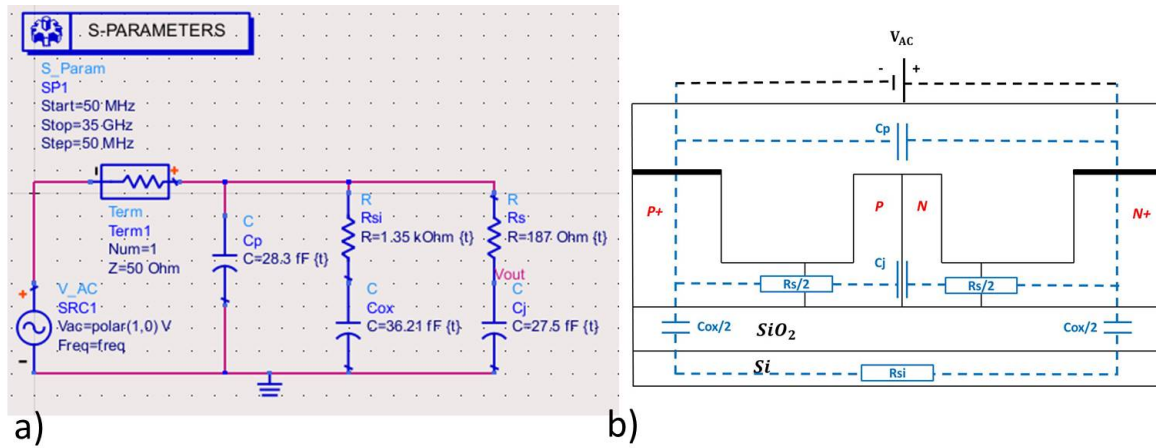


FIGURE 4.12: Small-signal circuit model for the ring modulator at 0 V bias

The access resistance R_s is extracted to be 180 Ω and the junction capacitance C_j is extracted to be 27.5 fF leading to an electrical bandwidth limitation of 32 GHz without parasitic components and 21 GHz with parasitic components.

The electrical bandwidth is also simulated with the electrical model developed in section 2.5.3 with the PN junction described in the figure 4.14. The doping levels shaping the PN junction are measured values from transmission line measurements (TLM) [106]. The access

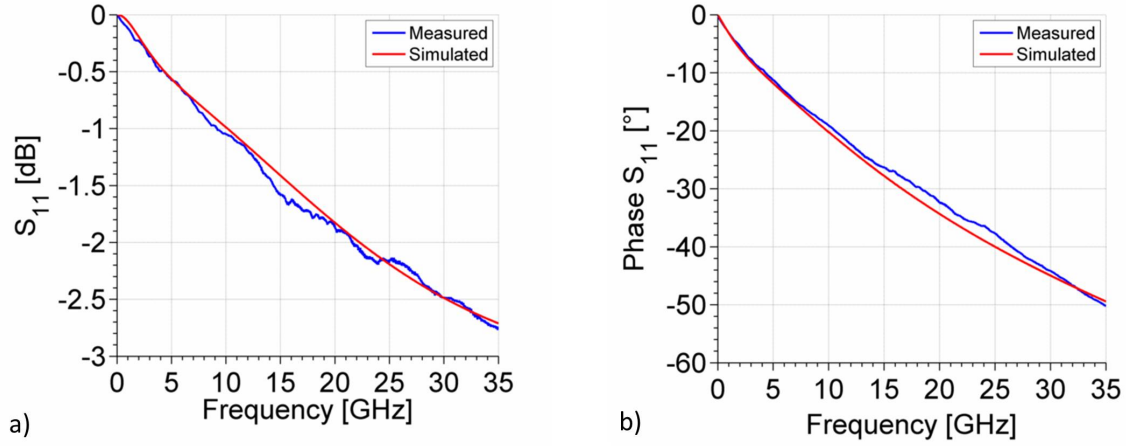


FIGURE 4.13: a) Comparison between measured S_{11} parameter and simulated electrical circuit for the ring resonator modulator in a) amplitude and b) phase

resistance and capacitance junction are simulated to be $93 \, \Omega$ and $22 \, \text{fF}$ respectively. The extracted junction capacitance from S_{11} is very close to the simulated junction capacitance but the simulation underestimates the access resistance (table 4.6). We suppose that the resistance error is due to lower doping concentrations in the slab sections. In fact, to reach an access resistance of $180 \, \Omega$, the slab doping concentrations must be around $1.2 \times 10^{17} \, \text{cm}^{-3}$.

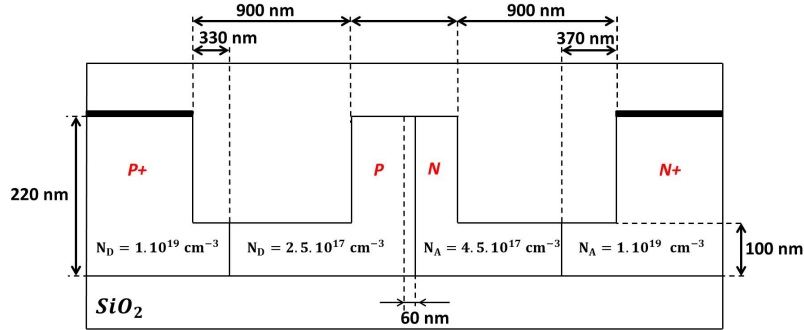


FIGURE 4.14: Schematic view of the PN junction

Table 4.6 sums up the access resistance, the junction capacitance and the electrical bandwidth obtained through the measured S_{11} extraction and through simulations. The doping implantation quality is very important. The measured electrical bandwidth is reduced by half compared with the simulations due to under-doped slab sections.

TABLE 4.6: Access resistance, junction capacitance and electrical bandwidth comparison

	Simulated	Simulated with under doped slab	Extracted from S11	Extracted from S11 with parasitic components
Access Resistance [Ω]	93	180	180	180
Junction Capacitance [fF]	22	23	27.5	27.5
Electrical Bandwidth [GHz]	74	39	32	21

4.3.3 Electro-optical bandwidth

We now evaluate the electro-optical bandwidth (E/O BW) through the electro-optical S21 parameter measurement. We expect an increase of E/O bandwidth when the laser wavelength is shifted from the resonant wavelength (see section 3.2.2). The measured RRM Q factor is 11600 leading to an optical bandwidth approximation of 16.6 GHz. With 21 GHz electrical bandwidth, the E/O bandwidth is expected to be 13 GHz in the worst case.

We measure the S21 parameter versus the laser wavelength. At the output of the tunable laser, the optical power is 12 mW. This high optical power was chosen in order to improve the signal to noise ratio. The VNA generates an electrical modulation power of 5 dBm leading to a voltage modulation amplitude of $1.1 V_{pp}$ applied to the RRM. Figure 4.15 a) and b) report the S21 and the OMA referenced to 1 GHz with a laser wavelength between 1538.98 nm (red) and 1538.74 nm (yellow) with a 0.02 nm measurement step. The RRM resonant wavelength without any voltage is equal to 1538.98 nm.

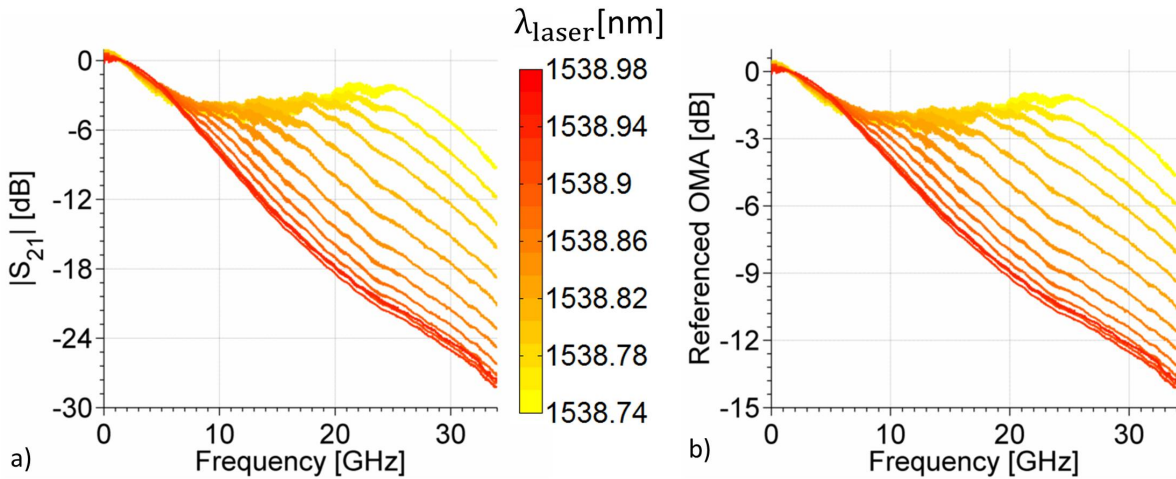


FIGURE 4.15: a) Measured electro-optical S21 parameter between 100 MHz and 35 GHz. a) Measured OMA between 100 MHz and 35 GHz. Measurements are done with a laser wavelength between 1538.98 nm in red and 1538.74 nm in yellow with a 0.02 nm step. They are referenced with their value at 1 GHz

From the OMA measurements (fig.4.15 b)), the -3 dB electro-optical bandwidth is extracted and reported in figure 4.16. As expected, the E/O bandwidth increases when the laser wavelength is shifted from the resonant wavelength. The smallest and the highest E/O bandwidths are 8 GHz and 30 GHz respectively. We also simulated the E/O bandwidth with the RRM transient model. The input parameters are those in table 4.5 with an electric bandwidth of 21 GHz. In the simulations, only 90 pm of laser wavelength shift from the RRM resonance wavelength is required to achieve 30 GHz E/O bandwidth instead of 200 pm for the measurements. This difference is explained by the self-heating effect. The measurements were made at high injected optical power, the RRM suffers from the self-heating which leads to a shift of the RRM resonant wavelength to higher wavelengths when the laser wavelength comes closer. Unfortunately, the self-heating effect was not taken into account in the simulation. To go further, new simulations taking into account self-heating could allow us to get a better comparison between the simulated and the measured E/O bandwidth.

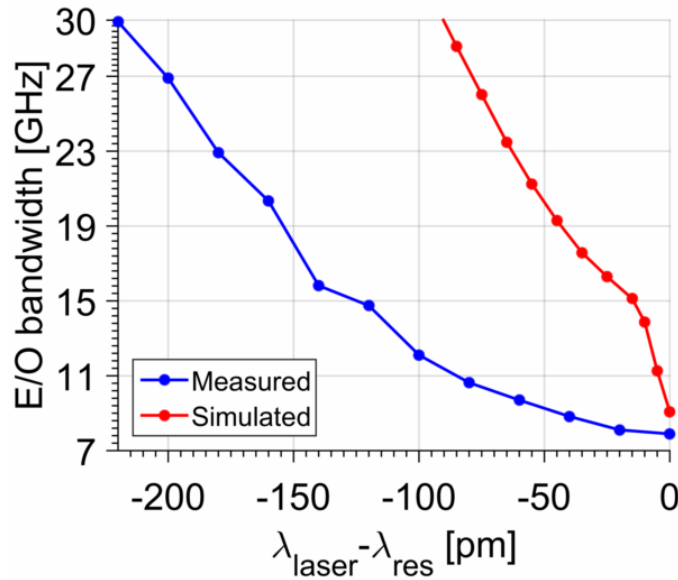


FIGURE 4.16: -3 dB E/O bandwidth comparison between extracted values from the OMA measurements and simulated values with the RRM parameters of table 4.5 and 21 GHz electric bandwidth

According to figure 4.16, we can suppose that the best laser wavelength position is when it is shifted at least of 200 pm from the resonant wavelength reaching 30 GHz electro-optical bandwidth. However, the extinction ratio decreases when the laser wavelength is shifted from the resonant wavelength as shown in figure 4.17. A trade-off has to be found between the E/O bandwidth, the insertion loss and the extinction ratio by choosing the correct wavelength laser position regarding to the RRM resonant wavelength. This trade-off will be further discussed in the next chapter.

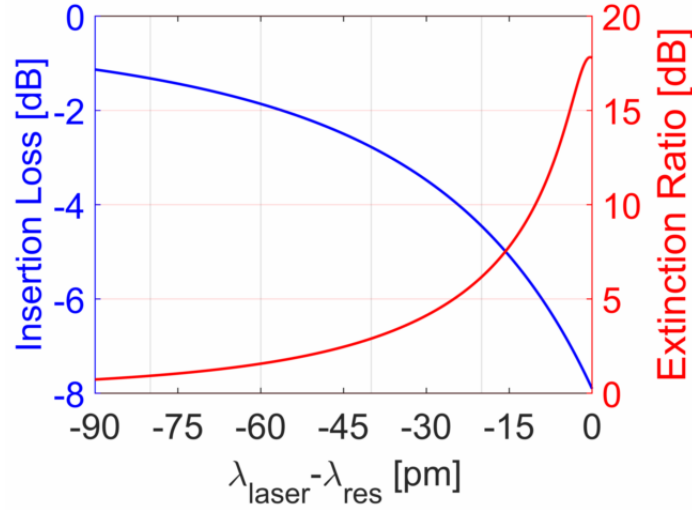


FIGURE 4.17: Simulated extinction ratio and insertion loss as a function of the laser wavelength position with $2 V_{pp}$ driving voltage

4.4 Chirp Quantification

We propose in this section to quantify the chirp of the [RRM](#). The chirp effect expresses the inherent phase modulation when achieving intensity modulation, which leads to a modification of the transmission properties after propagation over a few kilometers of dispersive single mode fibre ([SMF](#)). The chirp quantification is well known for Mach-Zehnder modulators and directly modulated lasers ([DMLs](#)) [107,108]. However, this has to be studied for [RRMs](#) in order to evaluate chirp penalties after a few kilometers of optical propagation [109].

We propose to use the fiber transfer function method [54, 110, 111] to quantify the chirp of the Si-RRM. This consists of measuring the E/O frequency response of the modulator after propagation over various lengths of [SMF](#). Figure 4.18 reports the measurements made for 10 km and 50 km fiber lengths. The measurements made for 20-30 km and 40 km are not reported for a sake of clarity. Measurements at 0 km are used as reference (figure 4.15). One can note that the dips move to higher frequency and their depths decrease when the laser wavelength is shifted from the [RRM](#) resonant wavelength.

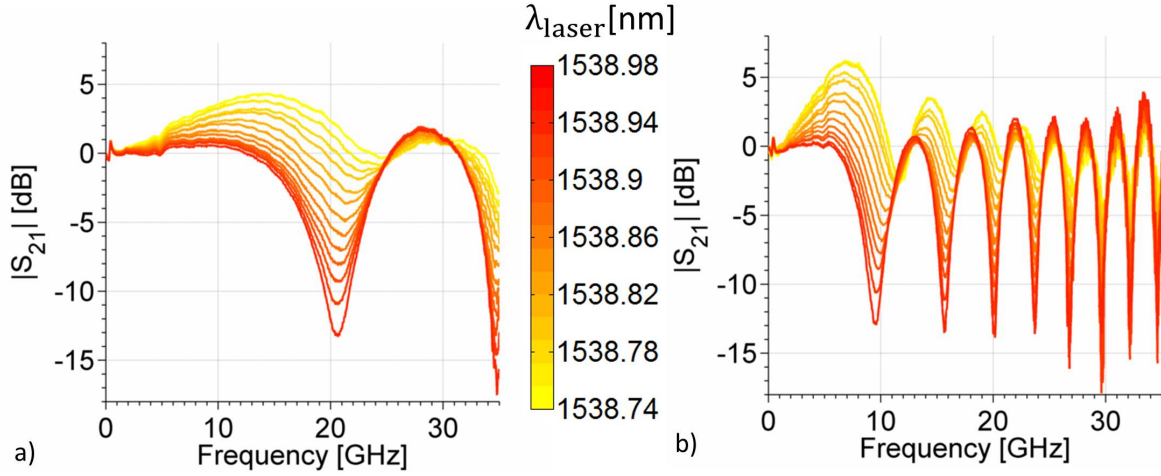


FIGURE 4.18: Measured electro-optical S21 parameter after 10 km a) and 50 km b) SMF. Measurements are done with a laser wavelength between 1538.98 nm in red and 1538.74 nm in yellow with a 0.02 nm step

In [111], authors have analytically modeled the transmission responses with a chirp parameter α which can be positive or negative, real and/or imaginary. The frequency position of the dips is essentially represented by the real part of the chirp parameter, and the dips depth is represented by its imaginary part. This is why, we use three different parameters (α_c , α_d and f_d) such as $\alpha = \alpha_c + i \cdot (\alpha_d + \frac{f_d}{f})$ to model the dips (Eq. 4.10).

$$\begin{aligned} S_{21} &= |\cos(\theta) - \sin(\theta) \cdot \alpha| \\ &= |\cos(\theta) - \sin(\theta) \cdot (\alpha_c + i \cdot (\alpha_d + \frac{f_d}{f}))| \end{aligned} \quad (4.10)$$

where θ is equal to $\theta = \frac{\pi L D \lambda^2 f^2}{c}$, D is the dispersion coefficient of the optical fiber equals to 16.5 pm/nm/km at 1.54 μm , L is the fiber length, λ is the laser wavelength and f is the modulation frequency.

We use the S21 measurement at 50 km to extract the three chirp parameters. Figure 4.19 reports the comparison between the measured S21 and the calculated S21 by the analytical model for three laser wavelength positions. The chirp parameters are obtained manually by changing them progressively to find the good fit.

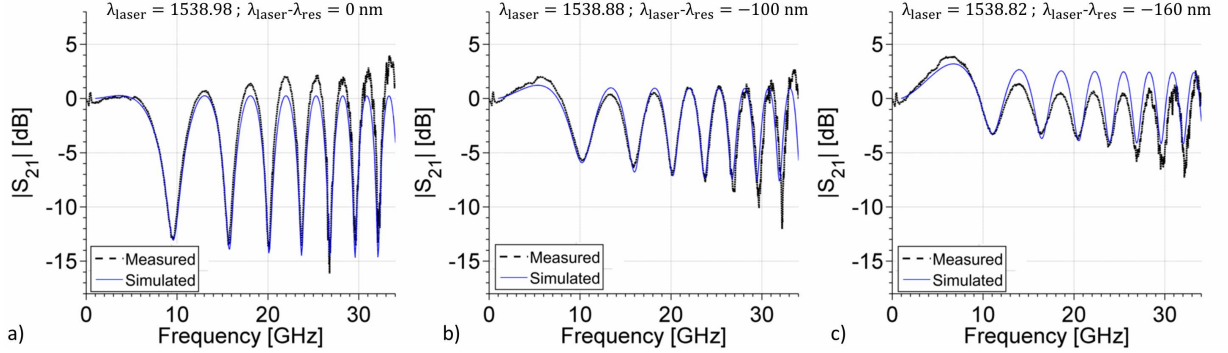


FIGURE 4.19: Comparison between measured and calculated S_{21} over 50 km SMF for three laser wavelength positions. The calculated S_{21} is obtained by analytic simulation with equation 4.10

This operation was repeated for all other laser wavelength positions. Figure 4.20 reports the extraction of the three chirp parameters against the laser wavelength position regarding to the **RRM** resonant wavelength. The real part of the chirp parameter α_c increases when the laser wavelength is shifted from the resonant wavelength, which is coherent with the increase in dip frequencies. It is worth noting that α_c is negative, leading to an increase in E/O bandwidth after the optical fiber [112]. The imaginary part of the chirp parameter α_i increases when the laser wavelength is shifted from the resonant wavelength which is coherent with the dip depth decrease.

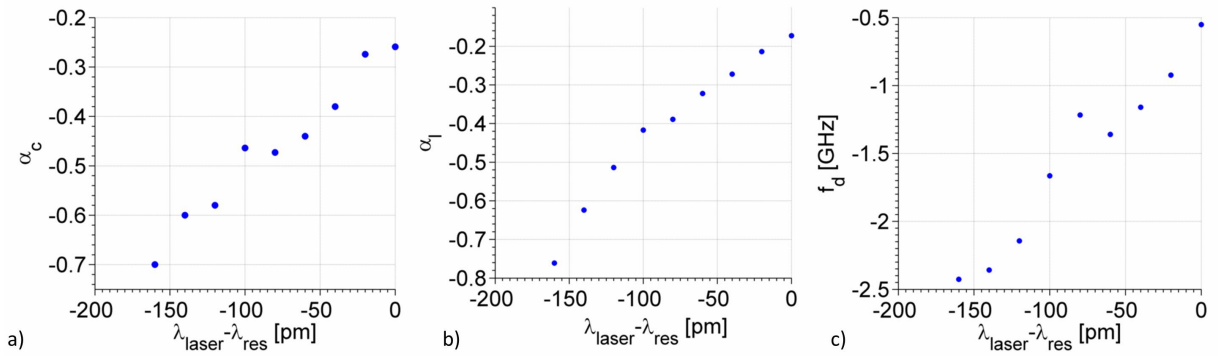


FIGURE 4.20: Extraction of the three chirp parameters through the measurements of S_{21} over 50 km SMF

We now simulate the S_{21} parameter after 50 km optical fiber with the **RRM** transient model (see chapter 3 for the model). Figure 4.21 reports simulations made from our model, indicating a better agreement with measurements. Thus, the developed **RRM** model is relevant for chirp considerations.

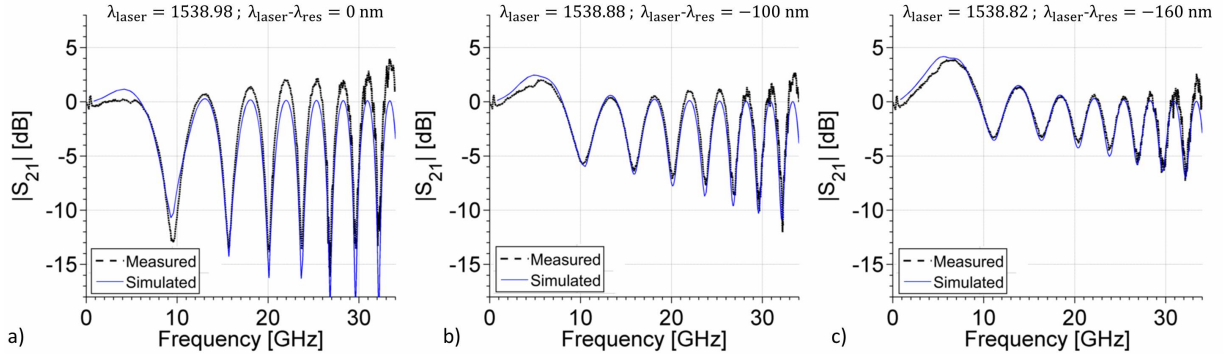


FIGURE 4.21: Comparison between measured and simulated S_{21} over 50 km SMF for three laser positions. The simulation is based on the RRM model

The study was done only for an over-coupled [RRM](#). To go further, it would be interesting to extend the analysis to under-coupled [RRM](#) [112].

For data center applications, the targeted communication distance is 2 kilometers. We have simulated the OMA after 2 km SMF optical propagation. The simulations are referenced with the simulated OMA at 0 km allowing to visualize the impact of the fiber dispersion and the chirp. The result is shown in figure 4.22. Over 2 km, at 33 GHz, 1.25 dB supplementary loss is added due to the fiber dispersion, so 50 Gbit/s optical communication is compromised. Following this comment, we conclude that the C-band (1.55 μm) is not the best choice for 2 km communication distances. In the next chapter, the C-band will no longer be considered, and we will work at 1.31 μm .

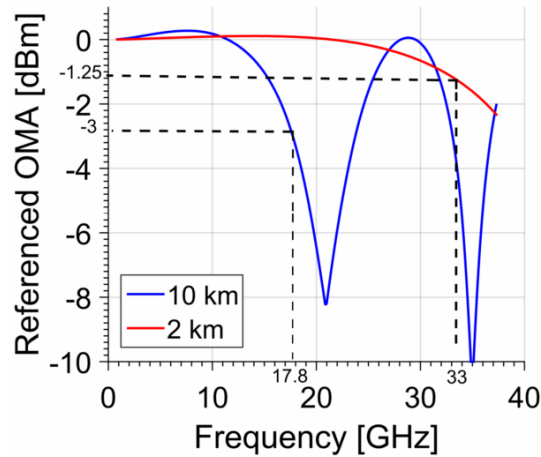


FIGURE 4.22: Simulated referenced OMA after 2 km SMF distance. The OMA is referenced with the OMA at 0 km

4.5 Conclusion

In summary, we have fully detailed the step-by-step comparison between simulated and measured **RRM** characteristics. Through DC and AC measurements, the **RRM** parameters have been extracted : the effective index n_{eff} , the group index n_g , the coupling factor κ_1 , the global loss within the ring, the self-heating parameter σ_{self} , the electrical bandwidth f_{elec} and the electro-optical bandwidth. This study allows the **RRM** model to be validated. Several trends have been highlighted for next **RRM** designs.

The quality of the resonator is defined by the coupling factor and the global loss within the ring. These two parameters have to be well known through simulations and measurements to choose the Q factor and to target the critical coupling condition.

The doping implantation quality is very important. Within the waveguide, it directly impacts the junction efficiency and the global loss. In the slab sections, it impacts the access resistance, and so the **RRM** electrical bandwidth. Indeed, we have seen through our measurements that the electrical bandwidth is reduced by half due to under-doped slab sections. The modulator performance improvement and its stability will be achieved through multi-test doping implantation conditions coupled with more realistic doping profiles simulations.

The laser wavelength position with respect to the resonant wavelength is a complex trade-off between the extinction ratio, the insertion loss and the electro-optical bandwidth. A laser wavelength position close to the resonant wavelength allows obtaining a very high extinction ratio at the cost of high insertion loss and low electro-optical bandwidth. In the following chapter, the Q factor will be chosen in order to fix the worst electro-optical bandwidth scenario, and the laser wavelength position will only depend on the couple [extinction ratio, insertion loss].

Chapter 5

Silicon Ring Resonator Modulator Optimization for 400 Gbit/s Applications

5.1 Introduction

In the previous chapters, we have developed an [RRM](#) model, which has been validated through measurements. Here, we propose to use this model in order to optimize modulators compliant with the 400 Gbit/s standard (see section 1.2). According to the technical specifications described in section 2.6, targeted [RRMs](#) will have an insertion loss (IL) lower than 3 dB and an extinction ratio (ER) higher than 8 dB. Two data rates are considered: 50 Gbaud with a [PAM-2](#) modulation and 25 Gbaud with a [PAM-4](#) modulation through 2 km optical transmission.

For a fixed voltage swing, the extinction ratio and the insertion loss depend on the performance of the junction, and on the laser wavelength position λ_{laser} compared to the resonant wavelength λ_{res} . A trade-off between the wavelength shift, the laser wavelength position and the Q factor is needed to achieve ERs higher than 8 dB and ILs lower than 3 dB with electro-optical bandwidths greater than the needs of the standards. In fact, optical bandwidth is affected by the Q factor and the laser wavelength position, while electrical bandwidth is affected by the doping levels of the PN junction, which is directly related to the performance of the junction.

In this chapter, we will consider a two-bus ring resonator modulator with a radius of 10 μm working at 1.31 μm and composed of a PN junction. The drop port is essential for the wavelength control against temperature variation (see section 3.3). Furthermore, by adding a second coupling coefficient, the Q factor can be chosen independently of the junction loss. The choice of the junction doping levels will be essential for the performance of the [RRM](#).

This chapter is divided into five parts. The main elements of the optimization methodology are explained in the first part, including the description of the cross sections chosen for the active region. The second section is focused on the PN junction optimization. Then, the main results are presented for a fixed Q factor equal to 13000 (equivalent to an approximated optical bandwidth of 18 GHz). This part also explains how to find an optimal operating point to achieve given performances. This approach is subsequently generalized for a full set of optical

bandwidths from 7 GHz to 38 GHz (Q factor between 6000 and 30000) in the third part. A compact model is derived with the related hypothesis and constitutive equations. In the fourth part, the compact model is used to find the best RRM responding to the desired performance at 50 Gbaud in PAM-2 format and at 25 Gbaud in PAM-4 format. The last part is dedicated to WDM transmitter considerations.

This chapter is based on the work published in [113] which has been carried out in close collaboration with Alexis Abraham.

5.2 Methodology

The numerical design flow is split into three steps, as shown in figure 5.1. In the first step, electro-optical simulations of the active region are performed for a matrix of doping levels leading to junction performances (Δn_{eff} , junction Loss α_j , $\Delta\alpha$, n_g , access resistance R and junction capacitance C). The purpose of these simulations is to provide data to be implemented in a ring resonator modulator model leading to modulator performances (OMA, ER, IL, f_{elec} , f_{opt}).

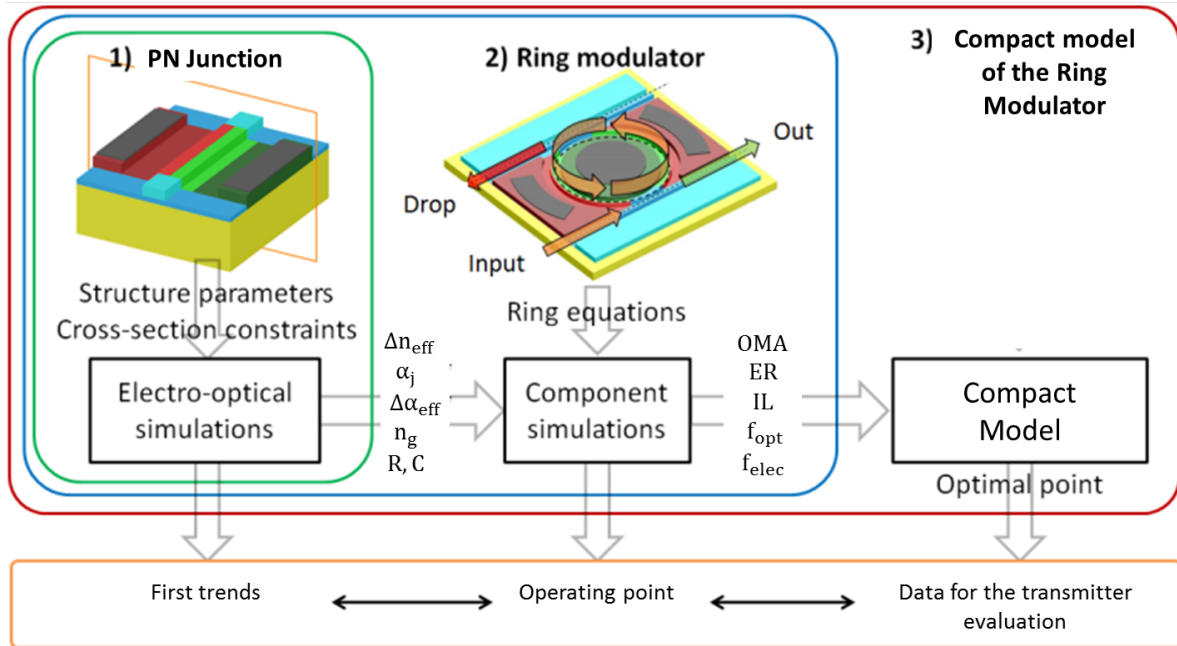


FIGURE 5.1: Numerical design flow for the design of the optimal ring resonator modulator

In the second step, the performances of the ring resonator modulator are simulated and we search to find the best OMA for each couple of $[\Delta n_{eff}(V), \Delta\alpha(V)]$ out of all the possibilities, by adjustment of the coupling factors $[\kappa_1^2, \kappa_2^2]$ and the laser wavelength relatively to the resonant wavelength. As explained in section 2.5.5, with the equation 2.34, for a fixed Q factor, the approximated optical bandwidth f_{opt} is the worst case (small peak to peak driving voltage and $\lambda_{laser} = \lambda_{res}$). We set some restrictions in the simulation: an insertion loss below 3 dB

and coupling factors κ_1^2 and κ_2^2 higher than 1% in intensity in order to guarantee a minimal coupling. A significant gain in computation time can be obtained if a first test is done at the critical coupling point with the laser wavelength aligned to the resonant wavelength. Indeed, in this condition the ER is maximum. If the IL is below 3 dB, the point is considered to be optimal. For other cases, we test all couples of $[\kappa_1^2, \kappa_2^2]$, setting the laser wavelength to a value which gives insertion loss lower than 3 dB and extract the extinction ratio. In step three, the compact-model of the RRM is build. The compact model will then be used to evaluate the performances of the transmitter.

5.3 PN junction optimization

5.3.1 PN junction cross-section description

This study is done for a 300/50 nm rib waveguide with 320 nm width which corresponds to the ST/CEA-Leti 1.3 μm fabrication platform ([72, 114]). The cross section of the PN structure is presented in figure 5.2. A slab height of 50 nm enables short bends, and a waveguide width of 320 nm ensures monomode behavior (section 2.2). Both doping levels are placed inside the rib for the PN junction. The position of the junction δ_{PN} ensuring the best efficiency for the PN modulator depends on each doping level, and will be taken into account during simulations. We assume a uniform doping profile for all doping regions in order to keep the most general approach. As the geometrical parameters are mainly fixed by the technology, we perform a parametric study depending on the doping level of the N side and the P side between 6×10^{16} and 5×10^{18} . The propagation loss due to technological imperfections (without losses due to the doping in the waveguide) are assumed to be 0.5 dB/mm (section 2.2).

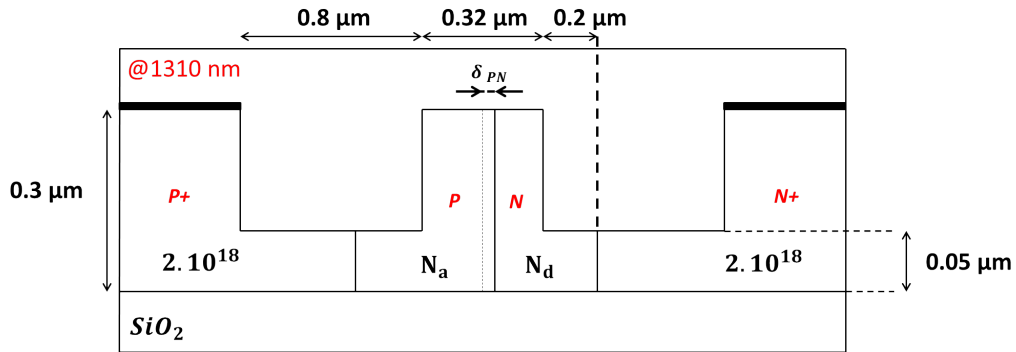


FIGURE 5.2: Cross-section of the PN modulator

As shown in previous chapters, the main modulation parameters to be computed are $\Delta n_{eff}(V_m)$ (indicating the efficiency of the modulator) and $\Delta \alpha(V_m)$ (associated with the optical losses). For more practicality, we use the following figures of merits: the optical losses are expressed in [dB/mm], and the phase modulation efficiency of the junction is expressed as $\Delta \phi$ between 0 V and V_m in [$^\circ/mm$].

5.3.2 PN junction simulation and optimization

Electro-optical simulations were done with Lumerical Solutions Software combining modal simulations and electrical simulations. Figures 5.3 reports the phase shift and the losses inside the ring against the doping levels. The PN junction is modulated between 0 V and -2V. For each possible phase shift, the optical losses can reach a minimal value: this expresses the limitation of the cross-section itself, and the best compromise between phase shift and losses, *i.e* highest efficiencies at lowest losses represented by a white line in figure 5.3 a).

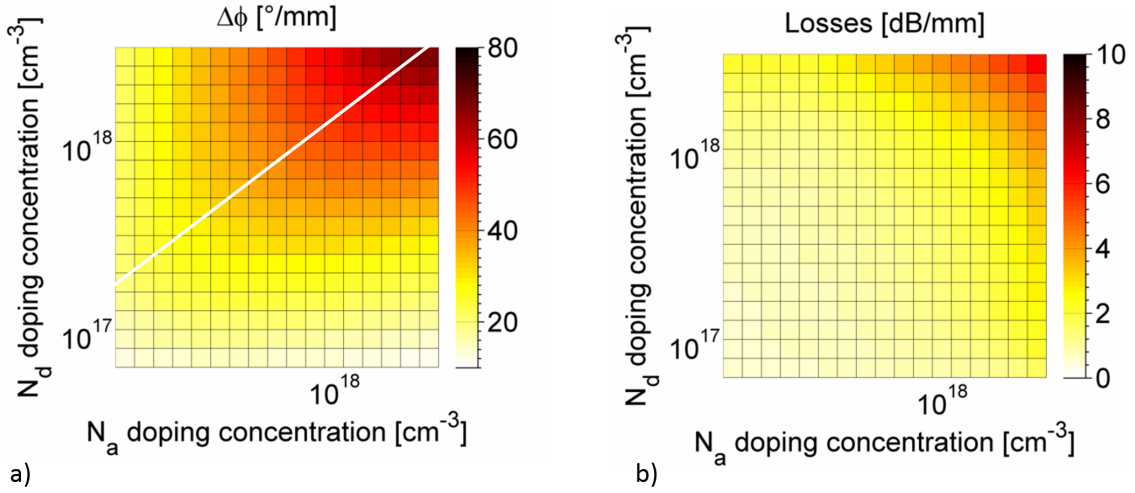


FIGURE 5.3: Phase shift a) and losses b) for the PN junction with $2V_{pp}$ against the doping concentration. The straight white lines correspond to the highest efficiencies at lowest losses

For all computed couples $[\Delta\phi, \text{Loss}]$ (blue zone in figure 5.4), there is a 2^{nd} order polynomial linking the highest efficiencies to the lowest losses (equation 5.1).

$$\text{Loss}_{min} [\text{dB/mm}] = 3 \times 10^{-3} \Delta\phi [^\circ/\text{mm}]^2 - 1.51 \times 10^{-1} \Delta\phi [^\circ/\text{mm}] + 2.69 \quad (5.1)$$

From here, we retrieve the junction parameters (N_a, N_d and δ_{PN}) corresponding to these minima (equation 5.2, 5.3 and 5.4). The junction position comes from the doping levels, and has to be approximated by a 4^{th} order polynomial. Figure 5.5 shows PN junction optimization curves. An example of use is reported with a phase shift equal to $50^\circ/\text{mm}$.

$$\log_{10}(N_a) [\text{cm}^{-3}] = -3.86 \cdot 10^{-4} \Delta\phi [^\circ/\text{mm}]^2 + 6.77 \cdot 10^{-2} \Delta\phi [^\circ/\text{mm}] + 15.54 \quad (5.2)$$

$$\log_{10}(N_d) [\text{cm}^{-3}] = 8.58 \cdot 10^{-6} \Delta\phi [^\circ/\text{mm}]^2 + 2.18 \cdot 10^{-2} \Delta\phi [^\circ/\text{mm}] + 17.00 \quad (5.3)$$

$$\delta_{PN} [\text{nm}] = 7.87 \cdot 10^{-5} \Delta\phi [^\circ/\text{mm}]^4 - 1.802 \cdot 10^{-2} \Delta\phi [^\circ/\text{mm}]^3 + 1.555 \Delta\phi [^\circ/\text{mm}]^2 - 60.69 \Delta\phi [^\circ/\text{mm}] + 932.12 \quad (5.4)$$

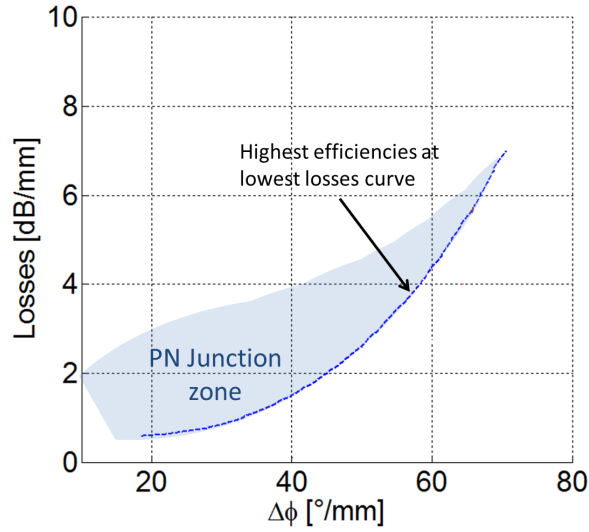


FIGURE 5.4: Diagram of optical losses at V_0 depending on the phase shift at V_m . The colored areas include all the couples of doping levels in the range $[7 \cdot 10^{16}; 5 \cdot 10^{18}]$. The line curve is the best case described by equation 5.1.

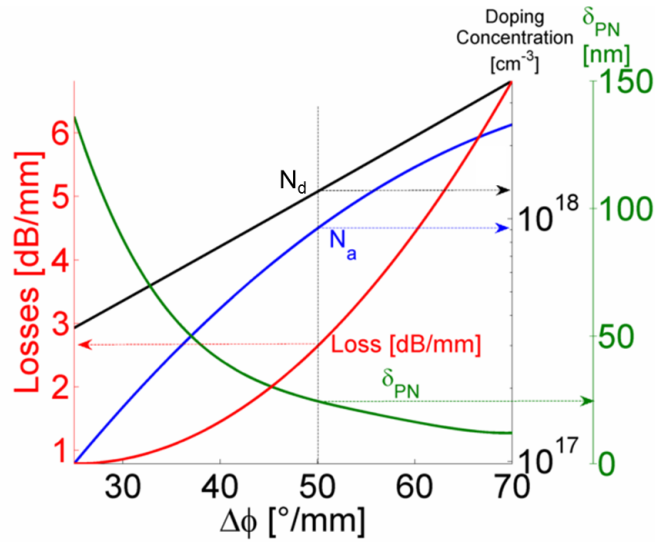


FIGURE 5.5: Optimization PN junction curves linked the highest efficiencies to the lowest losses and the corresponding junction parameters N_a , N_d and δ_{PN}

The electrical access resistance and junction capacitance are simulated as explained in section 2.5.3 considering uniform doping concentrations. Note that we will take into consideration the worst case of the junction capacitance, *i.e.* when the voltage is minimal (0 V in our case). Figure 5.6 reports the computed access resistance and junction capacitance for a 300/50/320 nm rib waveguide with δ_{PN} ensuring the best efficiency. We search low access resistance and low capacitance to achieve high electrical bandwidth. However, it is necessary to find a compromise as high doping concentrations mean low capacitance but high access resistance, while low doping concentrations mean low capacitance but high access resistance.

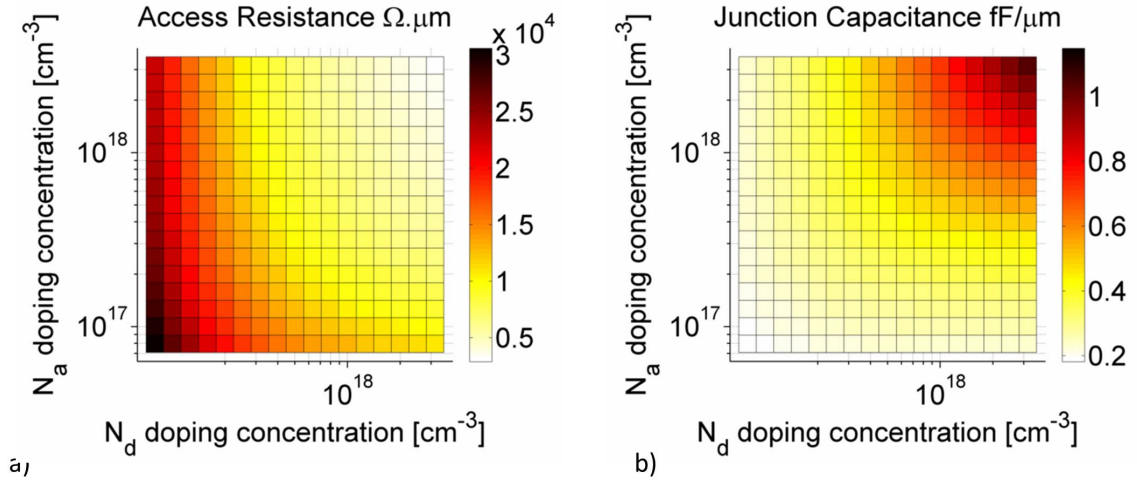


FIGURE 5.6: a) Access resistance and b) junction capacitance of the PN junction against the doping concentrations

The electrical bandwidth f_{elec} is estimated with the formula $1/(2\pi RC)$. It does not depend on the active length, which is why we can compute it now. Figure 5.7 reports the electrical bandwidth limitation for the PN junction. We can note that the electrical bandwidth is always higher than 50 GHz and will certainly be non-limiting.

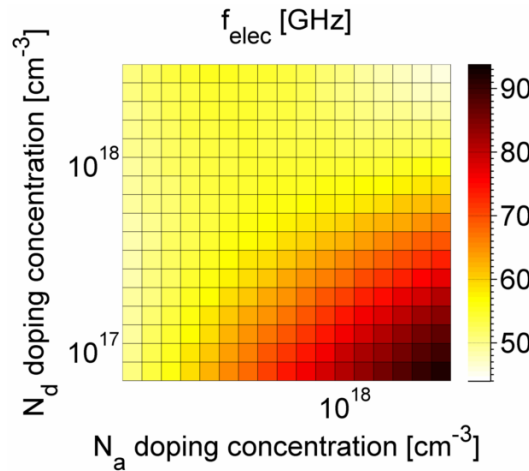


FIGURE 5.7: Electrical bandwidth for the PN junction

At this point the PN junction has been optimized. It has been checked that the electrical bandwidth is sufficient.

5.4 Ring resonator modulator optimization

In this section, we report the results of the parametric study as a function of the doping levels. The active region is implemented in a ring resonator modulator of $10 \mu\text{m}$ radius. This allows

to guarantee negligible bend loss compared to the propagation loss, estimated to be below 0.5 dB/mm (section 2.2). The ring modulator is a two-bus RRM. The absolute value of the swing voltage is set to [0 V, 2 V] in order to be compatible with 65 nm CMOS driver dynamic range.

The ring resonator modulator is implemented in a transmitter, which is shown schematically in figure 5.8. As previously reported in section 2.6, in order to be compliant with the 400 GBASE-LR8, the OMA at the output of the transmitter (OMA_{TX}) has to be higher than 0 dBm. We set the link loss (taking in to account grating coupler, packaging and propagation) identical to that in section 2.6 (also reported in table 5.1).

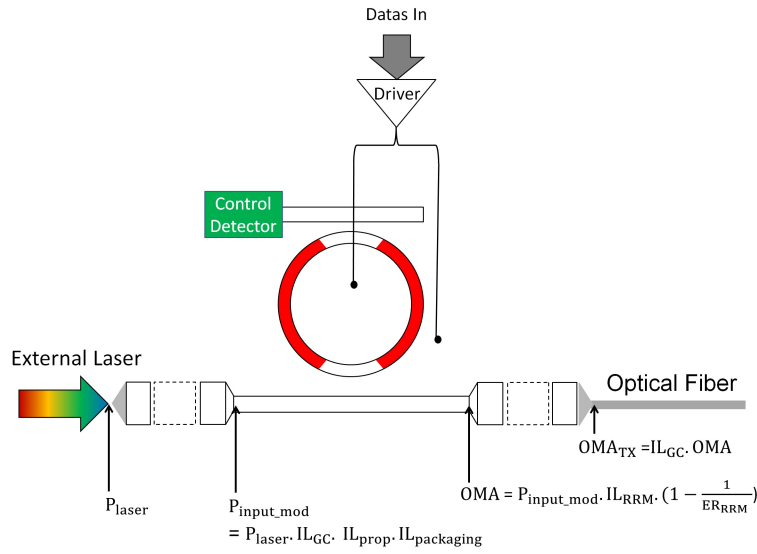


FIGURE 5.8: Simplified schematic representation of the transmitter using RRM

TABLE 5.1: Transmitter characteristics

Laser optical power P_{laser} [dBm]	Grating coupler loss IL_{GC} [dB]	Propagation loss IL_{prop} [dB]	Packaging loss $IL_{packaging}$ [dB]
10	2	0.7	1

According to section 2.6, the OMA_{TX} is higher than 0 dBm when the RRM exhibits an extinction ratio (ER_{RRM}) higher than 8 dB with an insertion loss (IL_{RRM}) lower than 3 dB. This section optimizes RRM against doping concentrations, which allows to get 8 dB extinction ratio to be achieved with a maximum insertion loss of 3 dB.

5.4.1 Ring resonator modulator optimization for Q factor=13000

The Q factor is set initially to 13000 in order to obtain an approximate bandwidth of 18 GHz. The radius of the ring is set to 10 μm with an active length of 80 % of the total perimeter of the ring. As explained in section 5.2, we use the PN junction simulations coupled with the ring modulator in order to maximize the extinction ratio with maximum insertion loss of 3 dB.

Figure 5.9 reports the result of the optimization for a ring modulator with 10 μm radius and a Q factor of 13000.

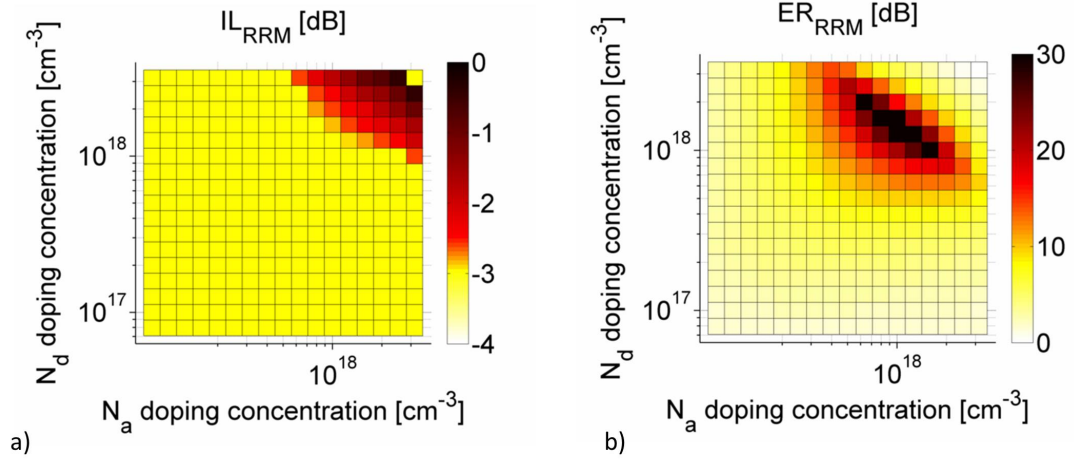


FIGURE 5.9: Optimal ER_{RRM}/IL_{RRM} with a modulated voltage between 0 and -2 V against the doping concentrations

Each couple of doping values $[N_a, N_d]$ corresponds to a set of optimal coupling factors $[\kappa_1, \kappa_2]$ and laser wavelength with respect to the ring resonance associated with ER_{RRM} and IL_{RRM} (figure 5.10).

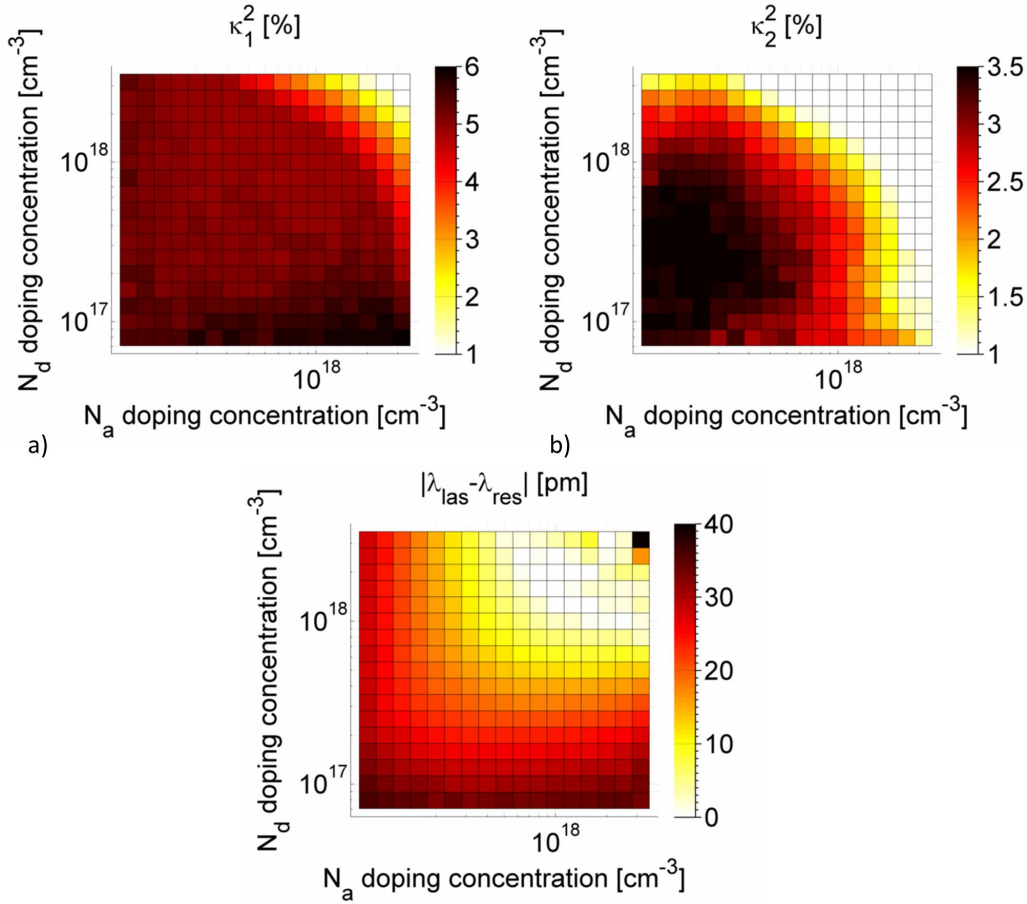


FIGURE 5.10: Associated κ_1^2, κ_2^2 and the λ_{laser} position against the doping concentrations

With the simulated ER_{RRM} and IL_{RRM} , the OMA_{TX} is computed and reported in figure 5.11. The $[N_a, N_d]$ range which leads to the acceptable performance values of ring modulators ($\text{ER}_{RRM} > 8$ dB and $\text{IL}_{RRM} < 3$ dB) is then represented in figure 5.11 by the " $\text{ER}_{RRM} > 8$ dB" region. This represents the doping region which has to be targeted. A large doping region achieves the target OMA of 0 dBm. In this region, a best point of design can be extracted and is given in table 5.2.

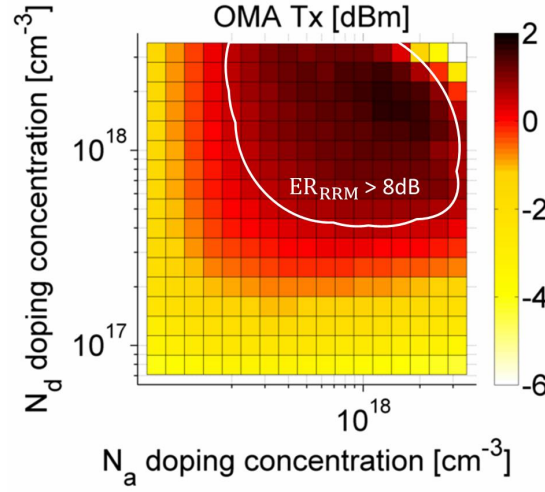


FIGURE 5.11: Optimal OMA at the output of the transmitter with a modulated voltage against the doping concentrations

TABLE 5.2: Best point of design for ring modulator with $Q=13000$

	OMA _{Tx} [dBm]	$N_A[\text{cm}^{-3}]$	$N_D[\text{cm}^{-3}]$	$\kappa_1^2[\%]$	$\kappa_2^2[\%]$	$ \lambda_{las} - \lambda_{res} [\text{pm}]$
PN Junction	1.49	$8.9 \cdot 10^{17}$	$1.4 \cdot 10^{18}$	4.83	1	0

5.4.2 Compact model

The optimization methodology described in 5.2 to design a ring modulator with 18 GHz electro-optical bandwidth can be applied to design ring modulators with different Q factors. This allows the optimization of the ring modulator for any optical bandwidth, and reveals the intrinsic limit of the PN junction that can be described by a compact model.

OMA function of the Q factor

For a Q factor between 6000 and 30000 (corresponding to an approximated optical bandwidth between 38 GHz and 7 GHz), we simulate the optimized ring modulators, and extract all best design points in terms of doping levels, coupling coefficients and laser wavelength shift. Figure 5.12 depicts the best OMA that can be achieved against the Q factor, without taking into account the electrical bandwidth. When adding the electrical bandwidth restriction, the points do not change since the electrical bandwidth is not limiting for the PN junction.

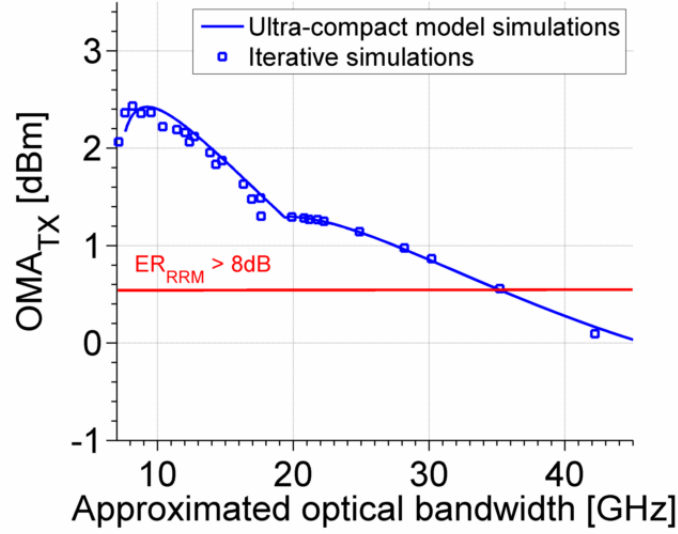


FIGURE 5.12: Optimal OMA_{TX} against the optical bandwidth for the PN junction without taking into account the electrical bandwidth limitation

The restriction of OMA_{TX} to values above 0 dBm for practical implementation is maintained for an optical bandwidth lower than 45 GHz. However, in order to keep an extinction ratio higher than 8 dB, the optical bandwidth has to be lower than 35 GHz (Q factor of 6500). This condition allows us to keep a margin for design exploration.

We emphasize here two essential points. First, we observe that among all the possible combinations depending on $(N_a, N_d, \lambda_{laser}, \kappa_1^2, \kappa_2^2)$, each optimal point is located on the curve $Loss_{min} = f(\Delta\phi)$ previously presented in equation 5.1. In other words, the optimization algorithm searches for points combining the highest efficiency with lowest losses, in order to achieve the critical coupling condition. Second, each dotted-series curve in figure 5.12 possesses different regimes, which are analytically modeled (plain curves), and constitute the basis of the compact model described in the next section.

Constitutive equations of the compact model

With a Lorentzian approximation of the transfer function of the ring resonator modulator (equation 5.5), some simplifications can be done in order to describe the optimal design points.

$$T_{out}(\lambda) = 1 - \left(1 - \frac{1}{ER_{on-off}}\right) \cdot \frac{1}{1 + 4 \cdot \left(Q \cdot \frac{\lambda - \lambda_{res}}{\lambda_{res}}\right)^2} \quad (5.5)$$

where ER_{on-off} is the maximum extinction ratio between a complete on and off resonance and Q is the Q factor.

We distinguish two different simplifications: high Q factor and low Q factor. The threshold is given by $Q_{threshold}$ expressed in equation 5.6. For our PN junction, $Q_{threshold}$ is equal to 1180.

$$Q_{threshold} = \frac{1}{4} \left(\frac{\lambda_{res}(V_0)}{\lambda_{res}(V_m)} \right)^2 \quad (5.6)$$

For low Q factor ($Q < Q_{threshold}$), the laser wavelength cannot be set to the resonant wavelength, because the insertion loss will be greater than 3 dB. Hence, the wavelength of the laser is set to 3 dB insertion loss, leading to equations 5.7 and 5.8.

$$\lambda_{laser} = \frac{\lambda_{res}(V_m)}{2 \cdot Q(V_m)} + \lambda_{res}(V_m) \quad (5.7)$$

$$\lambda_{res}(V_m) = \lambda_{res}(V_0) + \Delta\phi \frac{FSR}{360} \cdot L_{active} \quad (5.8)$$

Moreover, it is possible to achieve the critical coupling (with $\kappa_2^2 > 1\%$). Under this condition, equation 5.5 is simplified and gives a basic equations for the extinction ratio ER and the insertion loss IL (equation 5.9).

$$\begin{cases} IL = 0.5 \\ ER = \frac{1}{2} - \frac{1}{1 + 4 \cdot (Q(V_0) \cdot \frac{\lambda_{laser} - \lambda_{res}(V_0)}{\lambda_{res}(V_0)})^2} \end{cases} \quad (5.9)$$

For each Q factor, a corresponding $[\Delta\phi, (Loss)]$ related by equation 5.1 is given when the best $\Delta\phi$ and critical coupling is achieved. It leads to a fixed κ_2^2 equal to 1%.

For a high Q factor ($Q > Q_{threshold}$) given by equation 5.6, the laser wavelength will be set to the resonant wavelength $\lambda_{laser} = \lambda_{res}(V_0)$. When critical coupling can be achieved, $[Loss, \kappa_1^2, \kappa_2^2]$ can be deduced. The best $\Delta\phi$ is obtained for fixed κ_2^2 equal to 1%. This is described by equation 5.10. When the Q factor increases, the losses have to be decreased to achieve critical coupling.

$$\begin{cases} IL = 1 - \frac{1}{1 + 4 \cdot (Q(V_0) \cdot \frac{\lambda_{res} V_0 - \lambda_{res}(V_m)}{\lambda_{res}(V_m)})^2} \\ ER = +\infty \end{cases} \quad (5.10)$$

In figure 5.14, the compact model (straight line) is compared with the iterative simulated points. The model is in very good agreement with the simulated points. The iterative simulation gives one point for one Q factor in 3 hours computing time, while the compact model gives the entire curve in 1 minute (1000 Q factor computed values). Table 5.3 summarizes the different possible regimes with the various simplifications.

TABLE 5.3: Compact Model Summary

	Low Q factor $Q < Q_{threshold}$	High Q factor $Q >_{threshold}$
λ_{laser}	Eq. 5.7	$\lambda_{res}(V_0)$
Critical Coupling	Yes	Yes
κ_2^2	Minimum admissible 1%	Minimum admissible 1%
ER	Eq. 5.9	$+\infty$
IL	0.5	Eq. 5.10

Based on table 5.3, the ring modulator can be designed with the compact model. The choice of the Q factor sets the best regime of the ring modulator, resulting in specific values of coupling factors and loss. With equations 5.1, 5.2, 5.3 and 5.4, the corresponding phase shift, doping levels and junction position are then deduced. We will now use this procedure to obtain the best RRM for data rates equal to 50 Gbaud with PAM-2 modulation and 25 Gbaud with a PAM-4 modulation.

5.5 50 Gbaud PAM-2 and 25 Gbaud PAM-4 transmitter optimization

In this section, we will use the compact model presented previously in order to optimize RRM for 50 Gbaud PAM-2 and 25 Gbaud PAM-4 transmissions. For this optimization, the simulated transmitter is represented in figure 5.13. The ring resonator modulator is modulated by random PAM-2 or PAM-4 electric signal with 2 V_{pp}. The input laser power is set to 10 dBm. As in the previous section, the transmitter exhibits losses due the input and output fiber coupling, the light propagation in the waveguide and the packaging (table 5.1).

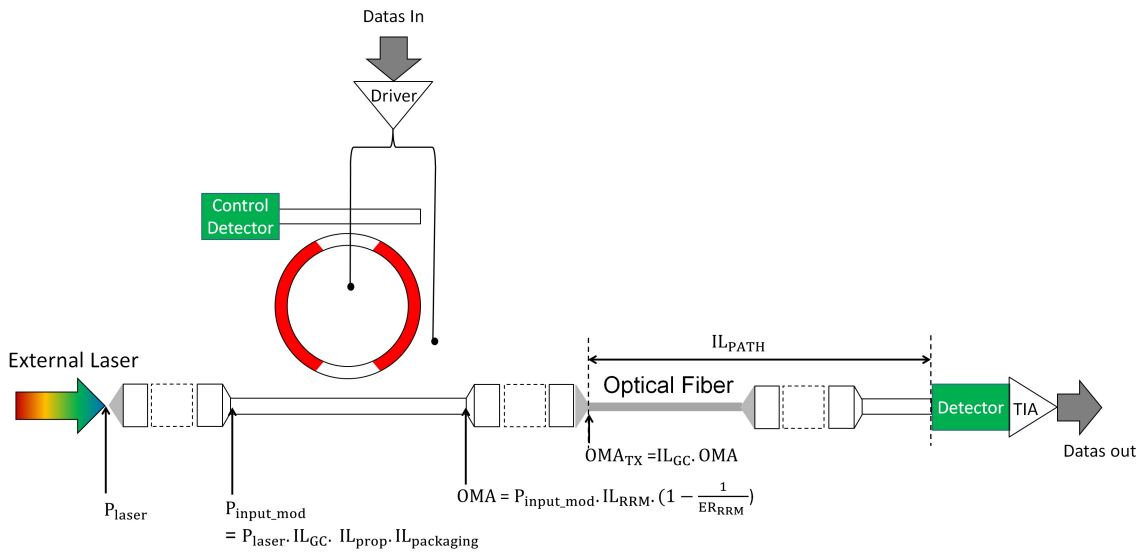


FIGURE 5.13: Simplified schematic of a high speed short reach link with the parameters used in the simulation

A link, which can be composed by combining multiplexers, grating couplers and fibers, is grouped into an optical loss IL_{PATH} . With 2 kilometers optical fiber, IL_{PATH} is evaluated at 7 dB (1 dB for the fiber propagation loss, 1 dB for the packaging, 3.25 dB for the 2 D grating coupler, 0.7 dB for the optical propagation in the optical chip, and 1 dB for demultiplexer). On the receiver side, the photo-receiver followed by a TIA has the characteristics reported in table 5.4.

TABLE 5.4: Receiver simulation characteristics

	Dark Current	Photo-diode responsivity	Bandwidth	NEP	Gain
	I_d [nA]	R_s [A/W]	f_c [GHz]	[pW/ $\sqrt{\text{Hz}}$]	[V/A]
For 50 Gbaud PAM-2	1	0.6	35	10	2000
For 25 Gbaud PAM-4	1	0.6	18	10	2000

As in [115], we aim to identify the best choice of RRM that maximizes the optical budget IL_{PATH} that the system can accommodate. If we can accommodate an IL_{PATH} higher than 7 dB, it means that the RRM can perform the 2 kilometers link. The targeted BER is 2×10^{-4} .

For each Q factor between 5000 and 25000 (which corresponds to an approximated optical bandwidth between 45 GHz and 10 GHz), we compute with the compact model the parameters of each optimized RRM ($\kappa_1, \kappa_2, \lambda_{laser} - \lambda_{res}$ and also N_a, N_d , and δ_{PN}). For each optimized RRM, we dynamically simulate the modulated transmitter output power. Before the photo-diode, we add some loss corresponding to IL_{PATH} . And after the photo-diode, we evaluate the BER using a gaussian model. Then, we select the IL_{PATH} values which give a BER of 2×10^{-4} .

Figure 5.14 shows the maximum IL_{PATH} for 50 Gbaud PAM-2 and 25 Gbaud PAM-4 data rates as a function of the optical bandwidth (*i.e.* Q factor).

A 25 Gbaud PAM-4 transmission is achieved with optical bandwidth higher than 20 GHz. A maximum IL_{PATH} of 12 dB is reached at 30 GHz optical bandwidth (Q factor=7500). For 50 Gbaud PAM-2 data rate, the optimum RRM is found to have an optical bandwidth of 32 GHz (Q factor=7000). For both, the corresponding RRM parameters are reported in table 5.5.

We can note that their optical bandwidths are lower than 35 GHz, which allows an extinction ratio higher than 8 dB, an insertion loss lower than 3 dB and an OMA_{TX} higher than 0 dBm (see section 5.4.2). It is possible to consider that PAM-4 modulation allows bandwidth requirement to be half while adding 3.5 dB modulation loss. Unfortunately, RRM are not perfectly linear modulators, thus leading to a high distortion in PAM-4 modulation. In the end, the same optical bandwidth is required, above 30 GHz. This is why a RRM based transmitter modulated at 50 Gbaud PAM-2 is better than the same transmitter modulated at 25 Gbaud with PAM-4 modulation.

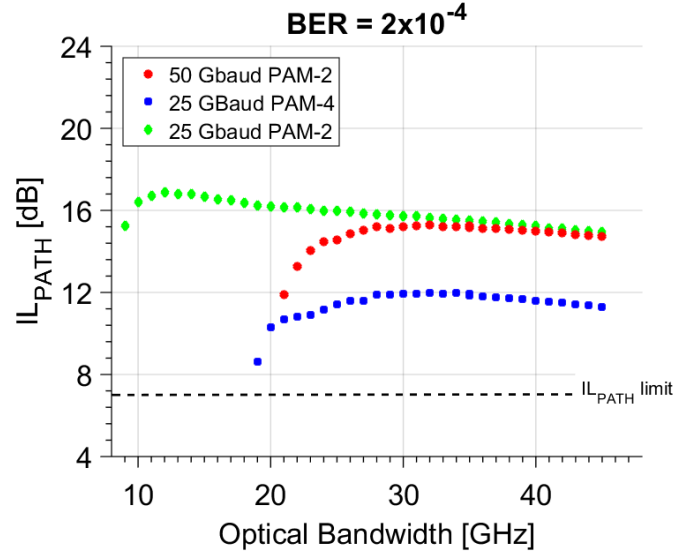


FIGURE 5.14: Maximum tolerable path loss as a function of optical bandwidth for $BER=2.4 \times 10^{-4}$

TABLE 5.5: 50 Gbaud PAM-2, 25 Gbaud PAM-4 RRM and 25 Gbaud PAM-2 designs

	50 Gbaud PAM-2	25 Gbaud PAM-4	25 Gbaud PAM-2
Q factor	7500	7000	13000
OMA [dBm]	-3.5	-3.6	-2.4
ER [dB]	9.8	8.8	Inf
IL [dB]	-3	-3	-2.4
$ \lambda_{laser} - \lambda_{res} $ [pm]	20	25	0
κ_1^2 [%]	8.1	8.65	4.00
κ_2^2 [%]	1	1	1
N_A [cm^{-3}]	$1.89 \cdot 10^{18}$	$2.04 \cdot 10^{18}$	$7.06 \cdot 10^{17}$
N_D [cm^{-3}]	$2.58 \cdot 10^{18}$	$2.85 \cdot 10^{18}$	$1.07 \cdot 10^{18}$
δ_{PN} [nm]	14	13	28
$\Delta\phi$ [$^\circ/\text{mm}$]	63.23	65.05	46.35
R [Ω]	67	64	93
C [fF]	50	52	32
f_{elec} [GHz]	47.6	48.1	53.8

In order to cope with this issue, PAM-4 modulation can be carried out optically *ie* based on the modulation of two modulators with two different PAM-2 electric signals which are then combined to create the PAM-4 in the optical domain [67, 116]. This architecture type is called optical digital to analog converter (DAC see in the next chapter). In this architecture each RRM has to maintain 25 Gbaud PAM-2 transmission to produce a 25 Gbaud PAM-4 transmission. The corresponding simulation is reported in figure 5.14, where the optimized optical bandwidth is 18 GHz (Q factor equal to 13000). The associated RRM parameters are reported in the table 5.5. One can note that in comparison to the previous design, the laser wavelength is set to the resonant wavelength and the doping levels are easier to achieve.

5.6 Conclusion

In summary, we have fully detailed the step-by-step design and modeling process of silicon ring modulators, starting from the fundamental opto-geometrical and electro-optical properties of PN junctions. The design was carried out for a 10 μm radius ring resonator modulator in the 300/50/320 nm technology at 1.31 μm . We have first described the design for ring resonator modulator with a Q factor equals to 13000 and then we have generalized for large range of Q factors. The optimized RRM is defined by the coupling coefficients (κ_1^2, κ_2^2), the position of the laser wavelength with respect to the resonant wavelength, and the doping levels associated with the junction position. An ultra compact model was proposed to easily link all RRM parameters to each other.

Using the ultra compact model, we have designed a RRM for 50 Gbaud PAM-2 and 25 Gbaud PAM-4 transmissions. The targeted design has to achieve an extinction ratio of 8 dB with insertion loss lower than 3 dB. Optimized points require very high efficiency (and consequently high doping levels) which are consistent with the recent measurements reported in [21]. Optimized RRM reach an OMA higher than 0 dBm at the output of the transmitter which is compliant with the 400 GBASE-FR8 standard. In order to achieve a 25 Gbaud PAM-4 or 50 Gbaud PAM-2, the RRM needs an optical bandwidth higher than 30 GHz. However, 50 Gbaud PAM-2 appears to be the best modulation approach since 3.5 dB supplementary optical margin is available.

Optimized points are at the limit of the PN performance. In order to cope with this issue, the use of the designed RRM for 25 Gbaud PAM-2 transmission allows the design complexity to be reduced, but at the same time also reduces the data rate per wavelength. Two solutions are possible. First, we can modulate the RRM in PAM-2 format and increase the number of wavelengths (from 8 to 16) to achieve a 400 Gbit/s transmission. One can note that this argument is coherent with the Open Computing solutions [19]. Alternatively, we can modulate the RRM in PAM-2 format at 25 Gbaud. With specific transmitter architectures, the PAM-4 transmission at 50 Gbit/s per channel is created with a digital to analog converter directly in the optical domain.

Chapter 6

Optical PAM-4 Digital to Analog Converter Transmitter Architectures

6.1 Introduction

So far, for obtaining 50 Gbit/s, we have evaluated two options: 50 Gbaud PAM-2 and 25 Gbaud PAM-4 using a single RRM. PAM-4 generation has been carried out in the electrical domain with a digital to analog converter (DAC). Due to the 4-level modulation, the required electro-optical bandwidth has to be higher than 30 GHz, though the transmission operates at 25 Gbaud. With the same electro-optical bandwidth, 50 Gbaud PAM-2 can be achieved and gives extra 5 dB margin loss.

In this chapter we investigate new architectures for implementing PAM-4 using two RRM_s modulated with two parallel bit streams. The DAC function is made in the optical domain. This presents immediate advantages:

- The required electro-optical bandwidth of the RRM can be reduced because each RRM is modulated over two levels only at 25 Gbaud PAM-2.
- The electrical DAC is no longer required. Hence, it reduces cost and power consumption as well as making it much simpler to implement in real time.

Optical DAC architectures were first proposed and implemented using MZMs. The MZM is segmented in a multi-electrode structure, where each electrode is driven by a different bit stream, as shown schematically in figure 6.1 [67,117,118]. Mazzini et al. [67] from Cisco Systems achieved 25 Gbaud PAM-4 through 10 km with a BER under 10^{-11} with a FEC encoder. The phase modulation structure used is a Silicon-Insulator-Silicon Capacitor (SISCAP) structure.

In this chapter, we propose to study two transmitter architectures using a RRM-based optical DAC to generate PAM-4 modulation. The first architecture is composed of two RRM_s arranged in series modulated with two bit streams. This was previously proposed by Seyedi et al. in [119]. The second architecture is composed of two RRM_s arranged in parallel (in a MZI) modulated with two bit streams [120]. We will explain the operating principles of both proposed transmitter architectures with associated simulated performances. Experiments will be reported to validate the architectures.

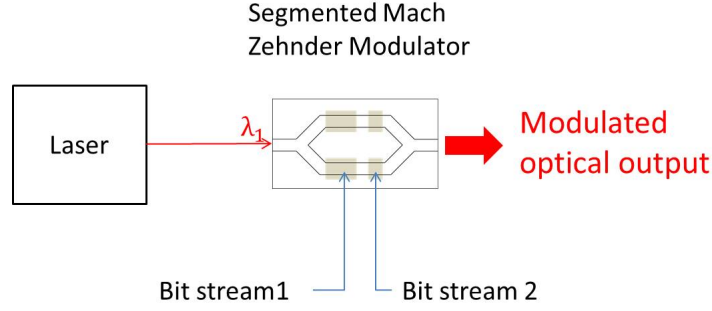


FIGURE 6.1: Segmented Mach Zehnder Architecture

6.2 Two ring resonator modulators arranged in series

6.2.1 Operating Principle

PAM-4 signal generation is achieved by modulating independently two identical RRM with two bit streams. These two RRM are arranged in series, as shown in figure 6.2. The two RRM exhibit the same transfer function but their resonant wavelengths are slightly shifted leading to different operating points *i.e.* different extinction ratios and insertion losses. Therefore, the response of two RRM arranged in series is the product of the transfer function of each RRM. This architecture is interesting as it can be used as a re-configurable transmitter because both modulators can be wavelength tuned in order to convert the single wavelength PAM-4 transmitter into a WDM PAM-2 transmitter.

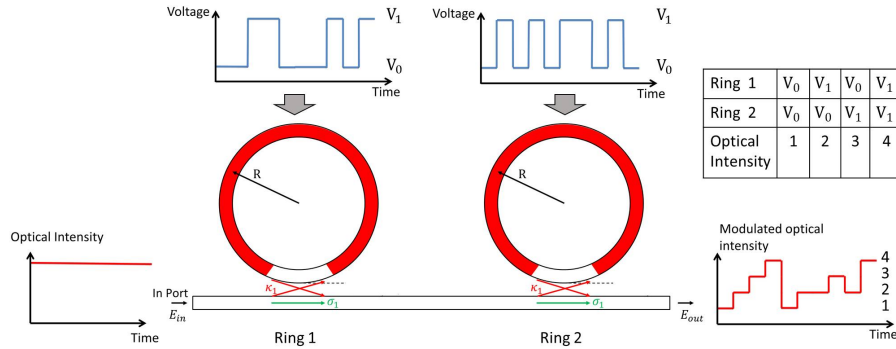


FIGURE 6.2: PAM-4 two ring resonator modulators arranged in series schematic

The separation between the laser wavelength and the resonant wavelength of the first RRM resonant wavelength ($\Delta\lambda_1$) and the distance between the laser wavelength and the second RRM resonant wavelength ($\Delta\lambda_2$) must be different to generate four output intensity levels, as shown in figure 6.3. This difference leads to different optical modulation amplitudes between the two RRM: OMA_1 for the first modulator and OMA_2 for the second modulator.

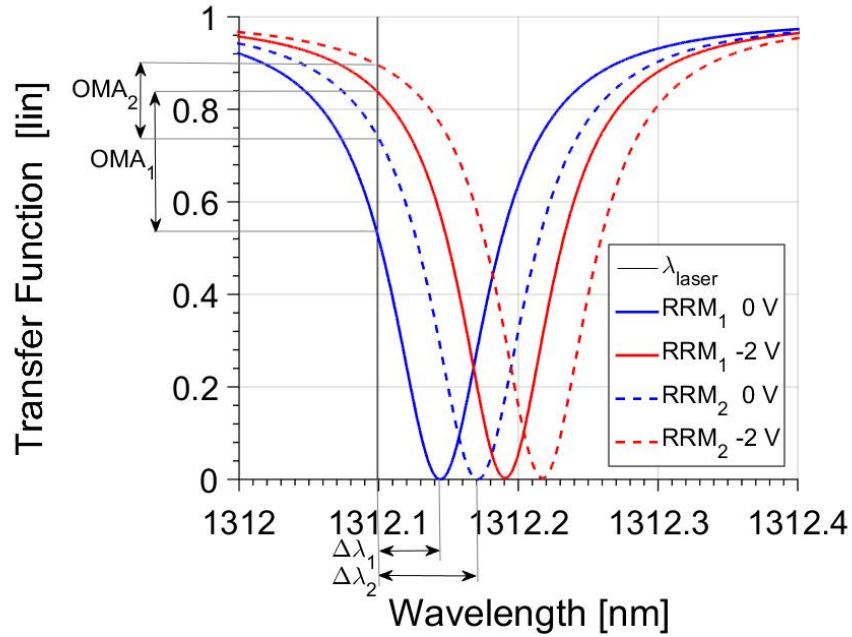
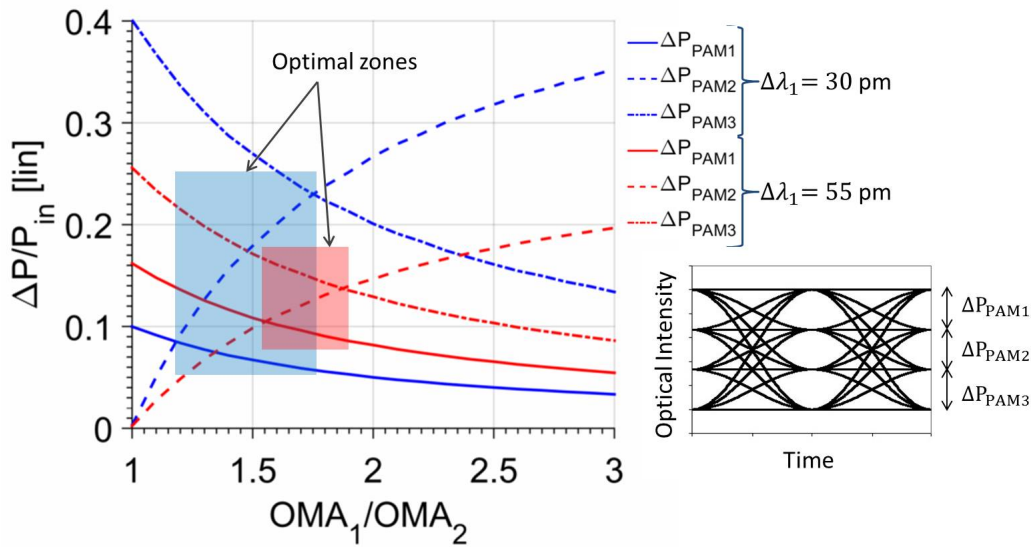


FIGURE 6.3: Transfer function representation of the two ring resonator modulators

We now simulate the two **RRM** arranged in series architecture with the previous optimized **RRM** of the chapter 5. The Q factor is set to 15500, which is the optimal Q factor for 25 Gbaud **PAM-2** modulation. Each **RRM** is modulated between 0 V and -2 V. In a first step, we fix $\Delta\lambda_1$ equal to 30 pm or 55 pm leading to a fix OMA_1 . $\Delta\lambda_2$ is used as a variable to change the ratio $\text{OMA}_1/\text{OMA}_2$. As a function of the ratio $\text{OMA}_1/\text{OMA}_2$, we plot in figure 6.4 the simulated differences between successive levels of the output **PAM-4** signal (ΔP_{PAM1} , ΔP_{PAM2} , ΔP_{PAM3}).

FIGURE 6.4: Simulated levels distribution as a function of $\text{OMA}_1/\text{OMA}_2$

Through the simulations, we observe that it is difficult to obtain equally spaced levels. Optimal zones are represented in figure 6.4 where ΔP_{PAM1} , ΔP_{PAM2} , and ΔP_{PAM3} are the closest. When the laser wavelength is shifted to the first resonant wavelength, the four levels become equally spaced, and the ratio OMA_1/OMA_2 tends to 2. However, if the laser wavelength is too far to the first resonant wavelength, ΔP_{PAM1} , ΔP_{PAM2} , and ΔP_{PAM3} begin to be too small, which leads to close the eye diagram. So, a trade-off between the amplitude of modulation (*i.e.* extinction ratio) and the four level distributions has to be found. This is a difficult trade-off to handle, which depends on the Q factor and the junction efficiency.

As a good compromise, we choose the operating point which allows an extinction ratio of 4.2 dB from the highest to the lowest level to be achieved with -1.5 dB insertion loss (figure 6.5). $\Delta\lambda_1$ is equal to 40 pm and $\Delta\lambda_2$ is equal to 60 pm leading to a ratio OMA_1/OMA_2 equal to 1.9.

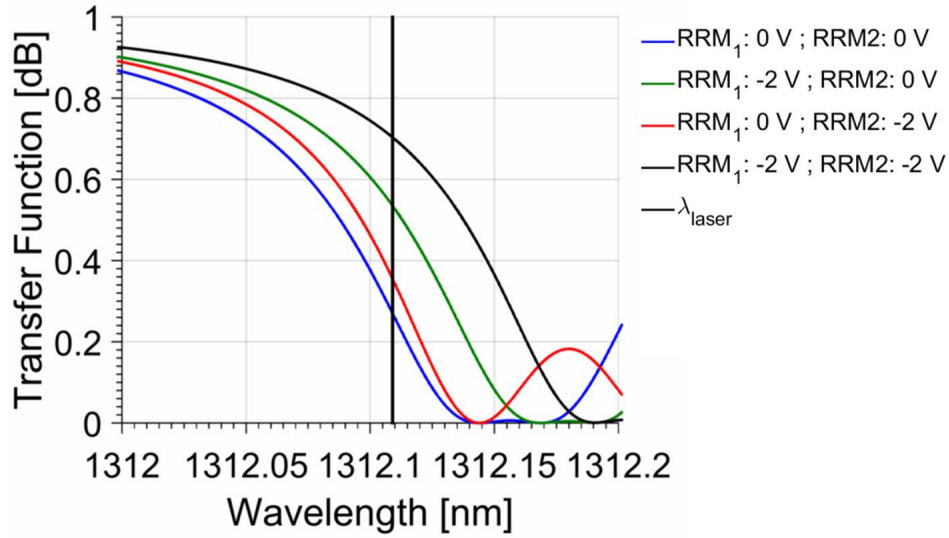


FIGURE 6.5: Simulated transfer function of a two RRM's arranged in series

6.2.2 Demonstrations

In this section, we propose to test the implementation of the transmitter architecture described above. The devices used for this experiment were fabricated in the 200 nm CEA-Leti process. They include two identical Si-RRMs of 15 μm radius coupled to the same bus waveguide with 450 nm gap. The rings are built with a 220/100 nm rib waveguide which includes a PN junction inside the 400 nm-wide waveguide. TiTiN resistive heaters are placed above the rings to compensate process and thermal variations and to enable the thermal tuning of the Si-RRM resonant wavelength. The Si-RRM spectra have been measured with 1 pm precision with different bias voltages and different heater voltages. The main parameters of the rings have been extracted and reported in table 7.8. The referenced transfer function of the Si-RRMs as a function of the wavelength is plotted for several negative bias voltages in figure 6.6.

TABLE 6.1: Parameters of the Si-RRMs

	Ring 1	Ring 2
$\lambda_{resonance}$ [nm]	1548.631	1549.117
FSR [nm]	6.66	6.66
$\frac{\partial \lambda_{res}}{\partial V_m}$ [pm/V]	14	14
$V_\pi L_\pi$ [V.cm]	1.9	1.9
$\Delta\phi$ between [0V,-2V] [$^\circ$ /mm]	-	-
Q factor at 0 V	11600	11400
Heater Efficiency [mW/FSR]	130	102

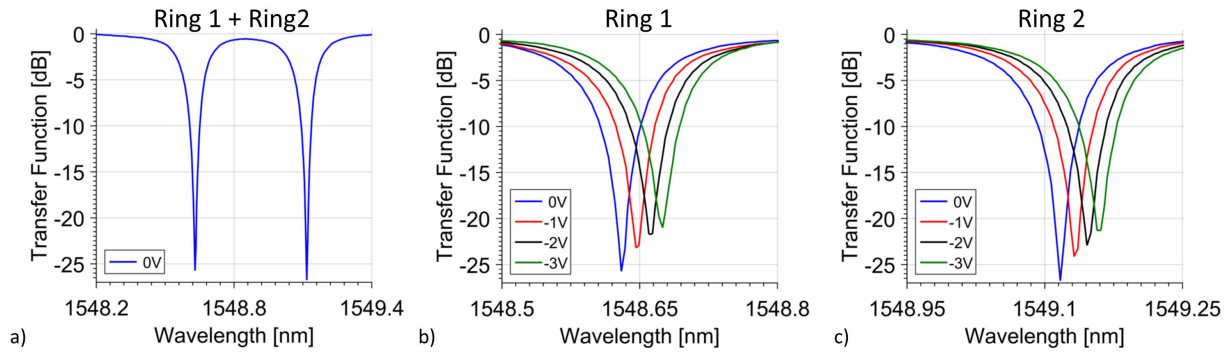


FIGURE 6.6: Referenced measured transfer function of the two rings modulators arranged in series at 0V a), and against bias voltage for ring modulator 1 b), and ring modulator 2 c)

Due to process variation, the two Si-RMMs are not totally wavelength-aligned. Indeed, the resonant wavelengths are separated by 486 pm and the Q factor is different. But, these variations are too small to impact a difference in the electro-optical bandwidth which is about 10 GHz. With 3.77 V (11 mW) applied on the heater above ring 1, the two ring resonances can be tuned to the same wavelength.

High Speed Experiments

To achieve a symmetric PAM-4 eye diagram, the modulation of each ring resonator must be accurately controlled with as sharp resonance as possible. In the ideal approach with our characteristics, the rings have to be detuned by 20-30 pm from each other. The detuning between the two rings depends on the electro-optical modulation efficiency and the Q factor. Higher efficiency means larger ring detuning. Therefore, the set point is a choice between insertion loss, extinction ratio and equally spaced eye diagrams. With our ring modulators, an ideal spacing is 24 pm. Indeed, with -1.5 V bias and 1 V_{pp} driving voltage applied to both rings, the first ring modulator achieves 2.32 dB extinction ratio and the second one achieves 1.1 dB extinction ratio. With this set point, the overall extinction ratio of the PAM-4 from the highest level to the lowest level is 3.4 dB and the insertion loss is 3.9 dB. Figure 6.7 represents the transfer function of both rings in their different states. The simulated eye diagram is also represented in figure 6.7 c), where four levels are clearly visible.

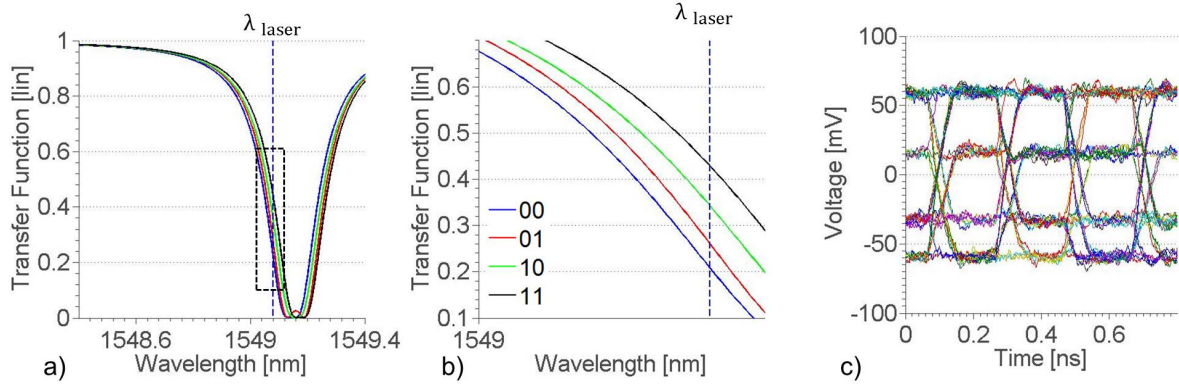


FIGURE 6.7: a) Simulated transfer function of the double ring modulators, b) Details of the simulated transfer function close to the laser wavelength position, c) Simulated PAM-4 eye diagram at 10 Gbit/s.

Figure 6.8 shows pictures of the setup measurement. On these pictures, we can see the in and out optical fibers, the RF probe which bring the electrical data, and three DC probes to set the operating points of the ring resonator modulators using heaters. The experimental setup for the PAM-4 transmission is shown in figure 6.9. Two channels of a M8020A Bit Error Rate Tester (BERT) launch two different PRBS patterns at 15 Gbit/s with 2^9-1 word lengths and modulate the ring modulators with $1 V_{pp}$. The BERT allows to decorrelate the two patterns by introducing a time delay. The optical output of the Si-RRMs is sent to a 12 GHz-photoreceiver followed by a TIA with a total gain of 650 V/W and a noise equivalent power of $20 \text{ pA}/\sqrt{\text{Hz}}$. It is followed by a wide bandwidth oscilloscope DCA-X 86100D in order to demonstrate the PAM-4 transmission link at 30 Gbit/s. Open eyes are clearly visible at 15 Gbit/s, 20 Gbit/s and 30 Gbit/s in figure 6.10. For the latter experiment, the 4 average voltage levels obtained are [-76 mV, -22 mV, 21 mV, 67 mV].

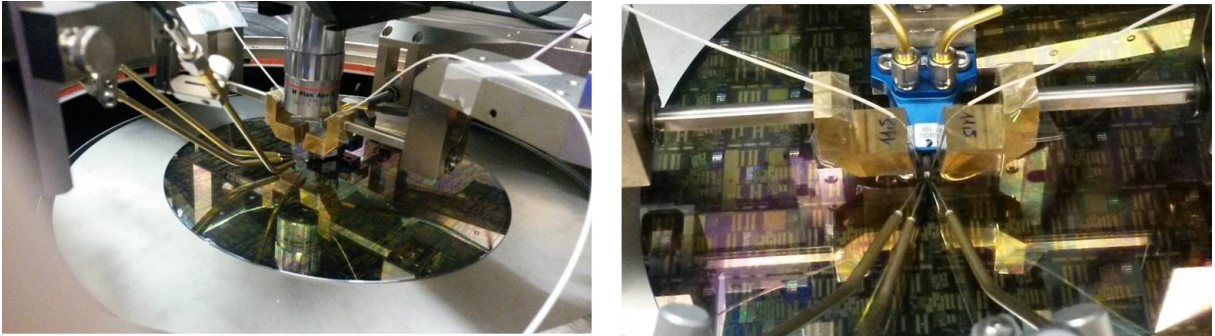


FIGURE 6.8: Pictures of the experimental setup

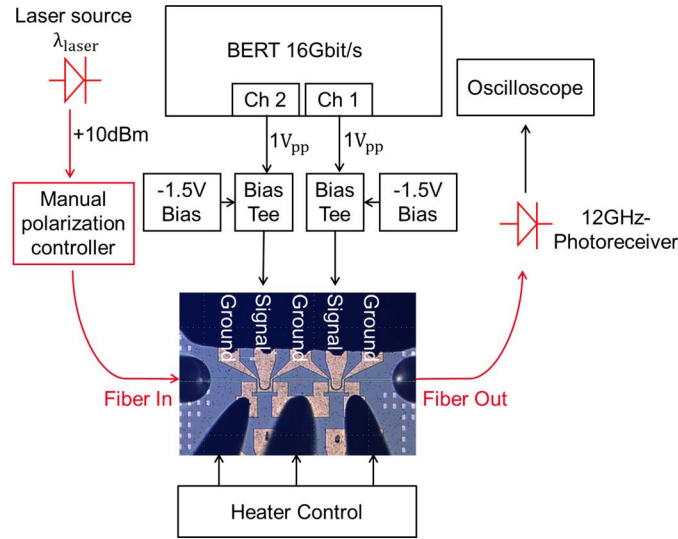


FIGURE 6.9: Experimental setup of PAM-4 transmission

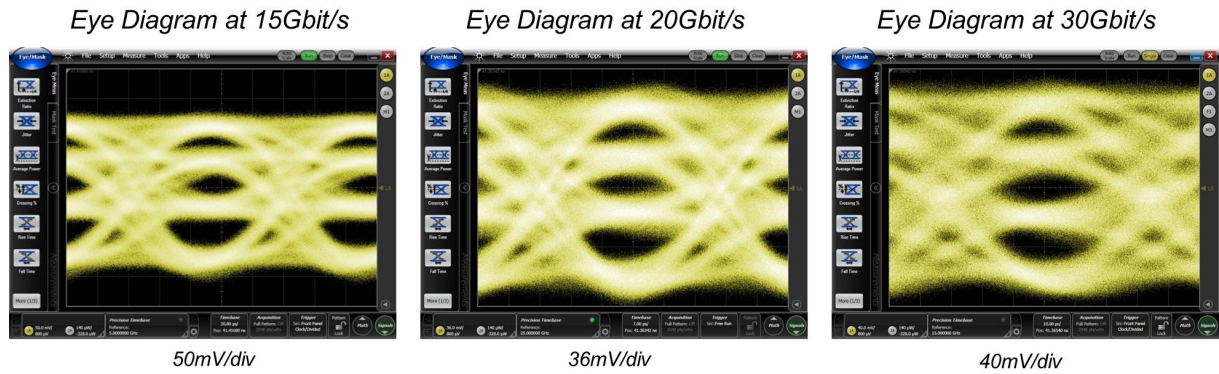


FIGURE 6.10: PAM-4 measured eye diagram at 15 Gbit/s a), 20 Gbit/s b) and 30 Gbit/s c)

To measure the BER of the link, the wide bandwidth oscilloscope is replaced by a real time oscilloscope DSA 72004C with 20 GHz bandwidth. In post-processing, the signal received is stored and then compared with the sent BERT signal. So, the BER is evaluated by error counting after sample acquisition in the scope. At 30 Gbit/s a BER of 4×10^{-3} with an average received power of 0 dBm is obtained (figure 6.11). Improved results are expected with a photo-receiver and an oscilloscope with a higher bandwidth. Unfortunately, we did not have a high bandwidth photo-diode with high gain at our disposal.

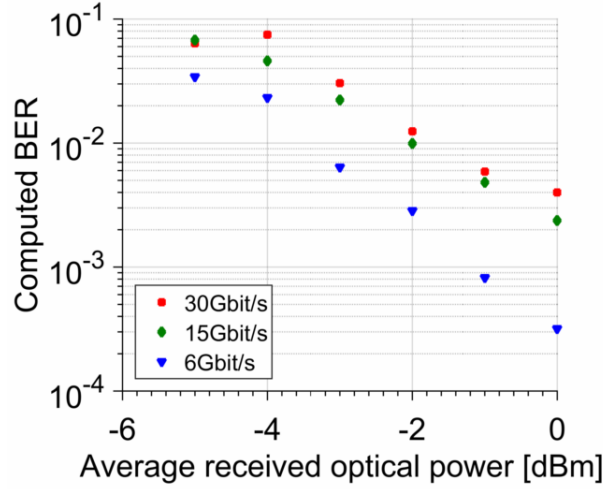


FIGURE 6.11: Measured BER with real time oscilloscope against the average received optical power at 6, 15 and 30Gbit/s

This work demonstrates a PAM-4 transmission link with a two silicon ring resonator modulator arranged in series. This approach enables the reduction of the complexity and energy consumption of CMOS drivers. A 30 Gbit/s transmission is demonstrated with only 1 V_{pp} driving voltage without any amplifiers.

6.3 Two ring resonator modulators arranged in parallel

6.3.1 Operating Principle

PAM-4 signal generation is achieved by modulating two identical RRM_s independently with two bit streams. These two RRM_s are placed in a Mach Zehnder interferometer (MZI). The two MZI arms are $\pi/2$ phase shifted, as shown in figure 6.12. The two RRM_s have to be set to the same operating point *i.e.* identical extinction ratio and insertion loss.

To explain the operating principle in a simple way, we make the approximation that RRM_s only modulate the optical field amplitude. We denote by coefficients A and B the modulation efficiency corresponding to the top and the bottom RRM_s respectively. At the output of the , optical fields are added up leading to equation 6.1 in field and in intensity.

$$\begin{cases} E_{out} = (A + B \cdot e^{i\pi/2}) \cdot \sqrt{\frac{I_{in}}{2}} \\ I_{out} = (A^2 + B^2) \cdot \frac{I_{in}}{4} \end{cases} \quad (6.1)$$

Equally spaced PAM-4 intensity levels are obtained when the coefficients take the values 0 or 1 for A, and 1 or $\sqrt{1/2}$ for B leading to four intensity levels reported in equation 6.2. This condition is respected when the top modulator is modulated by a voltage double that of the voltage applied to the bottom modulator. Due to the MZM architecture, 3 dB loss is added in the power budget.

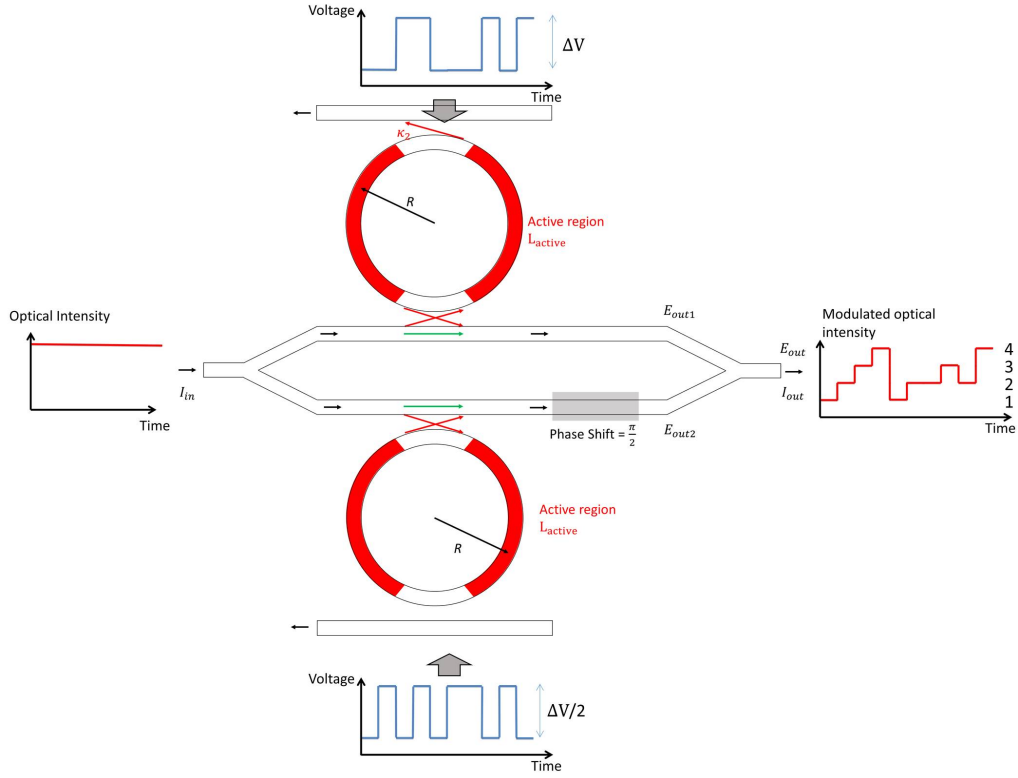


FIGURE 6.12: PAM-4 double ring resonator modulators in a MZI schematic

$$\begin{cases} I_{11} = I_{in}/2 \\ I_{10} = 0.75 \cdot I_{in}/2 \\ I_{01} = 0.5 \cdot I_{in}/2 \\ I_{00} = 0.25 \cdot I_{in}/2 \end{cases} \quad (6.2)$$

With the optimized RRM with a Q factor of 15500 (see section 5.5), we simulate the double RRM in a MZI architecture. The top RRM is modulated between 0 V and -2 V and the bottom RRM between -0.5 V and -1.5 V. The RRM resonant wavelength at 0 V and the laser wavelength are aligned. An extinction ratio of 12.3 dB from the highest to the lowest level is achieved with -5.4 dB of insertion loss.

Unfortunately, it is difficult to obtain perfect equally spaced levels due to the PN junction non-linearity but also due to the undesirable inherent phase modulation. But in compensation, the two RRM have to be set to the same working point allowing easy tuning of the RRM.

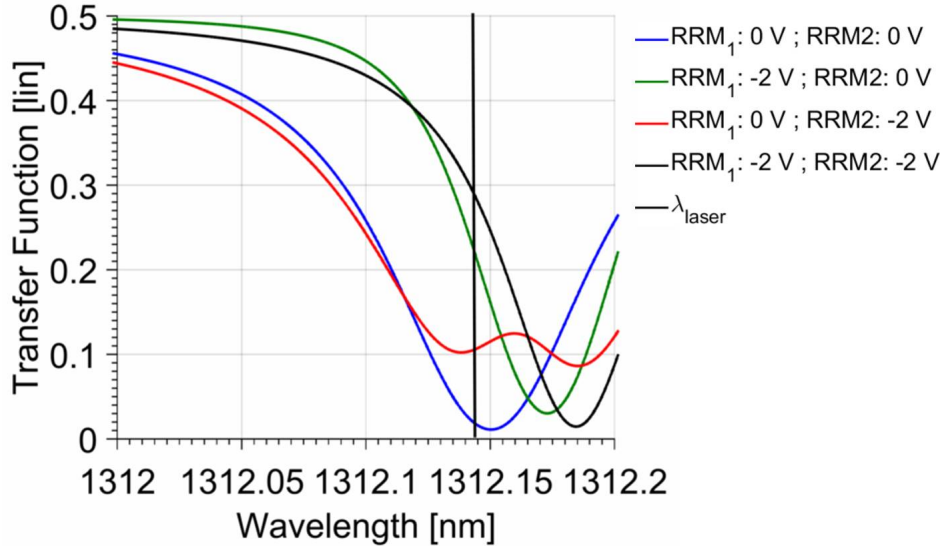


FIGURE 6.13: Simulated transfer function of a two RRM in a MZI

6.3.2 Demonstrations

In this section, we propose to show PAM-4 measurements with two RRM_s arranged in parallel architecture. The devices used for this experiment were fabricated in the 300 nm CEA-Leti and ST processes. They include two identical two-bus RRM_s of 25 μm radius coupled with different MZI arm waveguides with 300 nm gap. The rings are built with a 300/150 nm rib waveguide which includes a PN junction inside the 400 nm-wide waveguide. The drop ports are linked to output grating couplers allowing to measure independently each RRM. Doped resistive heaters are placed in the center of the rings to compensate process and thermal variations, and to enable thermal tuning of the RRM resonant wavelength. The RRM_s spectra have been measured with 1 pm precision for various bias voltages. The main parameters of the rings have been extracted and reported in table 6.2. The transfer function of the RRM_s as a function of the wavelength is plotted for several negative bias voltages in figure 6.14. In order to measure independently the transfer function of each RRM, the second modulator is tuned with its heater.

TABLE 6.2: Parameters of the Si-RRMs

	Ring 1	Ring 2
$\lambda_{\text{resonance}}$ [nm]	1315.750	1315.749
FSR [nm]	2.79	2.79
$\frac{\partial \lambda_{\text{res}}}{\partial V_m}$ [pm/V]	9.37	9.17
$V_\pi L_\pi$ [V.cm]	2.08	2.12
Q factor at 0 V	11250	10600
Heater Efficiency [mW/FSR]	320	320
Heater Resistance [Ω]	588	588

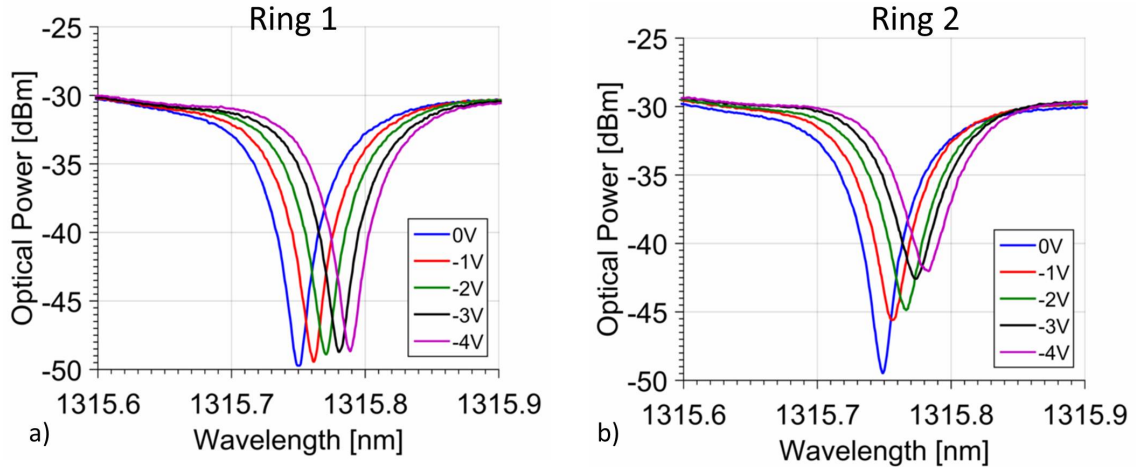


FIGURE 6.14: Measured transfer functions of the top ring modulator a), and the bottom ring modulator b) against bias voltage

The electro-optical bandwidth was then measured with a VNA for different laser wavelength positions. Because the test calibration was impossible, this measurements was not made for RRM in the PAM-4 architecture, but it was made for the same RRM design for a standalone version. As the RRM are identical and very closed on the wafer, we can suppose the same electro-optical bandwidths for the two RRM arranged in parallel. Figure 6.15 a) reports the measured electro-optical S21 for two laser wavelength positions and figure 6.15 b) reports the measured electro-optical bandwidth against the laser wavelength position. The measured electro-optical bandwidth is higher than 10 GHz and it increases quickly when the laser wavelength is shifted to the resonant wavelength. With an electro-optical bandwidth up to 20 GHz, we can expect a 50 Gbaud PAM4 optical transmission non-limited by the RRM.

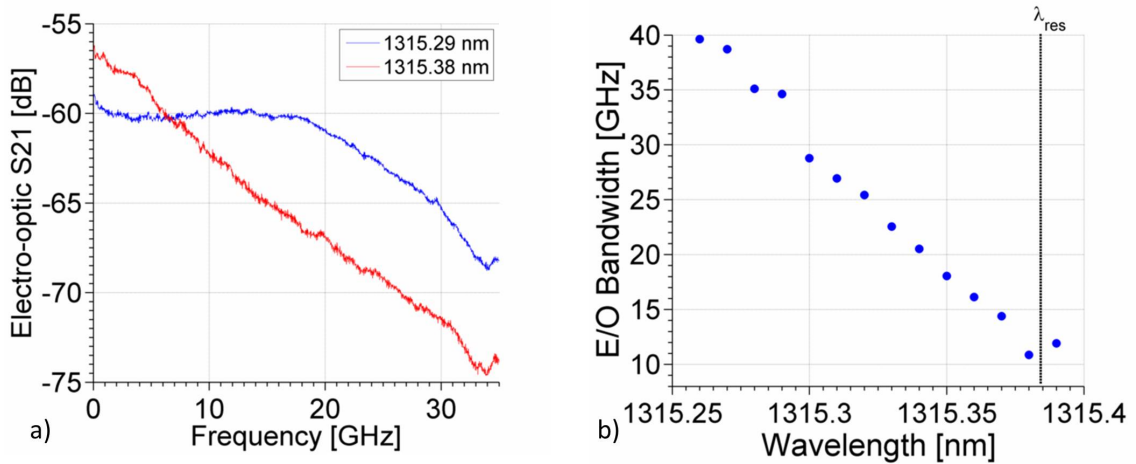


FIGURE 6.15: a) Measured electro-optical S21 for a single RRM near the PAM-4 system, b) Measured electro-optical bandwidth against the laser wavelength for a single RRM near the PAM-4 system

Doped resistive heaters were placed close to each arm of the MZI, thus allowing the phase shift between both arms to be chosen. In figure 6.16, for a laser wavelength close to 1315 nm, the optical power at the output of the MZI was measured as a function of the electrical power applied to the first heater (top arm) in blue and in the second heater (bottom arm) in red. The optical fields are $\pi/2$ phase shifted when the applied voltage on the bottom heater is equal to 4.54 V corresponding to an electrical power of 45 mW with an heater resistance of 450 Ω . This phase shift corresponds to the desired setting point to generate PAM-4 modulation.

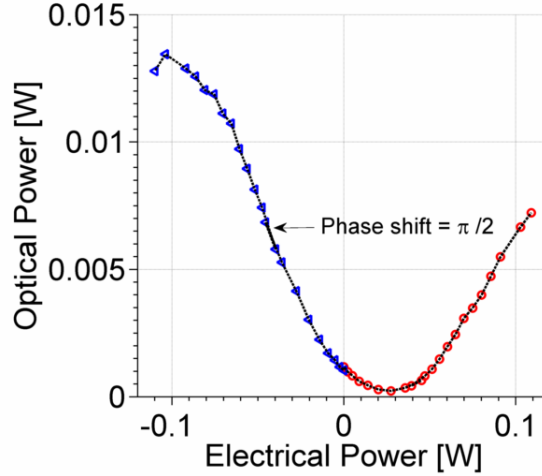


FIGURE 6.16: Measured optical power at the output of the MZI against the electrical power applied on the top heater (red) and bottom heater (blue)

High Speed Experiments

The experimental setup for the PAM-4 transmission link is shown in figure 6.17. Two channels of M8020A Bit Error Rate Tester (BERT) launch two different patterns at 15 Gbit/s with 2^{9-1} word lengths and modulate the ring modulators with ΔV_1 and ΔV_2 (maximum $1.2V_{pp}$). The optical output of the MZI is sent to a 12 GHz-photoreceiver followed by a TIA having a total gain of 650 V/W and a noise equivalent power of 20 pA/ $\sqrt{\text{Hz}}$. It is followed by a wide bandwidth oscilloscope DCA-X 86100D in order to demonstrate the PAM-4 transmission link at 30 Gbit/s.

TABLE 6.3: Experiment setting voltages

ΔV_1 [V_{pp}]	ΔV_2 [V_{pp}]	V_{bias1} [V]	V_{bias2} [V]	H_1 [V]	PsH_1 [V]	H_2 [V]	PsH_2 [V]
1.2	1.125	1.2	1.2	0	4.54	0.6	0

DC sources are used to control RRM resonant wavelength positions (H_1 and H_2). RRMs are set into the same working point when H_2 is equal to 0.6 V. To confirm, we measured 15 Gbit/s PAM-2 eye diagram of each RRM with the same voltage swing $1.2 V_{pp}$ and voltage bias -1.2 V. The two eye diagrams reported in figure 6.18 show that the two RRMs are at the same operating point.

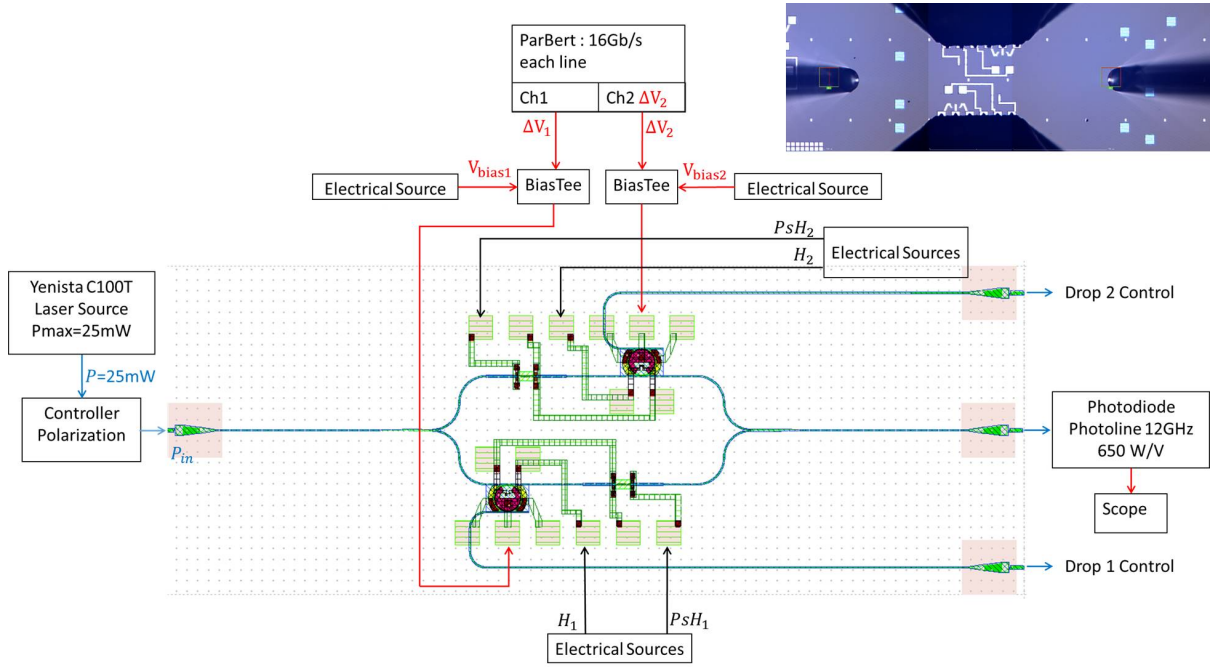


FIGURE 6.17: Experimental setup of PAM-4 Transmission

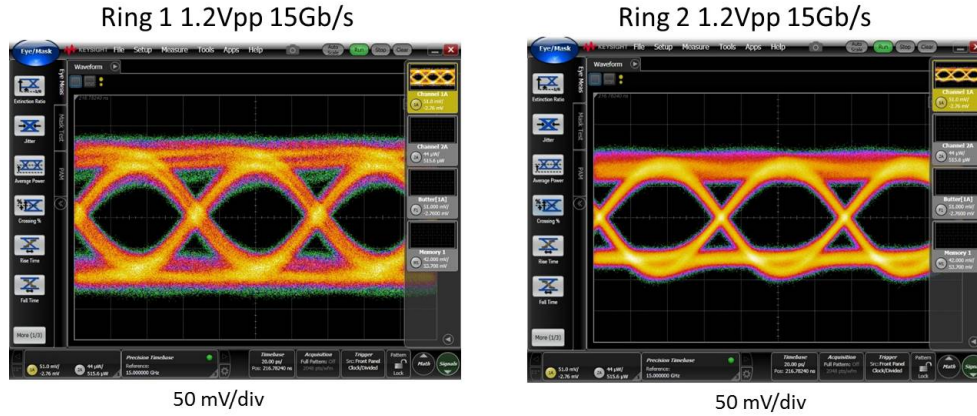


FIGURE 6.18: PAM-2 measured eye diagram at 15 Gbit/s by modulating consecutively each RRM

PAM-4 transmission is then carried out with aligned RRM and $\pi/2$ phase shift between the two arms. Table 6.3 reports the voltage settings for PAM-4 demonstration. Open eye diagrams are clearly visible at 30 Gbit/s in figure 6.19. To obtain equally spaced levels, we had to increase the voltage swing on the bottom RRM to $1.125 V_{pp}$ instead of $0.6 V_{pp}$. This is due to the different efficiencies between the two RRM and the non linearity of the PN junction. Unfortunately, BER measurements were not possible at the time of experiment, the real time scope was not available.

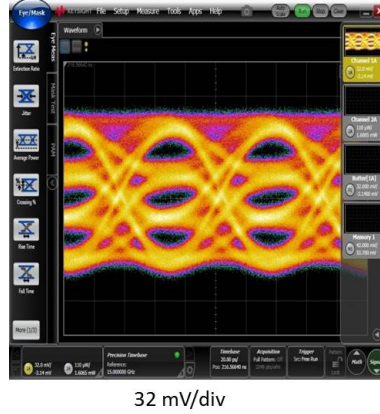


FIGURE 6.19: PAM-4 measured eye diagram at 30 Gbit/s

This work shows the capability of **RRMs** in **MZI** architecture to achieve **PAM-4** transmissions with a $1.2 V_{pp}$ driving voltage, compatible with CMOS drivers. A 30 Gbit/s transmission was achieved without any optical amplifier. Improved results are expected with receiver and the oscilloscope with higher bandwidth. Moreover, this demonstration exhibits 16 dB loss (propagation loss + grating coupler loss). We can expect in a future demonstrator to reduce the loss at 4 dB with lower propagation loss and new grating coupler design .

6.3.3 Complete demonstrator implementation

A demonstrator based on the double **RRMs** in **MZI** architecture is now being built. It will be composed of a photonic integrated circuit (**PIC**) interconnected with an electronic integrated circuit (**EIC**) using micropillars and under-bump metallization (**UBM**). The optical transmitter is implemented in **PIC** and it will be driven by the **EIC**. The EIC and the PIC will then be stacked together using flip-chip bonding. Figure 6.20 a) shows a typical stack. In [121], an electrical RLCG model of the micropillar and **UBM** are developed and reported values are the following: $R=45 \text{ m}\Omega$, $L=16 \text{ pH}$, $C=2 \text{ fF}$ and $G=9 \text{ mS/m}$. This interconnection will be bandwidth limiting for 25 GBaud **PAM-4** transmission.

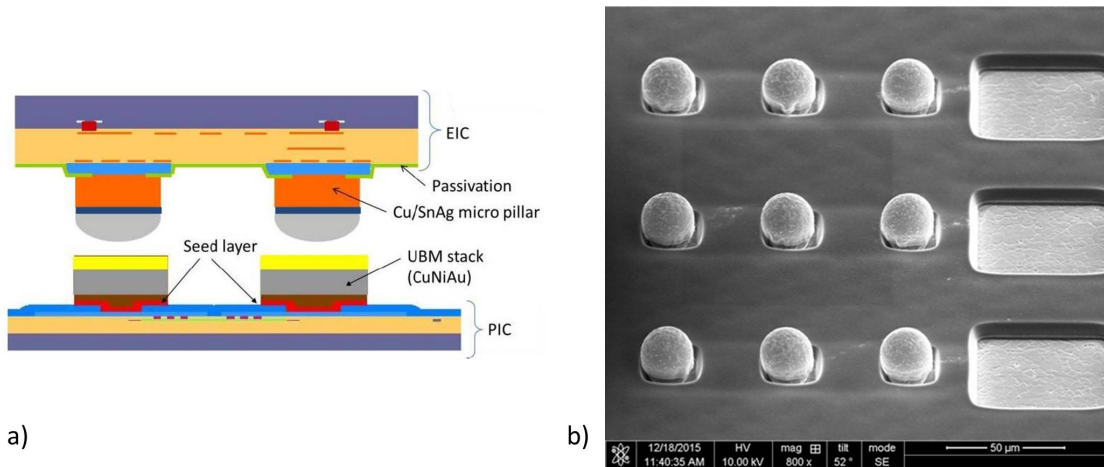


FIGURE 6.20: a) Cross section of PIC and EIC dies with UBM stack and micropillars from [121]. b) Fabricated micropillars picture

The demonstrator targets a PAM-4 optical transmission at 50 Gbit/s with two wavelengths. It is composed of four ring modulators, two different sets of ring resonator modulators in the Mach Zehnder interferometer. The demonstrator will reach 12.5 Gbaud PAM-4 with two wavelengths. The target RRM s have a radius of 15 μm and we target a junction efficiency below 1.5 V.cm with a Q factor of 14000. The EIC will generate the required driving voltages with a maximum voltage swing of 1.2 V_{pp} .

The photonic layout view of the demonstrator is reported in figure 6.21. The EIC will be flip-chipped using copperpillars for the interconnection with the PIC. Three electrical probes will be necessary to test the demonstrator. To the east, a first probe will bring the supply voltages. To the west, the second probe will allow the control of the heaters required for RRM alignment and phase shift setting. To the north, a third probe brings the data used by the EIC to generate the driving voltages. To the south, a multi-fibers probe will allow coupling and measurement of the light at both the input and output of the transmitter and to measure at the same time the optical intensity at the drop control outputs.

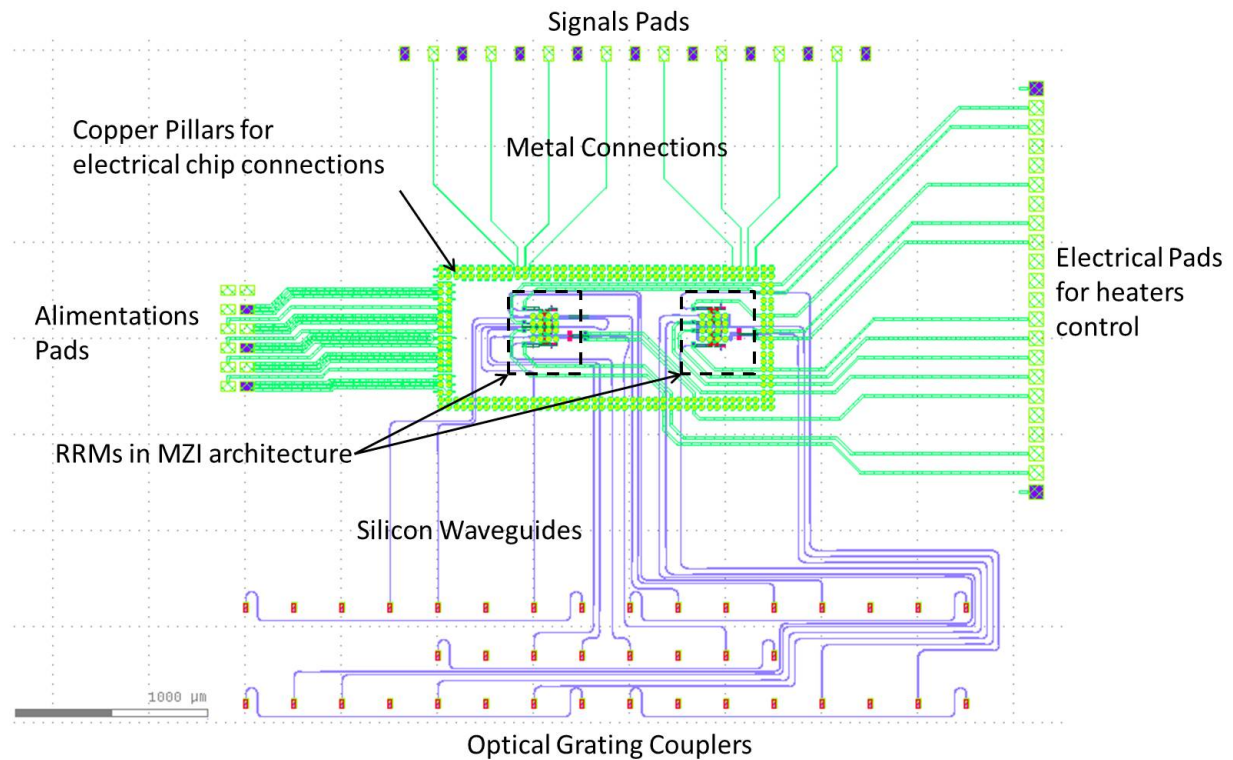


FIGURE 6.21: Layout view of the PAM-4 demonstrator

The EIC is fabricated and first measurements exhibit open eye diagrams at 10 Gbit/s (figure 6.22). At 12.5 Gbit/s, the eye diagram begins to close due to high jitter. We await for the fabricated PIC to propose complete demonstration.

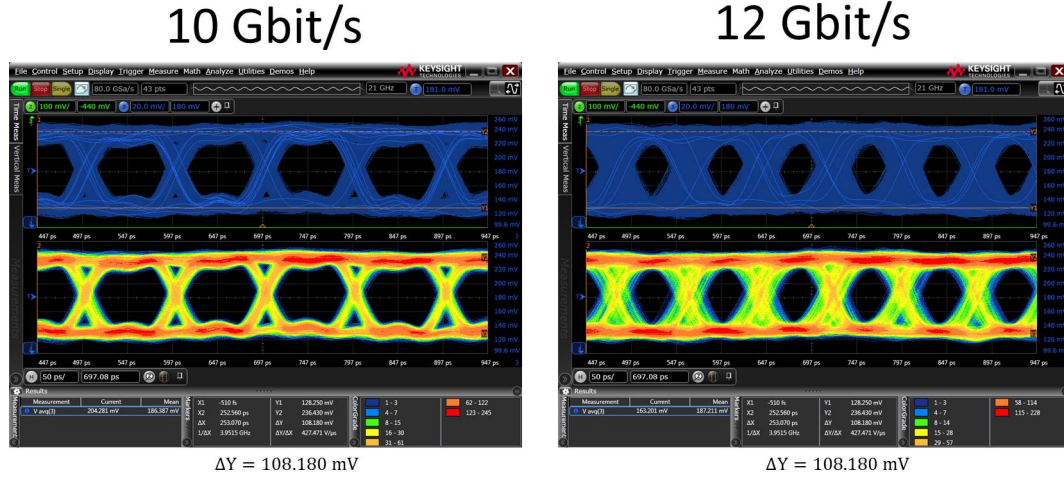


FIGURE 6.22: Measured electric eye diagram at the output of the EIC (probe at 1/10)

6.4 Conclusion

In summary, we proposed alternative transmitter architectures without the electrical DAC to generate the PAM-4 modulation. In this case, the complexity is moved to the optical domain. Thus, the transmitter circuit generates PAM-4 signals by combining two PAM-2 electrical signals.

This architecture is clearly of interest, since it allows the RRM requirements to be relaxed. The required optical bandwidth is thus reduced to 15 GHz ($Q=15500$). In this chapter, two architectures are reported.

The first is based on double RRM arranged in series. Based on the optimized RRM in the previous chapter, this transmitter can achieve an extinction ratio of 4.5 dB with 2.45 dB insertion loss. A demonstration is shown at 30 Gbit/s with 1 V_{pp} and no amplifier, thus validating this transmitter architecture. Its principal disadvantage resides in setting the resonant wavelengths to achieve equally spaced levels.

The second architecture is based on double RRM arranged in parallel in MZI. Both arms have to be $\pi/2$ phase shifted. Based on the optimized RRM from the previous chapter, this transmitter can achieve an extinction ratio of 12.3 dB with 5.4 dB insertion loss. A loss of 3 dB is due to the MZI. A demonstration is shown at 30 Gbit/s with 1.2 V_{pp} and 1.125 V_{pp} and no amplifier. This demonstration allows this architecture to be validated. The main advantage of this transmitter is that the resonant wavelengths of both RRM are equal. In addition, a complete transmitter implementation is under fabrication with dedicated EIC.

To go further, it can be interesting to do a comparative study between the power consumption, the complexity and the cost of PAM-4 architectures making use of electrical or optical DAC.

Conclusion

Silicon photonic has definitely made its way to **data center applications** as a commercial product. At the end of this thesis, it has emerged as the **lowest cost solution for 100 GBit/s optical communications between 100 m and 500 m**.

With the aim to address next **400 Gbit/s data-rate standard** over up to **2 kilometer links**, we have proposed and studied several transmitter architectures using silicon ring resonator modulator (**RRM**) and combining **wavelength division multiplexing (WDM)** and several modulation types: **two levels in pulse amplitude modulation (PAM-2)** or **four levels in pulse amplitude modulation (PAM-4)**.

A first part of this thesis was dedicated to the **measurements and modeling of the PN depleted Si-ring resonator modulator**. A transient model has been developed based on a coupled mode resolution, together with a PN-junction electrical model. It has been validated through **DC** and **AC** extensive measurements. From the measurements and modeling, the **dispersion of some key-RRM-characteristics** was pointed out. **Trade-offs on the RRM operating point** were clearly stated.

The **first issue**, shown by the measurements, is **the resonant wavelength drift** as a function of the temperature and the technological process variability. The resonant wavelength variation was evaluated at **0.1 nm/°C** for the temperature variation, and around **1 nm/nm** for waveguide width or waveguide height or slab height variation. We measured resonant wavelength variability higher than the free spectral range (**FSR**) in a 200 mm wafer. However, it is improved to less than **FSR/3** in the 300 mm fabrication platform. To cope with the resonant wavelength variability, a **feedback loop is necessary**. The best way is to use two-bus ring resonator modulators which allow a photo-diode to be coupled to the drop port measuring in real time the drift.

The **second issue** is to obtain a resonator with a **high depth resonance** (close to the critical coupling) with a **desired Q factor**. These parameters are defined by the coupling factor and the global loss within the ring. The coupling factor is well controlled: our simulations are very close to the measurements from 25% to 1 % coupling. But, the work on the global loss has to be continued in order to stabilize the roughness loss and the doping loss.

The **third issue** is the **accuracy of the doping implantations** defining the PN active region within the ring. It directly impacts the junction efficiency and the global loss. We measured a junction efficiency of 2 V.cm, while state of the art is less than 1 V.cm. The implantation in the slab section is also very important. Indeed, we have seen through our measurements that the electrical bandwidth is reduced by half due to under-doped slab sections. The modulator performance improvement and its stability can be achieved through multi-test doping implantation conditions coupled with more realistic doping profiles simulations.

The choice of the **RRM operating point** is decisive to target **high modulator performance levels** (*i.e.* high extinction ratio, low insertion loss, and high electro-optical bandwidth). **The depth of the resonance, the Q factor (the width of the resonance), the laser wavelength position with regard to the resonance wavelength position, and the junction efficiency** are the many parameters affecting the performance levels. We highlighted that the choice of the Q factor is important: low Q factor allows us to achieve high optical bandwidth (Q factor=6000 is equivalent to 35 GHz optical bandwidth). The second important choice is the laser wavelength position: the electro-optical bandwidth increases and the insertion loss decreases when the laser is shifted from the resonant wavelength, but at the cost of lower extinction ratio. Indeed, we measured electro-optical bandwidths from 8 GHz to 30 GHz when the laser wavelength is shifted from 0 to 210 pm from the resonant wavelength.

Following this thesis, the **RRM** parameters extraction associated with the **RRM** model can be used as a basis for other technology and/or design of **RRMs**.

Ring resonator modulator optimization for 400 Gbit/s applications

In a second time, the **RRM** optimization has been made targeting **400 Gbit/s communications over 2 kilometers-links**. The modulator is integrated into a transmitter with a laser power input of 10 dBm, and insertion loss evaluated at 5.7 dB including in and out fiber to chip coupling, propagation loss within the chip, and packaging loss. The goal was to achieve 50 Gbit/s per wavelength (per **RRM**) with an optical modulation amplitude higher than 0 dBm at the output of the transmitter. Two configurations have been studied: the **RRM** is modulated at **50 Gbaud PAM-2**, and the **RRM** is modulated at **25 Gbaud PAM-4** (four equally space voltage levels).

The optimization was made methodically:

- We first simulated the PN junction as a function of the doping implantation levels from $6 \cdot 10^{16}$ to $5 \cdot 10^{18} \text{ cm}^{-3}$. The results exhibit **an optimal curve representing the highest PN junction modulation efficiency at the lowest PN waveguide loss**.
- The **RRM** was then optimized as a function of the Q factor. The input parameters were the PN junction modulation efficiency and PN waveguide loss, the through/drop coupling factors to the ring, and the laser wavelength position from the resonant wavelength. The results led to a **compact model describing the best RRM performance levels** with an insertion loss always lower than 3 dB. An extinction ratio higher than 8 dB was achieved when the Q factor was higher than 6500 (corresponding to an approximated optical bandwidth lower than 35 GHz).
- The transmitter was then evaluated for the two modulation configurations: 50 Gbaud **PAM-2** and 25 Gbaud **PAM-4**. **The two configurations meet the preliminary specifications of the 'tentative standard' with a margin loss of 15 dB and 12 dB for the 50 Gbaud PAM-2 and 25 Gbaud PAM-4 respectively**. The required electro-optical bandwidths were quite similar between the two configurations with a required electro-optical bandwidth of 30 GHz and 32 GHz (Q factor equal to 7500 and 7000). Unfortunately, the required electro-bandwidth for the **PAM-4** configurations was not reduced compared to the **PAM-2**, and so **PAM-4** is thus of less interest as compared to **PAM-2**. The 50 Gbaud **PAM-2**

configuration is the best compromise to achieve 50 Gbit/s with a transmitter based on **RRM**.

To go further, the process optimization has to be validated through measurements, and the robustness of the performance levels has to be evaluated against fabrication process variability.

Proposal of new transmitter architectures

Finally, we have proposed **new transmitter architectures** for implementing **PAM-4**. The **PAM-4** is generated in the optical domain: **two bit streams are applied to two RRM**s arranged **in series or in parallel**, and generate an optical **PAM-4** signal. These architectures allow the required electro-optical bandwidth to be reduced, as each **RRM** is **PAM-2** modulated.

The first architecture is based on two identical **RRM**s arranged **in series**. **PAM-4** is generated by controlling the two **RRM**s at different operating points, *i.e.* different resonant wavelength positions with regard to the laser wavelength. **A demonstration was made at 30 Gbit/s with 1 V_{pp} and without any amplifiers**. The difficulty of this architecture is to **lock the resonant wavelengths** at the right position in order to achieve equally spaced intensity levels. One advantage of this architecture is that it is **re-configurable** allowing to go from a single wavelength **PAM-4** transmitter to a **WDM PAM-2** transmitter.

The second architecture is based on two identical **RRM**s arranged **in parallel** in a Mach Zehnder interferometer with $\pi/2$ phase shift between its two arms. **PAM-4** is generated by modulating the two **RRM**s at different voltages. This architecture is easily implemented, as **RRM**s have to be set to **the same operating point**, same laser wavelength position with regard to the resonant wavelength. The main drawback is an additional **3 dB loss** due to the interferometer structure. **A demonstration was made at 30 Gbit/s with 1.2 V_{pp} and 1.125 V_{pp} without any amplifiers**.

To finish, a demonstrator was designed and is being fabricated, targeting the integration with a dedicated driver made from FDSOI-28 nm technology. This circuit targets a data rate of 12.5 Gbaud **PAM-4** per wavelength with two **RRM**s arranged in parallel architecture. This demonstrator will combine an optical chip and an dedicated electrical driver chip designed by a Leti Electronics team. The electrical chip was fabricated and successfully tested. The optical chip was completed in November 2016. The assembly and the tests will be made in a first quarter of 2017.

French Summary

7.1 Contexte et motivations

Durant la dernière décennie, avec la diversification des périphériques connectés (PC, Tablets, téléviseurs, Smartphones), l'écosystème des communications par Internet s'est considérablement étendu. La croissance des connexions entre composants connectés est associée à une diversification des applications par Internet telle que le stockage Web, le partage de fichier ou la vidéo à la demande (figure 7.23) [1]). En outre, de plus en plus d'applications informatiques se déplacent vers le cloud *i.e.* les applications ne sont plus exécutées localement par l'utilisateur mais dans des centres de données. Ces services peuvent inclure le stockage de données ou de logiciels. Ils demandent un débit de données très élevé pour le transfert de fichiers entre les utilisateurs et les centres de données, entre centres de données, mais aussi dans les centres de données. Selon [2], le trafic IP global entre centres de données sera presque triplé entre 2015 et 2020, avec 77% du trafic concentré dans les centres de données eux-mêmes. Dans ce contexte, des communications à débit élevé sont nécessaires pour répondre à la demande, et les techniques de communication optique sont une réponse possible.

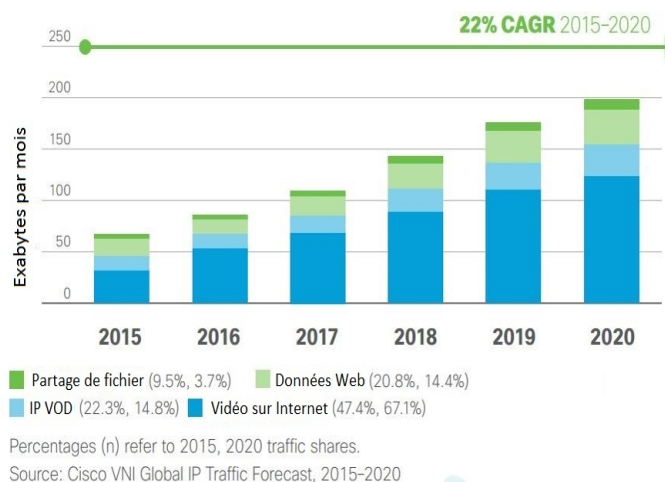


FIGURE 7.23: Prévisions de la croissance mondiale du trafic IP de 2015 à 2020 selon Cisco [1]

Cette thèse porte sur le segment de la communication optique pour les centres de données. Ces communications optiques sont des communications point à point comprises entre 500 mètres et 10 kilomètres à débit élevé. Au début de cette thèse, les normes de communication optique pour un débit binaire de 100 Gbit/s ont été définies par différents groupes comme le groupe IEEE 802.3 et le groupe Multi-Source Agreement (MSA). Par la suite, durant ces trois années de thèse, ces groupes ont commencé à étudier les prochaines normes d'optique pour un débit de 400 Gbit/s [3].

Dans ce qui suit, nous décrirons d'abord l'état actuel des communications optiques pour les centres de données à un débit de données de 100 Gbit/s. Nous détaillerons ensuite les voies d'évolution vers un débit de données de 400 Gbit/s.

7.1.1 Communications optiques à 100 Gbit/s

Pour atteindre un débit de données élevé, les normes de communications optiques jouent principalement avec quatre caractéristiques optiques principales, qui sont:

- le nombre de symboles par seconde
- le nombre de bits par symbole
- le nombre de paires de fibre optiques
- le nombre de longueurs d'ondes utilisées (multiplexage en longueur d'onde ([WDM](#)))

Comme la distance de communication est faible (entre 100 et 500 m) pour les communications entre/dans les centres de données, les émetteurs-récepteurs optiques avec détection directe sont préférés aux émetteurs-récepteurs cohérents. De plus, aucun amplificateur optique n'est implémenté dans cette application de transmission optique.

Le tableau 7.3 rapporte les principales normes pour les communications optiques à un débit de 100 Gbit/s. Toutes les normes ci-dessous sont caractérisées par un bit par symbole (modulation d'amplitude d'impulsion binaire [PAM-2](#)) [4–6]. Ces normes peuvent être divisées en trois segments différents selon la portée. Pour une faible portée, jusqu'à 500 m, le débit de données est obtenu en augmentant le nombre de fibres optiques, fibres multimodes ([MMF](#)) ou fibres monomodes ([SMF](#)). Pour le reste des normes et pour une plus grande portée, le débit de données est obtenu en multiplexant entre 4 et 10 longueurs d'onde différentes.

TABLE 7.3: Nomenclature et spécification des principales normes IEEE et MSA pour un débit de 100 Gbit/s. *standards MSA, fibre multimode (MMF), fibre monomode (SMF)

Nomenclature	Débit par canal [Gbit/s]	Type de fibre optique	Nombre de fibres optiques	Nombre de longueurs d'onde
Jusqu'à 500 m				
100GBASE-SR4	25	MMF at 0.85 μm	4	1
100GBASE-SR10	10		10	1
100GBASE-PSM4 *	25	SMF at 1.3 μm	4	1
Jusqu'à 2 km				
100GBASE-10x10 *	10	SMF at 1.55 μm	1	10 (Δλ=7.2nm)
100GBASE-CWDM4 *	25	SMF at 1.3 μm	1	4 (Δλ=20nm)
100GBASE-CLR4 *	25		1	4 (Δλ=20nm)
Jusqu'à 10 km				
100GBASE-10x10 *	10	SMF at 1.55 μm	1	10 (Δλ=7.2nm)
100GBASE-LR4	25	SMF at 1.3 μm	1	4 (Δf=800GHz)

Aujourd'hui, en fonction du coût et de la distance de communication, trois technologies différentes partagent le marché du 100 Gbit/s: la technologie utilisant le laser à émission de surface verticale (VCSEL), la technologie photonique sur silicium et la technologie utilisant le phosphore d'indium (InP). Selon Menezo et al. [7], la segmentation du marché est déterminée par le coût des fibres optiques et des émetteurs-récepteurs comme le montre la figure 7.24.

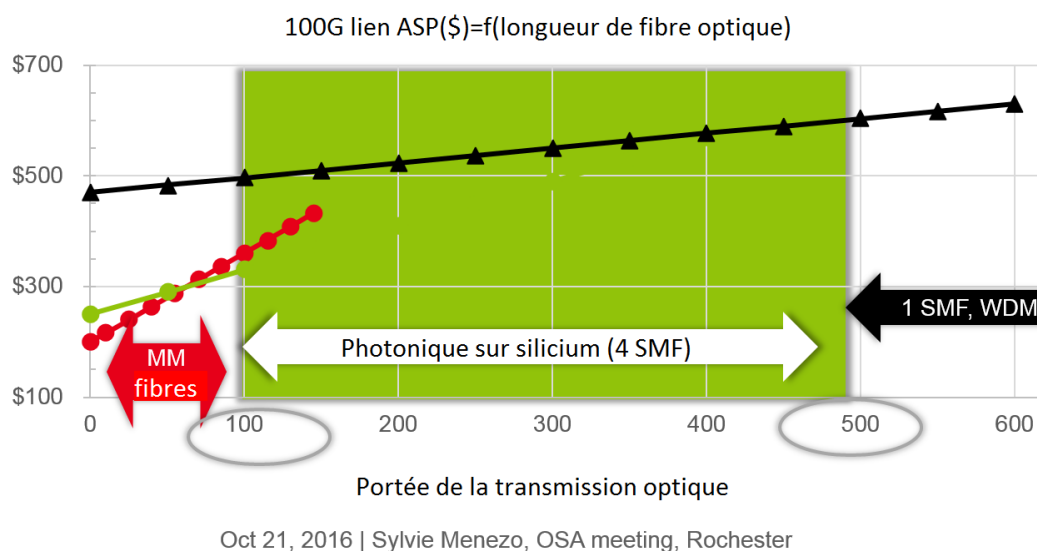


FIGURE 7.24: Normes de 100 Gbit/s en fonction de la portée de transmission selon [7]

Les émetteurs basés sur la technologie à laser à émission de surface à cavité verticale (VCSEL) représentent l'une des solutions satisfaisant la demande de débit à un faible coût [8–10]. Cependant, ils ne peuvent pas être utilisés à une portée de plus de 500 m de longueurs de transmission en raison de leurs natures multimodales. Les acteurs du marché ont donc décidé d'utiliser la technologie VCSEL pour des transmissions entre 0 et 100 m.

Pour les transmissions entre 100 et 500 m, les émetteurs-récepteurs basés sur la technologie photonique du silicium sont apparus comme la solution la moins coûteuse. Ils sont basés sur la norme MSA 100GBASE-PSM4 (figure 7.24). L'ensemble des 100 Gbit/s est réalisé avec 4 fibres monomodes en parallèle à 1,31 μm (25 Gbit/s par fibre). Deux produits sont déjà disponibles sur le marché: le LUX42604 QSFP28 proposé par Luxtera en 2015 et le 100G PSM4 QSFP28 proposé par Intel en 2016 utilisant un laser hybride intégré [11,12]. Le prix du marché pour les émetteurs-récepteurs photoniques sur silicium est évalué à 3 \$ par Gbit/s [7].

Pour les distances de transmission jusqu'à 10 km, les émetteurs-récepteurs optiques à base de phosphore d'indium (InP) restent prédominants. Par exemple, le module optique 100G Coarse WDM-4 QSFP2 de Finisar met en oeuvre quatre lasers à rétroaction répartie modulés chacun à 25 Gbit/s (DFB) [13]. Le prix de marché des émetteurs-récepteurs optiques InP est inférieur à 5 \$/Gbit/s [7]. La technologie photonique sur silicium commence aujourd'hui à devenir compétitive sur de telles distances et pourrait tendre à remplacer la technologie InP.

Pour exemple, en 2016, Intel a proposé un émetteur-récepteur optique Coarse WDM-4 100 Gbit/s basé sur la technologie photonique sur silicium [12].

Dans cette thèse, nous nous concentrons sur la technologie photonique sur silicium. Cette technologie permet l'intégration de tous les composants requis pour un émetteur-récepteur sur une même plate-forme technologique. Elle permet, avec une même technologie, de créer des émetteurs-récepteurs de grande complexité. Elle a de plus l'avantage d'avoir la possibilité de réduire les coûts par une production de masse. Nous avons concentré notre travail sur cette technologie dans le cadre des prochaines normes à un débit de 400 Gbit/s et nous allons viser une portée supérieure à 500 m afin de remplacer la technologie [InP](#).

7.1.2 Communications optiques à 400 Gbit/s

Afin d'atteindre un débit de données de 400 Gbit/s, différentes méthodes sont considérées par le groupe IEEE ou le groupe [MSA](#). Elles évaluent différentes combinaisons entre le débit de données par canaux, le nombre de longueurs d'onde et l'ordre de modulation (fig. 7.25) [14]. On peut noter qu'un format de modulation plus élevé, la modulation d'amplitude d'impulsion de 4 niveaux ([PAM-4](#)), est considéré comme remplaçant de la modulation d'amplitude d'impulsion à 2 niveaux ([PAM-2](#)). Selon le nombre de longueurs d'onde et le format de modulation, la vitesse de transmission sera différente ainsi que la bande passante requise du modulateur optique.

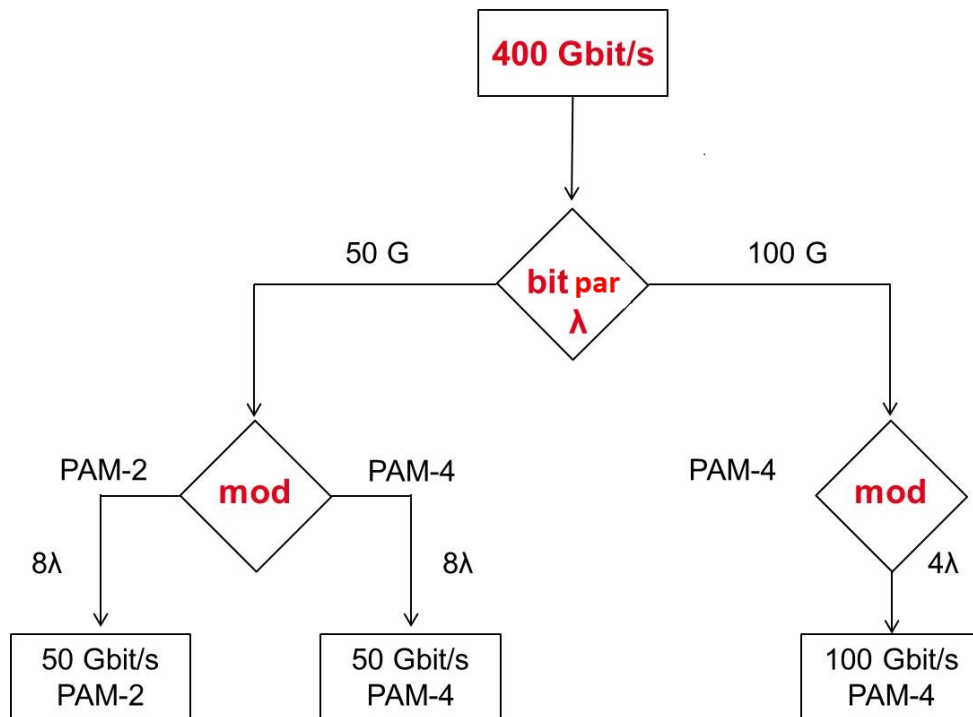


FIGURE 7.25: Architectures possibles pour une norme de 400 Gbit/s [14]. mod signifie modulation.

Le tableau 7.4, basé sur [15–19], résume les différentes spécifications possibles des normes 400 Gbit/s. Les solutions PAM-4 sont choisies pour les normes SR16, DR4, FR8 et LR8. On peut noter que la norme 400GBASE-DR4 équivaut à la norme MSA 100GBASE PSM4. Parce qu’aucun produit commercial n’est disponible, le choix n’est pas permanent et dépend essentiellement de la future évolution du marché. Une autre architecture est proposée par le consortium Open Compute Project [19]. Leur idée est de fonder leur solution sur une architecture de multiplexage en longueurs d’onde adaptative utilisant entre 4 et 32 canaux optiques espacés de 200 GHz. Cette solution ne considère pas le format de modulation PAM-4, mais plutôt un format de modulation PAM-2. Le projet Open Compute considère la bande C (1,55 μm) afin de maintenir une large bande de longueurs d’onde disponible entre 1511 et 1562 nm. Pour cette thèse, nous ne fixerons aucune longueur d’onde ainsi nous travaillerons à 1310 nm et 1550 nm.

TABLE 7.4: Spécifications des futures normes Ethernet à un débit de 400 Gbit/s

Nomenclature	Distance	Débit par canal [Gb/s]	Type de fibre optique	Nombre de fibres optiques	Nombre de longueurs d’onde	Modulation
400GBASE-SR16	100m	25	MMF at 0.85 μm	16	1	PAM-2
400GBASE-DR4	500m	50	SMF at 1.3 μm	4	1	PAM-4
400GBASE-FR8	2km	25	SMF at 1.3 μm	1	8 ($\Delta f=800\text{GHz}$)	PAM-4
Open Optics Solution	2km	25	SMF at 1.5 μm	1	16 ($\Delta f=200\text{GHz}$)	PAM-2
400GBASE-LR8	10km	25	SMF at 1.3 μm	1	8 ($\Delta f=800\text{GHz}$)	PAM-4

Notre objectif est de proposer des architectures d’émetteurs basées sur la technologie photonique sur silicium pour les transmissions optiques au sein des centres de données pouvant atteindre 2 kilomètres. Dans cette thèse, les formats de modulation binaires (PAM-2) et quaternaires (PAM-4) seront utilisés. Nous avons décidé de fixer le nombre de longueurs d’onde multiplexées à 8 avec un débit de données de 50 Gbit/s par longueur d’onde. La norme qui est proche de notre émetteur en terme de caractéristiques ciblées est la norme 400GBASE-FR8. Nous utiliserons cette norme comme base pour notre étude. Dans la section suivante, nous allons rapporter les spécifications de l’émetteur 400GBASE-FR8.

Spécifications de la norme 400GBASE-FR8

Plusieurs caractéristiques sont utilisées pour évaluer les performances d’un émetteur optique. Elles sont les suivantes :

- La puissance totale moyenne de sortie P_{out}

- La modulation optique d'amplitude (**OMA**) en sortie de l'émetteur. Elle est définie comme la différence entre la puissance optique modulée la plus élevée et la puissance optique modulée la plus basse à la sortie de l'émetteur. ($\text{OMA} = P_{out1}[\text{mW}] - P_{out0}[\text{mW}]$).
- Le taux d'extinction (**ER**). Il est défini comme le ratio entre la puissance optique modulée la plus élevée et la puissance optique modulée la plus basse à la sortie de l'émetteur. ($\text{ER} = P_{out1}[\text{mW}] / P_{out0}[\text{mW}]$)

La figure 7.26 représente les différentes puissances optiques à la sortie de l'émetteur pour un format de modulation **PAM-2**. Pour un format de modulation **PAM-4**, les caractéristiques précédentes restent valides. Deux puissances optiques doivent être ajoutées entre la puissance optique la plus élevée (P_{out1}) et la plus faible (P_{out0}) de manière à ce que les quatre niveaux soient créés de façon équidistante les uns des autres.

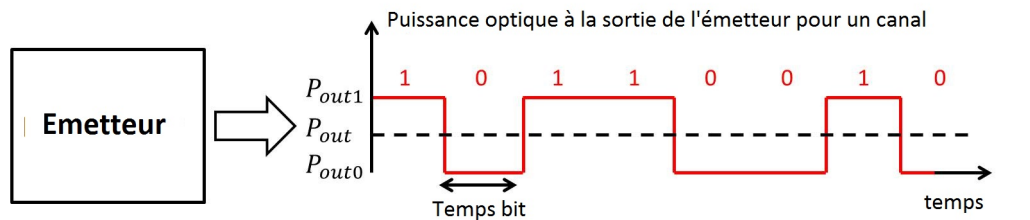


FIGURE 7.26: Puissances optiques en sortie de l'émetteur selon une modulation PAM-2

Pour évaluer la qualité de la liaison optique, le taux d'erreur (**BER**) est utilisé. Le **BER** est le rapport entre le nombre de bits erreurs et le nombre total de bits transmis. Aujourd'hui, le **BER** demandé est de 10^{-12} pour les liens optiques (*i.e.* en moyenne une erreur sur 10^{12} données transmises). Cependant, des pré-traitements et post-traitements tels que la correction d'erreur directe (**FEC**), y compris les correcteurs Reed-Solomon, permettent de relâcher l'exigence **BER**.

Pour le 400 GBASE-FR8, les spécifications sont présentées dans le tableau 7.5. Ces valeurs seront dans le manuscrit les valeurs de performances ciblées.

TABLE 7.5: Spécifications de la norme 400GBASE-FR8

Description	IEEE 400GBASE-FR8
Puissance moyenne totale P_{out} [dBm]	13.2
OMA à la sortie de l'émetteur [dBm]	[0 to 5.5]
Taux d'extinction ER [dB]	≥ 4.5
BER avec un FEC	$2 \cdot 10^{-4}$

Pour notre émetteur, nous avons décidé de ne pas fixer le format de modulation **PAM-2** ou **PAM-4**. Ces deux modulations seront évaluées avec l'émetteur qui sera développé dans ce manuscrit. Dans la section suivante, nous allons comparer brièvement les deux formats de modulation.

Explication des modulations d'amplitude d'impulsion binaire et quaternaire

La modulation d'amplitude à plusieurs niveaux est une modulation d'amplitude pure et appelée modulation d'amplitude d'impulsion selon M niveaux (PAM- M). Elle consiste à coder les bits sur M différents niveaux d'amplitude d'intensité. La figure 7.27 montre une modulation idéale PAM-2 et une modulation idéale PAM-4.

En prenant en compte un bruit gaussien, le BER peut être relié au rapport signal sur bruit (SNR) pour les deux types de modulation selon l'équation 7.3 (équation généralisée pour une modulation PAM- M) [20].

$$\text{BER}_{\text{PAM-M}} = \frac{M-1}{M \cdot \log_2(M)} \text{erfc} \left(\frac{Q}{(M-1)\sqrt{2}} \right) \quad (7.3)$$

dont M est le nombre de niveaux (dans notre cas 2 ou 4) et Q est le facteur de qualité relié au SNR par l'équation 7.4.

$$\text{SNR} = \frac{Q^2(M^2 - 1)}{6(M - 1)^2} \quad (7.4)$$

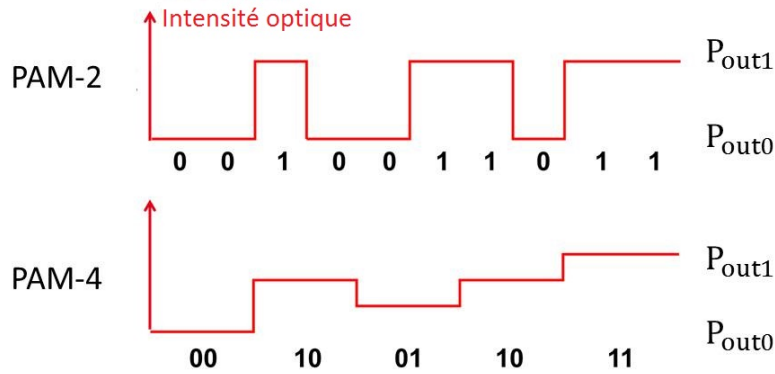


FIGURE 7.27: Modulations idéales PAM-2 et PAM-4

Comme on peut le remarquer, le nombre de symboles par période est multiplié par 2 entre les deux modulations (fig. 7.27). Cela signifie que pour un même débit en baud, la modulation PAM-4 code deux fois plus de données. Mais en contrepartie, les quatre niveaux doivent être espacés de façon équidistante, ce qui demande des générateurs de bits électriques linéaires et des modulateurs électro-optiques linéaires.

Pour la partie réception, le signal de modulation multi-niveaux est mesuré en détection directe avec un photo-détecteur suivi d'un amplificateur de transimpédance (TIA) et d'un circuit de décision de niveaux ($M-1$). Deux seuils de décision supplémentaires sont nécessaires pour démoduler un signal modulé PAM-4 comparé à un signal modulé PAM-2 tel qu'illustré sur la figure 7.28. Ceci entraîne une complexité supplémentaire de la partie réception.

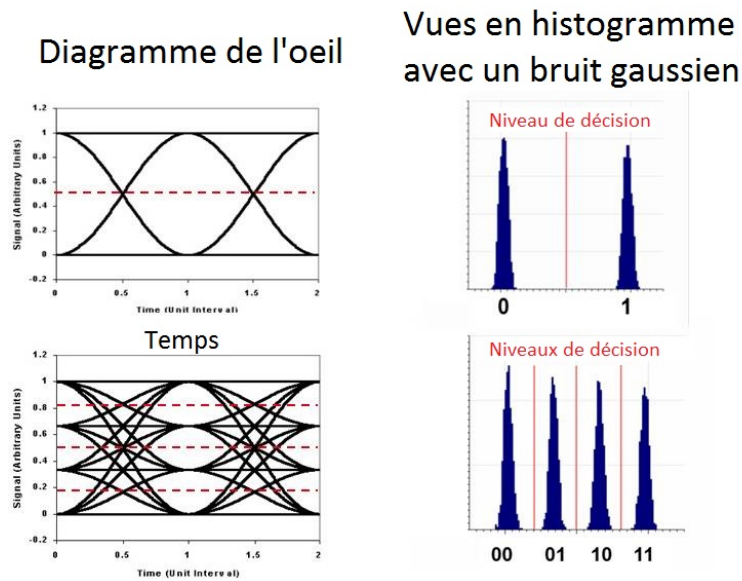


FIGURE 7.28: Diagrammes de l'oeil et vues en histogramme des modulations PAM-4 et PAM-2. Les lignes rouges représentent les seuils de décision

Afin d'atteindre un BER de 10^{-12} , le rapport signal sur bruit du PAM-4 par rapport à la modulation PAM-2 doit être augmenté de 4 dB comme indiqué dans la figure 7.29.

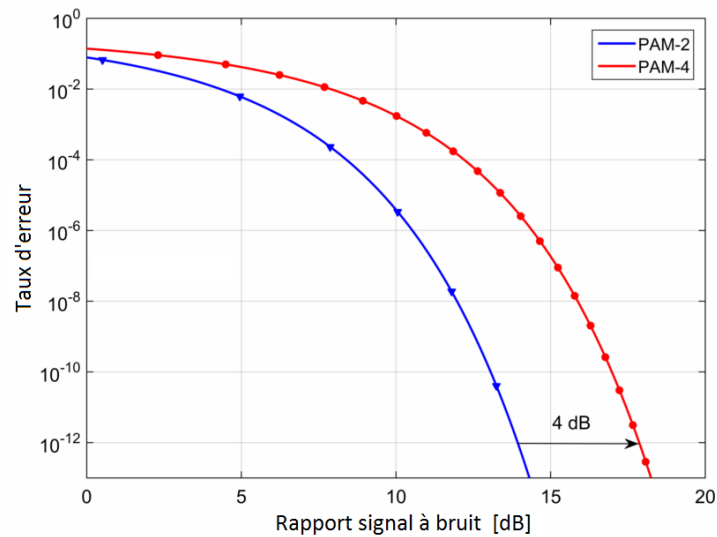


FIGURE 7.29: Taux d'erreur en fonction du rapport au signal sur bruit pour les formats de modulation PAM-2 et PAM-4

7.1.3 Organisation du résumé

Le but de ce travail est d'étudier les architectures d'émetteurs-récepteurs avec une technologie photonique sur silicium pour des transmissions de 400 Gbit/s. La thèse porte essentiellement sur les émetteurs composés de modulateurs en anneau résonant de silicium (RRMs). Il

s'agit d'un modulateur électro-optique très prometteur avec de nombreux avantages: faible taille, faible consommation d'énergie et faible diaphonie dans une architecture WDM. En outre, il a déjà été démontré à 50 Gbit/s avec une modulation PAM-2 [21]. Cependant, il présente un inconvénient majeur: une grande sensibilité à la température et une grande sensibilité aux variations de fabrication. Notre objectif est d'optimiser les performances du modulateur en anneau résonnant et d'évaluer son fonctionnement pour la norme Ethernet 400 GBASE. En particulier, nous ciblons les performances requises par le 400 GBASE-FR8.

Dans ce résumé, toutes les parties de cette thèse ne seront pas développées. Nous allons dans un premier temps focaliser le discours sur le principe de fonctionnement du modulateur en anneau et son état de l'art. Puis, nous proposerons deux architectures différentes utilisant le modulateur en anneau pour implémenter une modulation PAM-4.

7.2 Le modulateur en anneau

7.2.1 Principe de la modulation électro-optique

En technologie silicium, l'indice de réfraction effectif et le coefficient d'absorption d'un guide d'ondes en silicium peuvent être variés en modifiant la concentration des porteurs libres (trous et électrons) à l'intérieur de celui-ci [49]. Cet effet est appelé effet de dispersion plasma. Ces variations sont décrites par l'équation 2.17. Une architecture de jonction semi-conductrice à l'intérieur d'un guide d'ondes de silicium est généralement utilisée pour modifier la concentration des porteurs libres. Cette modification engendre une variation de l'indice de réfraction et du coefficient d'absorption.

$$\begin{aligned}
 \Delta n \text{ at } 1550 \text{ nm} &= -8.8 \cdot 10^{-22} \Delta N - 8.5 \cdot 10^{-18} \Delta P^{0.8} \\
 \Delta \alpha \text{ at } 1550 \text{ nm} &= 8.5 \cdot 10^{-18} \Delta N + 6 \cdot 10^{-18} \Delta P \\
 \Delta n \text{ at } 1310 \text{ nm} &= 6.2 \cdot 10^{-22} \Delta N - 6 \cdot 10^{-18} \Delta P^{0.8} \\
 \Delta \alpha \text{ at } 1310 \text{ nm} &= 6 \cdot 10^{-18} \Delta N + 4 \cdot 10^{-18} \Delta P
 \end{aligned} \tag{7.5}$$

où ΔN et ΔP sont les densités de porteurs d'électrons et de trous [cm^{-3}]. $\Delta \alpha$ est exprimé en [cm^{-1}].

Trois types de jonction sont utilisés dans la littérature: la jonction PN basée sur la déplétion des porteurs, la jonction PIN basée sur l'injection des porteurs et la jonction capacitive basée sur l'accumulation des porteurs [50]. Figure 7.30 représente les trois architectures de jonction dans un guide d'onde. La jonction PN doit être placée en configuration de polarisation inverse: la zone de charge varie en fonction de la tension. Au contraire, la jonction PIN doit être placée dans une configuration de polarisation directe afin d'injecter les électrons et les trous dans le guide d'onde. Pour la jonction capacitive, un oxyde est placé à l'intérieur du guide d'onde et en polarisation directe, les électrons et les trous se concentrent sur les parois d'oxyde. Ces architectures de jonction sont également évaluées dans des configurations entrelacées [51].

L'effet de dispersion du plasma n'est pas utilisé en modulation directe, car la variation du coefficient d'absorption est trop faible. C'est pourquoi, la jonction est placée à l'intérieur d'une structure interférométrique (par exemple l'interféromètre Mach-Zehnder ou le modulateur en

anneau). La variation de l'indice de réfraction conduit à une variation de l'indice effectif Δn_{eff} , qui est transformé en une variation de phase. Cette variation de phase est ensuite convertie en une variation de puissance optique par la structure interférométrique.

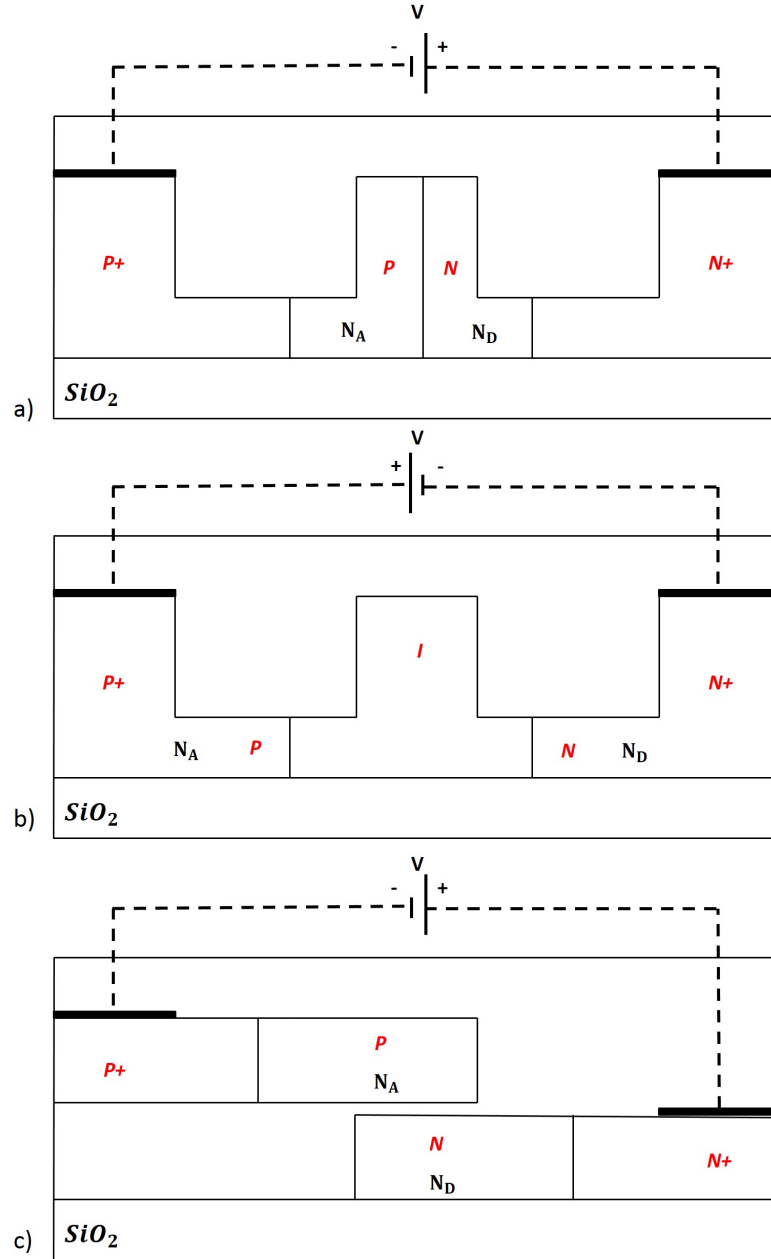


FIGURE 7.30: Architectures de jonction a) PN b) PIN et c) capacitif

7.2.2 Le modulateur en anneau résonant

Le modulateur en anneau résonant est composé d'un port d'entrée et de deux ports de sortie comme décrit dans la figure 7.31. Le champ optique d'entrée E_{in} est partiellement couplé à l'anneau selon un coefficient de couplage κ_1 , transmis dans un port de sortie $E_{t_{out}}$ (Through).

La figure 7.31 représente une vue schématique du modulateur en anneau à un bus et à deux bus. Le signal de sortie modulé est situé en sortie du port Through. Le port drop permet de contrôler le point de fonctionnement du modulateur afin de contrer les variations de température et de fabrication.

Puisqu'une jonction active est placée à l'intérieur de l'anneau, un effet électro-optique peut être produit par une tension de modulation V_m le long de la longueur active L_{active} , et donc les longueurs d'onde de résonances sont décalées. Afin de ne pas modifier le coefficient de couplage entre les bus et l'anneau, la longueur active ne se chevauche pas avec les zones de couplage.

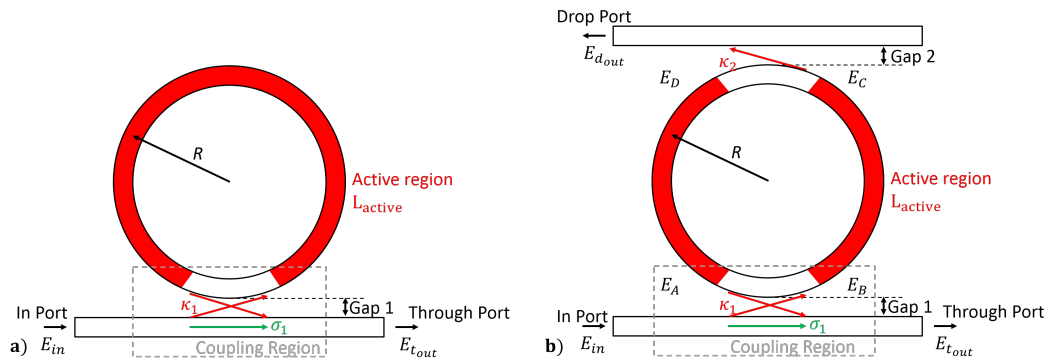


FIGURE 7.31: Schéma d'un modulateur à un bus a) et d'un modulateur à deux bus b)

La figure 7.32 représente la fonction de transfert d'un modulateur en anneau au port through pour deux valeurs de tension différentes. Cette figure met en exergue les caractéristiques statiques du modulateur en anneau : le taux d'extinction (ER) et les pertes d'insertion (IL). Lors de la conception d'un modulateur en anneau, on souhaite augmenter le taux d'extinction et diminuer les pertes d'insertion.

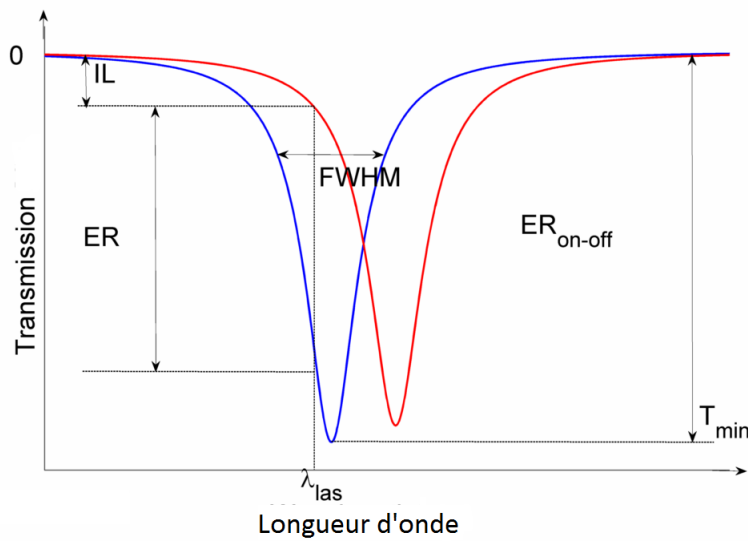


FIGURE 7.32: Fonction de transfert pour deux valeurs de tension différentes V_0 et V_m

Le taux d'extinction et les pertes d'insertion dépendent principalement de trois paramètres différents :

- l'efficacité de la jonction active
- le facteur de qualité directement relié à la largeur spectrale à mi-hauteur (FWHM)
- le positionnement de la longueur d'onde du laser par rapport à la longueur d'onde de résonance.

Bande passante électro-optique

Une dernière caractéristique, la bande passante électro-optique, représente les performances dynamiques du modulateur en anneau. La bande passante du modulateur est principalement limitée par deux contributions: une contribution électrique due à l'architecture de jonction et une contribution optique due à la durée de vie des photons τ_p . La durée de vie du photon est définie comme le temps nécessaire pour que l'intensité (nombre de photons) se décompose d'un facteur $1/e$.

En considérant que le RRM est au couplage critique et que la longueur d'onde du laser est positionnée sur la longueur d'onde de résonance, la bande passante optique est approximée par la largeur de ligne résonnante, donnée par l'équation 7.6 [74].

$$f_{opt} = \frac{1}{2\pi\tau_p} = \frac{c}{Q \cdot \lambda} \quad (7.6)$$

où c est la vitesse de la lumière dans le vide. Par l'équation 7.6, on peut observer qu'un facteur de qualité élevé réduit directement la bande passante optique. La figure 7.33 montre la bande passante optique en fonction du facteur de qualité Q à $1,31 \mu\text{m}$ et à $1,55 \mu\text{m}$.

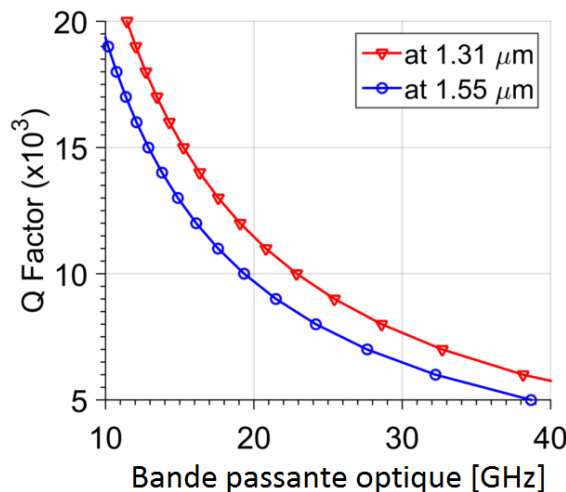


FIGURE 7.33: Facteur de qualité Q en fonction de la bande passante à $1,31 \mu\text{m}$ et $1,55 \mu\text{m}$

La bande passante électro-optique est aussi limitée par une contribution électrique due à l'architecture de la jonction. Un modèle semiconducteur de la jonction PN peut être utilisé pour représenter cette contribution. La limitation électrique est généralement supérieure à 50 GHz et donc non limitante pour notre débit de transmission.

7.2.3 Etat de l'art des modulateurs en anneau résonant

Le tableau 7.6 détaille l'état de l'art des modulateurs en anneau résonant. Ce tableau met en évidence les modulateurs en anneau avec un débit de données atteint supérieur à 40 Gbaud. Une très bonne performance a été obtenue par Pantouvaki et al. dans [21] avec la première démonstration à 56 Gbit/s. Aujourd'hui, des RRM à des débits de données inférieurs (entre 10 et 25 Gbaud) sont implémentés dans une puce électro-optique complète implémentant le multiplexage en longueurs d'onde [40,75].

TABLE 7.6: Résumé de l'état de l'art des modulateurs en anneau résonant (* avec une pré-structuration du signal modulant)

	Type de jonction	Rayon [μm] (FSR)	Q factor	$\Delta\lambda$ ($V_\pi L_\pi$) [V.cm]	IL [dB]	EO - BW [GHz]	Max débit [GBaud]	ER [dB] (V_{pp})
Injection de porteurs								
Baba [76] 2013	PIN	eq.10.7 (8.46nm)	4300	1.15nmV ⁻¹ (0.001)	5.2	1 (0.92V)	50*	4.58 (1.87)
Dépletion de porteurs								
Xiao [77] 2012	Zigzag PN	eq.10.6 (-)	8015	16pmV ⁻¹ (1.7)	-	25 (-)	40	3.01 (3)
Li [78] 2012	PN	5 (-)	4000	28pmV ⁻¹ (-)	-	28 (1V)	40	7 (2)
Malacarne [79] 2014	PN	30 (3.25nm)	7000	10pmV ⁻¹ (3)	2	15 (0V)	30	9.4 (3.75)
Timurdogan [80] 2014	Vertical PN	4.8 (-)	6500	250pmV ⁻¹ (-)	0.92	21 (0V)	44	8 (2.2)
Pantouvaki [21] 2015	PN	5 (19.3nm)	3500	62.9pmV ⁻¹ (0.48)	2.4	38 (0V)	56	4 (2.5)

7.3 Optimisation du modulateur en anneau pour un débit de 400 Gbit/s

Une conception optimisée des performances du modulateur en anneau a été conduite pendant cette thèse. Cette optimisation a été effectuée en ciblant une communication de 400 Gbit/s pour un lien d'une distance de 2 kilomètres. Le modulateur en anneau est intégré dans un émetteur avec en entrée une puissance laser de 10 dBm et des pertes d'insertion évaluées à 5,7 dB incluant le couplage entre les fibres et la puce optique, les pertes de propagation dans la puce et les pertes dues au packaging. L'objectif était d'atteindre 50 Gbit/s par longueur d'onde (par RRM) avec une amplitude de modulation optique supérieure à 0 dBm à la sortie

de l'émetteur. Deux configurations ont été étudiées: [RRM](#) modulé à 50 Gbaud [PAM-2](#) ou [RRM](#) modulé à 25 Gbaud [PAM-4](#).

L'optimisation a été faite méthodiquement:

- Nous avons tout d'abord simulé la jonction PN en fonction des niveaux de dopage implantés de 6.10^{16} à $5.10^{18} \text{ cm}^{-3}$. Les résultats présentent une courbe optimale représentant l'efficacité de modulation de la jonction PN la plus élevée présentant un minimum de perte de guide d'onde PN.
- Le [RRM](#) a ensuite été optimisé en fonction du facteur Q. Les paramètres d'entrée sont l'efficacité de modulation de jonction PN et la perte de guide d'onde PN, les facteurs de couplage et la position de la longueur d'onde laser par rapport à la longueur d'onde de résonance. Les résultats ont conduit à un modèle compact décrivant les meilleurs niveaux de performance du [RRM](#) avec une perte d'insertion toujours inférieure à 3 dB. Un taux d'extinction supérieur à 8 dB est atteint lorsque le facteur Q est supérieur à 6500 (correspondant à une bande passante approximative inférieure à 35 GHz).
- L'émetteur a ensuite été évalué pour les deux configurations de modulation: 50 Gbaud [PAM-2](#) et 25 Gbaud [PAM-4](#). Les deux configurations répondent aux spécifications de la norme 400GBASE-LR4 avec une perte de marge de 15 dB et 12 dB pour le lien modulé à 50 Gbaud [PAM-2](#) et le lien modulé à 25 Gbaud [PAM-4](#) respectivement. Les bandes électro-optiques requises sont assez similaires pour les deux configurations, avec une bande passante électro-optique requise de 30 GHz et 32 GHz (facteur Q égal à 7500 et 7000). Malheureusement, la bande passante requise pour la configuration [PAM-4](#) n'a pas été réduite par rapport au [PAM-2](#), et donc la solution [PAM-4](#) est ainsi moins intéressante. La configuration de 50 Gbaud [PAM-2](#) est le meilleur compromis pour atteindre 50 Gbit/s avec un émetteur constitué de modulateurs en anneau résonant.

Le détail de cette optimisation n'est malheureusement pas reporté dans ce résumé mais vous êtes vivement encouragés à lire le chapitre correspondant. Nous avons choisi dans ce résumé de détailler plus principalement de nouvelles architectures mettant en oeuvre la modulation [PAM-4](#) de manière optique qui ont été validées à travers des mesures.

7.4 Architectures PAM-4 optique

Jusqu'à présent, pour obtenir 50 Gbit/s, nous avons évalué deux options: 50 [PAM-2](#) and 25 Gbaud [PAM-4](#) en utilisant un seul [RRM](#). La génération du signal [PAM-4](#) était effectuée dans le domaine électrique avec un convertisseur numérique-analogique ([DAC](#)).

Dans cette section, nous étudions de nouvelles architectures pour implémenter la modulation [PAM-4](#) en utilisant deux [RRMs](#) modulés avec deux flux de bits parallèles. La fonction [DAC](#) est ainsi réalisée dans le domaine optique. Cela présente des avantages immédiats:

- La largeur de bande électro-optique requise du [RRM](#) peut être réduite parce que chaque [RRM](#) est modulé sur deux niveaux uniquement à 25 Gbaud [PAM-2](#).

- Le DAC électrique n'est plus nécessaire. Par conséquent, on peut envisager une réduction du coût de l'émetteur ainsi que de sa consommation, tout en la rendant plus simple à mettre en œuvre en temps réel.

Nous proposons d'étudier deux architectures d'émetteur. La première architecture est composée de deux RRM_s disposés en série modulés avec deux flux de bits différents. Cette architecture a été précédemment proposée par Seyedi et al. dans [119]. La deuxième architecture est composée de deux RRM_s disposés en parallèle (dans un interféromètre Mach-Zehnder) modulés avec deux flux de bits différents [120].

7.4.1 Architecture de deux modulateurs en anneau disposés en série

Principe de fonctionnement

La génération d'un signal PAM-4 est obtenue en modulant indépendamment deux RRM_s identiques avec deux flux de bits différents. Ces deux RRM_s sont disposés en série, comme le montre la figure 7.34. Les deux RRM_s possèdent la même fonction de transfert, mais leurs longueurs d'onde de résonances sont légèrement décalées conduisant à différents points de fonctionnement *i.e.* différents rapports d'extinction et pertes d'insertion. Par conséquent, la réponse de deux RRM_s disposés en série est le produit de la fonction de transfert de chaque RRM. Cette architecture est intéressante puisqu'elle peut être utilisée comme transmetteur reconfigurable car les deux modulateurs peuvent être décalés en longueur d'onde afin de convertir une modulation PAM-4 sur une longueur d'onde en une modulation PAM-2 sur deux longueurs d'onde.

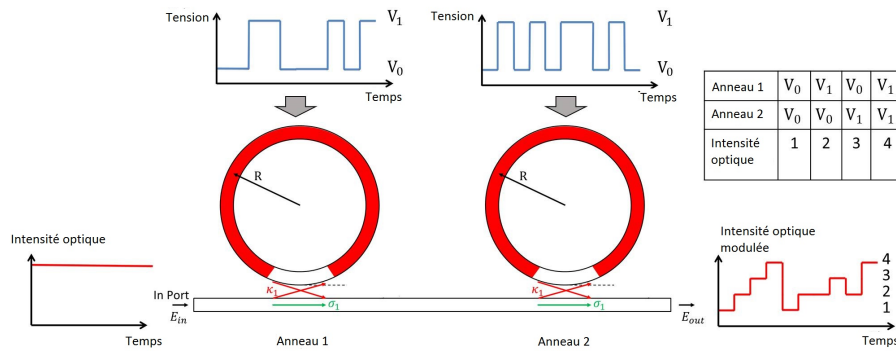


FIGURE 7.34: Schéma de l'architecture PAM-4 avec deux anneaux disposés en série

La distance entre la longueur d'onde du laser et la longueur d'onde de résonance du premier modulateur ($\Delta\lambda_1$) ainsi que la distance entre la longueur d'onde du laser et le deuxième modulateur ($\Delta\lambda_2$) doivent être différentes pour générer quatre niveaux d'intensité en sortie du dispositif, comme le montre la figure 7.35. Cette différence conduit à différentes amplitudes de modulation optique entre les deux RRM_s: OMA₁ pour le premier modulateur et OMA₂ pour le deuxième modulateur.

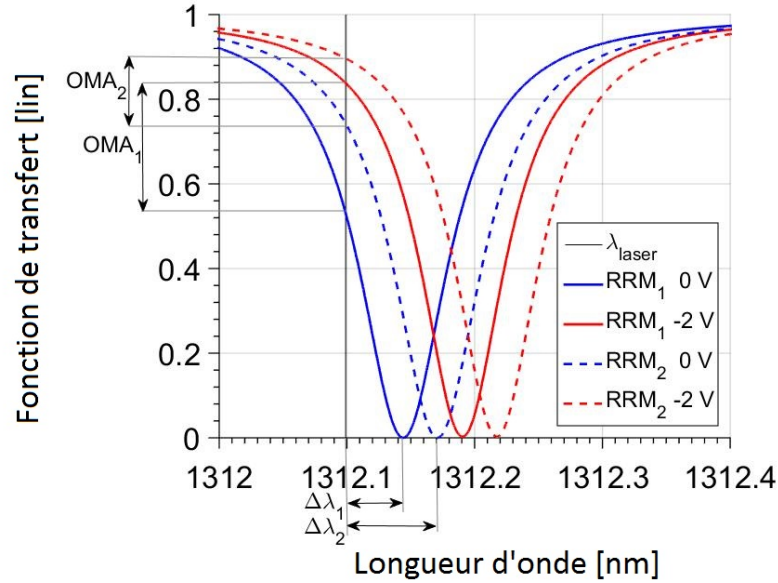
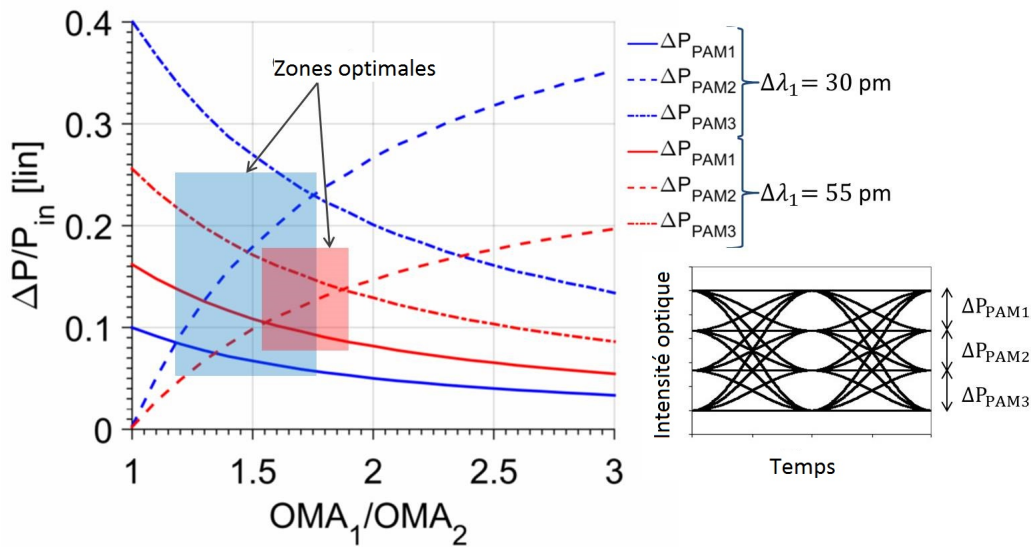


FIGURE 7.35: Représentation de la fonction de transfert des deux modulateurs

Nous simulons maintenant les deux **RRMs** disposés en série. Le facteur Q est choisi à 15500, facteur Q optimal pour une modulation de 25 Gbaud **PAM-2**. Chaque **RRM** est modulé entre 0 V et -2 V. Dans une première étape, nous fixons $\Delta\lambda_1$ égal à 30 pm ou 55 pm conduisant à fixer OMA_1 . $\Delta\lambda_2$ est utilisé comme variable pour modifier le ratio OMA_1/OMA_2 . En fonction du rapport OMA_1/OMA_2 , nous traçons dans la figure 7.36 les différences simulées entre les niveaux successifs du signal de sortie **PAM-4** signal (ΔP_{PAM1} , ΔP_{PAM2} , ΔP_{PAM3}).

FIGURE 7.36: Niveaux optiques simulés en fonction du rapport OMA_1/OMA_2

À travers les simulations, nous observons qu'il est difficile d'obtenir des niveaux également espacés. Les zones optimales sont représentées dans la figure 7.36 où ΔP_{PAM1} , ΔP_{PAM2} , et

ΔP_{PAM3} sont les plus proches. Lorsque la longueur d'onde du laser est décalée vers la première longueur d'onde résonance, les quatre niveaux sont également espacés et le rapport OMA_1/OMA_2 tend à 2. Cependant, si la longueur d'onde du laser est trop loin de la première longueur d'onde résonance, ΔP_{PAM1} , ΔP_{PAM2} , et ΔP_{PAM3} commencent à être très petits, conduisant à fermer le diagramme des yeux. Ainsi, un compromis entre l'amplitude de la modulation (*i.e.* le taux d'extinction) et la distribution des quatre niveaux doivent être trouvés. Il s'agit d'un compromis difficile à trouver, qui dépend du facteur Q et de l'efficacité de la jonction.

En tant que bon compromis, nous choisissons le point de fonctionnement qui permet d'obtenir un taux d'extinction de 4,2 dB entre le niveau le plus élevé et le niveau le plus bas avec une perte d'insertion de -1,5 dB (figure 7.37). $\Delta\lambda_1$ est égal à 40 pm et $\Delta\lambda_2$ est égal à 60 pm conduisant à un ratio OMA_1/OMA_2 égal à 1.9.

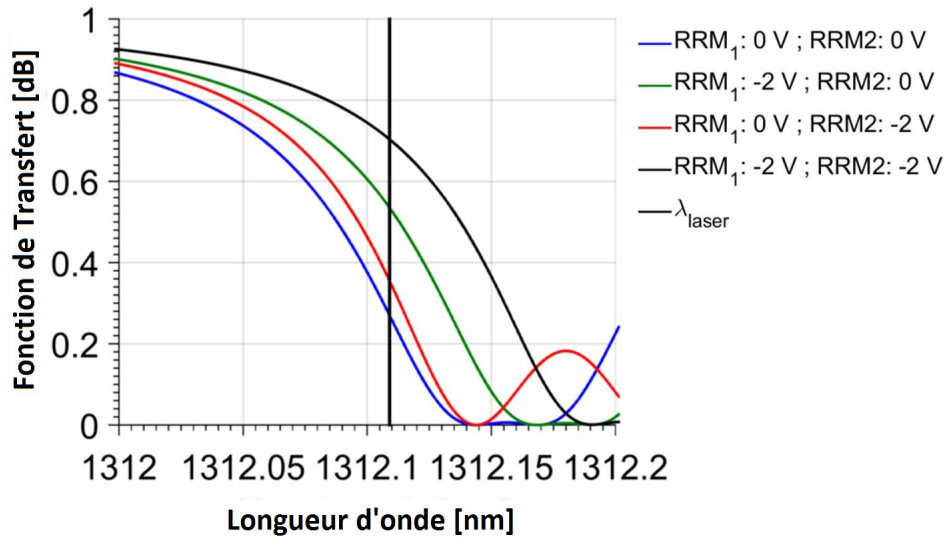


FIGURE 7.37: Simulation de la fonction de transfert des deux modulateurs disposés en série

Demonstrations

Dans cette section, nous proposons de tester la mise en œuvre de l'architecture de l'émetteur décrite précédemment. Les dispositifs utilisés pour cette expérience ont été fabriqués au sein du CEA-Leti 200 mm. Ils comprennent deux **RRMs** identiques de rayon 15 μm couplés au même guide d'onde selon un écart de 450 nm. Les anneaux sont construits selon une structure dite rib waveguide 220/100 nm comprenant une jonction PN. Des chaufferettes résistives en TiTiN sont placées au-dessus des anneaux pour compenser les variations de fabrication et les variations thermiques. La fonction de transfert des **RRMs** ont été mesurés avec une précision de 1 pm pour différentes tensions de polarisation et différentes tensions sur les chaufferettes. Les principales caractéristiques des anneaux ont été extraits et sont reportés dans le tableau 7.7. La fonction de transfert référencée des **RRMs** en fonction de la longueur d'onde est tracée pour plusieurs tensions de polarisation négatives dans la figure 7.38.

TABLE 7.7: Caractéristiques des modulateurs mesurés

	Anneau 1	Anneau 2
$\lambda_{resonance}$ [nm]	1548.631	1549.117
FSR [nm]	6.66	6.66
$\frac{\partial \lambda_{res}}{\partial V_m}$ [pm/V]	14	14
$V_\pi L_\pi$ [V.cm]	1.9	1.9
$\Delta\phi$ entre [0V,-2V] [$^\circ$ /mm]	-	-
facteur Q à 0 V	11600	11400
Efficacité de la chaufferette [mW/FSR]	130	102

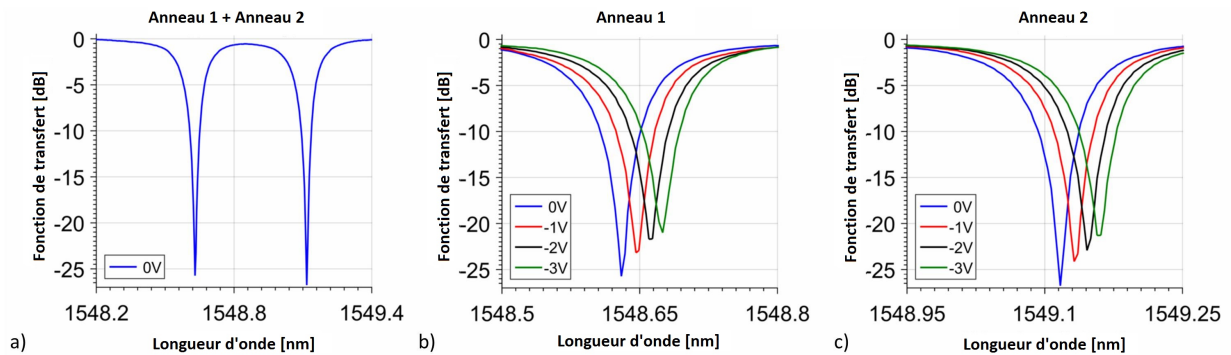


FIGURE 7.38: Fonctions de transfert mesurées et référencées des deux modulateurs en anneau disposés en série à 0V a) et en fonction de la tension de polarisation pour le modulateur en anneau 1 b) et le modulateur en anneau 2 c)

En raison de la variation de fabrication, les deux RMMs ne sont pas totalement alignés en longueur d'onde. En effet, les longueurs d'onde de résonances sont séparées par 486 pm et le facteur Q est différent. Mais, ces variations sont trop faibles pour influencer sur la bande passante électro-optique qui est d'environ 10 GHz. Avec 3,77 V (11 mW) appliqué sur la chaufferette au-dessus du modulateur 1, les deux résonances peuvent être réglées sur la même longueur d'onde.

Expériences à haut débit

Pour obtenir un diagramme de l'oeil PAM-4 symétrique, la modulation de chaque modulateur en anneau résonant doit être contrôlée avec précision. Dans l'approche idéale avec nos caractéristiques, les anneaux doivent être espacés entre 20 à 30 pm l'un de l'autre. Cette espacement dépend de l'efficacité de la modulation électro-optique et du facteur Q. Une efficacité plus élevée signifie un plus grand espacement. Par conséquent, le point de fonctionnement est un compromis entre la perte d'insertion, le taux d'extinction et un diagramme de l'oeil équitablement espacé. Avec nos modulateurs en anneau, un espacement de 24 pm est idéal. En effet, avec une polarisation de -1,5 V et une tension de modulation de $1 V_{pp}$ appliquée aux deux anneaux, le premier modulateur en anneau atteint un taux d'extinction de 2,32 dB et le second atteint un taux d'extinction de 1,1 dB. Avec ce point de fonctionnement, le taux d'extinction

global du PAM-4 entre le niveau le plus élevé et le niveau le plus bas est de 3,4 dB avec une perte d'insertion de 3,9 dB. La figure 7.39 représente la fonction de transfert des deux anneaux. Le diagramme de l'oeil est également simulé et représenté dans la figure 7.39 c), où quatre niveaux sont clairement visibles.

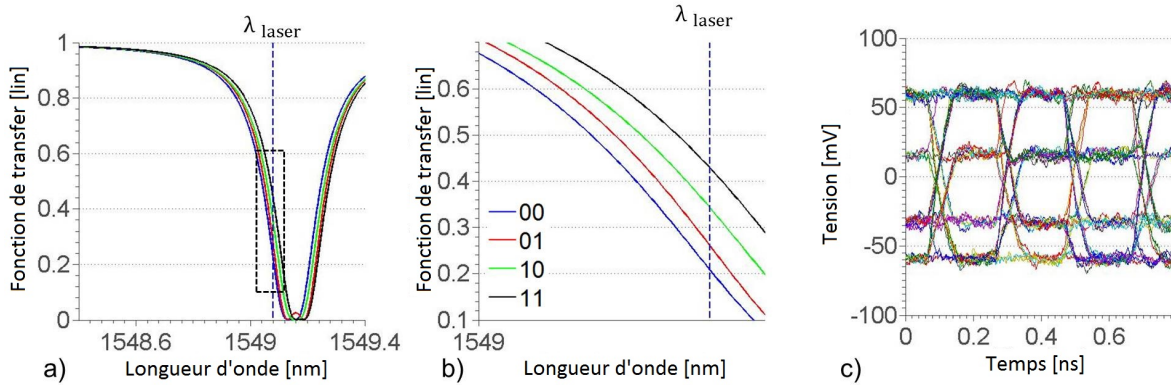


FIGURE 7.39: a) Fonction de transfert simulée des modulateurs, b) détails de la fonction de transfert simulée proche de la position de la longueur d'onde laser, c) diagramme de l'oeil simulé à 10 Gbit/s.

La figure 7.40 montre des images du procédé de mesure. Sur ces images, on peut voir les fibres optiques d'entrée et de sortie, la sonde RF qui apportent les données électriques et trois sondes DC pour régler les points de fonctionnement des modulateurs en anneau résonant à l'aide des chaufferettes. Le montage expérimental pour la transmission PAM-4 est illustrée dans la figure 7.41. Deux canaux d'un testeur de taux d'erreur M8020A (BERT) lancent deux motifs PRBS différents à 15 Gbit/s avec une séquence d'une longueur de $2^9 - 1$ et modulent les modulateurs en anneaux avec $1 V_{pp}$. Un délai entre les deux motifs PRBS permet de les décorréler. La sortie optique des RRM's est reçue par un photorécepteur avec une bande passante de 12 GHz suivi d'un TIA avec un gain de 650 V/W et une puissance équivalente au bruit de 20 pA/ $\sqrt{\text{Hz}}$. Celui-ci est suivi d'un oscilloscope à large bande passante DCA-X 86100D afin de démontrer la liaison de transmission PAM-4 à 30 Gbit/s. Des yeux ouverts sont clairement visibles à 15 Gbit/s, 20 Gbit/s et 30 Gbit/s dans la figure 7.42. Les 4 niveaux de tension sont évalués en moyenne à [-76 mV, -22 mV, 21 mV, 67 mV].

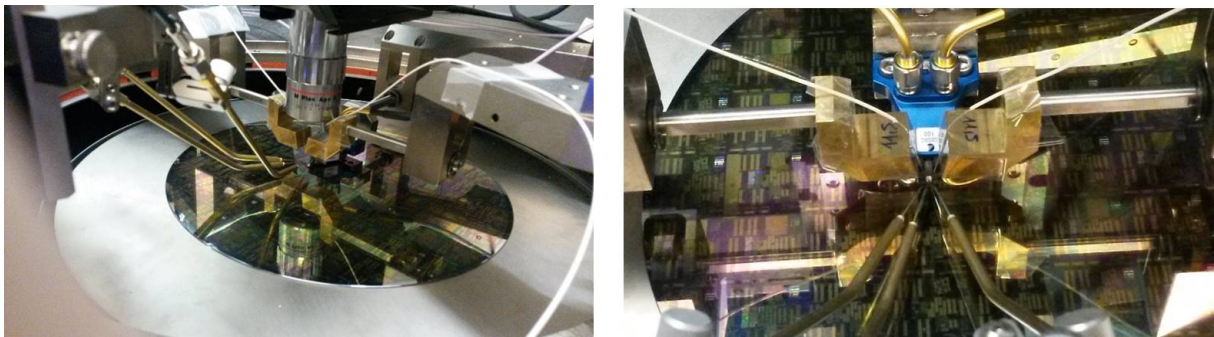


FIGURE 7.40: Images du procédé de mesure

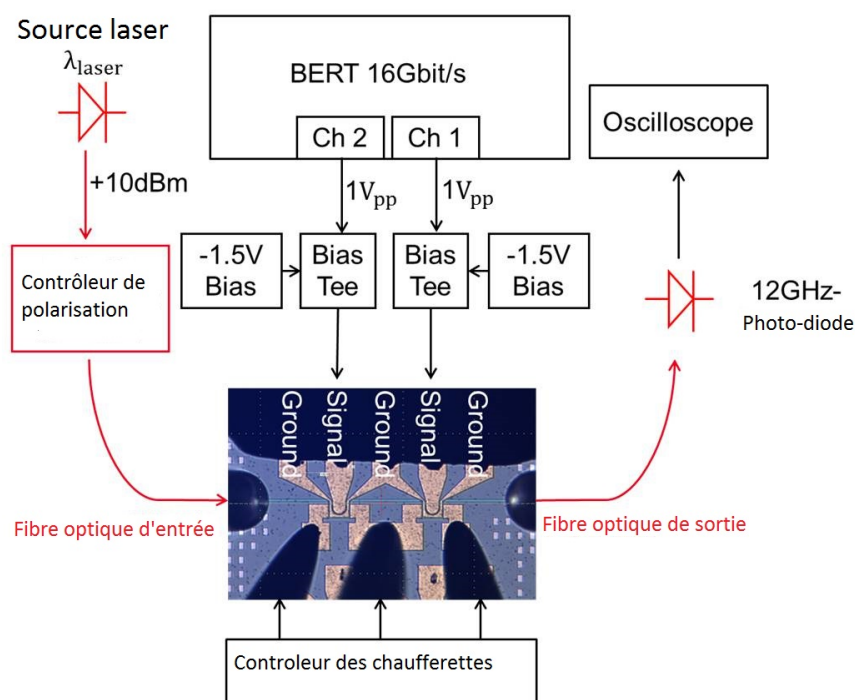


FIGURE 7.41: Configuration expérimentale



FIGURE 7.42: Diagrammes de l'oeil mesurés à 15 Gbit/s a), 20 Gbit/s b) et 30 Gbit/s c)

Pour mesurer le taux d'erreur du lien, l'oscilloscope à large bande passante est remplacé par un oscilloscope en temps réel DSA 72004C avec une bande passante de 20 GHz. En post-traitement, le signal reçu est mémorisé puis comparé au signal BERT envoyé. Le BER est ainsi évalué en comptant le nombre d'erreurs après l'acquisition de l'échantillon. À 30 Gbit/s, un BER de 4×10^{-3} avec une puissance moyenne reçue de 0 dBm est obtenu (figure 7.43). De meilleurs résultats sont attendus avec l'utilisation d'un photo-récepteur et d'un oscilloscope avec une bande passante plus élevée.

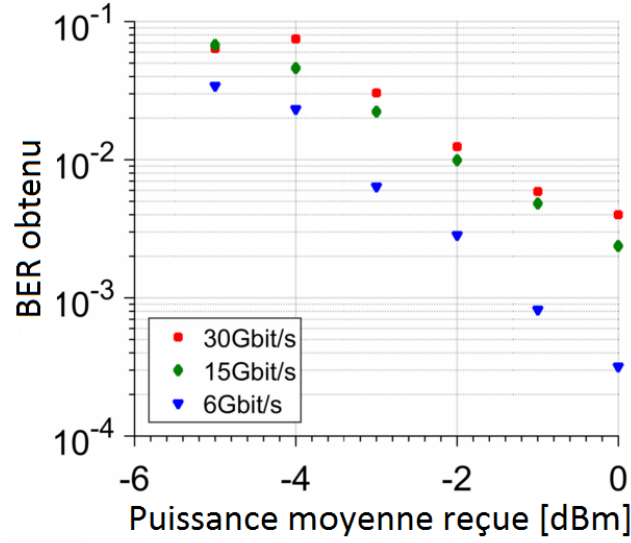


FIGURE 7.43: BER mesuré avec un oscilloscope en temps réel en fonction de la puissance optique moyenne reçue à 6, 15 et 30 Gbit/s

Ce travail démontre un lien de transmission PAM-4 avec deux modulateurs en anneau disposés en série. Cette approche permet de réduire la complexité et la consommation d'énergie de pilotes CMOS. Une transmission de 30 Gbit/s est démontrée avec seulement 1 V_{pp}.

Une deuxième architecture est proposée effectuant la même fonction, basée sur une structure où les deux modulateurs sont disposés en parallèle.

7.4.2 Architecture de deux modulateurs en anneau disposés en parallèle

Principe de fonctionnement

La génération d'un signal PAM-4 est obtenue en modulant deux RRM_s identiques avec deux flux de bits différents. Ces deux RRM_s sont placés dans un interféromètre Mach-Zehnder (MZI). Les deux bras MZI doivent être déphasés d'une valeur de $\pi/2$, comme représenté sur la figure 7.44. Les deux RRM_s doivent être positionnés au même point de fonctionnement *i.e.* avec un taux d'extinction et une perte d'insertion identiques.

Afin d'expliquer le principe de fonctionnement de manière simple, nous pouvons faire l'approximation que les RRM_s modulent seulement l'amplitude du champ optique. Et, nous désignons par les coefficients A et B l'efficacité de modulation correspondant respectivement au modulateur dans le bras haut et au modulateur dans le bras bas. À la sortie du MZI, les champs optiques sont ajoutés conduisant à l'équation 7.7 en champs et en intensité.

$$\begin{cases} E_{out} = (A + B \cdot e^{i\pi/2}) \cdot \sqrt{\frac{I_{in}}{2}} \\ I_{out} = (A^2 + B^2) \cdot \frac{I_{in}}{4} \end{cases} \quad (7.7)$$

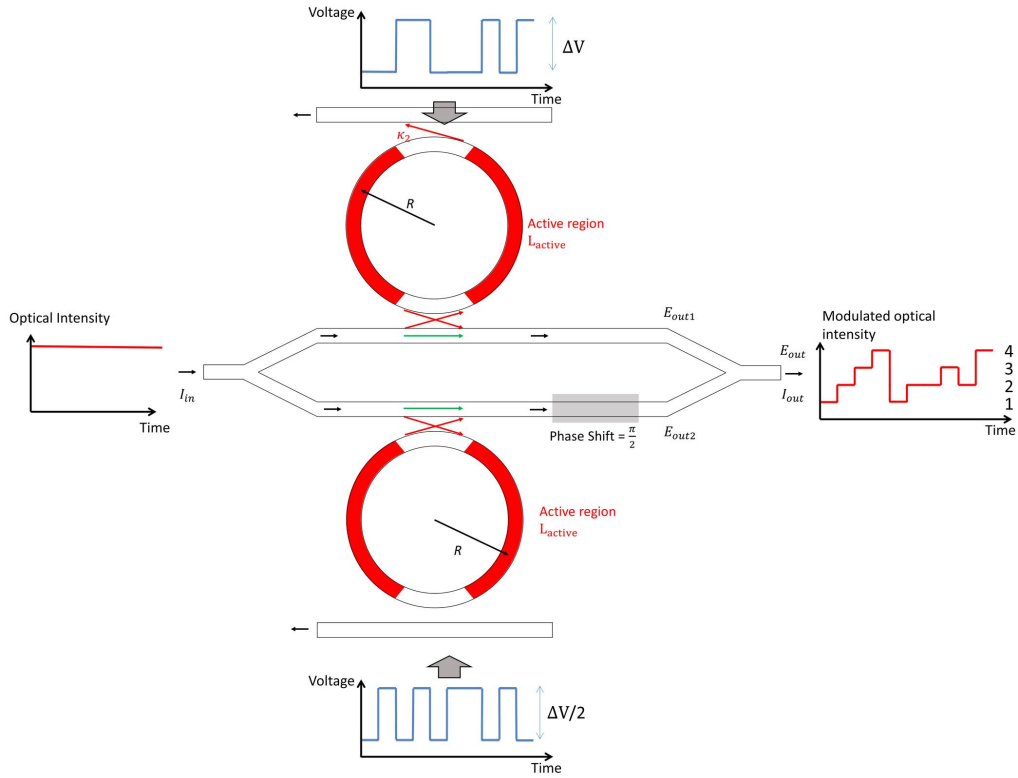


FIGURE 7.44: Schéma de l'architecture PAM-4 avec deux modulateurs en anneaux disposés en parallèle

Les quatre niveaux sont espacés équitablement lorsque les coefficients A et B prennent les valeurs 0 ou 1 pour A et 1 ou $\sqrt{1/2}$ pour B (équation 7.8). Cette condition est respectée lorsque le modulateur du haut est modulé par une tension double de celle de la tension appliquée au modulateur du bas. En raison de l'architecture spécifique en MZI, 3 dB de pertes sont ajoutés au bilan optique global.

$$\begin{cases} I_{11} = I_{in}/2 \\ I_{10} = 0.75 \cdot I_{in}/2 \\ I_{01} = 0.5 \cdot I_{in}/2 \\ I_{00} = 0.25 \cdot I_{in}/2 \end{cases} \quad (7.8)$$

Avec un RRM optimisé possédant un facteur de qualité de 15500, nous simulons l'architecture précédemment expliquée. Le modulateur du haut est modulé entre 0 V et -2 V et le modulateur du bas entre -0,5 V et -1,5 V. Les longueurs d'onde des deux modulateurs à 0 V sont alignées avec la longueur d'onde du laser. Un taux d'extinction de 12,3 dB entre le niveau le plus élevé et le plus bas est atteint avec -5,4 dB de pertes d'insertion.

Malheureusement, il est difficile d'obtenir quatre niveaux parfaitement espacés en raison de la non linéarité de la jonction PN, mais aussi en raison de la modulation de phase inhérente au modulateur en anneau. Mais en compensation pour cette architecture, les deux RRM sont configurés sur le même point de fonctionnement entraînant un réglage facile de l'architecture.

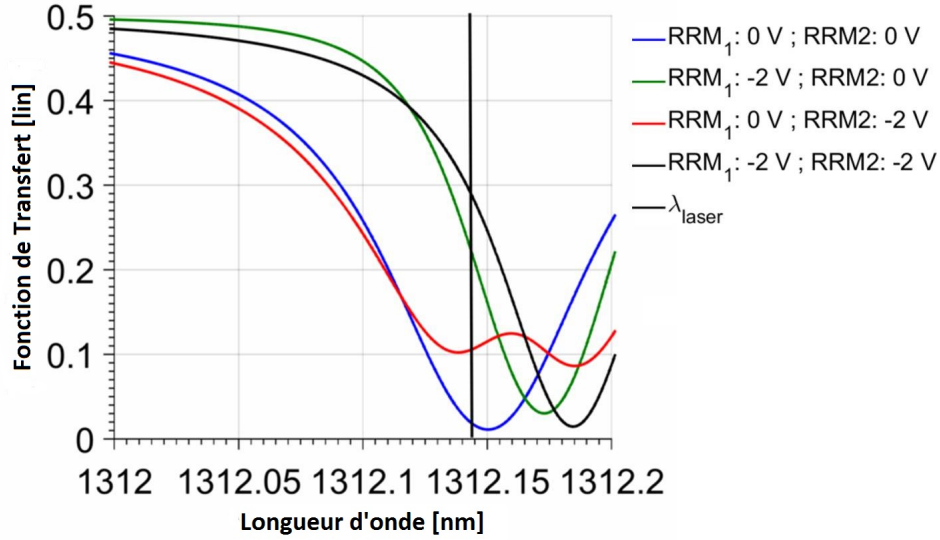


FIGURE 7.45: Simulation de la fonction de transfert de deux modulateurs en anneau mis en parallèle dans un MZI.

Demonstrations

Dans cette section, nous allons montrer des mesures PAM-4 avec l'architecture des deux RRM_s disposés en parallèle. Les composants utilisés pour cette expérience ont été fabriqués au sein des plateformes de fabrication du CEA-Leti et de ST Crolles en 300 nm. Ce dispositif comprend deux RRM_s de 25 μm de rayon couplés avec les différents bras du MZI avec un écart entre l'anneau et le guide d'onde de 300 nm. Les anneaux ont été construits avec un guide d'ondes dit de structure rib de dimension 300/150 nm qui comprend une jonction PN à l'intérieure du guide d'onde de 400 nm de largeur. Les ports de Drop des modulateurs sont reliés à des coupleurs de sortie permettant de mesurer indépendamment chaque RRM. Des chaufferettes résistives à dopage sont placées au centre des anneaux pour compenser les variations de fabrication et de températures, et elles permettent de positionner les RRM_s au bon point de fonctionnement. Les fonctions de transfert des deux modulateurs ont été mesurées avec une précision de 1 pm pour différentes tensions de polarisation. Les principaux paramètres des modulateurs en anneaux ont été extraits et ils sont reportés dans le tableau 7.8. La fonction de transfert de chaque RRM en fonction de la longueur d'onde est tracée pour plusieurs tensions de polarisation dans la figure 7.46. Afin de mesurer indépendamment la fonction de transfert de chaque RRM, le modulateur non mesuré est espacé en longueur d'onde avec sa chaufferette.

TABLE 7.8: Parametres des RRM's en silicium

	Anneau 1	Anneau 2
$\lambda_{resonance}$ [nm]	1315.750	1315.749
FSR [nm]	2.79	2.79
$\frac{\partial \lambda_{res}}{\partial V_m}$ [pm/V]	9.37	9.17
$V_\pi L_\pi$ [V.cm]	2.08	2.12
Q factor à 0 V	11250	10600
Efficacité des chaufferettes [mW/FSR]	320	320
Résistance des chaufferettes[Ω]	588	588

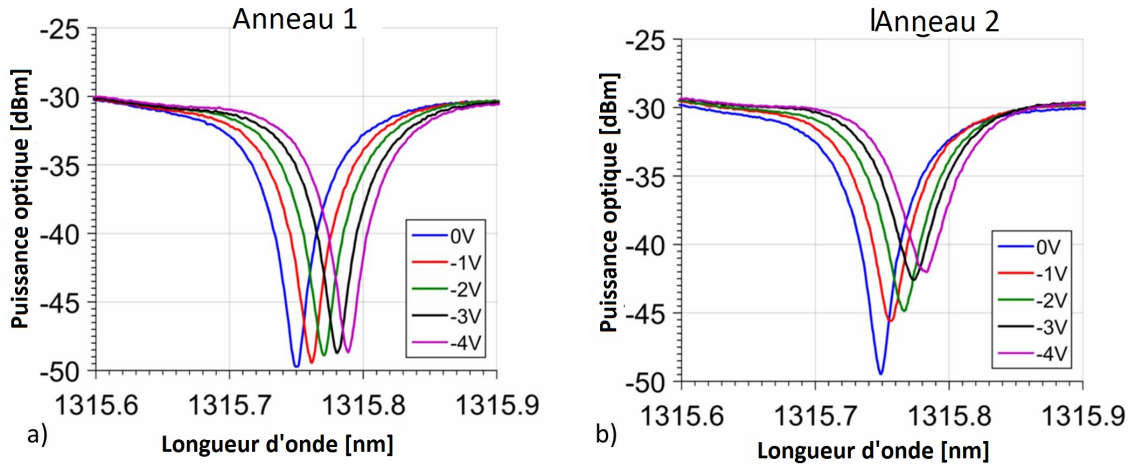


FIGURE 7.46: Fonctions de transfert mesurées du modulateur en anneau dans le bras du haut a) et du modulateur en anneau dans le bras du bas b) en fonction des tensions de polarisation

La bande passante électro-optique a ensuite été mesurée avec un VNA pour différentes positions de longueur d'onde laser. Étant donné que l'étalonnage du test était impossible, ces mesures n'ont pas été effectuées pour les RRM's placés dans l'architecture PAM-4, mais elles ont été faites pour un RRM de même dimension placé de façon autonome. Comme ces différents modulateurs sont de conceptions identiques et placés très proches sur le wafer, nous pouvons supposer que la bande passante mesurée sera celle disponible pour les des modulateurs de l'architecture. La figure 7.47 a) rapporte la mesure du S21 électro-optique pour deux positions de longueur d'onde laser et la figure 7.47 b) rapporte la bande passante électro-optique mesurée en fonction de la position de la longueur d'onde du laser. La bande passante électro-optique mesurée est supérieure à 10 GHz, et elle augmente rapidement lorsque la longueur d'onde du laser est décalée de la longueur d'onde de résonance du modulateur. Avec une bande passante électro-optique de 20 GHz, nous pouvons espérer atteindre une transmission optique à 50 Gbaud PAM-4.

Des chaufferettes résistives supplémentaires ont été placées près de chaque bras du MZI permettant de choisir le déphasage entre les deux bras. Dans la figure 7.48, pour une longueur d'onde laser proche de 1315 nm, la puissance optique à la sortie du MZI a été mesurée en

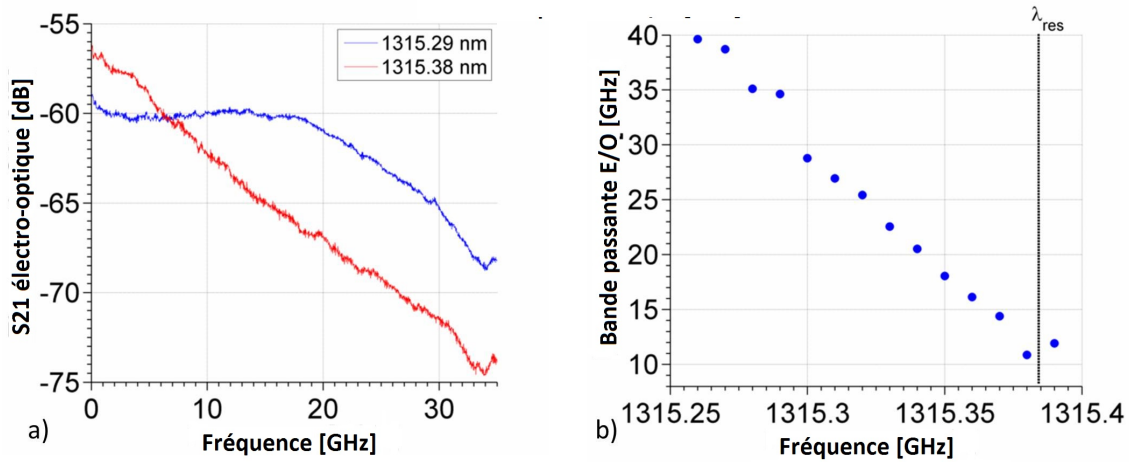


FIGURE 7.47: a) S21 électro-optique mesuré pour un modulateur unique proche de l'architecture PAM-4, b) bande passante électro-optique mesurée en fonction de la longueur d'onde du laser pour un modulateur unique proche de l'architecture PAM-4

fonction de la puissance électrique appliquée sur la première chaufferette placée dans le bras supérieur du MZI en bleu et dans la deuxième chaufferette placée dans le bras inférieur du MZI en rouge. Les champs optiques sont déphasés d'une valeur de $\pi/2$ lorsque la tension appliquée sur la chaufferette inférieure est égale à 4,54 V correspondant à une puissance électrique de 45 mW avec une résistance de 450 Ω . Ce décalage de phase correspond au point de fonctionnement souhaité pour générer la modulation PAM-4.

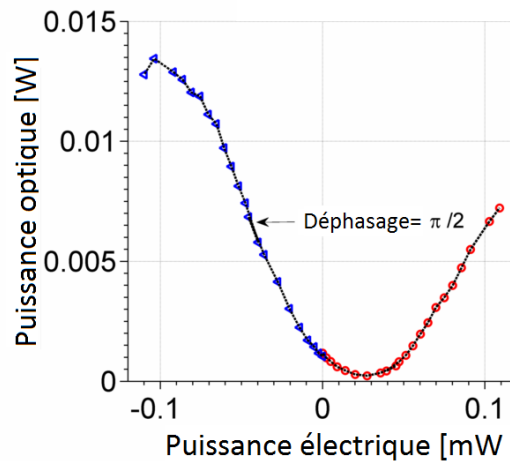


FIGURE 7.48: Puissance optique mesurée à la sortie du MZI en fonction la puissance électrique appliquée sur la chaufferette supérieure (rouge) et en fonction de la chaufferette inférieure (bleu)

Expériences à haut débit

La configuration expérimentale pour démontrer une transmission PAM-4 est illustrée par la figure 7.49. Deux canaux d'un testeur de taux de taux d'erreur M8020A (BERT) lancent deux motifs PRBS différents à 15 Gbit/s avec des séquences d'une longueur de $2^9 - 1$ qui modulent les

modulateurs en anneau résonant avec une tension ΔV_1 and ΔV_2 (maximum $1.2V_{pp}$). La sortie optique du MZI est reçue par un photorécepteur avec une bande passante de 12 GHz suivi d'un TIA possédant un gain de 650 V/W et une puissance équivalente de bruit de 20 pA/ $\sqrt{\text{Hz}}$. Ce dernier est suivi d'un oscilloscope à large bande passante DCA-X 86100D afin de démontrer une transmission PAM-4 à 30 Gbit/s.

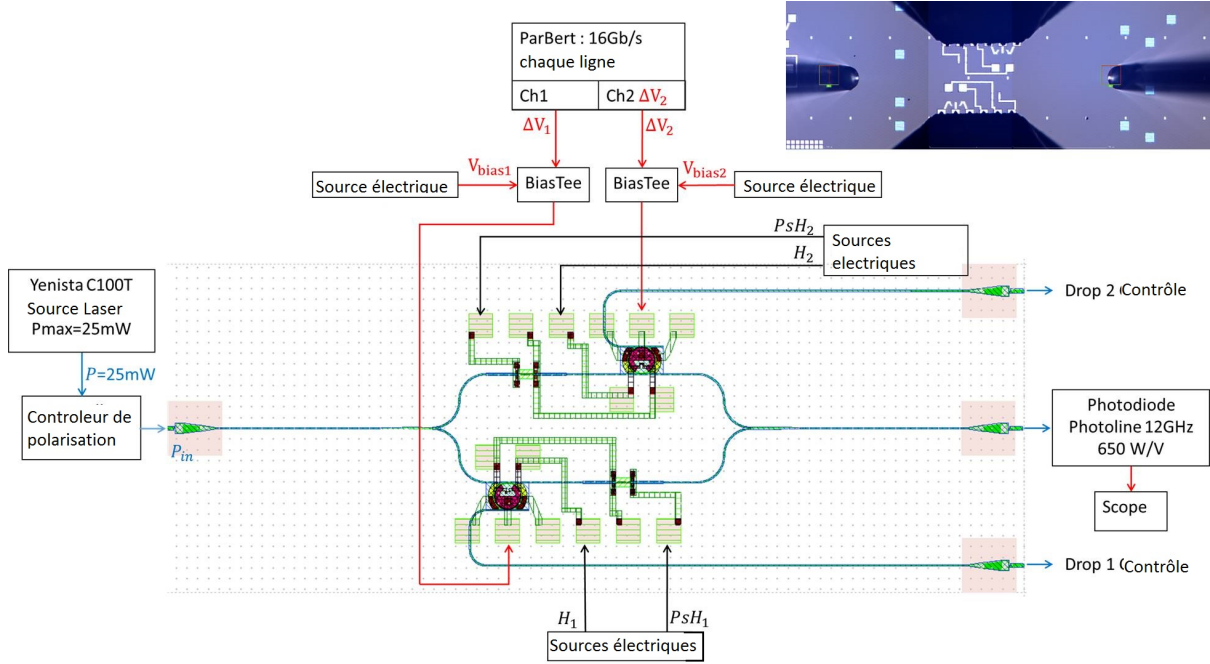


FIGURE 7.49: Configuration expérimentale

TABLE 7.9: Tensions de réglage de l'expérience

ΔV_1 [V _{pp}]	ΔV_2 [V _{pp}]	V_{bias1} [V]	V_{bias2} [V]	H_1 [V]	PsH_1 [V]	H_2 [V]	PsH_2 [V]
1.2	1.125	1.2	1.2	0	4.54	0.6	0

Des sources DCs sont utilisées pour contrôler les positions des longueurs d'onde résonance des RRM_s (H_1 et H_2). Les deux RRM_s sont positionnés au même point de fonctionnement lorsque H_2 est égal à 0,6 V. Pour confirmer ce point de fonctionnement, nous avons mesuré les diagrammes de l'oeil PAM-2 à 15 Gbit/s de chaque RRM avec la même tension de modulation à $1,2 V_{pp}$ et une tension de polarisation à -1,2 V. Les deux diagrammes de l'oeil représentés par la figure 7.50 démontrent que les deux RRM_s se trouvent au même point de fonctionnement.

La transmission PAM-4 est ensuite effectuée en alignant les RRM_s et avec un déphasage de $\pi/2$ entre les deux bras. Table 7.9 rapporte les paramètres de tension pour la démonstration PAM-4. Le diagramme de l'oeil mesuré de la figure 7.51 est clairement ouvert à 30 Gbit/s.

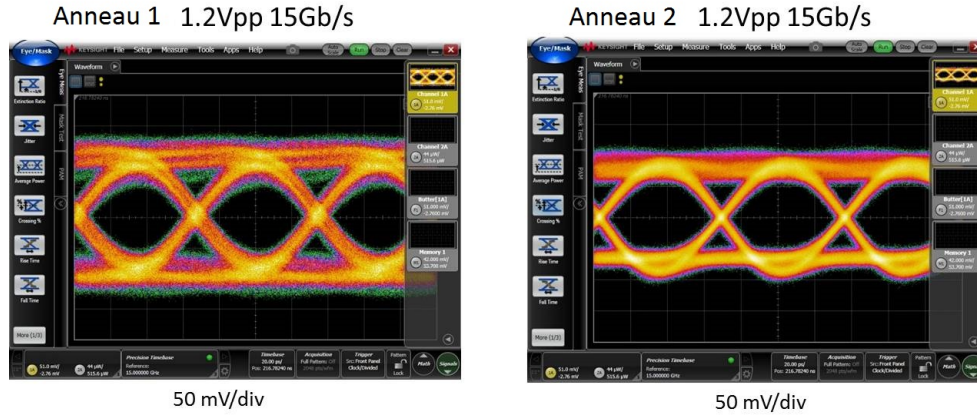


FIGURE 7.50: Diagrammes de l'oeil mesurés des deux modulateurs modulés de façon consecutive à 15 Gbit/s

Pour obtenir des niveaux également espacés, nous avons dû augmenter la tension de modulation du modulateur inférieur à $1.125 V_{pp}$ au lieu de $0.6 V_{pp}$. Cette différence peut-être expliquée par la différence d'efficacité entre les deux modulateurs et la non linéarité de la jonction PN. Malheureusement, les mesures de taux d'erreur n'étaient pas disponibles au moment de l'expérience.

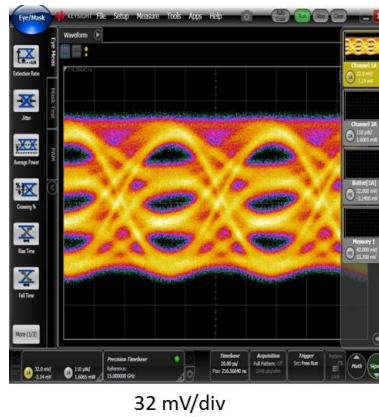


FIGURE 7.51: PAM-4 measured eye diagram at 30 Gbit/s

Cette démonstration montre la capacité de cette architecture, deux **RRMs** placés en parallèle, à réaliser des transmissions **PAM-4** avec une tension de modulation de $1.2 V_{pp}$, compatible avec les pilotes CMOS. Une transmission de 30 Gbit/s a été réalisée sans amplificateur optique. Des meilleurs résultats sont attendus avec le photorécepteur et un oscilloscope avec une bande passante plus importante.

7.5 Conclusion

L'objectif de cette thèse est de proposer et évaluer différentes architectures d'émetteurs en photonique sur silicium afin de répondre à la prochaine norme de débit de 400 Gbit/s sur des

transmissions de 2 kilomètres. Le modulateur électro-optique sélectionné est le modulateur silicium en anneau résonant. Il possède des avantages non négligeables: faibles dimensions, faible consommation énergétique, et il permet un multiplexage dense en longueurs d'ondes. Durant cette thèse, l'optimisation de l'émetteur optique a été faite de manière progressive: de la jonction active à l'émetteur complet. Cette première étude a identifié les différents compromis à faire sur les paramètres du modulateur en anneau afin de définir ses performances. Un modèle compact du modulateur a été créé, permettant d'optimiser le composant avec un temps de simulation très réduit.

Puis, tout en utilisant le modèle compact, deux architectures d'émetteur ont été étudiées basées sur une architecture classique de la littérature. Elles sont basées sur la mise en série de 8 modulateurs en anneau, chacun modulant une longueur d'onde différente. La différence entre les deux architectures vient du format de modulation utilisé: la première est modulée avec des signaux sur deux niveaux électriques au format d'impulsions en amplitude (PAM-2) à 50 Gbaud ; alors que la deuxième est modulée avec des signaux sur 4 niveaux électriques au format d'impulsions en amplitude (PAM-4). Les deux solutions répondent aux demandes de performances de la norme 400 Gbit/s avec les mêmes points de compromis de fonctionnement.

Finalement, de nouvelles architectures d'émetteur optique ont été proposées permettant de réaliser la modulation PAM-4. Contrairement aux solutions précédemment étudiées, ces architectures utilisent en entrée deux flux de bits en parallèle qui sont optiquement combinés pour générer en sortie une modulation PAM-4. Une première solution est basée sur la mise en série de deux modulateurs en anneaux. Cette architecture a été validée avec des caractérisations de transmission optique à 30 Gbit/s avec seulement 1 V_{pp} de tension de commande. Une seconde solution a ensuite été proposée, elle est basée sur l'utilisation de deux modulateurs mis en parallèle dans un interféromètre Mach Zehnder. De la même manière, des caractérisations de transmission à 30 Gbit/s avec 1.2 V_{pp} de tension de commande ont permis de valider le fonctionnement de l'architecture.

Plusieurs perspectives sont possibles à la suite de cette thèse. Nous souhaitons intégrer nos propositions d'architecture avec des pilotes électriques dédiés à notre application. Mais aussi, un management des points de fonctionnement des modulateurs semblent une voie nécessaire pour construire une architecture complète d'émetteur.

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List of Abbreviations

AWG	Arrayed Waveguide Grating
BER	Bit Error Rate
DDPPMZM	Dual Drive Push Pull Mach Zehnder Modulator
DFE	Decision Feedback Equalizer
DML	Direct Modulated Laser
DMT	Discrete MultiTone
DSP	Digital Signal Processing
EIC	Electronic Integrated Circuit
ER	Extinction Ratio
FEC	Forward Error Correction
DFE	Feed-Forward Equalizer
IFFT	Fast Fourier Transform
FWHM	Full Width at Half Maximum
IFFT	Inverse Fast Fourier Transform
LSB	Least Significant Bit
MMF	Multi Mode Fiber
MMI	Multi-mode Interferometer
MSA	Multi Source Agreement
MSB	Most Significant Bit
MZM	Mach Zehnder Modulator
MZI	Mach Zehnder Interferometer
NEP	Noise Equivalent Power
NRZ	Non Return to Zero
OMA	Optical Amplitude Modulation
OFDM	Orthogonal Frequency-Division Multiplexing
PAM	Pulse Amplitude Modulation
PIC	Photonic Integrated Circuit
PSK	Phase Shift Keying
PTAT	Proportional To Absolute Temperature
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
QSFP	Quad Small Form Factor
RRF	Ring Resonator Filter
RRM	Ring Resonator Modulator
SDPPMZM	Single Drive Push Pull Mach Zehnder Modulator
SEM	Scanning Electron Microscopy
SISCAP	Silicon-Insulator Silicon-Capacitor
SMF	Single Mode Fiber
S-MZM	Segmented Mach Zehnder Modulator

TE	Transverse Electric
TDP	Transmitter and Dispersion Penalty
TIA	TransImpedance Amplifier
TM	Transverse Magnetic
TW-MZM	Traveling Wave Mach Zehnder Modulator
TW-DDPPMZM	Traveling Wave Dual Drive Push Pull Mach Zehnder Modulator
TW-SDPPMZM	Traveling Wave Single Drive Push Pull Mach Zehnder Modulator
UBM	Under-Bump Metallization
VCSEL	Vertical Cavity Surface Emitting Laser
WDM	Wavelength Division Multiplexing

List of Publications

Journal

1. A. Abraham, T. Anfray, **O. Dubray**, D. Fowler, S. Olivier, D. Marris-Morini & B. Charbonnier, "Optimization of Silicon MZM Fabrication Parameters for High Speed Short Reach Interconnects at 1310 nm", *Applied Sciences*, vol. 6, no. 12, pp.395 , 2016
2. **O. Dubray**, A. Abraham, K. Hassan, S. Olivier, D. Marris-Morini, L. Vivien, I. O'Connor and S. Menezo, "Electro-Optical Ring Modulator : An Ultracompact Model for the Comparison and Optimization of pn, pin, and Capacitive Junction", *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 22, no. 6, pp. 84-98, 2016.
3. S. Menezo, E. Temporiti, J. Lee, **O. Dubray**, M. Fournier, D. Baldi, B. Blampey, G. Minoia, M. Repossi, S. Messaoud, L. Carroll, S. Abrate, R. Gaudino, S. Member, P. O. Brien, and B. Charbonnier, "Transmitter Made up of a Silicon Photonic IC and its Flip-Chipped IC and its Flip-Chipped CMOS IC Driver Targeting Implementation in FDMA-PON", *Journal of Lightwave Technology*, vol. 34, no. 10, pp. 2391–2397, 2016.
4. **O. Dubray** and S. Menezo, "Simulation and Measurements of Chirp Penalties for Silicon Ring Resonator Modulators", *IEEE Photonics Technology Letters*, vol. 28, no. 3, pp. 280-283, 2016

Conferences

5. **O. Dubray**, A. Seyedi, B. Charbonnier, A. Descos, M. Fiorentino, C. Hui Chen, R.G. Beausoleil, S. Menezo, "30Gbit/s PAM-4 Transmission by modulating a dual Silicon Ring Resonator Modulator", *Optical Interconnects Conference*, 2016
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9. G. Beninca De Farias, S. Menez, **O. Dubray**, D. Marris-Morini, L. Vivien, A. Myko, B. Blampey, "Up to 64-QAM modulation of a Silicon-Ring-Resonator-Modulator", *Optical Fiber Communications Conference and Exhibition*, 2014
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Patents

11. **O. Dubray**, B. Blampey, S. Menez, "Dispositif de génération d'une modulation d'un signal optique", French patent submitted in November 2015
12. S. Menez, **O. Dubray**, "Dispositif de modulation d'amplitude d'un signal optique", French patent submitted in December 2014
13. S. Menez, G. Benica de Farias, and **O. Dubray**, "Device for generating a modulation of an optical signal", - EP 2843468 A3," 2014